
SECTION 8

ELECTROLYTES, ELECTROMOTIVE FORCE, AND CHEMICAL EQUILIBRIUM

8.1 ACTIVITY COEFFICIENTS	8.2
Table 8.1 Individual Activity Coefficients of Ions in Water at 25°C	8.3
Table 8.2 Approximate Effective Ionic Radii in Aqueous Solutions at 25°C	8.4
Table 8.3 Constants of the Debye-Hückel Equation from 0 to 100°C	8.5
Table 8.4 Individual Ionic Activity Coefficients at Higher Ionic Strengths at 25°C	8.5
8.2 EQUILIBRIUM CONSTANTS	8.6
Table 8.5 Ionic Product Constant of Water	8.6
Table 8.6 Solubility Product Constants	8.6
8.2.1 Proton-Transfer Reactions	8.17
Table 8.7 Proton Transfer Reactions of Inorganic Materials in Water at 25°C	8.18
Table 8.8 pK_s Values of Organic Materials in Water at 25°C	8.24
Table 8.9 Selected Equilibrium Constants in Aqueous Solution at Various Temperatures	8.73
Table 8.10 Properties of Common Acid-Base Solvents	8.80
Table 8.11 pK_s Values for Proton-Transfer Reactions in Nonaqueous Solvents	8.81
8.2.2 Formation Constants of Metal Complexes	8.82
Table 8.12 Cumulative Formation Constants for Metal Complexes with Inorganic Ligands	8.83
Table 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands	8.88
8.3 BUFFER SOLUTIONS	8.104
8.3.1 Standard Reference pH Buffer Solutions	8.104
Table 8.14 National Bureau of Standards (U.S.) Reference pH Buffer Solutions	8.105
Table 8.15 Compositions of Standard pH Buffer Solutions [National Bureau of Standards (U.S.)]	8.106
8.3.2 Standards for pH Measurement of Blood and Biological Media	8.106
Table 8.16 Composition and pH Values of Buffer Solutions	8.107
Table 8.17 Standard Reference Values pH_s^* for the Measurement of Acidity in 50 Weight Percent Methanol-Water	8.109
Table 8.18 pH^* Values for Buffer Solutions in Alcohol-Water Solvents at 25°C	8.109
8.3.3 Buffer Solutions Other Than Standards	8.110
Table 8.19 pH Values of Biological and Other Buffers for Control Purposes	8.110
8.4 REFERENCE ELECTRODES	8.113
Table 8.20 Potentials of Reference Electrodes in Volts as a Function of Temperature	8.113
Table 8.21 Potentials of Reference Electrodes (in Volts) at 25°C for Water–Organic Solvent Mixtures	8.114
8.4.1 Electrometric Measurement of pH	8.115
Table 8.22 Values of $2.3026RT/F$ at Several Temperatures	8.115
8.5 INDICATORS	8.116
Table 8.23 Indicators for Aqueous Acid-Base Titrations	8.116
Table 8.24 Mixed Indicators	8.118

Table 8.25	Fluorescent Indicators	8.120
Table 8.26	Selected List of Oxidation-Reduction Indicators	8.122
8.6	ELECTRODE POTENTIALS	8.124
Table 8.27	Potentials of the Elements and Their Compounds at 25°C	8.124
Table 8.28	Potentials of Selected Half-Reactions at 25°C	8.137
Table 8.29	Overpotentials for Common Electrode Reactions at 25°C	8.140
Table 8.30	Half-Wave Potentials of Inorganic Materials	8.141
Table 8.31	Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C	8.146
8.7	CONDUCTANCE	8.157
Table 8.32	Limiting Equivalent Ionic Conductances in Aqueous Solutions	8.157
Table 8.33	Standard Solutions for Calibrating Conductivity Vessels	8.160
Table 8.34	Electrical Conductivity of Various Pure Liquids	8.161
Table 8.35	Equivalent Conductivities of Electrolytes in Aqueous Solutions at 18°C	8.163
Table 8.36	Conductivity of Very Pure Water at Various Temperatures and the Equivalent Conductances of Hydrogen and Hydroxyl Ions	8.168
8.7.1	Common Conductance Relations	8.168

8.1 ACTIVITY COEFFICIENTS

Although it is not possible to measure an individual ionic activity coefficient, f_i , it may be estimated from the following equation of the Debye-Hückel theory:

$$-\log f_i = \frac{Az_i^2\sqrt{I}}{1 + B\hat{a}_i\sqrt{I}}$$

where I is the ionic strength of the medium, and $\hat{a}$ is the ion-size parameter—the effective ionic radius (Table 8.2). The values of A and B vary with the temperature and dielectric constant of the solvent; values from 0 to 100°C for aqueous medium ($\hat{a}$ in angstrom units) are listed in Table 8.3. Corresponding values of A and B for unit weight of solvent (when employing molality) can be obtained by multiplying the corresponding values for unit volume (molarity units) by the square root of the density of water at the appropriate temperature.

The ionic strength can be estimated from the summation of the product molarity times ionic charge squared for all the ionic species present in the solution, i.e., $I = 0.5(c_1z_1^2 + c_2z_2^2 + \cdots + c_iz_i^2)$.

Values for the activity coefficients of ions in water at 25°C are given in Table 8.1 in terms of their effective ionic radii.

At moderate ionic strengths a considerable improvement is effected by subtracting a term bI from the Debye-Hückel expression; b is an adjustable parameter which is 0.2 for water at 25°C. Table 8.4 gives the values of the ionic activity coefficients (for z_i from 1 to 6) with $\hat{a}$ taken to be 4.6Å.

In general, the mean ionic activity coefficient is given by

$$f_{\pm} = {}^{(x+y)}\sqrt{f_+^x f_-^y}$$

TABLE 8.1 Individual Activity Coefficients of Ions in Water at 25°C

Effective Ionic Radii $\bar{a}$ (in Å)	f_i at Ionic Strength of				
	0.001	0.005	0.01	0.05	0.1
Univalent Ions					
9	0.967	0.933	0.914	0.86	0.83
8	0.966	0.931	0.912	0.85	0.82
7	0.965	0.930	0.909	0.845	0.81
6	0.965	0.929	0.907	0.835	0.80
5	0.964	0.928	0.904	0.83	0.79
4	0.964	0.928	0.902	0.82	0.775
3.5	0.964	0.926	0.900	0.81	0.76
3	0.964	0.925	0.899	0.805	0.755
2.5	0.964	0.924	0.898	0.80	0.75
Divalent Ions					
8	0.872	0.755	0.69	0.52	0.45
7	0.872	0.755	0.685	0.50	0.425
6	0.870	0.749	0.675	0.485	0.405
5	0.868	0.744	0.67	0.465	0.38
4.5	0.868	0.741	0.663	0.45	0.36
4	0.867	0.740	0.660	0.445	0.355
Trivalent Ions					
6	0.731	0.52	0.415	0.195	0.13
5	0.728	0.51	0.405	0.18	0.115
4	0.725	0.505	0.395	0.16	0.095
Tetravalent Ions					
11	0.588	0.35	0.255	0.10	0.065
5	0.57	0.31	0.20	0.048	0.021
Pentavalent Ions					
9	0.43	0.18	0.105	0.020	0.009

where f_+ , f_- are the individual ionic activity coefficients, and x, y are the charge numbers (z_+ , z_-) of the respective ions. In binary electrolyte solution.

$$f_{\pm} = \sqrt{f_+ f_-}$$

In ternary electrolytes, e.g., BaCl_2 or K_2SO_4 ,

$$f_{\pm} = \sqrt[3]{f_+ f_-^2} \quad \text{or} \quad f_{\pm} = \sqrt[3]{f_+^2 f_-}$$

In quaternary electrolytes, e.g., LaCl_3 or $\text{K}_3[\text{Fe}(\text{CN})_6]$,

$$f_{\pm} = \sqrt[4]{f_+ f_-^3} \quad \text{or} \quad f_{\pm} = \sqrt[4]{f_+^3 f_-}$$

TABLE 8.2 Approximate Effective Ionic Radii in Aqueous Solutions at 25°C

$\text{\AA}$ (in $\text{\AA}$)	Inorganic Ions	$\text{\AA}$ (in $\text{\AA}$)	Organic Ions
2.5.....	Rb ⁺ , Cs ⁺ , NH ₄ ⁺ , Tl ⁺ , Ag ⁺	3.5.....	HCOO ⁻ , H ₂ Cit ⁻ , CH ₃ NH ₃ ⁺ , (CH ₃) ₂ NH ₂ ⁺
3.....	K ⁺ , Cl ⁻ , Br ⁻ , I ⁻ , CN ⁻ , NO ₂ ⁻ , NO ₃ ⁻	4.....	H ₃ N ⁺ CH ₂ COOH, (CH ₃) ₃ NH ⁺ , C ₂ H ₅ NH ₃ ⁺
3.5.....	OH ⁻ , F ⁻ , SCN ⁻ , OCN ⁻ , HS ⁻ , ClO ₃ ⁻ , ClO ₄ ⁻ , BrO ₃ ⁻ , IO ₄ ⁻ , MnO ₄ ⁻	4.5.....	CH ₃ COO ⁻ , ClCH ₂ COO ⁻ , (CH ₃) ₄ N ⁺ , (C ₂ H ₅) ₂ NH ₂ ⁺ , H ₂ NCH ₂ COO ⁻ , oxalate ²⁻ , HCit ²⁻
4.....	Na ⁺ , CdCl ⁺ , Hg ₂ ²⁺ , ClO ₂ ⁻ , IO ₃ ⁻ , HCO ₃ ⁻ , H ₂ PQ ⁻ , HSQ ⁻ , H ₂ AsO ₄ ⁻ , SO ₄ ²⁻ , S ₂ O ₃ ²⁻ , S ₂ O ₈ ²⁻ , SeO ₄ ²⁻ , CrO ₄ ²⁻ , HPO ₄ ²⁻ , S ₂ O ₆ ²⁻ , PO ₄ ³⁻ , Fe(CN) ₆ ³⁻ , Cr(NH ₃) ₆ ³⁺ , Co(NH ₃) ₆ ³⁺ , Co(NH ₃) ₅ H ₂ O ³⁺	5.....	Cl ₂ CHCOO ⁻ , Cl ₃ COO ⁻ , (C ₂ H ₅) ₃ NH ⁺ , C ₃ H ₇ NH ₃ ⁺ , Cit ³⁻ , succi- nate ²⁻ , malonate ²⁻ , tartrate ²⁻
4.5.....	Pb ²⁺ , CO ₃ ²⁻ , SO ₃ ²⁻ , MoO ₄ ²⁻ , Co(NH ₃) ₅ Cl ²⁺ , Fe(CN) ₅ NO ²⁻	6.....	benzoate ⁻ , hydroxybenzoate ⁻ , chlorobenzoate ⁻ , phenylace- tate ⁻ , vinylacetate ⁻ , (CH ₃) ₂ C=CHCOO ⁻ , (C ₂ H ₅) ₄ N ⁺ , (C ₃ H ₇) ₂ NH ₂ , phthalate ²⁻ , glutarate ²⁻ , adipate ²⁻
5.....	Sr ²⁺ , Ba ²⁺ , Ra ²⁺ , Cd ²⁺ , Hg ²⁺ , S ²⁻ , S ₂ O ₄ ²⁻ , WO ₄ ²⁻ , Fe(CN) ₆ ⁴⁻	7.....	trinitrophenolate ⁻ , (C ₃ H ₇) ₃ NH ⁺ , methoxybenzoate ⁻ , pime- late ²⁻ , suberate ²⁻ , Congo red anion ²⁻
6.....	Li ⁺ , Ca ²⁺ , Cu ²⁺ , Zn ²⁺ , Sn ²⁺ , Mn ²⁺ , Fe ²⁺ , Ni ²⁺ , Co ²⁺ , Co(en) ₃ ³⁺ , Co(S ₂ O ₃)(CN) ₅ ⁴⁻	8.....	(C ₆ H ₅) ₂ CHCOO ⁻ , (C ₃ H ₇) ₄ N ⁺
8.....	Mg ²⁺ , Be ²⁺		
9.....	H ⁺ , Al ³⁺ , Fe ³⁺ , Cr ³⁺ , Sc ³⁺ , Y ³⁺ , La ³⁺ , In ³⁺ , Ce ³⁺ , Pr ³⁺ , Nd ³⁺ , Sm ³⁺ , Co(SO ₃) ₂ (CN) ₅ ⁵⁻		
11.....	Th ⁴⁺ , Zr ⁴⁺ , Ce ⁴⁺ , Sn ⁴⁺		

TABLE 8.3 Constants of the Debye-Hückel Equation from 0 to 100°C

$$-\log f_i = \frac{Az_i^2\sqrt{I}}{1 + B\hat{a}\sqrt{I}}$$

Temp., °C	Unit Volume of Solvent		Temp., °C	Unit Volume of Solvent	
	A	B		A	B
0	0.4918	0.3248	55	0.5432	0.3358
5	0.4952	0.3256	60	0.5494	0.3371
10	0.4989	0.3264	65	0.5558	0.3384
15	0.5028	0.3273	70	0.5625	0.3397
20	0.5070	0.3282	75	0.5695	0.3411
25	0.5115	0.3291	80	0.5767	0.3426
30	0.5161	0.3301	85	0.5842	0.3440
35	0.5211	0.3312	90	0.5920	0.3456
40	0.5262	0.3323	95	0.6001	0.3471
45	0.5317	0.3334	100	0.6086	0.3488
50	0.5373	0.3346			

The values for unit weight of solvent (molality scale) can be obtained by multiplying the corresponding values for unit volume by the square root of the density of water at the appropriate temperature.

TABLE 8.4 Individual Ionic Activity Coefficients at Higher Ionic Strengths at 25°C

The values were calculated from the modified Debye-Hückel equation utilizing the modifications proposed by Robinson and by Guggenheim and Bates:

$$-\frac{\log f_i}{z_i^2} = \frac{0.511I}{1 + 1.5I} - 0.2I$$

where I is the ionic strength and $\hat{a}$ is assumed to be 4.6 Å.

I	$-\frac{\log_{10} f_i}{z_i^2}$	f_i for $z_i =$					
		1	2	3	4	5	6
0.05	0.0756	0.840	0.498	0.209	0.0617	0.0129	0.00190
0.1	0.0896	0.814	0.438	0.156	0.0369	0.00576	0.000595
0.2	0.0968	0.800	0.410	0.138	0.0283	0.00380	0.000328
0.3	0.0936	0.806	0.422	0.144	0.0318	0.00457	0.000427
0.4	0.0858	0.821	0.454	0.169	0.0424	0.00716	0.000815
0.5	0.0753	0.841	0.500	0.210	0.0624	0.0131	0.00195
0.6	0.0631	0.865	0.559	0.270 ₅	0.0978	0.0265	0.00535
0.7	0.0496	0.892	0.633	0.358	0.161	0.0575 ₅	0.0164
0.8	0.0352	0.922	0.723	0.482	0.273	0.132	0.0541
0.9	0.0201	0.955	0.831	0.659	0.477	0.314	0.189
1.0	0.0044	0.900	0.960	0.913	0.850	0.776	0.694

8.2 EQUILIBRIUM CONSTANTS

TABLE 8.5 Ionic Product Constant of Water

This table gives values of pKw on a molal scale, where Kw is the ionic activity product constant of water. Values are from W. L. Marshall and E. U. Franck, *J. Phys. Chem. Ref. Data*, **10**:295 (1981).

Temp., °C	pKw	Temp., °C	pKw	Temp., °C	pKw
0	14.938	45	13.405	95	12.345
5	14.727	50	13.275	100	12.264
10	14.528	55	13.152	125	11.911
15	14.340	60	13.034	150	11.637
18	14.233	65	12.921	175	11.431
20	14.163	70	12.814	200	11.288
25	13.995	75	12.711	225	11.207
30	13.836	80	12.613	250	11.192
35	13.685	85	12.520	275	11.251
40	13.542	90	12.431	300	11.406

TABLE 8.6 Solubility Product Constants

The data refer to various temperatures between 18 and 25°C, and were compiled from values cited by Bjerrum, Schwarzenbach, and Sillen, *Stability Constants of Metal Complexes*, part II, Chemical Society, London, 1958, and values taken from publications of the IUPAC Solubility Data Project: *Solubility Data Series*, International Union of Pure and Applied Chemistry, Pergamon Press, Oxford, 1979–1992; H. L. Clever, and F. J. Johnston, *J. Phys. Chem. Ref. Data*, **9**:751 (1980); Y. Marcus, *Ibid.* **9**:1307 (1980); H. L. Clever, S. A. Johnson, and M. E. Derrick, *Ibid.* **14**:631 (1985), and **21**:941 (1992).

In the table, “L” is the abbreviation of the organic ligand.

Compound	Formula	pK _{sp}	K _{sp}
Actinium hydroxide	Ac(OH) ₃	15	1 × 10 ^{−15}
Aluminum arsonate	AlAsO ₄	15.80	1.6 × 10 ^{−16}
cupferrate	AlL ₃	18.64	2.3 × 10 ^{−19}
hydroxide	Al(OH) ₃	32.89	1.3 × 10 ^{−33}
phosphate	AlPO ₄	20.01	9.84 × 10 ^{−21}
8-quinolinolate	AlL ₃	29.00	1.00 × 10 ^{−29}
selenide	Al ₂ Se ₃	24.4	4 × 10 ^{−25}
sulfide	Al ₂ S ₃	6.7	2 × 10 ^{−7}
Americium (III) hydroxide	Am(OH) ₃	19.57	2.7 × 10 ^{−20}
(IV) hydroxide	Am(OH) ₄	56	1 × 10 ^{−56}
Ammonium uranyl arsenate	NH ₄ UO ₂ AsO ₄	23.77	1.7 × 10 ^{−24}
Arsenic (III) sulfide	As ₂ S ₃	21.68	2.1 × 10 ^{−22}

TABLE 8.6 Solubility Product Constants (*Continued*)

Compound	Formula	pK_{sp}	K_{sp}
Barium			
arsenate	$Ba_3(AsO_4)_2$	50.11	8.0×10^{-51}
bromate	$Ba(BrO_3)_2$	5.50	2.43×10^{-4}
carbonate	$BaCO_3$	8.59	2.58×10^{-9}
chromate	$BaCrO_4$	9.93	1.17×10^{-10}
ferricyanide 6-hydrate	$Ba_2[Fe(CN)_6] \cdot 6H_2O$	7.49	3.2×10^{-8}
fluoride	BaF_2	6.74	1.84×10^{-7}
hexafluorosilicate	$BaSiF_6$	6	1×10^{-6}
hydrogen phosphate	$BaHPO_4$	6.49	3.2×10^{-7}
hydroxide 8-hydrate	$Ba(OH)_2 \cdot 8H_2O$	3.59	2.55×10^{-4}
iodate hydrate	$Ba(IO_3)_2 \cdot H_2O$	8.40	4.01×10^{-9}
molybdate	$BaMoO_4$	7.45	3.54×10^{-8}
niobate	$Ba(NbO_3)_2$	16.50	3.2×10^{-17}
nitrate	$Ba(NO_3)_2$	2.33	4.64×10^{-3}
oxalate	BaC_2O_4	6.79	1.6×10^{-7}
oxalate hydrate	$BaC_2O_4 \cdot H_2O$	7.64	2.3×10^{-8}
permanganate	$Ba(MnO_4)_2$	9.61	2.5×10^{-10}
perrhenate	$Ba(ReO_4)_2$	1.28	5.2×10^{-2}
phosphate	$Ba_3(PO_4)_2$	22.47	3.4×10^{-23}
pyrophosphate	$Ba_2P_2O_7$	10.50	3.2×10^{-11}
8-quinolinolate	BaL_2	8.30	5.0×10^{-9}
selenate	$BaSeO_4$	7.47	3.40×10^{-8}
sulfate	$BaSO_4$	9.97	1.08×10^{-10}
sulfite	$BaSO_3$	9.30	5.0×10^{-10}
thiosulfate	BaS_2O_3	4.79	1.6×10^{-5}
Beryllium			
carbonate 4-hydrate	$BeCO_3 \cdot 4H_2O$	3	1×10^{-3}
hydroxide (amorphous)	$Be(OH)_2$	21.16	6.92×10^{-22}
molybdate	$BeMoO_4$	1.49	3.2×10^{-2}
niobate	$Be(NbO_3)_2$	15.92	1.2×10^{-16}
Bismuth			
arsenate	$BiAsO_4$	9.35	4.43×10^{-10}
cupferrate	BiL_3	27.22	6.0×10^{-28}
hydroxide	$Bi(OH)_3$	30.4	6.0×10^{-31}
iodide	BiI_3	18.11	7.71×10^{-19}
oxide bromide	$BiOBr$	6.52	3.0×10^{-7}
oxide chloride	$BiOCl$	30.75	1.8×10^{-31}
oxide hydroxide	$BiO(OH)$	9.4	4×10^{-10}
oxide nitrate	$BiO(NO_3)$	2.55	2.82×10^{-3}
oxide nitrite	$BiO(NO_2)$	6.31	4.9×10^{-7}
oxide thiocyanate	$BiO(SCN)$	6.80	1.6×10^{-7}
phosphate	$BiPO_4$	22.89	1.3×10^{-23}
sulfide	Bi_2S_3	97	1×10^{-97}
Cadmium			
anthranilate	CdL_2	8.27	5.4×10^{-9}
arsenate	$Cd_3(AsO_4)_2$	32.66	2.2×10^{-33}
benzoate 2-hydrate	$CdL_2 \cdot 2H_2O$	2.7	2×10^{-3}
borate, <i>meta</i>	$Cd(BO_2)_2$	8.64	2.3×10^{-9}
carbonate	$CdCO_3$	12.0	1.0×10^{-12}
cyanide	$Cd(CN)_2$	8.0	1.0×10^{-8}
ferrocyanide	$Cd_2[Fe(CN)_6]$	16.49	3.2×10^{-17}
fluoride	CdF_2	2.19	6.44×10^{-3}

TABLE 8.6 Solubility Product Constants (Continued)

Compound	Formula	pK _{sp}	K _{sp}
hydroxide	Cd(OH) ₂ fresh	14.14	7.2 × 10 ⁻¹⁵
iodate	Cd(IO ₃) ₂	7.60	2.5 × 10 ⁻⁸
oxalate 3-water	CdC ₂ O ₄ · 3H ₂ O	7.85	1.42 × 10 ⁻⁸
phosphate	Cd ₃ (PO ₄) ₂	32.60	2.53 × 10 ⁻³³
quinaldate	CdL ₂	12.30	5.0 × 10 ⁻¹³
sulfide	CdS	26.10	8.0 × 10 ⁻²⁷
tungstate	CdWO ₄	5.7	2 × 10 ⁻⁶
Calcium			
acetate 3-water	Ca(OAc) ₂ · 3H ₂ O	2.4	4 × 10 ⁻³
arsenate	Ca ₃ (AsO ₄) ₂	18.17	6.8 × 10 ⁻¹⁹
benzoate 3-water	CaL ₂ · 3H ₂ O	2.4	4 × 10 ⁻³
carbonate	CaCO ₃	8.54	2.8 × 10 ⁻⁹
carbonate (calcite)	CaCO ₃	8.47	3.36 × 10 ⁻⁹
carbonate (aragonite)	CaCO ₃	8.22	6.0 × 10 ⁻⁹
carbonatomagnesium	Ca[Mg(CO ₃) ₂] dolomite	11	1 × 10 ⁻¹¹
chromate	CaCrO ₄	3.15	7.1 × 10 ⁻⁴
fluoride	CaF ₂	8.28	5.3 × 10 ⁻⁹
hexafluorosilicate	Ca[SiF ₆]	3.09	8.1 × 10 ⁻⁴
hydrogen phosphate	CaHPO ₄	7.0	1.0 × 10 ⁻⁷
hydroxide	Ca(OH) ₂	5.26	5.5 × 10 ⁻⁶
iodate 6-water	Ca(IO ₃) ₂ · 6H ₂ O	6.15	7.10 × 10 ⁻⁷
molybdate	CaMoO ₄	7.84	1.46 × 10 ⁻⁸
niobate	Ca(NbO ₃) ₂	17.06	8.7 × 10 ⁻¹⁸
oxalate hydrate	CaC ₂ O ₄ · H ₂ O	8.63	2.32 × 10 ⁻⁹
phosphate	Ca ₃ (PO ₄) ₂	28.68	2.07 × 10 ⁻²⁹
8-quinolinolate	CaL ₂	11.12	7.6 × 10 ⁻¹²
selenate	CaSeO ₄	3.09	8.1 × 10 ⁻⁴
selenite	CaSeO ₃	5.53	8.0 × 10 ⁻⁶
silicate, meta	CaSiO ₃	7.60	2.5 × 10 ⁻⁸
sulfate	CaSO ₄	4.31	4.93 × 10 ⁻⁵
sulfate dihydrate	CaSO ₄ · 2H ₂ O	4.50	3.14 × 10 ⁻⁵
sulfite	CaSO ₃	7.17	6.8 × 10 ⁻⁸
sulfite 0.5-water	CaSO ₃ · 0.5H ₂ O	6.51	3.1 × 10 ⁻⁷
tartrate dihydrate	CaL · 2H ₂ O	6.11	7.7 × 10 ⁻⁷
tungstate	CaWO ₄	8.06	8.7 × 10 ⁻⁹
Cerium			
(III) fluoride	CeF ₃	15.1	8 × 10 ⁻¹⁶
(III) hydroxide	Ce(OH) ₃	19.80	1.6 × 10 ⁻²⁰
(IV) hydroxide	Ce(OH) ₄	47.7	2 × 10 ⁻⁴⁸
(III) iodate	Ce(IO ₃) ₃	9.50	3.2 × 10 ⁻¹⁰
(IV) iodate	Ce(IO ₃) ₄	16.3	5 × 10 ⁻¹⁷
(III) oxalate 9-water	Ce ₂ (C ₂ O ₄) ₃ · 9H ₂ O	25.50	3.2 × 10 ⁻²⁶
(III) phosphate	CePO ₄	23	1 × 10 ⁻²³
(III) selenite	Ce ₂ (SeO ₃) ₃	24.43	3.7 × 10 ⁻²⁵
(III) sulfide	Ce ₂ S ₃	10.22	6.0 × 10 ⁻¹¹
(III) tartrate	Ce ₂ L ₃	19.0	1.0 × 10 ⁻¹⁹
Cesium			
bromate	CsBrO ₃	1.7	5 × 10 ⁻²
chlorate	CsClO ₃	1.4	4 × 10 ⁻²
cobaltihexanitrite	Cs ₃ [Co(NO ₂) ₆]	15.24	5.7 × 10 ⁻¹⁶
hexachloroplatinate(IV)	Cs ₂ [PtCl ₆]	7.50	3.2 × 10 ⁻⁸
hexafluoroplatinate(IV)	Cs ₂ [PtF ₆]	5.62	2.4 × 10 ⁻⁶
hexafluorosilicate	Cs ₂ [SiF ₆]	4.90	1.3 × 10 ⁻⁵

TABLE 8.6 Solubility Product Constants (*Continued*)

Compound	Formula	pK_{sp}	K_{sp}
perchlorate	$CsClO_4$	2.40	3.95×10^{-3}
periodate	$CsIO_4$	5.29	5.16×10^{-6}
permanganate	$CsMnO_4$	4.08	8.2×10^{-5}
perrhanate	$CsReO_4$	3.40	4.0×10^{-4}
tetrafluoroborate	$Cs[BF_4]$	4.7	5×10^{-5}
Chromium(II)			
hydroxide	$Cr(OH)_2$	15.7	2×10^{-16}
Chromium(III)			
arsenate	$CrAsO_4$	20.11	7.7×10^{-21}
fluoride	CrF_3	10.18	6.6×10^{-11}
hydroxide	$Cr(OH)_3$	30.20	6.3×10^{-31}
phosphate 4-water	$CrPO_4 \cdot 4H_2O$ green	22.62	2.4×10^{-23}
	violet	17.00	1.0×10^{-17}
Cobalt			
anthranilate	CoL_2	9.68	2.1×10^{-10}
arsenate	$Co_3(AsO_4)_2$	28.17	6.80×10^{-29}
carbonate	$CoCO_3$	12.84	1.4×10^{-13}
ferrocyanide	$Co_2[Fe(CN)_6]$	14.74	1.8×10^{-15}
hydrogen phosphate	$CoHPO_4$	6.7	2×10^{-7}
(II) hydroxide	$Co(OH)_2$ fresh	14.23	5.92×10^{-15}
(III) hydroxide	$Co(OH)_3$	43.80	1.6×10^{-44}
iodate	$Co(IO_3)_2$	4.0	1.0×10^{-4}
phosphate	$Co_3(PO_4)_2$	34.69	2.05×10^{-35}
selenite	$CoSeO_3$	6.80	1.6×10^{-7}
quinaldate	CoL_2	10.80	1.6×10^{-11}
8-quinolinolate	CoL_2	24.80	1.6×10^{-25}
sulfide	α -CoS	20.40	4.0×10^{-21}
	β -CoS	24.70	2.0×10^{-25}
Copper(I)			
azide	CuN_3	8.31	4.9×10^{-9}
bromide	$CuBr$	8.20	6.27×10^{-9}
chloride	$CuCl$	6.76	1.72×10^{-7}
cyanide	$CuCN$	19.46	3.47×10^{-20}
hydroxide	$CuOH$	14	1×10^{-14}
iodide	CuI	11.90	1.27×10^{-12}
sulfide	Cu_2S	47.60	2.5×10^{-48}
tetraphenylborate	CuL	8.0	1.0×10^{-8}
thiocyanate	$CuSCN$	12.75	1.77×10^{-13}
Copper(II)			
anthranilate	CuL_2	13.22	6.0×10^{-14}
arsenate	$Cu_3(AsO_4)_2$	35.10	7.95×10^{-36}
azide	$Cu(N_3)_2$	9.20	6.3×10^{-10}
carbonate	$CuCO_3$	9.86	1.4×10^{-10}
chromate	$CuCrO_4$	5.44	3.6×10^{-6}
dithiooxamide	CuL	15.12	7.67×10^{-16}
ferrocyanide	$Cu_2[Fe(CN)_6]$	15.89	1.3×10^{-16}
hydroxide	$Cu(OH)_2$	19.66	2.2×10^{-20}
iodate	$Cu(IO_3)_2$	7.16	6.94×10^{-8}
oxalate	CuC_2O_4	9.35	4.43×10^{-10}
phosphate	$Cu_3(PO_4)_2$	36.85	1.40×10^{-37}
pyrophosphate	$Cu_2P_2O_7$	15.08	8.3×10^{-16}
quinaldate	CuL_2	16.80	1.6×10^{-17}
8-quinolinolate	CuL_2	29.70	2.0×10^{-30}

TABLE 8.6 Solubility Product Constants (Continued)

Compound	Formula	pK_{sp}	K_{sp}
selenite	$CuSeO_3$	7.68	2.1×10^{-8}
sulfide	CuS	35.20	6.3×10^{-36}
Dysprosium			
chromate 10-water	$Dy_2(CrO_4)_3 \cdot 10H_2O$	8	1×10^{-8}
hydroxide	$Dy(OH)_3$	21.85	1.4×10^{-22}
Erbium			
hydroxide	$Er(OH)_3$	23.39	4.1×10^{-24}
Europium			
hydroxide	$Eu(OH)_3$	23.03	9.38×10^{-24}
Gadolinium			
hydrogen carbonate	$Gd(HCO_3)_3$	1.7	2×10^{-2}
hydroxide	$Gd(OH)_3$	22.74	1.8×10^{-23}
Gallium			
ferrocyanide	$Ga_4[Fe(CN)_6]_3$	33.82	1.5×10^{-34}
hydroxide	$Ga(OH)_3$	35.14	7.28×10^{-36}
8-quinolinolate	GaL_3	40.80	1.6×10^{-41}
Germanium			
oxide	GeO_2	57.0	1.0×10^{-57}
Gold(I)			
chloride	$AuCl$	12.70	2.0×10^{-13}
iodide	AuI	22.80	1.6×10^{-23}
Gold(III)			
chloride	$AuCl_3$	24.50	3.2×10^{-25}
hydroxide	$Au(OH)_3$	45.26	5.5×10^{-46}
iodide	AuI_3	46	1×10^{-46}
oxalate	$Au_2(C_2O_4)_3$	10	1×10^{-10}
Hafnium			
hydroxide	$Hf(OH)_3$	25.40	4.0×10^{-26}
Holmium			
hydroxide	$Ho(OH)_3$	22.3	5.0×10^{-23}
Indium			
ferrocyanide	$In_4[Fe(CN)_6]_3$	43.72	1.9×10^{-44}
hydroxide	$In(OH)_3$	33.2	6.3×10^{-34}
quinolinolate	InL_3	31.34	4.6×10^{-32}
selenite	$In_2(SeO_3)_3$	32.60	4.0×10^{-33}
sulfide	In_2S_3	73.24	5.7×10^{-74}
Iron(II)			
carbonate	$FeCO_3$	10.50	3.13×10^{-11}
fluoride	FeF_2	5.63	2.36×10^{-6}
hydroxide	$Fe(OH)_2$	16.31	4.87×10^{-17}
oxalate dihydrate	$FeC_2O_4 \cdot 2H_2O$	6.50	3.2×10^{-7}
sulfide	FeS	17.20	6.3×10^{-18}
Iron(III)			
arsenate	$FeAsO_4$	20.24	5.7×10^{-21}
ferrocyanide	$Fe_4[Fe(CN)_6]_3$	40.52	3.3×10^{-41}
hydroxide	$Fe(OH)_3$	38.55	2.79×10^{-39}
phosphate dihydrate	$FePO_4 \cdot 2H_2O$	15.00	9.91×10^{-16}
quinaldate	FeL_3	16.89	1.3×10^{-17}
selenite	$Fe_2(SeO_3)_3$	30.70	2.0×10^{-31}
Lanthanum			
bromate 9-water	$La(BrO_3)_3 \cdot 9H_2O$	2.50	3.2×10^{-3}
fluoride	LaF_3	16.2	7×10^{-17}

TABLE 8.6 Solubility Product Constants (*Continued*)

Compound	Formula	pK_{sp}	K_{sp}
hydroxide	$\text{La}(\text{OH})_3$	18.70	2.0×10^{-19}
iodate	$\text{La}(\text{IO}_3)_3$	11.12	7.50×10^{-12}
molybdate	$\text{La}_2(\text{MoO}_4)_3$	20.4	4×10^{-21}
oxalate 9-water	$\text{La}_2(\text{C}_2\text{O}_4)_3$	26.60	2.5×10^{-27}
phosphate	LaPO_4	22.43	3.7×10^{-23}
sulfide	La_2S_3	12.70	2.0×10^{-13}
tungstate trihydrate	$\text{La}_2(\text{WO}_4)_3 \cdot 3\text{H}_2\text{O}$	3.90	1.3×10^{-4}
Lead			
acetate	$\text{Pb}(\text{OAc})_2$	2.75	1.8×10^{-3}
anthranilate	PbL_2	9.81	1.6×10^{-10}
arsenate	$\text{Pb}_3(\text{AsO}_4)_3$	35.39	4.0×10^{-36}
azide	$\text{Pb}(\text{N}_3)_2$	8.59	2.5×10^{-9}
borate, <i>meta</i>	$\text{Pb}(\text{BO}_2)_3$	10.78	1.6×10^{-11}
bromate	$\text{Pb}(\text{BrO}_3)_2$	1.70	2.0×10^{-2}
bromide	PbBr_2	6.82	6.60×10^{-6}
carbonate	PbCO_3	13.13	7.4×10^{-14}
chloride	PbCl_2	4.77	1.70×10^{-5}
chloride fluoride	PbClF	8.62	2.4×10^{-9}
chlorite	$\text{Pb}(\text{ClO}_2)_2$	8.4	4×10^{-9}
chromate	PbCrO_4	12.55	2.8×10^{-13}
ferrocyanide	$\text{Pb}_2[\text{Fe}(\text{CN})_6]$	14.46	3.5×10^{-15}
fluoride	PbF_2	7.48	3.3×10^{-8}
fluoride iodide	PbFI	8.07	8.5×10^{-9}
hydrogen phosphate	PbHPO_4	9.90	1.3×10^{-10}
hydrogen phosphite	PbHPO_3	6.24	5.8×10^{-7}
hydroxide	$\text{Pb}(\text{OH})_2$	14.84	1.43×10^{-15}
hydroxide bromide	PbOHBr	14.70	2.0×10^{-15}
hydroxide chloride	PbOHCl	13.7	2×10^{-14}
hydroxide nitrate	PbOHNO_3	3.55	2.8×10^{-4}
iodate	$\text{Pb}(\text{IO}_3)_2$	12.43	3.69×10^{-13}
iodide	PbI_2	8.01	9.8×10^{-9}
molybdate	PbMoO_4	13.00	1.0×10^{-13}
niobate	$\text{Pb}(\text{NbO}_3)_2$	16.62	2.4×10^{-17}
oxalate	PbC_2O_4	9.32	4.8×10^{-10}
phosphate	$\text{Pb}_3(\text{PO}_4)_2$	42.10	8.0×10^{-43}
quinaldate	PbL_2	10.60	2.5×10^{-11}
selenate	PbSeO_4	6.84	1.37×10^{-7}
selenite	PbSeO_3	11.50	3.2×10^{-12}
sulfate	PbSO_4	7.60	2.53×10^{-8}
sulfide	PbS	27.10	8.0×10^{-28}
thiocyanate	$\text{Pb}(\text{SCN})_2$	4.70	2.0×10^{-5}
thiosulfate	PbS_2O_3	6.40	4.0×10^{-7}
tungstate	PbWO_4	6.35	4.5×10^{-7}
Lead(IV)			
hydroxide	$\text{Pb}(\text{OH})_4$	65.50	3.2×10^{-66}
Lithium			
carbonate	Li_2CO_3	1.60	2.5×10^{-2}
fluoride	LiF	2.74	1.84×10^{-3}
phosphate	Li_3PO_4	10.63	2.37×10^{-11}
uranylarsenate	$\text{LiUO}_2\text{AsO}_4$	18.82	1.5×10^{-19}
Lutetium			
hydroxide	$\text{Lu}(\text{OH})_3$	23.72	1.9×10^{-24}

TABLE 8.6 Solubility Product Constants (Continued)

Compound	Formula	pK _{sp}	K _{sp}
Magnesium			
ammonium phosphate	MgNH ₄ PO ₄	12.60	2.5 × 10 ⁻¹³
arsenate	Mg ₃ (AsO ₄) ₂	19.68	2.1 × 10 ⁻²⁰
carbonate	MgCO ₃	5.17	6.82 × 10 ⁻⁶
carbonate trihydrate	MgCO ₃ · 3H ₂ O	5.62	2.38 × 10 ⁻⁶
fluoride	MgF ₂	10.29	5.16 × 10 ⁻¹¹
hydroxide	Mg(OH) ₂	11.25	5.61 × 10 ⁻¹²
iodate 4-water	Mg(IO ₃) ₂ · 4H ₂ O	2.50	3.2 × 10 ⁻³
niobate	Mg(NbO ₃) ₂	16.64	2.3 × 10 ⁻¹⁷
oxalate dihydrate	MgC ₂ O ₄ · 2H ₂ O	5.32	4.83 × 10 ⁻⁶
phosphate	Mg ₃ (PO ₄) ₂	23.98	1.04 × 10 ⁻²⁴
8-quinolinolate	MgL ₂	15.40	4.0 × 10 ⁻¹⁶
selenite	MgSeO ₃	4.89	1.3 × 10 ⁻⁵
sulfite	MgSO ₃	2.50	3.2 × 10 ⁻³
Manganese			
anthranilate	MnL ₂	6.75	1.8 × 10 ⁻³
arsenate	Mn ₃ (AsO ₄) ₂	28.72	1.9 × 10 ⁻²⁹
carbonate	MnCO ₃	10.63	2.34 × 10 ⁻¹¹
ferrocyanide	Mn ₂ [Fe(CN) ₆]	12.10	8.0 × 10 ⁻¹³
iodate	Mn(IO ₃) ₂	6.36	4.37 × 10 ⁻⁷
hydroxide	Mn(OH) ₂	12.72	1.9 × 10 ⁻¹³
oxalate dihydrate	MnC ₂ O ₄ · 2H ₂ O	6.77	1.70 × 10 ⁻⁷
8-quinolinolate	MnL ₂	21.70	2.0 × 10 ⁻²²
selenite	MnSeO ₃	6.90	1.3 × 10 ⁻⁷
sulfide	MnS amorphous	9.60	2.5 × 10 ⁻¹⁰
	MnS crystalline	12.60	2.5 × 10 ⁻¹³
Mercury(I)			
azide	Hg ₂ (N ₃) ₂	9.15	7.1 × 10 ⁻¹⁰
bromide	Hg ₂ Br ₂	22.19	6.40 × 10 ⁻²³
carbonate	Hg ₂ CO ₃	16.44	3.6 × 10 ⁻¹⁷
chloride	Hg ₂ Cl ₂	17.84	1.43 × 10 ⁻¹⁸
cyanide	Hg ₂ (CN) ₂	39.3	5 × 10 ⁻⁴⁰
chromate	Hg ₂ CrO ₄	8.70	2.0 × 10 ⁻⁹
ferricyanide	(Hg ₂) ₃ [Fe(CN) ₆] ₂	20.07	8.5 × 10 ⁻²¹
fluoride	Hg ₂ F ₂	5.51	3.10 × 10 ⁻⁶
hydrogen phosphate	Hg ₂ HPO ₄	12.40	4.0 × 10 ⁻¹³
hydroxide	Hg ₂ (OH) ₂	23.70	2.0 × 10 ⁻²⁴
iodate	Hg ₂ (IO ₃) ₂	13.71	2.0 × 10 ⁻¹⁴
iodide	Hg ₂ I ₂	28.72	5.2 × 10 ⁻²⁹
oxalate	Hg ₂ C ₂ O ₄	12.76	1.75 × 10 ⁻¹³
quinaldate	Hg ₂ L ₂	17.90	1.3 × 10 ⁻¹⁸
selenite	Hg ₂ SeO ₃	14.20	8.4 × 10 ⁻¹⁵
sulfate	Hg ₂ SO ₄	6.19	6.5 × 10 ⁻⁷
sulfite	Hg ₂ SO ₃	27.0	1.0 × 10 ⁻²⁷
sulfide	Hg ₂ S	47.0	1.0 × 10 ⁻⁴⁷
thiocyanate	Hg ₂ (SCN) ₂	19.49	3.2 × 10 ⁻²⁰
tungstate	Hg ₂ WO ₄	16.96	1.1 × 10 ⁻¹⁷
Mercury(II)			
bromide	HgBr ₂	19.21	6.2 × 10 ⁻²⁰
hydroxide	Hg(OH) ₂	25.52	3.2 × 10 ⁻²⁶
iodate	Hg(IO ₃) ₂	12.49	3.2 × 10 ⁻¹³
iodide	HgI ₂	28.54	2.9 × 10 ⁻²⁹
1,10-phenanthroline	HgL ₂	24.70	2.0 × 10 ⁻²⁵

TABLE 8.6 Solubility Product Constants (*Continued*)

Compound	Formula	pK_{sp}	K_{sp}
quinaldate	HgL_2	16.80	1.6×10^{-17}
selenite	$HgSeO_3$	13.82	1.5×10^{-14}
sulfide	HgS red	52.4	4×10^{-53}
	HgS black	51.80	1.6×10^{-52}
Neodymium			
carbonate	$Nd_2(CO_3)_3$	32.97	1.08×10^{-33}
hydroxide	$Nd(OH)_3$	21.49	3.2×10^{-22}
Neptunyl(VI)			
hydroxide	$NpO_2(OH)_2$	21.60	2.5×10^{-22}
Nickel			
ammine perrhenate	$[Ni(NH_3)_6][ReO_4]_2$	3.29	5.1×10^{-4}
anthranilate	NiL_2	9.09	8.1×10^{-10}
arsenate	$Ni_3(AsO_4)_2$	25.51	3.1×10^{-26}
carbonate	$NiCO_3$	6.85	1.42×10^{-7}
ferrocyanide	$Ni_2[Fe(CN)_6]$	14.89	1.3×10^{-15}
hydrazine sulfate	$[Ni(N_2H_4)_3]SO_4$	13.15	7.1×10^{-15}
hydroxide	$Ni(OH)_2$ fresh	15.26	5.48×10^{-16}
iodate	$Ni(IO_3)_2$	4.33	4.71×10^{-5}
oxalate	NiC_2O_4	9.4	4×10^{-10}
phosphate	$Ni_3(PO_4)_2$	31.32	4.74×10^{-32}
pyrophosphate	$Ni_2P_2O_7$	12.77	1.7×10^{-13}
quinaldate	NiL_2	10.1	8×10^{-11}
8-quinolinolate	NiL_2	26.1	8×10^{-27}
selenite	$NiSeO_3$	5.0	1.0×10^{-5}
α -sulfide	α -NiS	18.50	3.2×10^{-19}
β -sulfide	β -NiS	24.0	1.0×10^{-24}
γ -sulfide	γ -NiS	25.70	2.0×10^{-26}
Palladium			
(II) hydroxide	$Pd(OH)_2$	31.0	1.0×10^{-31}
(IV) hydroxide	$Pd(OH)_4$	70.20	6.3×10^{-71}
quinaldate	PdL_2	12.90	1.3×10^{-13}
thiocyanate	$Pd(SCN)_2$	22.36	4.39×10^{-23}
Platinum			
(IV) bromide	$PtBr_4$	40.50	3.2×10^{-41}
(II) hydroxide	$Pt(OH)_2$	35	1×10^{-35}
Plutonium			
(III) fluoride	PuF_3	15.60	2.5×10^{-16}
(IV) fluoride	PuF_4	19.20	6.3×10^{-20}
(IV) hydrogen phosphate	$Pu(HPO_4)_2 \cdot xH_2O$	27.7	2×10^{-28}
(III) hydroxide	$Pu(OH)_3$	19.70	2.0×10^{-20}
(IV) hydroxide	$Pu(OH)_4$	55	1×10^{-55}
(IV) iodate	$Pu(IO_3)_4$	12.3	5×10^{-13}
(VI) carbonate	PuO_2CO_3	12.77	1.7×10^{-13}
(V) hydroxide	$PuO_2(OH)$	9.3	5×10^{-10}
(VI) hydroxide	$PuO_2(OH)_2$	24.7	2×10^{-25}
Polonium			
sulfide	PoS	28.26	5.6×10^{-29}
Potassium			
hexabromoplatinate	$K_2[PtBr_6]$	4.20	6.3×10^{-5}
hexachloropalladate	$K_2[PdCl_6]$	5.22	6.0×10^{-6}
hexachloroplatinate	$K_2[PtCl_6]$	5.13	7.48×10^{-6}
hexafluoroplatinate	$K_2[PtF_6]$	4.54	2.9×10^{-5}

TABLE 8.6 Solubility Product Constants (Continued)

Compound	Formula	pK _{sp}	K _{sp}
hexafluorosilicate	K ₂ [SiF ₆]	6.06	8.7 × 10 ⁻⁷
hexafluorozirconate	K ₂ [ZrF ₆]	3.3	5 × 10 ⁻⁴
iodate	KIO ₄	3.43	3.74 × 10 ⁻⁴
perchlorate	KClO ₄	1.98	1.05 × 10 ⁻²
sodium cobaltinitrite hydrate	K ₂ Na[Co(NO ₂) ₆] · H ₂ O	10.66	2.2 × 10 ⁻¹¹
tetraphenylborate	K[B(C ₆ H ₅) ₄]	7.66	2.2 × 10 ⁻⁸
uranyl arsenate	K[UO ₂ AsO ₄]	22.60	2.5 × 10 ⁻²³
uranyl carbonate	K ₄ [UO ₂ (CO ₃) ₃]	4.20	6.3 × 10 ⁻⁵
Praseodymium hydroxide	Pr(OH) ₃	23.45	3.39 × 10 ⁻²⁴
Promethium hydroxide	Pm(OH) ₃	21	1 × 10 ⁻²¹
Radium iodate	Ra(IO ₃) ₂	8.94	1.16 × 10 ⁻⁹
sulfate	RaSO ₄	10.44	3.66 × 10 ⁻¹¹
Rhodium hydroxide	Rh(OH) ₃	23	1 × 10 ⁻²³
Rubidium cobaltinitrite	Rb ₃ [Co(NO ₂) ₆]	14.83	1.5 × 10 ⁻¹⁵
hexachloroplatinate	Rb ₂ [PtCl ₆]	7.20	6.3 × 10 ⁻⁸
hexafluoroplatinate	Rb ₂ [PtF ₆]	6.12	7.7 × 10 ⁻⁷
hexafluorosilicate	Rb ₂ [SiF ₆]	6.30	5.0 × 10 ⁻⁷
perchlorate	RbClO ₄	2.52	3.0 × 10 ⁻³
periodate	RbIO ₄	3.26	5.5 × 10 ⁻⁴
Ruthenium hydroxide	Ru(OH) ₃	36	1 × 10 ⁻³⁶
Samarium hydroxide	Sm(OH) ₃	22.08	8.3 × 10 ⁻²³
Scandium fluoride	ScF ₃	23.24	5.81 × 10 ⁻²⁴
hydroxide	Sc(OH) ₃	30.65	2.22 × 10 ⁻³¹
Silver acetate	AgOAc	2.71	1.94 × 10 ⁻³
arsenate	Ag ₃ AsO ₄	21.99	1.03 × 10 ⁻²²
azide	AgN ₃	8.54	2.8 × 10 ⁻⁹
bromate	AgBrO ₃	4.27	5.38 × 10 ⁻⁵
bromide	AgBr	12.27	5.35 × 10 ⁻¹³
carbonate	Ag ₂ CO ₃	11.07	8.46 × 10 ⁻¹²
chloride	AgCl	9.75	1.77 × 10 ⁻¹⁰
chlorite	AgClO ₂	3.70	2.0 × 10 ⁻⁴
chromate	Ag ₂ CrO ₄	11.95	1.12 × 10 ⁻¹²
cobaltinitrite	Ag ₃ [Co(NO ₂) ₆]	20.07	8.5 × 10 ⁻²¹
cyanamide	Ag ₂ CN ₂	10.14	7.2 × 10 ⁻¹¹
cyanate	AgOCN	6.64	2.3 × 10 ⁻⁷
cyanide	AgCN	16.22	5.97 × 10 ⁻¹⁷
dichromate	Ag ₂ Cr ₂ O ₇	6.70	2.0 × 10 ⁻⁷
dicyanamide	AgN(CN) ₂	8.85	1.4 × 10 ⁻⁹
ferrocyanide	Ag ₄ [Fe(CN) ₆]	40.81	1.6 × 10 ⁻⁴¹
hydroxide	AgOH	7.71	2.0 × 10 ⁻⁸
hyponitrite	Ag ₂ N ₂ O ₂	18.89	1.3 × 10 ⁻¹⁹
iodate	AgIO ₃	7.50	3.17 × 10 ⁻⁸

TABLE 8.6 Solubility Product Constants (*Continued*)

Compound	Formula	pK_{sp}	K_{sp}
iodide	AgI	16.07	8.52×10^{-17}
molybdate	Ag ₂ MoO ₄	11.55	2.8×10^{-12}
nitrite	AgNO ₂	3.22	6.0×10^{-4}
oxalate	Ag ₂ C ₂ O ₄	11.27	5.40×10^{-12}
phosphate	Ag ₃ PO ₄	16.05	8.89×10^{-17}
quinaldate	AgL	16.89	1.3×10^{-17}
perrhenate	AgReO ₄	4.10	8.0×10^{-5}
selenate	Ag ₂ SeO ₄	7.25	5.7×10^{-8}
selenite	Ag ₂ SeO ₃	15.00	1.0×10^{-15}
selenocyanate	AgSeCN	15.40	4.0×10^{-16}
sulfate	Ag ₂ SO ₄	4.92	1.20×10^{-5}
sulfite	Ag ₂ SO ₃	13.82	1.50×10^{-14}
sulfide	Ag ₂ S	49.20	6.3×10^{-50}
thiocyanate	AgSCN	11.99	1.03×10^{-12}
vanadate	AgVO ₃	6.3	5×10^{-7}
tungstate	Ag ₂ WO ₄	11.26	5.5×10^{-12}
Sodium			
ammonium cobaltinitrite	Na(NH ₄) ₂ [Co(NO ₂) ₆]	10.66	2.2×10^{-11}
antimonate	Na[Sb(OH) ₆]	7.4	4×10^{-8}
hexafluoroaluminate	Na ₂ [AlF ₆]	9.39	4.0×10^{-10}
uranyl arsenate	NaUO ₂ AsO ₄	21.87	1.3×10^{-22}
Strontium			
arsenate	Sr ₃ (AsO ₄) ₂	18.37	4.29×10^{-19}
carbonate	SrCO ₃	9.25	5.60×10^{-10}
chromate	SrCrO ₄	4.65	2.2×10^{-5}
fluoride	SrF ₂	8.36	4.33×10^{-9}
iodate	Sr(IO ₃) ₂	6.94	1.14×10^{-7}
iodate hydrate	Sr(IO ₃) ₂ · H ₂ O	6.42	3.77×10^{-7}
molybdate	SrMoO ₄	6.7	2×10^{-7}
niobate	Sr(NbO ₃) ₂	17.38	4.2×10^{-18}
oxalate hydrate	SrC ₂ O ₄ · H ₂ O	6.80	1.6×10^{-7}
phosphate	Sr ₃ (PO ₄) ₂	27.39	4.0×10^{-28}
8-quinolinolate	SrL ₂	9.3	5×10^{-10}
selenate	SrSeO ₄	3.09	8.1×10^{-4}
selenite	SrSeO ₃	5.74	1.8×10^{-6}
sulfate	SrSO ₄	6.46	3.44×10^{-7}
sulfite	SrSO ₃	7.4	4×10^{-8}
tungstate	SrWO ₄	9.77	1.7×10^{-10}
Terbium			
hydroxide	Tb(OH) ₃	21.70	2.0×10^{-22}
Tellurium			
hydroxide	Te(OH) ₄	53.52	3.0×10^{-54}
Thallium(I)			
azide	TlN ₃	3.66	2.2×10^{-4}
bromate	TlBrO ₃	4.96	1.10×10^{-5}
bromide	TlBr	5.43	3.71×10^{-6}
chloride	TlCl	3.73	1.86×10^{-4}
chromate	Tl ₂ CrO ₄	12.06	8.67×10^{-13}
ferrocyanide dihydrate	Tl ₄ [Fe(CN) ₆] · 2H ₂ O	9.3	5×10^{-10}
hexachloroplatinate	Tl ₂ [PtCl ₆]	11.40	4.0×10^{-12}
iodate	TlIO ₃	5.51	3.12×10^{-6}
iodide	TlI	7.26	5.54×10^{-8}

TABLE 8.6 Solubility Product Constants (Continued)

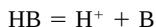
Compound	Formula	pK _{sp}	K _{sp}
oxalate	Tl ₂ C ₂ O ₄	3.7	2 × 10 ⁻⁴
selenate	Tl ₂ SeO ₄	4.00	1.0 × 10 ⁻⁴
selenite	Tl ₂ SeO ₃	38.7	2 × 10 ⁻³⁹
sulfide	Tl ₂ S	20.30	5.0 × 10 ⁻²¹
thiocyanate	TlSCN	3.80	1.57 × 10 ⁻⁴
Thallium(III)			
hydroxide	Tl(OH) ₃	43.77	1.68 × 10 ⁻⁴⁴
8-quinolinolate	TlL ₃	32.40	4.0 × 10 ⁻³³
Thorium			
hydrogen phosphate	Th(HPO ₄) ₂	20	1 × 10 ⁻²⁰
hydroxide	Th(OH) ₄	44.40	4.0 × 10 ⁻⁴⁵
iodate	Th(IO ₃) ₄	14.60	2.5 × 10 ⁻¹⁵
oxalate	Th(C ₂ O ₄) ₂	22	1 × 10 ⁻²²
phosphate	Th ₃ (PO ₄) ₄	78.60	2.5 × 10 ⁻⁷⁹
Thulium			
hydroxide	Tm(OH) ₃	23.48	3.3 × 10 ⁻²⁴
Tin			
(II) hydroxide	Sn(OH) ₂	27.26	5.45 × 10 ⁻²⁸
(IV) hydroxide	Sn(OH) ₄	56	1 × 10 ⁻⁵⁶
(II) sulfide	SnS	25.00	1.0 × 10 ⁻²⁵
Titanium			
(III) hydroxide	Ti(OH) ₃	40	1 × 10 ⁻⁴⁰
(IV) oxide hydroxide	TiO(OH) ₂	29	1 × 10 ⁻²⁹
Uranium(IV)			
fluoride 2.5-water	UF ₄ · 2.5H ₂ O	21.24	5.7 × 10 ⁻²²
Uranyl(VI)(2+)			
carbonate	UO ₂ CO ₃	11.73	1.8 × 10 ⁻¹²
ferrocyanide	UO ₂ [Fe(CN) ₆]	13.15	7.1 × 10 ⁻¹⁴
hydrogen arsenate	UO ₂ HAsO ₄	10.50	3.2 × 10 ⁻¹¹
hydrogen phosphate	UO ₂ HPO ₄	10.67	2.1 × 10 ⁻¹¹
hydroxide	UO ₂ (OH) ₂	21.95	1.1 × 10 ⁻²²
iodate hydrate	UO ₂ (IO ₃) ₂ · H ₂ O	7.50	3.2 × 10 ⁻⁸
oxalate trihydrate	UO ₂ C ₂ O ₄ · 3H ₂ O	3.7	2 × 10 ⁻⁴
phosphate	(UO ₂) ₃ (PO ₄) ₂	46.7	2 × 10 ⁻⁴⁷
sulfite	UO ₂ SO ₃	8.58	2.6 × 10 ⁻⁹
thiocyanate	(UO ₂)(SCN) ₂	3.4	4 × 10 ⁻⁴
Vanadium			
(IV) hydroxide	VO(OH) ₂	22.13	5.9 × 10 ⁻²³
(III) phosphate	(VO ₂) ₃ PO ₄	24.1	8 × 10 ⁻²⁵
Ytterbium			
hydroxide	Yt(OH) ₃	23.60	2.5 × 10 ⁻²⁴
Yttrium			
carbonate	Y ₂ (CO ₃) ₃	2.99	1.03 × 10 ⁻³
fluoride	YF ₃	20.06	8.62 × 10 ⁻²¹
hydroxide	Y(OH) ₃	22.00	1.00 × 10 ⁻²²
iodate	Y(IO ₃) ₃	9.95	1.12 × 10 ⁻¹⁰
oxalate	Y ₂ (C ₂ O ₄) ₃	28.28	5.3 × 10 ⁻²⁹
Zinc			
anthranilate	ZnL ₂	9.23	5.9 × 10 ⁻¹⁰
arsenate	Zn ₃ (AsO ₄) ₂	27.55	2.8 × 10 ⁻²⁸
borate hydrate	Zn(BO ₂) ₂ · H ₂ O	10.18	6.6 × 10 ⁻¹¹
carbonate	ZnCO ₃	9.94	1.46 × 10 ⁻¹⁰
ferrocyanide	Zn ₂ [Fe(CN) ₆]	15.40	4.0 × 10 ⁻¹⁵

TABLE 8.6 Solubility Product Constants (*Continued*)

Compound	Formula	pK_{sp}	K_{sp}
fluoride	ZnF_2	1.52	3.04×10^{-2}
hydroxide	$Zn(OH)_2$	16.5	3×10^{-17}
iodate dihydrate	$Zn(IO_3)_2 \cdot 2H_2O$	5.37	4.1×10^{-6}
oxalate dihydrate	$ZnC_2O_4 \cdot 2H_2O$	8.86	1.38×10^{-9}
phosphate	$Zn_3(PO_4)_2$	32.04	9.0×10^{-33}
quinaldate	ZnL_2	13.80	1.6×10^{-14}
8-quinolinolate	ZnL_2	24.30	5.0×10^{-25}
selenide	$ZnSe$	25.44	3.6×10^{-26}
selenite hydrate	$ZnSeO_3 \cdot H_2O$	6.80	1.57×10^{-7}
sulfide	$\alpha\text{-ZnS}$	23.80	1.6×10^{-24}
	$\beta\text{-ZnS}$	21.60	2.5×10^{-22}
Zirconium			
oxide hydroxide	$ZrO(OH)_2$	48.20	6.3×10^{-49}
phosphate	$Zr_3(PO_4)_4$	132	1×10^{-132}

8.2.1 Proton-Transfer Reactions

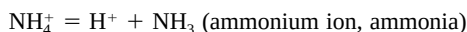
The pK_a values listed in Tables 8.7 and 8.8 are the negative (decadic) logarithms of the acidic dissociation constant, i.e., $-\log_{10} K_a = pK_a$. For the general proton-transfer reaction



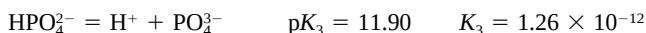
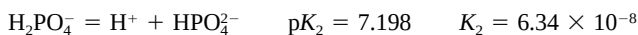
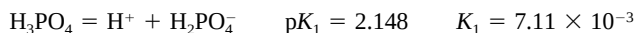
the acidic dissociation constant is formulated as follows:

$$K_a = \frac{[H^+][B]}{[HB]}$$

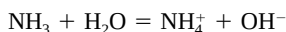
The most common charge types for the acid HB and its conjugate base B are



Acids which have more than one acidic hydrogen ionize in steps, as shown for phosphoric acid:



If the basic dissociation constant K_b for the equilibrium such as



is required, pK_b may be calculated from the relationship

$$pK_b = pK_w - pK_a$$

TABLE 8.7 Proton Transfer Reactions of Inorganic Materials in Water at 25°C

Substance	Formula or remarks	p <i>K</i> ₁	p <i>K</i> ₂
Aluminic acid	H ₃ AlO ₃	11.2	
Aluminum ion (aquo)	Al ³⁺ (aquo)	4.98(4)	
Americium(III) ion	Am ³⁺ (aquo) μ = 0.1	5.92	
Ammonium ion	NH ₄ ⁺	9.246(2)	
Ammonium- <i>d</i> ₃	ND ₃ H ⁺	9.757	
Antimonic acid	HSb(OH) ₆ = Sb(OH) ₆ [−] + H ⁺ μ = 0.5	2.55	
Antimony(III) ion	SbO ⁺ + H ₂ O = Sb(OH) ₃ + H ⁺ μ = 1.0	1.42	
Barium ion	p <i>K</i> _b of Ba(OH) ⁺ μ = 0.1	0.64	
Berkelium(III) ion	p <i>K</i> for hydrolysis of Bk ³⁺ μ = 0.1	5.66	
Beryllium(II) ion	Be ²⁺ (aquo) = BeOH ⁺ + H ⁺ μ = 1.0	6.5	
Bismuth(III) ion	Bi ³⁺ = BiOH ²⁺ + H ⁺ μ = 3.0	1.58	
Boric acid, tetra-	H ₂ B ₄ O ₇	4	9
Bromine	Br ₂ + H ₂ O = HBrO + H ⁺ + Br [−]	7.92	
Cadmium ion	Cd ²⁺ (aquo) hydrolysis	9.2(1)	
Calcium ion	Ca ²⁺ (aquo) hydrolysis	12.67(3)	
Californium(III) ion	Cf ³⁺ (aquo) hydrolysis μ = 0.1	5.62	
Carbon dioxide	CO ₂ (aquo)	6.352(1)	10.329
	CO ₂ in D ₂ O	6.77	10.93
Cerium(III) ion	Ce ³⁺ (aquo) hydrolysis	ca. 9.3	
Cerium(IV) ion	Hydrolysis to Ce(OH) ³⁺ and Ce(OH) ₂ ⁺	− 1.15	0.82
Chromium(III) ion	Cr ³⁺ (aquo) hydrolysis	3.95	
Cobalt(II) ion	Co ²⁺ (aquo) hydrolysis	8.9	
Cobalt(III) ion	Co ³⁺ (aquo) hydrolysis <i>m</i> = 1	1.75	
Copper(II) ion	Cu ²⁺ (aquo) hydrolysis	7.34	
Curium(III) ion	Cm ³⁺ (aquo) hydrolysis <i>m</i> = 0.1	6.00(5)	
Deuterium oxide	D ₂ O (molal scale)	14.956(1)	
Dysprosium(III) ion	Dy ³⁺ (aquo) hydrolysis	8.10	
Erbium(III) ion	Er ³⁺ (aquo) hydrolysis μ = 3	9.0	
Europium(III) ion	Eu ³⁺ (aquo) hydrolysis	8.03	
Fermium(III) ion	Fm ³⁺ hydrolysis μ = 0.1	3.8	
Gadolinium(III) ion	Gd ³⁺ hydrolysis	8.27	
Gallium(III) ion	Ga ³⁺ (successive values for hydrolysis)	2.92	3.77
		p <i>K</i> ₃ 4.75	
Gold(III) hydroxide	H ₃ AuO ₃	< 11.7	13.36
Hafnium(IV) ion	Hf ⁴⁺ hydrolysis μ = 1	− 0.12	0.23
Hexaminotriphosphazene	N ₃ P ₃ (NH ₂) ₆	< 3.2	7.68(3)
Holmium(III) ion	Ho ³⁺ hydrolysis μ = 0.3	8.04	

TABLE 8.7 Proton Transfer Reactions of Inorganic Materials in Water at 25°C (*Continued*)

Substance	Formula or remarks	p <i>K</i> ₁	p <i>K</i> ₂
Hydrazinium(2+) ion	⁺ H ₃ N—NH ₃ ⁺	0.27	7.94(3)
Hydrogen amidodisulfonate	HNSO(OH) ₂	p <i>K</i> ₃ 8.50	
Hydrogen amidophosphate	H ₂ NPO(OH) ₂ (26°C)	2.739	8.102
Hydrogen arsenate	H ₃ AsO ₄	2.223	6.760
Hydrogen- <i>d</i> ₃ arsenate	D ₃ AsO ₄	2.596	
Hydrogen arsenite	HAsO ₂	9.28(10)	
Hydrogen azide	HN ₃	4.62	
Hydrogen- <i>d</i> azide	DN ₃ (in D ₂ O)	5.115	
Hydrogen borate (3−)	H ₃ BO ₃	9.236	
Hydrogen bromate	HBrO ₃ (in formamide)	1.02	
Hydrogen bromide	HBr	−8.72(15)	
Hydrogen chlorate	HClO ₃ (theoretical prediction)	−2.7	
Hydrogen chloride	HCl	−6.2(1)	
Hydrogen- <i>d</i> chloride	DCl (in dimethylformamide)	3.58	
Hydrogen chlorite	HClO ₂	1.94	
Hydrogen chromate	H ₂ CrO ₄	0.74	6.488
Hydrogen cyanate	HOCN	3.46	
Hydrogen cyanide	HCN	9.21	
Hydrogen- <i>d</i> cyanide	DCN (in D ₂ O) $\mu = 0.11$	8.97	
Hydrogen diamidophosphate	(NH ₂)PO(OH) ₂ (30°C)	1.279(+ 1)	4.889
Hydrogen diamidothiophosphate	(NH ₂)PO(SH) ₂ (20°C)	2.0(+ 1)	4.3
Hydrogen diimidotriphosphate	(HO) ₂ PO(NH)PO(OH)(NH)PO(OH) ₂ $\mu = 0.1$	~1	~2
		p <i>K</i> ₃ 3.03	p <i>K</i> ₄ 6.61
		p <i>K</i> ₅ 9.84	
Hydrogen diphosphate	H ₄ P ₂ O ₇	0.91	2.10
		p <i>K</i> ₃ 6.70	p <i>K</i> ₄ 9.35
		−12	−8
Hydrogen disulfate	H ₂ S ₂ O ₇ (theoretical prediction)	−3.4	−0.2
Hydrogen dithionate	H ₂ S ₂ O ₆	0.35	2.45
Hydrogen dithionite	H ₂ S ₂ O ₄	3.20(4)	
Hydrogen fluoride	H ₂ F ₂	9.01	12.30
Hydrogen germanate	H ₂ GeO ₄		1.92
Hydrogen hexafluorosilicate	H ₂ SiF ₆		2.50
Hydrogen hydrosulfite	H ₂ S ₂ O ₄	0.35	
Hydrogen hypobromite	HBrO	8.55	
Hydrogen hypochlorite	HClO	7.537	
Hydrogen hypoiodite	HIO	10.5(5)	
Hydrogen hyponitrite	H ₂ N ₂ O ₂	7.21	11.45(10)
Hydrogen iodate	HIO ₃	0.804	

Source: J. J. Christensen, L. D. Hansen, and R. M. Izatt, *Handbook of Proton Ionization Heats and Related Thermodynamic Quantities*, Wiley-Interscience, New York, 1976; D. D. Perrin, *Ionisation Constants of Inorganic Acids and Bases in Aqueous Solution*, 2d ed., Pergamon Press, 1982.

TABLE 8.7 Proton Transfer Reactions of Inorganic Materials in Water at 25°C (*Continued*)

Substance	Formula or remarks	pK ₁	pK ₂
Hydrogen- <i>d</i> iodate	DIO ₃ (in D ₂ O)	1.15	
Hydrogen iodide	HI	−8.56	
Hydrogen manganate(VI)	H ₂ MnO ₄ (35°C) $\mu = 0.1$		10.15
Hydrogen nitrate	HNO ₃	−1.37(7)	
Hydrogen nitrite	HNO ₂	3.14(1)	
Hydrogen perchlorate	HClO ₄	−1.6	
Hydrogen periodate	HIO ₄	1.64	
Hydrogen peroxide	H ₂ O ₂	11.64(2)	
Hydrogen peroxophosphate	H ₃ PO ₅ $\mu = 0.2$	1.1	5.5
		pK ₃ 12.8	
Hydrogen peroxosulfate	H ₂ SO ₅	1.0	9.86
Hydrogen perrhenate	HReO ₄	−1.25	
Hydrogen pertechnetate	HTcO ₄	0.3	
Hydrogen perthiocarbonate	H ₂ CS ₄	3.54	7.24
Hydrogen perxenate	H ₄ XeO ₆	pK ₃ 10.5	
Hydrogen phosphate(3−)	H ₃ PO ₄	2.148(20)	7.198(10)
		pK ₃ 12.32(6)	
Hydrogen- <i>d</i> ₂ phosphate	D ₂ PO ₄ (in D ₂ O)	7.780	
Hydrogen phosphinate	H ₂ PHO ₂	1.23	
Hydrogen phosphonate	H ₂ PHO ₃	1.43	6.68(14)
Hydrogen selenate	H ₂ SeO ₄		1.66
Hydrogen selenide	H ₂ Se $\mu = 0.03$	3.89	11.0
Hydrogen selenite	H ₂ SeO ₃	2.62	8.30(15)
Hydrogen silicate(4−)	H ₄ SiO ₄	9.60(10)	11.8(1)
Hydrogen sulfamate	H ₂ NSO ₃ H	0.99	
Hydrogen sulfate	H ₂ SO ₄		1.99(1)
Hydrogen sulfide	H ₂ S	6.97	12.90
Hydrogen sulfite	SO ₂ + H ₂ O = HSO ₃ [−] = H ⁺	1.89	7.205
Hydrogen tellurate	H ₆ TeO ₆	7.65(5)	11.00(5)
Hydrogen telluride	H ₂ Te (18°C)	2.64	11–12
Hydrogen tellurite	H ₂ TeO ₃ (20°C)	6.27	8.43
Hydrogen tetrafluoroborate	HBF ₄	0.5	
Hydrogen tetracyanonickelate	H ₂ Ni(CN) ₄	4.69	6.59
Hydrogen tetraperoxo chromate	H ₃ CrO ₈ (30°C) $\mu = 3$	7.16	
Hydrogen tetrapolyphosphate	H ₄ P ₄ O ₁₃ $\mu = 0.034$	1.99	2.64
		pK ₃ 6.62	pK ₄ 8.2

TABLE 8.7 Proton Transfer Reactions of Inorganic Materials in Water at 25°C (*Continued*)

Substance	Formula or remarks	pK ₁	pK ₂
Hydrogen tetrathiophosphate	H ₃ PS ₄	1.5	3.5
Hydrogen thiocyanate	HSCN $\mu = 3$	pK ₃ 6.6	
Hydrogen thiophosphate	H ₃ PO ₃ S	−1.8	
		1.788	5.427
Hydrogen thiosulfate	H ₂ S ₂ O ₃	pK ₃ 10.08	
Hydrogen tripolyphosphate	H ₃ P ₃ O ₉	0.6	1.74
		~1	1.7
		pK ₃ 2.00(10)	
		pK ₄ 5.83(7)	
		pK ₅ 8.51(6)	
Hydrogen triselenocarbonate	H ₂ CSe ₃	1.16	7.70
Hydrogen trithiocarbonate	H ₂ CS ₃ (20°C)	2.68	8.18
Hydrogen tungstate	H ₂ WO ₄	2.20	3.70
Hydrogen vanadate(−1)	HVO ₃	3.80	
Hydrogen vanadate(3−)	H ₃ VO ₄	3.78	7.78(4)
Hydroxylamine- <i>N,N</i> -disulfonic acid	HON(SO ₃ H) ₂ $\mu = 1.6$	pK ₃ 11.85	
Hydroxylamine <i>O</i> -sulfonate	⁺ H ₃ NOSO ₃ [−] $\mu = 1$	1.48	
Imidodiphosphoric acid	(HO) ₂ PO(NH)PO(OH) ₂ $\mu = 0.2$	~2	2.85
		pK ₃ 7.08	pK ₄ 9.72
Indium(III) ion	In ³⁺ hydrolysis	3.54	4.28
Iridium(III) ion	Ir ³⁺ hydrolysis $\mu = 1$	4.37	5.20
Iron(II) ion	Fe ²⁺ hydrolysis $\mu = 1$	6.8	
Iron(III) ion	Fe ³⁺ hydrolysis	2.19	
Lanthanum(III) ion	La ³⁺ hydrolysis	9.06	
Lead(II) ion	Pb ²⁺ hydrolysis $\mu = 0.3$	7.8	
Lead(IV) ion	Pb ⁴⁺ hydrolysis	1.8	3.2
Lithium(I) ion	Li ⁺	13.8	
Lutetium(III) ion	Lu ³⁺ hydrolysis	7.94	
Magnesium(II) ion	Mg ²⁺ hydrolysis	11.41	
Manganese(II) ion	Mn ²⁺ hydrolysis	10.59	
Manganese(III) ion	Mn ³⁺ hydrolysis	0.4	
Mercury(I) ion	Hg ₂ ²⁺ hydrolysis $\mu = 0.5$	5.0	
Mercury(II) ion	Hg ²⁺ hydrolysis $\mu = 0.5$	3.70	2.65
Neodymium(III) ion	Nd ³⁺ hydrolysis $\mu = 3$	9.0(5)	
Neptunium(III) ion	Np ³⁺ hydrolysis $\mu = 0.3$	7.43	
Neptunium(IV) ion	Np ⁴⁺ hydrolysis $\mu = 2$	2.30	
Neptunium(V) ion	NpO ₂ ⁺ hydrolysis	8.90(2)	
Nickel(II) ion	Ni ²⁺ hydrolysis	9.86	
Osmium tetroxide	OsO ₄ hydrolysis $\mu = 1$	12.1	
Palladium(II) ion	Pd ²⁺ (stepwise pK _b values)	13.0	12.8
Pentacyanoaquoferrate(II) ion	Fe(CN) ₅ (H ₂ O) ^{3−} $\mu = 0.1$	2.63	

TABLE 8.7 Proton Transfer Reactions of Inorganic Materials in Water at 25°C (*Continued*)

Substance	Formula or remarks	pK ₁	pK ₂
Plutonium(III) ion	Pu ³⁺ hydrolysis $\mu = 0.07$	7.2(2)	
Plutonium(IV) ion	Pu ⁴⁺ hydrolysis $\mu = 2$	1.26	
Plutonium(V) ion	PuO ₂ ⁺ hydrolysis $\mu = 0.003$	9.7	
Plutonium(VI) ion	PuO ₂ ²⁺ hydrolysis	3.33	4.05
Polonium(IV) ion	Po ⁴⁺ hydrolysis	0.48	2.74
		pK ₃ 5.58	
Praseodymium(III) ion	Pr ³⁺ hydrolysis $\mu = 0.3$	8.55	
Protoactinium(IV) ion	Pa ⁴⁺ hydrolysis $\mu = 3$	0.14	0.38
Protoactinium(V) ion	Pa ⁵⁺ hydrolysis $\mu = 3$	1.05	
Scandium(III) ion	Sc ³⁺ hydrolysis $\mu = 0.05$	4.58(3)	
Silver(I) ion	Ag ⁺ hydrolysis	> 11.1	
Sodium ion	Na ⁺ (aquo)	14.67(10)	
Strontium ion	Sr ²⁺ (aquo)	13.18	
Terbium(III) ion	Tb ³⁺ hydrolysis $\mu = 0.3$	8.16	
Thallium(I) ion	Tl ⁺	13.36(15)	
Thallium(III) ion	Tl ³⁺ hydrolysis $\mu = 3$	1.14	
Thorium(IV) ion	Th ⁴⁺ hydrolysis $\mu = 0.5$	3.89	4.20
Tin(II) ion	Sn ²⁺ hydrolysis $\mu = 3$	3.81(10)	
Titanium(III)	Ti ³⁺ hydrolysis $\mu = 3$	2.55	
Titanium(IV)	TiO ₂ ²⁺ + H ₂ O = TiO(OH) ⁺ + H	1.3	
Tritium oxide	pK _w for T ₂ O = T ⁺ + OH ⁻	15.21	
Uranium(IV) ion	U ⁴⁺ hydrolysis	0.68	
Uranyl(VI) ion	UO ₂ ²⁺ $\mu = 0.035$	5.82	
Vanadium(II) ion	V ²⁺ hydrolysis	6.85	
Vanadium(III) ion	V ³⁺ hydrolysis	2.92	3.5
Vanadyl(IV) ion	VO ²⁺ hydrolysis	6.86(10)	
Vanadyl(V) ion	VO ₂ ⁺ (20°C) $\mu = 0.1$	1.83	
Xenon trioxide	XeO ₃ + H ₂ O = HXeO ₄ ⁻ + H ⁺	10.5	
Ytterbium(III) ion	Yb ³⁺ hydrolysis	7.99(6)	
Yttrium(III) ion	Y ³⁺ hydrolysis $\mu = 0.3$	8.34	
Zinc ion	Zn ²⁺ hydrolysis	8.96	
Zirconium(IV) ion	Zr ⁴⁺ hydrolysis $\mu = 1$	-0.32	0.06
		pK ₃ 0.35	

If a desired organic acid is not entered in Table 8.8, a useful estimate of its pK_a value can sometimes be obtained by making a comparison with recognizably similar compounds for which pK_a values are known: (1) alkyl chains, alicyclic rings, or saturated carbocyclic rings fused to aromatic or heterocyclic rings can be replaced by methyl or ethyl groups; (2) acid-strengthening inductive and mesomeric effects of a nitro group attached to an aromatic ring are very similar to those of a nitrogen atom located at the same position in a heteroaromatic ring (e.g., 3-hydroxypyridine and 3-nitrophenol).

Hammett and Taft substituent constants and, in particular, Tables 9.1 through 9.4 may also prove useful for estimating pK_a values.

8.2.1.1 Calculation of the Approximate pH Value of Solutions

Strong acid: $pH = -\log [\text{acid}]$

Strong base: $pH = 14.00 + \log [\text{base}]$

Weak acid: $pH = \frac{1}{2}pK_a - \frac{1}{2} \log [\text{acid}]$

Weak base: $pH = 14.00 - \frac{1}{2}pK_b + \frac{1}{2} \log [\text{base}]$

Salt formed by a weak acid and a strong base:

$$pH = 7.00 + \frac{1}{2}pK_a + \frac{1}{2} \log [\text{salt}]$$

Acid salts of a dibasic acid:

$$pH = \frac{1}{2}pK_1 + \frac{1}{2}pK_2 - \frac{1}{2} \log [\text{salt}] + \frac{1}{2} \log (K_1 + [\text{salt}])$$

Buffer solution consisting of a mixture of a weak acid and its salt:

$$pH = pK_a + \log \left(\frac{[\text{salt}] + [\text{H}_3\text{O}^+] - [\text{OH}^-]}{[\text{acid}] - [\text{H}_3\text{O}^+] + [\text{OH}^-]} \right)$$

8.2.1.2 Calculation of Concentrations of Species Present at a Given pH

$$\begin{aligned}\alpha_0 &= \frac{[\text{H}^+]^n}{[\text{H}^+]^n + K_1[\text{H}^{+}]^{n-1} + K_1K_2[\text{H}^{+}]^{n-2} + \cdots + K_1K_2 \cdots K_n} = \frac{[\text{H}_n\text{A}]}{C_{\text{acid}}} \\ \alpha_1 &= \frac{K_1[\text{H}^+]^{n-1}}{[\text{H}^+]^n + K_1[\text{H}^{+}]^{n-1} + K_1K_2[\text{H}^{+}]^{n-2} + \cdots + K_1K_2 \cdots K_n} = \frac{[\text{H}_{n-1}\text{A}^-]}{C_{\text{acid}}} \\ \alpha_2 &= \frac{K_1K_2[\text{H}^+]^{n-2}}{[\text{H}^+]^n + K_1[\text{H}^{+}]^{n-1} + K_1K_2[\text{H}^{+}]^{n-2} + \cdots + K_1K_2 \cdots K_n} = \frac{[\text{H}_{n-2}\text{A}^{2-}]}{C_{\text{acid}}} \\ &\vdots \\ \alpha_n &= \frac{K_1K_2 \cdots K_n}{[\text{H}^+]^n + K_1[\text{H}^{+}]^{n-1} + K_1K_2[\text{H}^{+}]^{n-2} + \cdots + K_1K_2 \cdots K_n} = \frac{[\text{A}^{n-}]}{C_{\text{acid}}}\end{aligned}$$

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C

Ionic strength μ is zero unless otherwise indicated. Protonated cations are designated by (+1), (+ 2), etc., after the pK_a value; neutral species by (0), if not obvious; and negatively charged acids by (−1), (−2), etc.

Substance	pK_1	pK_2	pK_3	pK_4
Abietic acid	7.62			
Acetamide	− 0.37(+1)			
Acetamidine	1.60(+1)			
<i>N</i> -(2-Acetamido)-2-aminoethane-sulfonic acid (20°C)	6.88			
2-Acetamidobenzoic acid	3.63			
3-Acetamidobenzoic acid	4.07			
4-Acetamidobenzoic acid	4.28			
2-(Acetamido)butanoic acid	3.716			
<i>N</i> -(2-Acetamido)iminodiacetic acid (20°C)	6.62			
3-Acetamidopyridine	4.37(+1)			
Acetanilide	0.4(+1)	13.39(0) ^{40°C}		
Acetic acid	4.756			
Acetic acid- <i>d</i> (in D ₂ O)	5.32			
Acetoacetic acid (18°C)	3.58			
Acetohydrazine	3.24(+1)			
Acetone oxime	12.2			
2-Acetoxybenzoic acid (acetylsalicyclic acid)	3.48			
3-Acetoxybenzoic acid	4.00			
4-Acetoxybenzoic acid	4.38			
Acetylacetic acid (18°C)	3.58			
<i>N</i> -Acetyl- α -alanine	3.715			
<i>N</i> -Acetyl- β -alanine	4.455			
2-Acetylaminobutanoic acid	3.72			
3-Acetylaminopropionic acid	4.445			
2-Acetylbenzoic acid	4.13			
3-Acetylbenzoic acid	3.83			
4-Acetylbenzoic acid	3.70			
2-Acetylcyclohexanone	14.1			
<i>N</i> -Acetylcysteine (30°C)	9.52			
Acetylenedicarboxylic acid	1.75	4.40		
<i>N</i> -Acetylglycine	3.670			
<i>N</i> -Acetylguanidine	8.23(+1)			
<i>N</i> - α -Acetyl- L -histidine	7.08			
Acetylhydroxamic acid (20°C)	9.40			
<i>N</i> -Acetyl-2-mercaptoethylamine	9.92(SH)			
4-Acetyl- β -mercaptoisoleucine (30°C)	10.30			
2-Acetyl-1-naphthol (30°C)	13.40			
<i>N</i> -Acetylpenicillamine (30°C)	9.90			
2-Acetylphenol	9.19			
4-Acetylphenol	8.05			
2-Acetylpyridine	2.643(+1)			
3-Acetylpyridine	3.256(+1)			
4-Acetylpyridine	3.505(+1)			
Aconitine	8.11(+1)			
Acridine	5.60(+1)			
Acrylic acid	4.26			
Adenine	4.17(+1)	9.75(0)		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Adeninedeoxyriboside-5'-phosphoric acid	—	4.4	6.4	
Adenine- <i>N</i> -oxide	2.69(+1)	8.49(0)		
Adenosine	3.5(+1)	12.34(0)		
Adenosine-5'-diphosphoric acid	—	4.2(−1)	7.20(−2)	
Adenosine-2'-phosphoric acid	3.81(+1)	6.17(0)		
Adenosine-3'-phosphoric acid	3.65(0)	5.88(−1)		
Adenosine-5'-phosphoric acid	3.74(0)	6.05(−1)	13.06(−2)	
Adenosine-5'-triphosphoric acid	—	4.00(−1)	6.48(−2)	
Adipamic acid (adipic acid monoamide)	4.629			
Adipic acid	4.418	5.412		
α -Alanine	2.34(+1)	9.69(0)		
β -Alanine	3.55(+1)	10.238(0)		
α -Alanine, methyl ester ($\mu = 0.10$)	7.743(+1)			
β -Alanine, methyl ester ($\mu = 0.10$)	9.170(+1)			
<i>N</i> - D -Alanyl- α - D -alanine ($\mu = 0.1$)	3.32(+1)	8.13(0)		
<i>N</i> - L -Alanyl- α - L -alanine ($\mu = 0.1$)	3.32(+1)	8.13(0)		
<i>N</i> - L -Alanyl- α - D -alanine	3.12(+1)	8.30(0)		
<i>N</i> - α -Alanylglycine	3.11(+1)	8.11(0)		
Alanylglycylglycine	3.190(+1)	8.15(0)		
β -Alanylhistidine	2.64	6.86	9.40	
Albumin (bovine serum ($\mu = 0.15$))	10–10.3			
2-Aldoxime pyridine	3.42(+1)	10.22(0)		
Alizarin Black SN	5.79	12.8		
Alizarin-3-sulfonic acid	5.54	11.01		
Allantoin	8.96			
Allothreonine	2.108(+1)	9.096(0)		
Alloxanic acid	6.64			
Allylacetic acid	4.68			
Allylamine	9.69(+1)			
5-Allylbarbituric acid	4.78(+1)			
5-Allyl-5-(methylbutyl)barbituric acid	8.08			
2-Allylphenol	10.28			
1-Allylpiperidine	9.65(+1)			
2-Allylpropionic acid	4.72			
3-Amidotetrazoline	3.95(+1)			
2-Aminoacetamide	7.95(+1)			
Aminoacetonitrile	5.34(+1)			
9-Aminoacridine (20°C)	9.95(+1)			
4-Aminoantipyrine	4.94(+1)			
2-Aminobenzenesulfonic acid	2.459(0)			
3-Aminobenzenesulfonic acid	3.738(0)			
4-Aminobenzenesulfonic acid	3.227(0)			
2-Aminobenzoic acid	2.09(+1)	4.79(0)		
3-Aminobenzoic acid	3.07(+1)	4.79(0)		
4-Aminobenzoic acid	2.41(+1)	4.85(0)		
2-Aminobenzoic acid, methyl ester	2.36(+1)			
3-Aminobenzoic acid, methyl ester	3.58(+1)			
4-Aminobenzoic acid, methyl ester	2.45(+1)			
3-Aminobenzonitrile	2.75(+1)			
4-Aminobenzonitrile	1.74(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
4-Aminobenzophenone	2.15(+1)			
2-Aminobenzothiazole (20°C)	4.48(+1)			
2-Aminobenzoylhydrazide	1.85	3.47	12.80	
2-Aminobiphenyl	3.78(+1)			
3-Aminobiphenyl	4.18(+1)			
4-Aminobiphenyl	4.27(+1)			
4-Amino-3-bromomethylpyridine	7.47(+1)			
4-Amino-3-bromopyridine (20°C)	7.04(+1)			
2-Aminobutanoic acid	2.286(+1)	9.830(0)		
3-Aminobutanoic acid	—	10.14(0)		
4-Aminobutanoic acid	4.031(+1)	10.556(0)		
2-Aminobutanoic acid, methyl ester ($\mu = 0.1$)	7.640(+1)			
4-Aminobutanoic acid, methyl ester ($\mu = 0.1$)	9.838(+1)			
D-(+)-2-Amino-1-butanol	9.52(+1)			
3-Amino-N-butyl-3-methyl-2-butanone oxime	9.09(+1)			
4-Aminobutylphosphonic acid	2.55	7.55	10.9	
2-Amino-N-carbamoylbutanoic acid	3.886(+1)			
4-Amino-N-carbamoylbutanoic acid	4.683(+1)			
2-Amino-N-carbamoyl-2-methylpropanoic acid	4.463			
1-Amino-1-cycloheptanecarboxylic acid	2.59(+1)	10.46(0)		
1-Amino-1-cyclohexanecarboxylic acid	2.65(+1)	10.03(0)		
2-Amino-1-cyclohexanecarboxylic acid	3.56(+1)	10.21(0)		
1-Aminocyclopentane	10.65(+1)			
1-Aminocyclopropane	9.10(+1)			
10-Aminodecylphosphonic acid	—	8.0	11.25	
10-Aminodecylsulfonic acid	2.65(+1)			
1-Amino-2-di(aminomethyl)butane	3.58(+3)	8.59(+2)	9.66(+1)	
2-Amino-N,N-dihydroxyethyl-2-hydroxyl-1,3-propanediol	6.484(+1)			
2-Amino-N,N-dimethylbenzoic acid	1.63(+1)	8.42(0)		
4-Amino-2,5-dimethylphenol	5.28(+1)	10.40(0)		
4-Amino-3,5-dimethylpyridine (20°C)	9.54(+1)			
12-Aminododecanoic acid	4.648(+1)			
2-Aminoethane-1-phosphoric acid	5.838	10.64		
1-Aminoethanesulfonic acid	-0.33	9.06		
2-Aminoethanesulfonic acid	1.5	9.061		
2-Aminoethanethiol (cysteamine) ($\mu = 0.01$)	8.23(+1)			
2-Aminoethanol (ethanolamine)	9.50(+1)			
2-[2-(2-Aminoethyl)amino-ethyl]pyridine	3.50	6.59	9.51	
2-Amino-2-ethyl-1-butanol	9.82(+1)			
3-(2-Aminoethyl)indole	—	10.2		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
3-Amino- <i>N</i> -ethyl-3-methyl-2-butanone oxime	9.23(+1)			
<i>N</i> -(2-Aminoethyl)morpholine	4.06(+2)	9.15(+1)		
<i>p</i> -(2-Aminoethyl)phenol	9.3	10.9		
2-Aminoethylphosphonic acid	2.45(+1)	7.0(0)	10.8(−1)	
<i>N</i> -(2-Aminoethyl)piperidine (30°C)	6.38	9.89		
2-(2-Aminoethyl)pyridine ($\mu = 0.5$)	4.24(+2)	9.78(+1)		
4-Amino-3-ethylpyridine (20°C)	9.51(+1)			
<i>N</i> -(2-Aminoethyl)pyrrolidine (30°C)	6.56(+2)	9.74(+1)		
2-Aminofluorine	10.34(+1)			
2-Amino- D - β -glucose ($\mu = 0.05$)	2.20(+1)	9.08(0)		
2-Amino- <i>N</i> -glycylbutanoic acid	3.155(+1)	8.331(0)		
7-Aminoheptanoic acid	4.502			
2-Aminohexanoic acid	2.335(+1)	9.834(0)		
6-Aminohexanoic acid	4.373(+1)	10.804(0)		
<i>C</i> -Amino- <i>C</i> -hydrazinocarbonylmethane	2.38(+2)	7.69(+1)		
2-Amino-3-hydroxybenzoic acid	2.5(+1)	5.192(0)	10.118(OH)	
L -2-Amino-3-hydroxybutanoic acid (threonine)	2.088(+1)	9.100(0)		
DL -2-Amino-4-hydroxybutanoic acid ($\mu = 0.1$)	2.265(+1)	9.257(0)		
DL -4-Amino-3-hydroxybutanoic acid ($\mu = 0.1$)	3.834(+1)	9.487(0)		
2-Amino-2'-hydroxydiethyl sulfide	9.27(+1)			
4-Amino-2-hydroxypyrimidine (cytosine)	4.58(+1)	12.15(0)		
3-Amino- <i>N</i> -isopropyl-3-methyl-2-butanone oxime	9.09(+1)			
4-Amino-3-isopropylpyridine (20°C)	9.54(+1)			
1-Aminoisoquinoline (20°C, $\mu = 0.01$)	7.62(+1)			
3-Aminoisoquinoline (20°C, $\mu = 0.005$)	5.05(+1)			
4-Aminoisoxazolidine-3-one	7.4(+1)			
Aminomalonic acid	3.32(+1)	9.83(0)		
DL -2-Amino-4-mercaptobutanoic acid	2.22(+1)	8.87(0)	10.86(SH)	
2-Amino-3-mercapto-3-Methylbutanoic acid	1.8(+1)	7.9(0)	10.5(SH)	
2-Amino-6-methoxybenzothiazole	4.50(+1)			
3-Amino-4-methylbenzenesulfonic acid	3.633			
4-Amino-3-methylbenzenesulfonic acid	3.125			
2-Amino-4-methylbenzothiazole	4.7(+1)			
1-Amino-3-methylbutane	10.64(+1)			
3-Amino-3-methyl-2-butanone oxime	9.09(+1)			
3-Amino- <i>N</i> -methyl-3-methyl-2-butanone oxime	9.23(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Amino-3-methylpentanoic acid	2.320(+1)	9.758(0)	10.8	
3-Aminomethyl-6-methylpyridine (30°C)	8.70(+1)			
Aminomethylphosphonic acid	2.35	5.9		
2-Amino-2-methyl-1,3-propanediol	8.801			
2-Amino-2-methyl-1-propanol	9.694(+1)			
2-Amino-2-methylpropanoic acid	2.357(+1)	10.205(0)		
(2-Aminomethyl(pyridine ($\mu = 0.5$))	2.31(+2)	8.79(+1)		
2-Amino-3-methylpyridine	7.24(+1)			
4-Amino-3-methylpyridine	9.43(+1)			
2-Amino-4-methylpyridine	7.48(+1)			
2-Amino-5-methylpyridine	7.22(+1)		8.66	
2-Amino-6-methylpyridine	7.41(+1)			
2-Amino-4-methylpyrimidine (20°C)	4.11(+1)			
Aminomethylsulfonic acid	5.57(+1)			
N-Aminomorpholine	4.19(+1)			
4-Amino-1-naphthalenesulfonic acid	2.81			
1-Amino-2-naphthalenesulfonic acid	1.71			
1-Amino-3-naphthalenesulfonic acid	3.20			
1-Amino-5-naphthalenesulfonic acid	3.69			
1-Amino-6-naphthalenesulfonic acid	3.80			
1-Amino-7-naphthalenesulfonic acid	3.66			
1-Amino-8-naphthalenesulfonic acid	5.03			
2-Amino-1-naphthalenesulfonic acid	2.35			
2-Amino-4-naphthalenesulfonic acid	3.79			
2-Amino-6-naphthalenesulfonic acid	3.79	8.94		
2-Amino-8-naphthalenesulfonic acid	3.89			
3-Amino-1-naphthoic acid	2.61	4.39		
4-Amino-2-naphthoic acid	2.89	4.46		
8-Amino-2-naphthol	4.20(+1)			
DL-2-Aminopentanoic acid (DL-norvaline)	2.318(+1)	9.808		
3-Aminopentanoic acid	4.02(+1)	10.399(0)		
4-Aminopentanoic acid	3.97(+1)	10.46(0)		
5-Aminopentanoic acid	4.20(+1)	9.758(0)		
5-Aminopentanoic acid, ethyl ester	10.151			
2-Aminophenol	9.28	9.72		
3-Aminophenol	9.83	9.87		
4-Aminophenol	8.50	10.30		
4-Aminophenylacetic acid (20°C)	3.60	5.26		
2-Aminophenylarsonic acid	ca 2	3.77		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
3-Aminophenylarsonic acid	ca 2	4.02	8.92	9.82(−2)
4-Aminophenylarsonic acid	ca 2	4.02	8.62	
3-Aminophenylboric acid	4.46	8.81		
4-Aminophenylboric acid	3.71	9.17		
4-Aminophenyl (4-chlorophenyl) sulfone	1.38			
2-Aminophenylphosphonic acid	—	4.10	7.29	
3-Aminophenylphosphonic acid	—	—	7.16	
4-Aminophenylphosphonic acid	—	—	7.53	
1-Amino-1,2,3-propanetricarboxylic acid ($\mu = 2.2$)	2.10(+1)	3.60(0)	4.60(−1)	
3-Aminopropanoic acid	3.551(+1)	10.235(0)		
1-Amino-1-propanol	9.96(+1)			
DL-2-Amino-1-propanol	9.469(+1)			
3-Amino-1-propanol	9.96(+1)			
3-Aminopropene	9.691(+1)			
3-Amino- <i>N</i> -propyl-3-methyl-2-butanone oxime	9.09(+1)			
2-Aminopropylsulfonic acid	—	9.15		
2-Aminopyridine	6.71(+1)			
3-Aminopyridine	6.03(+1)			
4-Aminopyridine	9.114(+1)			
2-Aminopyridine-1-oxide	2.58(+1)			
3-Aminopyridine-1-oxide	1.47(+1)			
4-Aminopyridine-1-oxide	3.54(+1)			
8-Aminoquinaldine	4.86(+1)			
2-Aminoquinoline (20°C, $\mu = 0.01$)	7.34(+1)			
3-Aminoquinoline (20°C, $\mu = 0.01$)	4.95(+1)			
4-Aminoquinoline (20°C, $\mu = 0.01$)	9.17(+1)			
5-Aminoquinoline (20°C, $\mu = 0.01$)	5.46(+1)			
6-Aminoquinoline (20°C, $\mu = 0.01$)	5.63(+1)			
8-Aminoquinoline (20°C, $\mu = 0.01$)	3.99(+1)			
4-Aminosalicyclic acid	1.991(+1)	3.917(0)	13.74	
5-Aminosalicyclic acid	2.74(+1)	5.84(0)		
2-Amino-3-sulfopropanoic acid	1.89(+1)	8.70(0)		
4-Amino-2,3,5,6-tetramethylpyridine (20°C)	10.58(+1)			
5-Amino-1,2,3,4-tetrazole (20°C)	1.76	6.07		
2-Aminothiazole (20°C)	5.36(+1)			
1-Amino-3-thiobutane (30°C)	9.18(+1)			
5-Amino-3-thio-1-pentanol (30°C)	9.12(+1)			
2-Aminothiophenol	<2(+1)	7.90(0)		
2-Amino-4,4,4-trifluorobutanoic acid		8.171(0)		
3-Amino-4,4,4-trifluorobutanoic acid		5.831(0)		
3-Amino-2,4,6-trinitrobenzene		9.5(+1)		
Angiotensin II	10.37			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
Anhydroplatynecine	9.40			
Aniline	4.60(+1)			
2-Anilinoethylsulfonic acid	3.80(+1)			
3-Anilinoethylsulfonic acid	4.85(+1)			
Anthracene-1-carboxylic acid	3.68			
Anthracene-2-carboxylic acid	4.18			
Anthracene-9-carboxylic acid	3.65			
Anthraquinone-1-carboxylic acid (20°C)	3.37			
Anthraquinone-2-carboxylic acid (20°C)	3.42			
9,10-Anthraquinone monoxime	9.78			
9,10-Anthraquinone-1-sulfonic acid	0.27			
9,10-Anthraquinone-2-sulfonic acid	0.38			
Antipyrine	1.45(+1)			
Apomorphine (15°C)		8.92		
D-(−)-Arabinose	12.34			
L-(+)-Arginine	2.17	9.04(+1)	12.47(−1)	
Arsenazo III [pK_5 10.5(−4); pK_6 12.0(−5)]		1.2	2.7	7.9(−3)
Arsenoacetic acid		4.67	7.68	
Arsenoacrylic acid		4.23	8.60	
Arsenobutanoic acid		4.92	7.64	
2-Arsenocrotonic acid		4.61	8.75	
3-Arsenocrotonic acid		4.03	8.81	
Arsenopentanoic acid		4.89	7.75	
L-(+)-Ascorbic acid (vitamin C)	4.17	11.57		
L-(+)-Asparagine	2.01(0)	8.80(+1)		
L-Asparaginyglycine		4.53	9.07	
D-Aspartic acid	1.89(0)	3.65	9.60	
Aspartic diamide ($\mu = 0.2$)	7.00			
Aspartylaspartic acid		3.40	4.70	8.26
α -Aspartylhistidine (38°C, $\mu = 0.1$)		3.02	6.82	7.98
β -Aspartylhistidine (38°C, $\mu = 0.1$)		2.95	6.93	8.72
N-Aspartyl-p-tyrosine ($\mu = 0.01$)		3.57	8.92	10.23(OH)
Aspidospermine	7.65			
Atropine (17°C)	4.35(+1)			
1-Azacycloheptane	11.11(+1)			
1-Azacyclooctane	11.1(+1)			
Azetidine	11.29(+1)			
Aziridine	8.04(+1)			
Barbituric acid		8.372(0)		
m-Benzetaine	3.217(+1)			
p-Benzetaine	3.245(+1)			
Benzenearsonic acid (22°C)		8.48(−1)		
Benzene-1-arsonic acid-4-carboxylic acid		4.22 (COOH)	5.59	
Benzeneboronic acid	13.7			
Benzene-1-carboxylic acid-2-phosphoric acid		3.78	9.17	
Benzene-1-carboxylic acid-3-phosphoric acid		4.03	7.03	

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Benzene-1-carboxylic acid-4-phosphoric acid	1.50	3.95	6.89	
Benzenediazine	11.08(+1)			
1,3-Benzenedicarboxylic acid (isophthalic acid)	3.62(0)	4.60(−1)		
1,4-Benzenedicarboxylic acid (terephthalic acid)	3.54(0)	4.46(−1)		
1,3-Benzenedicarboxylic acid mononitrile	3.60(0)			
1,4-Benzenedicarboxylic acid mononitrile	3.55(0)			
Benzenhexacarboxylic acid (pK_5 6.32; pK_6 7.49)	0.68	2.21	3.52	5.09
Benzenepentacarboxylic acid (pK_5 6.46)	1.80	2.73	3.96	5.25
Benzenesulfinic acid	1.50			
Benzenesulfonic acid	2.554			
1,2,3,4-Benzenetetracarboxylic acid	2.05	3.25	4.73	6.21
1,2,3,5-Benzenetetracarboxylic acid	2.38	3.51	4.44	5.81
1,2,4,5-Benzenetetracarboxylic acid	1.92	2.87	4.49	5.63
1,2,3-Benzenetricarboxylic acid	2.88	4.75	7.13	
1,2,4-Benzenetricarboxylic acid	2.52	3.84	5.20	
1,3,5-Benzenetricarboxylic acid	2.12	4.10	5.18	
Benzil- α -dioxime	12.0			
Benzilic acid	3.09			
Benzimidazole	5.53(+1)	12.3(0)		
Benzohydroxamic acid (20°C)	8.89(0)			
Benzoic acid	4.204			
5,6-Benzoquinoline (20°C)	5.00(+1)			
7,8-Benzoquinoline (20°C)	4.15(+1)			
1,4-Benzoquinone monoxime	6.20			
Benzosulfonic acid	0.70			
1,2,3-Benzotriazole	8.38(+1)			
1-Benzoylacetone	8.23			
Benzoylamine	9.34(+1)			
2-Benzoylbenzoic acid	3.54			
Benzoylglutamic acid	3.49	4.99		
N-Benzoyglycine (hippuric acid)	3.65			
Benzoylhydrazine	3.03(+2)	12.45(+1)		
Benzoylpyruvic acid	6.40	12.10		
3-Benzoyl-1,1,1-trifluoroacetone	6.35			
Benzylamine	9.35(+1)			
Benzylamine-4-carboxylic acid	3.59	9.64		
2-Benzyl-2-phenylsuccinic acid (20°C)	3.69	6.47		
2-Benzylpyridine	5.13(+1)			
4-Benzylpyridine-1-oxide	−1.018(+1)			
1-Benzylpyrrolidine	9.51(+1)			
2-Benzylpyrrolidine	10.31(+1)			
Benzylsuccinic acid (20°C)	4.11	5.65		
3-(Benzylthio)propanoic acid	4.463			
Berberine (18°C)	11.73(+1)			
Betaine	1.832(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
Biguanide	2.96(+2)	11.51(+1)		
2,2'-Biimidazolyl ($\mu = 0.3$)	5.01(+1)			
2-Biphenylcarboxylic acid	3.46			
(1,1'-Biphenyl)-4,4'-diamine	3.63(+2)	4.70(+1)		
Bis(2-aminoethyl) ether (30°C)	8.62(+2)	9.59(+1)		
<i>N,N'</i> -Bis(2-aminoethyl)-ethylenedi- amine (20°C)	3.32(+4)	6.67(+3)	9.20(+2)	9.92(+1)
<i>N,N</i> -Bis(2-hydroxyethyl)-2-ami- noethane sulfonic acid (BES) (20°C)	7.15			
<i>N,N</i> -Bis(2-hydroxyethyl)glycine (bicine) (20°C)	8.35			
Bis(2-hydroxyethyl)iminotris (hy- droxymethyl)methane (bis-tris)	6.46(+1)			
1,3-Bis[tris(hydroxymethyl)methyl- lamino]propane (20°C)	6.80(+1)			
Bromoacetic acid	2.902			
2-Bromoaniline	2.53(+1)			
3-Bromoaniline	3.53(+1)			
4-Bromoaniline	3.88(+1)			
2-Bromobenzoic acid	2.85			
3-Bromobenzoic acid	3.810			
4-Bromobenzoic acid	3.99			
2-Bromobutanoic acid (35°C)	2.939			
<i>erythro</i> -2-Bromo-3-chlorosuccinic acid (19°C, $\mu = 0.1$)	1.4	2.6		
<i>threo</i> -2-Bromo-chlorosuccinic acid (19°C, $\mu = 0.1$)	1.5	2.8		
<i>trans</i> -2-Bromocinnamic acid	4.41			
3-Bromo-4-(dimethylam- ino)pyridine (20°C)	6.52(+1)			
2-Bromo-4,6-dinitroaniline	-6.94(+1)			
3-Bromo-2-hydroxymethylbenzoic acid (20°C)	3.28			
6-Bromo-2-hydroxymethylbenzoic acid (20°C)	2.25			
7-Bromo-8-hydroxyquinoline-5- sulfonic acid	2.51	6.70		
3-Bromomandelic acid	3.13			
3-Bromo-4-methylaminopyridine (20°C)	7.49(+1)			
(2-Bromomethyl)butanoic acid	3.92			
Bromomethylphosphonic acid	1.14	6.52		
2-Bromo-6-nitrobenzoic acid	1.37			
2-Bromophenol	8.452			
3-Bromophenol	9.031			
4-Bromophenol	9.34			
2-(2'-Bromophenoxy)acetic acid	3.12			
2-(3'-Bromophenoxy)acetic acid	3.09			
2-(4'-Bromophenoxy)acetic acid	3.13			
2-Bromo-2-phenylacetic acid	2.21			
2-(Bromophenyl) acetic acid	4.054			
4-(Bromophenyl)acetic acid	4.188			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
4-Bromophenylarsonic acid	3.25	8.19		
4-Bromophenylphosphinic acid (17°C)	2.1			
2-Bromophenylphosphonic acid	1.64	7.00		
3-Bromophenylphosphonic acid	1.45	6.69		
4-Bromophenylphosphonic acid	1.60	6.83		
3-Bromophenylselenic acid	4.43			
4-Bromophenylselenic acid	4.50			
2-Bromopropanoic acid	2.971			
3-Bromopropanoic acid	3.992			
Bromopropynoic acid	1.855			
2-Bromopyridine	0.71(+1)			
3-Bromopyridine	2.85(+1)			
4-Bromopyridine	3.71(+1)			
3-Bromoquinoline	2.69(+1)			
Bromosuccinic acid	2.55	4.41		
2-Bromo- <i>p</i> -tolylphosphonic acid	1.81	7.15		
Brucine (15°C)	2.50(+2)	8.16(+1)		
2-Butanamine (<i>sec</i> -butylamine)	10.56(+1)			
1,2-Butanediamine	6.399(+2)	9.388(+1)		
1,4-Butanediamine	9.35(+2)	10.82(+1)		
2,3-Butanediamine	6.91(+2)	10.00(+1)		
1,2,3,4-Butanetetra-carboxylic acid	3.43	4.58	5.85	7.16
<i>cis</i> -2-Butenoic acid (<i>isocrotonic</i> acid)	4.44			
<i>trans</i> -2-Butenoic acid (<i>trans-cro-</i> <i>tonic</i> acid) (35°C)	4.676			
3-Butenoic acid (<i>vinylacetic</i> acid)	4.68			
3-Butoxybenzoic acid (20°C)	4.25			
Butylamine	10.64(+1)			
<i>tert</i> -Butylamine	10.685(+1)			
4- <i>tert</i> -Butylaniline	3.78(+1)			
<i>N-tert</i> -Butylaniline	7.10(+1)			
Butylarsonic acid (18°C)	4.23	8.91		
2- <i>tert</i> -Butylbenzoic acid	3.57			
3- <i>tert</i> -Butylbenzoic acid	4.199			
4- <i>tert</i> -Butylbenzoic acid	4.389			
<i>N</i> -Butylethylenediamine	7.53(+2)	10.30(+1)		
<i>N</i> -Butylglycine	2.35(+1)	10.25(0)		
<i>tert</i> -Butylhydroperoxide	12.80			
1-(<i>tert</i> -Butyl)-2-hydroxybenzene	10.62			
1-(<i>tert</i> -Butyl)-3-hydroxybenzene	10.119			
1-(<i>tert</i> -Butyl)-4-hydroxybenzene	10.23			
Butylmethylamine	10.90(+1)			
2-Butyl-1-methyl-2-pyrroline	11.84(+1)			
4- <i>tert</i> -Butylphenylactic acid	4.417			
Butylphosphinic acid	3.41			
<i>tert</i> -Butylphosphinic acid	4.24			
<i>tert</i> -Butylphosphonic acid	2.79	8.88		
1-Butylpiperidine ($\mu = 0.02$)	10.43(+1)			
2- <i>tert</i> -Butylpyridine	5.76(+1)			
3- <i>tert</i> -Butylpyridine	5.82(+1)			
4- <i>tert</i> -Butylpyridine	5.99(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
2- <i>tert</i> -Butylthiazole ($\mu = 0.1$)	3.00(+1)			
4- <i>tert</i> -Butylthiazole ($\mu = 0.1$)	3.04(+1)			
2-Butyn-1,4-dioic acid	1.75	4.40		
2-Butynoic acid (tetrolic acid)	2.620			
Butyric acid	4.817			
4-Butyrobetaine (20°C)	3.94(+1)			
Caffeine (40°C)	10.4			
Calcein ($pK_5 > 12$)	<4		9.0	10.5
Calmagite	8.14	12.35		
D-Camphoric acid	4.57	5.10		
Canaline	2.40	3.70	9.20	
Canavanine	2.50(+2)	6.60(+1)	9.25(0)	
N-Carbamoylacetic acid	3.64			
N-Carbamoyl- α -D-alanine	3.89(+1)			
N-Carbamoyl- β -alanine	4.99(+1)			
DL-N-Carbamoylalanine	3.892(+1)			
N-Carbamoylglycine	3.876			
2-Carbamoylpyridine (20°C)	2.10(+1)			
3-Carbamoylpyridine	3.328(+1)			
4-Carbamoylpyridine (20°C)	3.61(+1)			
β -Carboxymethylaminopropanoic acid	3.61(+1)	9.46(0)		
Chloroacetic acid	2.867			
N-(2'-Chloroacetyl)glycine	3.38(0)			
cis-3-Chloroacrylic acid (18°C, $\mu = 0.1$)	3.32			
trans-3-chloroacrylic acid (18°C, $\mu = 0.1$)	3.65			
2-Chloroaniline	2.64(+1)			
3-Chloroaniline	3.52(+1)			
4-Chloroaniline	3.99(+1)			
2-Chlorobenzoic acid	2.877			
3-Chlorobenzoic acid	3.83			
4-Chlorobenzoic acid	3.986			
2-Chlorobutanoic acid	2.86			
3-Chlorobutanoic acid	4.05			
4-Chlorobutanoic acid	4.50			
2-Chloro-3-butenic acid	2.54			
3-Chlorobutylarsonic acid (18°C)	3.95	8.85		
trans-2'-Chlorocinnamic acid	4.234			
trans-3'-Chlorocinnamic acid	4.294			
trans-4'-Chlorocinnamic acid	4.413			
2-Chlorocrotonic acid	3.14			
3-Chlorocrotonic acid	3.84			
Chlorodifluoroacetic acid	0.46			
1-Chloro-1,2-dihydroxybenzene	8.522			
1-Chloro-2,6-dimethyl-4-hydroxy-benzene	9.549			
4-Chloro-2,6-dinitrophenol	2.97			
2-Chloroethylarsonic acid	3.68	8.37		
3-Chlorohexyl-1-arsonic acid (18°C)	3.51	8.31		
2-Chloro-3-hydroxybutanoic acid	2.59			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
3-Chloro-2-(hydroxy-methyl)benzoic acid (20°C)	3.27			
6-Chloro-2-(hydroxy-methyl)benzoic acid (20°C)	2.26			
7-Chloro-8-hydroxyquinoline-5-sulfonic acid	2.92	6.80		
2-Chloroisocrotonic acid	2.80			
3-Chloroisocrotonic acid	4.02			
3-Chlorolactic acid	3.12			
3-Chloromandelic acid	3.237			
3-Chloro-4-methoxyphenyl-phosphonic acid	2.25	6.7		
3-Chloro-4-methylaniline	4.05(+1)			
4-Chloro- <i>N</i> -methylaniline	3.9(+1)			
4-Chloro-3-methylphenol	9.549			
Chloromethylphosphonic acid	1.40	6.30		
2-Chloro-2-methylpropanoic acid	2.975			
2-Chloro-6-nitroaniline	-2.41(+1)			
4-Chloro-2-nitroaniline	-1.10(+1)			
2-Chloro-3-nitrobenzoic acid	2.02			
2-Chloro-4-nitrobenzoic acid	1.96			
2-Chloro-5-nitrobenzoic acid	2.17			
2-Chloro-6-nitrobenzoic acid	1.342			
4-Chloro-2-nitrophenol	6.48			
2-Chlorophenol	8.55			
3-Chlorophenol	9.10			
4-Chlorophenol	9.43			
(4-Chloro-3-nitrophenoxy)acetic acid	2.959			
2-Chloro-4-nitrophenylphosphonic acid	1.12	6.14		
3-Chloropentyl-1-arsonic acid (18°C)	3.71	8.77		
2-Chlorophenoxyacetic acid	3.05			
3-Chlorophenoxyacetic acid	3.07			
4-Chlorophenoxyacetic acid	3.10			
4-Chlorophenoxy-2-methylacetic acid	3.26			
2-Chlorophenylacetic acid	4.066			
3-Chlorophenylacetic acid	4.140			
4-Chlorophenylacetic acid	4.190			
2-Chlorophenylalanine	2.23(+1)	8.94(0)		
3-Chlorophenylalanine	2.17(+1)	8.91(0)		
DL -4-Chlorophenylalanine	2.08(+1)	8.96(0)		
4-Chlorophenylarsonic acid	3.33	8.25		
2-Chlorophenylphosphonic acid	1.63	6.98		
3-Chlorophenylphosphonic acid	1.55	6.65		
4-Chlorophenylphosphonic acid	1.66	6.75		
3-(2'-Chlorophenyl)propanoic acid	4.577			
3-(3'-Chlorophenyl)propanoic acid	4.585			
3-(4'-Chlorophenyl)propanoic acid	4.607			
3-Chlorophenylselenic acid	4.47			
4-Chlorophenylselenic acid	4.48			
4-Chloro-1,2-phthalic acid	1.60			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Chloropropanoic acid	2.84			
3-Chloropropanoic acid	3.992			
2-Chloropropylarsonic acid (18°C)	3.76	8.39		
3-Chloropropylarsonic acid (18°C)	3.63	8.53		
Chloropropynoic acid	1.854			
2-Chloropyridine	0.49(+1)			
3-Chloropyridine	2.84(+1)			
4-Chloropyridine	3.83(+1)			
7-Chlorotetracycline	3.30(+1)	7.44	9.27	
4-Chloro-2-(2'-thiazolylazo)phenol	7.09			
4-Chlorothiophenol	5.9			
<i>N</i> -Chloro- <i>p</i> -toluenesulfonamide	4.54(+1)			
3-Chloro- <i>o</i> -toluidine	2.49(+1)			
4-Chloro- <i>o</i> -toluidine	3.385(+1)			
5-Chloro- <i>o</i> -toluidine	3.85(+1)			
6-Chloro- <i>o</i> -toluidine	3.62(+1)			
Chrome Azurol S	2.45	4.86	11.47	
Chrome Dark Blue	7.56	9.3	12.4	
Cinchonine	5.85(+2)	9.92(+1)		
<i>cis</i> -Cinnamic acid	3.879			
<i>trans</i> -Cinnamic acid	4.438			
Citraconic acid	2.29(0)	6.15(−1)		
Citric acid	3.128	4.761	6.396	
L-(+)-Citrulline	2.43(+1)	9.41(0)		
Cocaine	8.41(+1)			
Codeine	7.95(+1)			
Colchicine	1.65(+1)			
Coniine ($\mu = 0.5$)	11.24(+1)			
Creatine (40°C)	3.28(+1)			
Creatinine	3.57(+1)			
<i>o</i> -Cresol	10.26			
<i>m</i> -Cresol	10.00			
<i>p</i> -Cresol	10.26			
Cumene hydroperoxide	12.60			
Cupreine	7.63(+1)			
Cyanamide	10.27			
Cyanoacetic acid	2.460			
Cyanoacetohydrazide	2.34(+2)	11.17(+1)		
2-Cyanobenzoic acid	3.14			
3-Cyanobenzoic acid	3.60			
4-Cyanobenzoic acid	3.55			
4-Cyanobutanoic acid	4.44			
<i>trans</i> -1-Cyanocyclohexane-2-carboxylic acid	3.865			
4-Cyano-2,6-dimethylphenol	8.27			
4-Cyano-3,5-dimethylphenol	8.21			
2-Cyanoethylamine	7.7(+1)			
<i>N</i> -(2-Cyano)ethylnorcodeine	5.68(+1)			
Cyanomethylamine	5.34(+1)			
2-Cyano-2-methyl-2-phenylacetic acid	2.290			
1-Cyanomethylpiperidine	4.55(+1)			
2-Cyano-2-methylpropanoic acid	2.422			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
3-Cyanophenol	8.61			
<i>o</i> -Cyanophenoxyacetic acid	2.98			
<i>m</i> -Cyanophenoxyacetic acid	3.03			
<i>p</i> -Cyanophenoxyacetic acid	2.93			
2-Cyanopropanoic acid	2.37			
3-Cyanopropanoic acid	3.99			
2-Cyanopyridine	-0.26(+1)			
3-Cyanopyridine	1.45(+1)			
4-Cyanopyridine	1.90(+1)			
Cyanuric acid	6.78			
Cyclobutanecarboxylic acid	4.785			
1,1-Cyclobutanedicarboxylic acid	3.13	5.88		
<i>cis</i> -1,2-Cyclobutanedicarboxylic acid	3.90	5.89		
<i>trans</i> -1,2-Cyclobutanedicarboxylic acid	3.79	5.61		
<i>cis</i> -1,3-Cyclobutanedicarboxylic acid	4.04	5.31		
<i>trans</i> -1,3-Cyclobutanedicarboxylic acid	3.81	5.28		
Cyclohexanecarboxylic acid	4.90			
1,1-Cyclohexanediactic acid	3.49	6.96		
<i>cis</i> -1,2-Cyclohexanediactic acid (20°C)	4.42	5.45		
<i>trans</i> -1,2-Cyclohexanediactic acid (20°C)	4.38	5.42		
<i>cis</i> -1,2-Cyclohexanediamine	6.43(+2)	9.93(+1)		
<i>trans</i> -1,2-Cyclohexanediamine	6.34(+2)	9.74(+1)		
1,1-Cyclohexanedicarboxylic acid	3.45	4.11		
<i>cis</i> -1,2-Cyclohexanedicarboxylic acid (20°C)	4.34	6.76		
<i>trans</i> -1,2-Cyclohexanedicarboxylic acid (20°C)	4.18	5.93		
<i>cis</i> -1,3-Cyclohexanedicarboxylic acid (16°C)	4.10	5.46		
<i>trans</i> -1,3-Cyclohexanedicarboxylic acid (19°C)	4.31	5.73		
<i>trans</i> -1,4-Cyclohexanedicarboxylic acid (16°C)	4.18	5.42		
1,3-Cyclohexanedione	5.26			
<i>cis,cis</i> -1,3,5-Cyclohexanetriamine	6.9(+3)	8.7(+2)	10.4(+1)	
Cyclohexanimine	9.15			
<i>cis</i> -4-Cyclohexene-1,2-dicarboxylic acid (20°C)	3.89	6.79		
<i>trans</i> -4-Cyclohexene-1,2-dicarboxylic acid (20°C)	3.95	5.81		
Cyclohexylacetic acid	4.51			
Cyclohexylamine	10.64(+1)			
2-(Cyclohexylamino)ethanesulfonic acid (CHES) (20°C)	9.55			
3-Cyclohexylamino-1-propanesulfonic acid (CAPS) (20°C)	10.40			
4-Cyclohexylbutanoic acid	4.95			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
Cyclohexylcyanoacetic acid	2.367			
1,2-Cyclohexylenedinitriloacetic acid ($\mu = 0.1$)	2.4	3.5	6.16	12.35
3-Cyclohexylpropanoic acid	4.91			
2-Cyclohexylpyrrolidine	10.76(+1)			
2-Cyclohexyl-2-pyrroline	7.91(+1)			
Cyclohexylthioacetic acid	3.488			
Cyclopentanecarboxylic acid	4.905			
cis-Cyclopentane-1-carboxylic acid-2-acetic acid	4.40	5.79		
trans-Cyclopentane-1-carboxylic acid-2-acetic acid	4.39	5.67		
Cyclopentane-1,2-diamine- <i>N,N',N'</i> -tetraacetic acid ($\mu = 0.1$)	—	—	—	10.20
Cyclopentane-1,1-dicarboxylic acid	3.23	4.08		
cis-Cyclopentane-1,2-dicarboxylic acid	4.43	6.67		
trans-Cyclopentane-1,2-dicarboxylic acid	3.96	5.85		
cis-Cyclopentane-1,3-dicarboxylic acid	4.26	5.51		
trans-Cyclopentane-1,3-dicarboxylic acid	4.32	5.42		
Cyclopentylamine	10.65(+1)			
1,1-Cyclopentylldiacetic acid	3.80	6.77		
cis-Cyclopentyl-1,2-diacetic acid	4.42	5.42		
trans-Cyclopentyl-1,2-diacetic acid	4.43	5.43		
Cyclopropanecarboxylic acid	4.827			
Cyclopropane-1,1-dicarboxylic acid	1.82	5.43		
cis-Cyclopropane-1,2-dicarboxylic acid	3.33	6.47		
trans-Cyclopropane-1,2-dicarboxylic acid	3.65	5.13		
Cyclopropylamine	9.10(+1)			
5-Cyclopropyl-1,2,3,4-tetrazole	4.90(+1)			
L-Cysteic acid (3-sulfo-L-alanine)	1.89(+1)	8.7(0)		
L-(+)-Cysteine	1.96	8.18	10.29(SH)	
L-(+)-Cysteine, ethyl ester	6.69	9.17(SH)		
	(NH_3^+)			
L-(+)-Cysteine, methyl ester	6.56	8.99(SH)		
	(NH_3^+)			
L-Cysteinyl-L-asparagine	2.97	7.09	8.47	
L-Cystine (35°C)	1.6(+2)	2.1(+1)	8.02(0)	8.71(−1)
Cystinylglycylglycine (35°C)	3.12	3.21	6.01	6.87
Cytidine	4.08(+1)	12.24(0)		
Cytidine-2'-phosphoric acid	0.8(+1)	4.36(0)	6.17(−1)	
Cytidine-3'-phosphoric acid	0.80(+1)	4.31(0)	6.04(−1)	13.2(sugar)
Cytidine-5'-phosphoric acid	—	4.39(0)	6.62(−1)	
Cytosine	4.58(+1)	12.15(0)		
Decanedioic acid (sebacic acid)	4.59	5.59		
Dehydroascorbic acid (20°C)	3.21	7.92	10.3	
2'-Deoxyadenosine ($\mu = 0.1$)	3.8(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Deoxycholic acid	6.58			
2-Deoxyglucose	12.52			
2-Deoxyguanosine ($\mu = 0.1$)	2.5(+1)			
5-Desoxypyridoxal ($\mu = 0$)	4.17(+1)	8.14(OH)		
1,1-Diacetic acid semicarbazide (30°C, $\mu = 0.1$)	2.96	4.04		
Diacetylacetone	7.42			
Diallylamine ($\mu = 0.02$)	9.29(+1)			
5,5-Diallylbarbituric acid	7.78(0)			
1,3-Diamino-2-aminomethylpropane	6.44(+3)	8.56(+2)	10.38(+1)	
3,5-Diaminobenzoic acid	5.30			
1,3-Diamino- <i>N,N'</i> -bis-(2-aminoethyl)propane ($\mu = 0.5$)	6.01(+4)	7.26(+3)	9.49(+2)	10.23(+1)
2,4-Diaminobutanoic acid (20°C)	1.85(+2)	8.24(+1)	10.40(0)	
2,2'-Diaminodiethyl sulfide (30°C)	8.84(+2)	9.64(+1)		
1,8-Diamino-3,6-dithiooctane (30°C)	8.43(+2)	9.31(+1)		
2,7-Diaminooctanedioic acid (20°C, $\mu = 0.1$)	1.84(+2)	2.64(+1)	9.23(0)	9.89(−1)
1,8-Diamino-3,6-octanedione (30°C)	8.60(+2)	9.57(+1)		
1,8-Diamino-3-oxa-6-thiooctane	8.54(+2)	9.46(+1)		
2,3-Diaminopropanoic acid ($\mu = 0.1$)	1.33(+2)	6.674(+1)	9.623(0)	
2,3-Diaminopropanoic acid, methyl ester ($\mu = 0.1$)	4.412(+1)	8.250(0)		
1,3-Diamino-2-propanol (20°C)	7.93(+2)	9.69(+1)		
2,5-Diaminopyridine (20°C)	2.13(+2)	6.48(+1)		
1,4-Diazabicyclo[2.2.2]octane	2.90(+2)	8.60(+1)		
Dibenzylamine	8.52(+1)			
Dibenzylsuccinic acid (20°C)	3.96	6.66		
Dibromoacetic acid	1.39			
3,5-Dibromoaniline	2.35(+1)			
3,5-Dibromophenol	8.056			
2,2-Dibromopropanoic acid	1.48			
2,3-Dibromopropanoic acid	2.33			
<i>rac</i> -2,3-Dibromosuccinic acid (20°C)	1.43	2.24		
<i>meso</i> -2,3-Dibromosuccinic acid (20°C)	1.51	2.71		
3,5-Dibromo- <i>p</i> -L-tyrosine	2.17(+1)	6.45(0)	7.60(−1)	
Dibutylamine	11.25(+1)			
Di- <i>sec</i> -butylamine	10.91(+1)			
2,6-Di- <i>tert</i> -butylpyridine	3.58(+1)			
<i>rac</i> -2,3-Di- <i>tert</i> -butylsuccinic acid ($\mu = 0.1$)	3.58	10.2		
1,12-Dicarboxydodecaborane	9.07	10.23		
Dichloroacetic acid	1.26			
Dichloroacetylacetic acid	2.11			
3,5-Dichloroaniline	2.37(+1)			
1,3-Dichloro-2,5-dihydroxybenzene ($\mu = 0.65$)	7.30	9.99		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
2,5-Dichloro-3,6-dihydroxy- <i>p</i> -benzoquinone	1.09	2.42		
Dichloromethylphosphonic acid	1.14	5.61		
2,4-Dichloro-6-nitroaniline	−3.00(+1)			
2,5-Dichloro-4-nitroaniline	−1.74(+1)			
2,6-Dichloro-4-nitroaniline	−3.31(+1)			
2,3-Dichlorophenol	7.44			
2,4-Dichlorophenol	7.85			
2,6-Dichlorophenol	6.78			
3,4-Dichlorophenol	8.630			
3,5-Dichlorophenol	8.179			
2,4-Dichlorophenoxyacetic acid (2,4-D)	2.64			
4,6-Dichlorophenoxy-2-methylacetic acid	3.13			
3,6-Dichlorophthalic acid	1.46			
2,2-Dichloropropanoic acid	2.06			
2,3-Dichloropropanoic acid	2.85			
<i>rac</i> -2,3-Dichlorosuccinic acid (20°C)	1.43	2.81		
<i>meso</i> -2,3-Dichlorosuccinic acid	1.49	2.97		
3,5-Dichloro- <i>p</i> -tyrosine	2.12	6.47	7.62	
2-Dicyanoethylamine	5.14(+1)			
2,2-Dicyanopropanoic acid	−2.8			
Dicyclohexylamine	11.25(+1)			
Dicyclopentylamine	10.93(+1)			
Didodecylamine	10.99(+1)			
Diethanolamine	8.88(+1)			
Di(ethoxyethyl)amine	8.47(+1)			
3,5-Diethoxyphenol	9.370			
3-(Diethoxyphosphinyl)benzoic acid	3.65			
4-(Diethoxyphosphinyl)benzoic acid	3.60			
3-(Diethoxyphosphinyl)phenol	8.66			
4-(Diethoxyphosphinyl)phenol	8.28			
Diethylamine	10.8(+1)			
2-(Diethylamino)ethyl-4-aminobenzoate	8.85(+1)			
α -(Diethylamino)toluene	9.44(+1)			
<i>N,N</i> -Diethylaniline	6.56(+1)			
5,5-Diethylbarbituric acid (veronal)	8.020(0)			
<i>N,N</i> -Diethylbenzylamine	9.48(+1)			
Diethylbiguanide (30°C)	2.53(+1)	11.68(0)		
Diethylenetriamine	4.42(+3)	9.21(+2)	10.02(+1)	
Diethylenetriaminepentaacetic acid (pK_5 , 10.58)	1.80(0)	2.55(−1)	4.33(−2)	8.60(−3)
<i>N,N</i> -Diethylethylenediamine	7.70(+2)	10.46(+1)		
2,2-Diethylglutaric acid	3.62	7.12		
<i>N,N</i> -Diethylglycine	2.04(+1)	10.47(0)		
Diethylglycolic acid (18°C)	3.804			
Diethylmalonic acid	2.151	7.417		
Diethylmethylethylamine	10.43(+1)			
<i>rac</i> -2,3-Diethylsuccinic acid	3.63	6.46		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
<i>meso</i> -2,3-Diethylsuccinic acid	3.54	6.59		
<i>N,N</i> -Diethyl- <i>o</i> -toluidine	7.18(+1)			
Difluoroacetic acid	1.33			
3,3-Difluoroacrylic acid	3.17			
Diglycolic acid	2.96			
Diguanidine	12.8			
Dihexylamine	11.0(+1)			
Dihydroarecaidine	9.70			
Dihydroarecaidine, methyl ester	8.39			
Dihydrocodeine	8.75(+1)			
Dihydroergonovine	7.38(+1)			
α -Dihydrolysergic acid	3.57	8.45		
γ -Dihydrolysergic acid	3.60	8.71		
α -Dihydrolysergol	8.30			
β -Dihydrolysergol	8.23			
Dihydromorphine	9.35			
3,4-Dihydroxyalanine	2.32(+1)	8.68(0)	9.87(-1)	
1,2-Dihydroxyanthraquinone-3-sulfonic acid (alizarin-3-sulfonic acid)	—	5.54(-1)	11.01(-2)	
3,4-Dihydroxybenzaldehyde	7.55			
1,2-Dihydroxybenzene (pyrocatechol) ($\mu = 0.1$)	9.356(0)	12.98(-1)		
1,3-Dihydroxybenzene (resorcinol)	9.44(0)	12.32(-1)		
1,4-Dihydroxybenzene (hydroquinone)	9.91(0)	12.04(-1)		
4,5-Dihydroxybenzene-1,3-disulfonic acid	—	—	7.66(-2)	12.6(-3)
2,3-Dihydroxybenzoic acid (30°C)	2.98	10.14		
2,4-Dihydroxybenzoic acid (β -resorcylic acid)	3.29	8.98		
2,5-Dihydroxybenzoic acid	2.97	10.50		
2,6-Dihydroxybenzoic acid	1.30			
3,4-Dihydroxybenzoic acid	4.48	8.67	11.74	
3,5-Dihydroxybenzoic acid	4.04			
2,5-Dihydroxy- <i>p</i> -benzoquinone	2.71	5.18		
3,4-Dihydroxy-3-cyclobutene-1,2-dione	0.541	3.480		
2,3-Dihydroxy-2-cyclopenten-1-one (20°C)	4.72			
1,4-Dihydroxy-2,6-dinitrobenzene	4.42	9.14		
Di(2,2'-hydroxyethyl)amine	8.8(+1)			
<i>N,N</i> -Di(2-hydroxyethyl)glycine	8.333			
Dihydroxymaleic acid	1.10			
Dihydroxymalic acid	1.92			
1,3-Dihydroxy-2-methylbenzene ($\mu = 0.65$)	10.05	11.64		
2,2-Di(hydroxymethyl)-3-hydroxypropanoic acid	4.460			
2,4-Dihydroxy-5-methylpyrimidine	9.90			
2,4-Dihydroxy-6-methylpyrimidine	9.52			
1,4-Dihydroxynaphthalene (26°C, $\mu = 0.65$)	9.37	10.93		
1,2-Dihydroxy-3-nitrobenzene	6.68			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
1,2-Dihydroxy-4-nitrobenzene ($\mu = 0.1$)	6.701			
2,4-Dihydroxy-1-phenylazobenzene ($\mu = 0.1$)	11.98			
2,4-Dihydroxyoxazolidine	6.11(+1)			
2,4-Dihydroxypteridine	<1.3	7.92		
2,6-Dihydroxypurine	7.53(0)	11.84(−1)		
2,4-Dihydroxypyridine (20°C)	1.37(+1)	6.45(0)	13(−1)	
Dihydroxytartaric acid	1.95	4.00		
1,4-Dihydroxy-2,3,5,6-tetramethyl- benzene ($\mu = 0.65$)	11.25	12.70		
3,5-Diiodoaniline	2.37(+1)			
2,5-Diiodohistamine	2.31(+2)	8.20(+1)	10.11(0)	
2,5-Diiodohistidine ($\mu = 0.1$)	2.72	8.18	9.76	
3,5-Diiodophenol	8.103			
3,5-Diiodotyrosine	2.117(+1)	6.479(0)	7.821(−1)	
Diisopropylmalonic acid	2.124	8.848		
Dilactic acid	2.955			
<i>threo</i> -1,4-Dimercapto-2,3-butane- diol	8.9			
<i>meso</i> -2,3-Dimercaptosuccinic acid	2.71	3.48	8.89(SH)	10.79(SH)
3,5-Dimethoxyaniline	3.86(+1)			
2,6-Dimethoxybenzoic acid	3.44			
1,10-Dimethoxy-3,8-dimethyl-4,7- phenanthroline	7.21			
Di(2-methoxyethyl)amine	9.51(+1)			
3,5-Dimethoxyphenol	9.345			
(3,4-Dimethoxy)phenylacetic acid	4.333			
Dimethylamine	10.77(+1)			
4-Dimethylaminobenzaldehyde	1.647(+1)			
<i>N,N</i> -Dimethylaminocyclohexane	10.72(+1)			
4-Dimethylamino-2,3-dimethyl-1- phenyl-3-pyrazolin-5-one	4.18(+1)			
4-Dimethylamino-3,5-dimethylpyr- idine (20°C)	8.15(+1)			
2-(Dimethylamino)ethanol	9.26(+1)			
2-[2-(Dimethyl- amino)ethyl]pyridine	3.46(+2)	8.75(+1)		
3-(Dimethylaminoethyl)pyridine	4.30(+2)	8.86(+1)		
4-(Dimethylaminoethyl)pyridine	4.66(+2)	8.70(+1)		
4-(Dimethylamino)-3-ethylpyridine (20°C)	8.66(+1)			
4-(Dimethylamino)-3-isopropylpyr- idine (20°C)	8.27(+1)			
2-(Dimethylaminomethyl)pyridine	2.58(+2)	8.12(+1)		
3-(Dimethylaminomethyl)pyridine	3.17(+2)	8.00(+1)		
4-(Dimethylaminomethyl)pyridine	3.39(+2)	7.66(+1)		
4-(Dimethylamino)-3-methylpyri- dine (20°C)	8.68(+1)			
4-(Dimethylamino- phenyl)phosphonic acid	2.0(+1)	4.2	7.35	
3-(Dimethylamino)propanoic acid	9.85(+1)			
4-(Dimethylamino)pyridine (20°C)	6.09(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
<i>N,N</i> -Dimethylaniline	5.15(+1)			
2,3-Dimethylaniline	4.70(+1)			
2,4-Dimethylaniline	4.89(+1)			
2,5-Dimethylaniline	4.53(+1)			
2,6-Dimethylaniline	3.95(+1)			
3,4-Dimethylaniline	5.17(+1)			
3,5-Dimethylaniline	4.765(+1)			
<i>N,N</i> -Dimethylaniline-4-phosphonic acid (17°C)	2.0(+1)	4.2	7.39	
Dimethylarsinic acid (cacodylic acid)	1.67	6.273		
1,3-Dimethylbarbituric acid	4.68(+1)			
2,3-Dimethylbenzoic acid	3.771			
2,4-Dimethylbenzoic acid	4.217			
2,5-Dimethylbenzoic acid	3.990			
2,6-Dimethylbenzoic acid	3.362			
3,4-Dimethylbenzoic	4.41			
3,5-Dimethylbenzoic acid	4.302			
<i>N,N</i> -Dimethylbenzylamine	9.02(+1)			
Dimethylbiguanide	2.77(+1)	11.52		
2,2-Dimethylbutanoic acid (18°C)	5.03			
Dimethylchlorotetracycline ($\mu = 0.01$)	3.30(+1)			
2,6-Dimethyl-4-cyanophenol	8.27			
3,5-Dimethyl-4-cyanophenol	8.21			
5,5-Dimethyl-1,3-cyclohexanedione	5.15			
<i>cis</i> -3,3-Dimethyl-1,2-cyclopropane-dicarboxylic acid	2.34	8.31		
<i>trans</i> -3,3-Dimethyl-1,2-cyclopropanedicarboxylic acid	3.92	5.32		
3,5-Dimethyl-4-(dimethylamino)-pyridine (20°C)	8.12(+1)			
2,2-Dimethyl-1,3-dioxane-4,6-dione	5.1			
1,1-Dimethylethanethiol ($\mu = 0.1$)	11.22			
<i>N,N</i> -Dimethylethylenediamine- <i>N,N</i> -diacetic acid	6.63	9.53		
<i>N,N'</i> -Dimethylethylenediamine- <i>N,N'</i> -diacetic acid	7.40	10.16		
<i>N,N</i> -Dimethylethylenediamine- <i>N,N'</i> -diacetic acid	5.99	9.97		
<i>N,N</i> -Dimethylglycine	2.146(+1)	9.940(0)		
Dimethylglycolic acid (18°C)	4.04			
<i>N,N</i> -Dimethylglycylglycine	3.11(+1)	8.09(0)		
Dimethylglyoxime	10.60			
5,5-Dimethyl-2,4-hexanedione	10.01			
5,5-Dimethylhydantoin	9.19			
2,4-Dimethyl-8-hydroxyquinoline	6.20(+1)	10.60(0)		
3,4-Dimethyl-8-hydroxyquinoline	5.80(+1)	10.05(0)		
2,4-Dimethyl-8-hydroxyquinoline-7-sulfonic acid	3.20	10.14(OH)		
	(NH ⁺)			
Dimethylhydroxytetracycline	7.5	9.4		
2,4-Dimethylimidazole	8.38(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Dimethylmalic acid	3.17	6.06		
2,2-Dimethylmalonic acid	3.17	6.06		
3,5-Dimethyl-4-(methylamino) pyridine (20°C)	9.96(+1)			
2,3-Dimethylnaphthalene-1-carboxylic acid	3.33			
2,6-Dimethyl-4-nitrophenol	7.190			
3,5-Dimethyl-4-nitrophenol	8.245			
α,α -Dimethylaloacetic acid	1.77	4.62		
3,3-Dimethylpentanedioic acid	3.70	6.34		
2,2-Dimethylpentanoic acid	4.969			
4,4-Dimethylpentanoic acid (18°C)	4.79			
2,3-Dimethylphenol	10.50			
2,4-Dimethylphenol	10.58			
2,5-Dimethylphenol	10.22			
2,6-Dimethylphenol	10.59			
3,4-Dimethylphenol	10.32			
3,5-Dimethylphenol	10.15			
2,6-Dimethylphenoxyacetic acid	3.356			
Dimethylphenylsilylacetic acid	5.27			
<i>N,N'</i> -Dimethylpiperazine	4.630(+2)	8.539(+1)		
1,2-Dimethylpiperidine	10.22			
<i>cis</i> -2,6-Dimethylpiperidine	11.07(+1)			
2,2-Dimethylpropanoic acid (pivalic acid)	5.031			
2,2'-Dimethylpropylphosphonic acid	2.84	8.65		
2,4-Dimethylpyridine (2,4-lutidine)	6.74(+1)			
2,5-Dimethylpyridine (2,5-lutidine)	6.43(+1)			
2,6-Dimethylpyridine (2,6-lutidine)	6.71(+1)			
3,4-Dimethylpyridine (3,4-lutidine)	6.47(+1)			
3,5-Dimethylpyridine (3,5-lutidine)	6.09(+1)			
2,4-Dimethylpyridine-1-oxide	1.627(+1)			
2,5-Dimethylpyridine-1-oxide	1.208(+1)			
2,6-Dimethylpyridine-1-oxide	1.366(+1)			
3,4-Dimethylpyridine-1-oxide	1.493(+1)			
3,5-Dimethylpyridine-1-oxide	1.181(+1)			
2,3-Dimethylquinoline	4.94(+1)			
2,6-Dimethylquinoline	5.46(+1)			
<i>meso</i> -2,2-Dimethylsuccinic acid	3.77	5.936		
<i>rac</i> -2,2-Dimethylsuccinic acid	3.93	6.20		
D -2,3-Dimethylsuccinic acid	3.82	5.93		
<i>meso</i> -2,3-Dimethylsuccinic acid	3.67	5.30		
<i>rac</i> -2,3-Dimethylsuccinic acid	3.94	6.20		
2,4-Dimethylthiazole ($\mu = 0.1$)	3.98			
2,5-Dimethylthiazole ($\mu = 0.1$)	3.91			
4,5-Dimethylthiazole ($\mu = 0.1$)	3.73			
<i>N,N</i> -Dimethyl- <i>o</i> -toluidine	5.86(+1)			
<i>N,N</i> -Dimethyl- <i>p</i> -toluidine	7.24(+1)			
2,4-Dinitroaniline	-4.25(+1)			
2,6-Dinitroaniline	-5.23(+1)			
3,5-Dinitroaniline	0.229(+1)			
2,3-Dinitrobenzoic acid	1.85			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2,4-Dinitrobenzoic acid	1.43			
2,5-Dinitrobenzoic acid	1.62			
2,6-Dinitrobenzoic acid	1.14			
3,4-Dinitrobenzoic acid	2.82			
3,5-Dinitrobenzoic acid	2.85			
1,1-Dinitrobutane (20°C)	5.90			
1,1-Dinitrodecane	3.60			
1,1-Dinitroethane (20°C)	5.21			
Dinitromethane (20°C)	3.60			
1,1-Dinitropentane	5.337			
2,4-Dinitrophenol	4.08			
2,5-Dinitrophenol	5.216			
2,6-Dinitrophenol	3.713			
3,4-Dinitrophenol	5.424			
3,5-Dinitrophenol	6.732			
2,4-Dinitrophenylacetic acid	3.50			
1,1-Dinitropropane (20°C)	5.5			
2,6-Dioxo-1,2,3,6-tetrahydro-4-pyr- imidinecarboxylic acid (orotic acid)	1.8(+1)	9.55(0)		
Diphenylacetic acid	3.939			
Diphenylamine	0.9(+1)			
2,2-Diphenylglutaric acid (20°C)	3.91	5.38		
1,3-Diphenylguanidine	10.12			
2,2-Diphenylheptanedioic acid (20°C)	4.28	5.39		
2,2-Diphenylhexanedioic acid (20°C)	4.17	5.40		
3,3-Diphenylhexanedioic acid	4.22	5.19		
Diphenylhydroxyacetic acid (35°C)	3.05			
Diphenylketimine	6.82			
2,2-Diphenylnonanedioic acid (20°C)	4.33	5.38		
<i>meso</i> -2,2-Diphenylsuccinic acid	3.48			
<i>rac</i> -2,2-Diphenylsuccinic acid	3.58			
2,2-Diphenylsuccinic acid, 1- methyl ester (20°C)	4.47			
2,2-Diphenylsuccinic acid, 4- methyl ester (20°C)	3.900			
Diphenylthiocarbazone	4.50	15		
Dipropylamine	10.91(+1)			
Dipropyleneetriamine	7.72(+3)	9.56(+2)	10.65(+1)	
2,2-Dipropylglutaric acid	3.688	7.31		
Dipropylmalonic acid	2.04	7.51		
2,2'-Dipyridyl	-0.52(+2)	4.352(+1)		
2,3'-Dipyridyl (20°C)	1.52(+2)	4.42(+1)		
2,4'-Dipyridyl (20°C)	1.19(+2)	4.77(+1)		
3,3'-Dipyridyl (20°C, $\mu = 0.2$)	3.0(+2)	4.60(+1)		
3,4'-Dipyridyl (20°C, $\mu = 0.2$)	3.0(+2)	4.85(+1)		
4,4'-Dipyridyl	3.17(+2)	4.82(+1)		
Dithiodiacetic acid (18°C)	3.075	4.201		
1,4-Dithioerythritol	9.5			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Dithiooxamide (rubeanic acid)	10.89			
Dulcitol	13.46			
Ecgonine	10.91			
Emetine	7.36(+1)	8.23(0)		
Epinephrine enantiomorph	9.39(+1)			
Epinephrine, pseudo	9.53(+1)			
Ergometrinine	7.32(+1)			
Ergonovine	6.73(+1)			
Eriochrome Black T	6.3	11.55		
1,2-Ethanediamine	6.85(+2)	9.92(+1)		
Ethane-1,2-diamino- <i>N,N'</i> -dimethyl- <i>N,N'</i> -diacetic acid (20°C)	6.047(0)	10.068(−1)		
1,2-Ethanedithiol	8.96	10.54		
Ethanethiol ($\mu = 0.015$)	10.61			
Ethoxyacetic acid (18°C)	3.65			
2-Ethoxyaniline (<i>o</i> -phenetidine)	4.47(+1)			
3-Ethoxyaniline	4.17(+1)			
4-Ethoxyaniline	5.25(+1)			
2-Ethoxybenzoic acid (20°C)	4.21			
3-Ethoxybenzoic acid (20°C)	4.17			
4-Ethoxybenzoic acid (20°C)	4.80			
Ethoxycarbonylethylamine	9.13(+1)			
2-Ethoxyethanethiol	9.38			
2-Ethoxyethylamine	6.26(+1)			
2-Ethoxyphenol	10.109			
3-Ethoxyphenol	9.655			
(4-Ethoxyphenyl)phosphonic acid	2.06	7.28		
4-Ethoxypyridine	6.67(+1)			
Ethyl acetoacetate	10.68			
3-Ethylacrylic acid	4.695			
<i>N</i> -Ethylalanine	2.22(+1)	10.22(0)		
Ethylamine	10.63(+1)			
(3-Ethylamino)phenylphosphonic acid	1.1(+1)	4.90(0)	7.24(−1)	
<i>N</i> -Ethylaniline	5.11(+1)			
2-Ethylaniline	4.42(+1)			
3-Ethylaniline	4.70(+1)			
4-Ethylaniline	5.00(+1)			
Ethylarsonic acid (18°C)	3.89	8.35		
Ethylbarbituric acid	3.69(+1)			
2-Ethylbenzimidazole ($\mu = 0.16$)	6.27(+1)			
2-Ethylbenzoic acid	3.79			
4-Ethylbenzoic acid	4.35			
Ethylbiguanide	2.09(+1)	11.47(0)		
2-Ethylbutanoic acid (20°C)	4.710			
<i>S</i> -Ethyl- L -cysteine ($\mu = 0.1$)	2.03(+1)	8.60(0)		
Ethylenebiguanide (30°C)	1.74	2.88	11.34	11.76
Ethylenebis(thioacetic acid) (18°C)	3.382(0)	4.352(−1)		
Ethylenediamine- <i>N,N'</i> -diacetic acid	6.42	9.46		
Ethylenediamine- <i>N,N</i> -dimethyl- <i>N',N'</i> -diacetic acid	6.047	10.068		
Ethylenediamine- <i>N,N</i> -dipropanoic acid (30°C)	6.87	9.60		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Ethylenediamine- <i>N,N,N',N'</i> -tetra-acetic acid ($\mu = 0.1$)	1.99	2.67	6.16	10.26
Ethylenediamine- <i>N,N,N',N'</i> -tetra-propanoic acid (30°C)	3.00	3.43	6.77	9.60
Ethylene glycol	14.22			
Ethyleneimine	8.04(+1)			
<i>cis</i> -Ethylene oxide dicarboxylic acid	1.93	3.92		
<i>trans</i> -Ethylene oxide dicarboxylic acid	1.93	3.25		
<i>N</i> -Ethylethylenediamine	7.63(+2)	10.56(+1)		
<i>N</i> -Ethylglycine ($\mu = 0.1$)	2.34(+1)	10.23(0)		
3-Ethylglutaric acid	4.28	5.33		
Ethyl hydroperoxide	11.80			
Ethyl hydrogen malonate	3.55			
3-Ethyl-2-hydroxypyridine	5.00(+1)			
Ethylmalonic acid	2.90(0)	5.55(−1)		
<i>N</i> -Ethyl mercaptoacetamide	8.14(SH)			
Ethyl 2-mercaptoacetate	7.95(SH)			
Ethyl 3-mercaptopropanoate	9.48(SH)			
3-Ethyl-4-(methylamino)pyridine (20°C)	9.90(+1)			
5-Ethyl-5-(1-methylbutyl)barbituric acid	8.11(0)			
Ethyl methyl ketoxime	12.45			
Ethylmethylmalonic acid	2.86(0)	6.41(−1)		
1-Ethyl-2-methylpiperidine	10.66(+1)			
3-Ethyl-6-methylpyridine (20°C)	6.51(+1)			
3-Ethyl-4-methylpyridine-1-oxide	−1.534(+1)			
5-Ethyl-2-methylpyridine-1-oxide	−1.288(+1)			
1-Ethyl-2-methyl-2-pyrroline	11.84(+1)			
Ethylmorphine (15°C)	8.08			
Ethyl nitroacetate	5.85			
3-Ethylpentane-2,4-dione	11.34			
2-Ethylpentanoic acid (18°C)	4.71			
5-Ethyl-5-pentylbarbituric acid	7.960			
2-Ethylphenol	10.2			
3-Ethylphenol	10.07			
4-Ethylphenol	10.0			
4-Ethylphenylacetic acid	4.373			
5-Ethyl-5-phenylbarbituric acid	7.445			
Ethylphosphinic acid	3.29			
Ethylphosphonic acid	2.43	8.05		
1-Ethylpiperidine ($\mu = 0.01$)	10.45(+1)			
2,2-Ethylpropylglutaric acid	3.511			
Ethylpropylmalonic acid	3.14	7.43		
2-Ethylpyridine	5.89(+1)			
3-Ethylpyridine (20°C)	5.80(+1)			
4-Ethylpyridine	5.87(+1)			
Ethyl 3-pyridinecarboxylate	3.35(+1)			
Ethyl 4-pyridinecarboxylate	3.45(+1)			
2-Ethylpyridine-1-oxide	−1.19(+1)			
3-Ethylpyridine-1-oxide	−0.965(+1)			
Ethylpyrrolidine	10.43(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Ethyl-2-pyrroline	7.87(+1)			
Ethylsuccinic acid	4.08(0)			
S-Ethylthioacetic acid	5.06			
N-Ethyl- <i>o</i> -toluidine	4.92(+1)			
N-Ethylveratramine	7.40(+1)			
β -Eucaine	9.35(+1)			
Fluoroacetic acid	2.586			
2-Fluoroacrylic acid	2.55			
2-Fluoroaniline	3.20(+1)			
3-Fluoroaniline	3.58(+1)			
4-Fluoroaniline	4.65(+1)			
2-Fluorobenzoic acid	3.27			
3-Fluorobenzoic acid	3.865			
4-Fluorobenzoic acid	4.14			
Fluoromandelic acid	4.244			
2-Fluorophenol	8.73			
3-Fluorophenol	9.29			
4-Fluorophenol	9.89			
2-Fluorophenoxyacetic acid	3.08			
3-Fluorophenoxyacetic acid	3.08			
4-Fluorophenoxyacetic acid	3.13			
4-Fluorophenylacetic acid	4.25			
2'-Fluorophenylalanine	2.14(+1)	9.01(0)		
3'-Fluorophenylalanine	2.10(+1)	8.98(0)		
4-Fluorophenylalanine	2.13(+1)	9.05(0)		
2-Fluorophenylphosphonic acid	1.64	6.80		
3-Fluorophenylselenic acid	4.34			
4-Fluorophenylselenic acid	4.50			
2-Fluoropyridine	-0.44(+1)			
3-Fluoropyridine	2.97(+1)			
5-Fluorouracil	8.00(0)	ca 13(-1)		
Folic acid (pteroylglutamic acid)	8.26			
Formic acid	3.751			
N-Formylglycine	3.43			
2-Formyl-3-hydroxypyridine (20°C)	3.40(+1)	6.95(OH)		
4-Formyl-3-hydroxypyridine	4.05(+1)	6.77(OH)		
2-Formyl-3-methoxypyridine (20°C)	3.89(+1)	12.95		
Formyl-3-methoxypyridine (20°C)	4.45(+1)	11.7		
D-(-)-Fructose	12.03			
Fumaric acid	3.10	4.60		
2-Furancarboxylic acid (2-furoic acid)	3.164			
D-(+)-Galactose	12.35			
Galactose-1-phosphoric acid	1.00	6.17		
Glucoscorbic acid	4.26	11.58		
D-Gluconic acid	3.86			
α -D-(+)-Glucose	12.28			
α -D-Glucose-1-phosphate	1.11(0)	6.504(-1)		
trans-Glutaconic acid	3.77	5.08		
D-(-)-Glutamic acid	2.162(+1)	4.272(0)	9.358(-1)	
L-Glutamic acid	2.19(+1)	4.25(0)	9.67(-1)	

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Glutamic acid, 1-ethyl ester	3.85(+1)	7.84(0)		
Glutamic acid, 5-ethyl ester	2.15(+1)	9.19(0)		
L-Glutamine ($\mu = 0.2$)	2.17(+1)	9.13(0)		
Glutaric acid	3.77	6.08		
Glutaric acid monoamide	4.600(0)			
Glutarimide	11.43			
Glutathione	2.12(+1)	3.53(0)	8.66	9.12
DL-Glyceric acid	3.64			
Glycerol	14.15			
Glyceryl-1-phosphoric acid	—	6.656(−1)		
Glyceryl-2-phosphoric acid	1.335(0)	6.650(−1)		
Glycine	2.341(+1)	9.60(0)		
Glycine amide	8.03(+1)			
Glycine, ethyl ester	7.66(+1)			
Glycine hydroxamic acid	7.10	9.10		
Glycine, methyl ester	7.59(+1)			
Glycine-O-phenylphosphorylserine	2.96	8.07		
Glycolic acid	3.831			
N-Glycl- α -alanine	3.15(+1)	8.33(0)		
Glycylalanylalanine	3.38(+1)	8.10(0)		
N-Glycylasparagine	2.942			
Glycylaspartic acid	2.81(+1)	4.45(0)	8.60(−1)	
Glycyl-DL-glutamine (18°C)	2.88(+1)	8.33(0)		
N-Glycylglycine	3.126(+1)	8.252(0)		
Glycylglycylcysteine (35°C)	2.71	2.71	7.94	7.94
Glycylglycylglycine	3.225(+1)	8.090(0)		
Glycyl-L-histidine ($\mu = 0.16$)	6.79	8.20		
Glycylisoleucine	8.00			
N-Glycyl-L-leucine	3.180(+1)	8.327(0)		
Glycyl-O-phosphorylserine	2.90	6.02	8.43	
L-Glycylproline ($\mu = 0.1$)	2.81(+1)	8.65(0)		
N-Glycylsarcosine ($\mu = 0.1$)	2.98(+1)	8.55(0)		
N-Glycylserine	2.98(+1)	8.38(0)		
Glycylserylglycine	3.32	7.99		
Glycyltyrosine	2.93	8.45	10.49	
Glycylvaline	3.15	8.18		
Glyoxaline	7.03(+1)			
Glyoxylic acid	3.30(0)			
Guanidineacetic acid	2.82(+1)			
Guanine	3.3(+1)	9.2	12.3	
Guanine deoxyriboside-3'-phosphoric acid	—	2.9	6.4	9.7
Guanosine	1.9(+1)	9.25(0)	12.33(OH)	
Guanosine-5'-diphosphoric acid ($\mu = 0.1$; pK_5 9.6)	—	—	2.9	6.3
Guanosine-3'-phosphoric acid	0.7	2.3	5.92	9.38
Guanosine-5'-phosphoric acid ($\mu = 0.1$)	—	2.4	6.1	9.4
Guanosine-5'-triphosphoric acid [$\mu = 0.1$; pK_5 7.10(−3); pK_6 9.3(−4)]	—	—	—	3.0(−2)
Guanylurea	1.80	8.20		
Harmine (20°C)	7.61(+1)			
Heptafluorobutanoic acid	0.17			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
4,4,5,5,6,6,6-Heptafluorohexanoic acid	4.18			
4,4,5,5,6,6,6-Heptafluoro-2-hexenoic acid	3.23			
Heptanedioic acid (pimelic acid)	4.484	5.424		
2,4-Heptanedione	8.43(keto); 9.15(enol)			
Heptanoic acid	4.893			
Heroin	7.6(+1)			
2,4-Hexadienoic acid (sorbic acid)	4.77			
1,1,1,3,3,3-Hexafluoro-2,2-propanediol	8.801			
1,1,1,3,3,3-Hexafluoro-2-propanol	9.42			
Hexahydroazepine	11.07			
Hexamethyldisilazine	7.55			
1,2,3,8,9,10-Hexamethyl-4,7-phenanthroline (20°C)	7.26			
1,6-Hexanediamine	9.830(+2)	10.930(+1)		
1,6-Hexanedioic acid	4.418	5.412		
2,4-Hexanedione	8.49 (enol); 9.32 (keto)			
2,2',4,4',6,6'-Hexanitrodiphenylamine	5.42(+1)			
Hexanoic acid (20°C)	4.849			
<i>trans</i> -2-Hexenoic acid	4.74			
<i>trans</i> -3-Hexenoic acid	4.72			
3-Hexen-4-oic acid	4.58			
4-Hexen-5-oic acid	4.74			
Hexylamine	10.64(+1)			
Hexylarsonic acid	4.16	9.19		
Hexylphosphonic acid	2.6	7.9		
DL -Histidine	1.82(+2)	6.00(+1)	9.16(0)	
Histidine amide ($\mu = 0.2$)	5.78(+2)	7.64(+1)		
Histidine, methyl ester ($\mu = 0.1$)	5.01(+2)	7.23(+1)		
Histidylglycine	2.40(+2)	5.80(+1)	7.82(0)	
Histidylhistidine ($\mu = 0.16$)	5.40(+2)	6.80(+1)	7.95(0)	
DI -Homatropine	9.7(+1)			
DI -Homocysteine	2.222(+1)	8.87	10.86	
Homocysteine ($\mu = 0.1$)	1.593(+2)	2.523(+1)	8.676(0)	9.413(-1)
Hydantoin	9.12			
Hydrastine	6.23(+1)			
Hydrazine- <i>N,N</i> -diacetic acid	<0.1	2.8	3.8	
Hydrazine- <i>N'</i> - <i>N'</i> -diacetic acid	2.40	3.12	7.32	
4-Hydrazinocarbonylpyridine (20°C)	1.82	3.52	10.79	
<i>N</i> -Hydroxyacetamide	9.40			
2'-Hydroxyacetophenone	9.90			
3'-Hydroxyacetophenone	9.19			
4'-Hydroxyacetophenone	8.05			
1-Hydroxyacridine (15°C)	5.72			
2-Hydroxyacridine (15°C)	5.62			
3-Hydroxyacridine (15°C)	5.30			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
α -Hydroxyasparagine	2.28(+1)	7.20(0)	9.11(−1)	
β -Hydroxyasparagine	2.09(+1)	8.29(0)		
Hydroxyaspartic acid	1.91(+1)	3.51(0)		
2-Hydroxybenzaldehyde (salicyl- aldehyde)	8.34		12.11	
3-Hydroxybenzaldehyde	9.00			
4-Hydroxybenzaldehyde	7.620			
2-Hydroxybenzaldehyde oxime	1.37(+1)	9.18		
2-Hydroxybenzamide	8.36			
2-Hydroxybenzenemethanol (2-hy- droxybenzyl alcohol)	9.92			
3-Hydroxybenzenemethanol	9.83			
4-Hydroxybenzenemethanol	9.82			
4-Hydroxybenzenesulfonic acid	—	9.055(−1)		
2-Hydroxybenzohydroxamic acid	5.19			
2-Hydroxybenzoic acid (salicylic acid)	2.98	12.38	9.93	
3-Hydroxybenzoic acid	4.076	9.85		
4-Hydroxybenzoic acid	4.582	9.23		
4-Hydroxybenzonitrile	7.95			
2-Hydroxy-5-bromobenzoic acid	2.61			
2-Hydroxybutanoic acid (30°C)	3.65			
L-3-Hydroxybutanoic acid (30°C)	4.41			
4-Hydroxybutanoic acid (30°C)	4.71			
2-Hydroxy-5-chlorobenzoic acid	2.63			
<i>trans</i> -2'-Hydroxycinnamic acid	4.614			
<i>trans</i> -3'-Hydroxycinnamic acid	4.40			
10-Hydroxycodeine	7.12			
<i>cis</i> -2-Hydroxycyclohexane-1-car- boxylic acid	4.796			
<i>trans</i> -2-Hydroxycyclohexane-1- carboxylic acid	4.682			
<i>cis</i> -3-Hydroxycyclohexane-1-car- boxylic acid	4.602			
<i>trans</i> -3-Hydroxycyclohexane-1- carboxylic acid	4.815			
<i>cis</i> -4-Hydroxycyclohexane-1-car- boxylic acid	4.836			
<i>trans</i> -4-Hydroxycyclohexane-1- carboxylic acid	4.687			
1-Hydroxy-2,4-dihydroxymethyl- benzene	9.79			
<i>N</i> -(Hydroxyethyl)biguanide	2.8(+2)	11.53(+1)		
<i>N</i> -(2-Hydroxy- ethyl)ethylenediamine	7.21(+2)	10.12(+1)		
<i>N'</i> -(2-Hydroxyethyl)ethylenediam- ine- <i>N,N,N'</i> -triacetic acid	2.39	5.37		
<i>N</i> -(2-Hydroxyethyl)iminodiacetic acid ($\mu = 0.1$)	2.2	8.65		
<i>N</i> -(2-Hydroxyethyl)piperazine- <i>N'</i> - ethansulfonic acid (20°C)	7.55			
4'-(2-Hydroxyethyl)-1'-piperazine- propanesulfonic acid (20°C)	8.00			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Hydroxyethyltrimethylamine	8.94(+1)			
L- β -Hydroxyglutamic acid	2.09	4.18	9.20	
1-Hydroxy-4-hydroxymethylbenzene	9.84			
5-Hydroxy-2-(hydroxymethyl)-4H-pyran-4-one	7.90	8.03		
3-Hydroxy-2-hydroxymethylpyridine (20°C, $\mu = 0.2$)	5.00(+1)	9.07(OH)		
3-Hydroxy-4-hydroxymethylpyridine (20°C, $\mu = 0.2$)	5.00(+1)	8.95(OH)		
8-Hydroxy-7-iodoquinoline-5-sulfonic acid	2.51(0)	7.417(−1)		
Hydroxylysine (38°C, $\mu = 0.1$)	2.13(+2)	8.62(+1)	9.67(0)	
2-Hydroxy-3-methoxybenzaldehyde	7.912			
3-Hydroxy-4-methoxybenzaldehyde (isovanillin)	8.889			
4-Hydroxy-3-methoxybenzaldehyde (vanillin)	7.396			
4-Hydroxy-3-methoxybenzoic acid	4.355			
1-Hydroxy-2-methoxybenzylamine	8.70(+1)	10.52(0)		
2-Hydroxy-1-methoxybenzylamine	8.89(+1)	10.52(0)		
3-Hydroxy-2-methoxybenzylamine	8.94(+1)	10.42(0)		
2-Hydroxymethyl-2-benzeneacetic acid	4.12			
(2-Hydroxy-5-methylbenzene)-methanol	10.15			
2-Hydroxy-3-methylbenzoic acid	2.99			
2-Hydroxy-4-methylbenzoic acid	3.17			
2-Hydroxy-5-methylbenzoic acid	4.08			
2-Hydroxy-6-methylbenzoic acid	3.32			
2-Hydroxy-2-methylbutanoic acid (18°C)	3.991			
3-Hydroxy-2-methylbutanoic acid (18°C)	4.648			
4-Hydroxy-4-methylpentanoic acid (18°C)	4.873			
1-Hydroxymethylphenol	9.95			
Hydroxymethylphosphoric acid	1.91	7.15		
2-Hydroxy-2-methylpropanoic acid ($\mu = 0.1$)	3.717			
2-Hydroxy-4-methylpyridine	4.529(+1)			
8-Hydroxy-2-methylquinoline	5.55(+1)	10.31(0)		
8-Hydroxy-4-methylquinoline	5.56(+1)	10.00(0)		
8-Hydroxy-2-methylquinoline-5-sulfonic acid	4.80(0)	9.30(−1)		
8-Hydroxy-4-methylquinoline-7-sulfonic acid	4.78(0)	10.01(−1)		
8-Hydroxy-6-methylquinoline-5-sulfonic acid	4.20(0)	8.7(−1)		
2-Hydroxy-1-naphthoic acid (20°C)	3.29	9.68		
2-Hydroxy-2-nitrobenzoic acid	2.23			
2-Hydroxy-3-nitrobenzoic acid	1.87			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Hydroxy-5-nitrobenzoic acid	2.12			
2-Hydroxy-6-nitrobenzoic acid	2.24			
2-Hydroxy-4-nitrophenylphosphonic acid	1.22	5.39		
8-Hydroxy-7-nitroquinoline-5-sulfonic acid	1.94(0)	5.750(−1)		
3-Hydroxy-4-nitrotoluene ($\mu = 0.1$)	7.41			
4-Hydroxypentanoic acid (18°C)	4.686			
4-Hydroxy-3-pentenoic acid	4.30			
3-Hydroxyphenazine (15°C)	2.67			
4-Hydroxyphenylarsonic acid	3.89	8.37 (phenol)	10.05	
3-Hydroxyphenylboric acid	8.55	10.84		
2-Hydroxy-2-phenylpropanoic acid	3.532			
2-(2-Hydroxyphenyl)pyridine (20°C)	4.19(+1)	10.64		
<i>trans</i> -4-Hydroxyproline	1.818(+1)	9.662(0)		
Hydroxypropanedioic acid (tartronic acid)	2.37	4.74		
2-Hydroxypropanoic acid	3.858			
1-Hydroxy-2-propylbenzene	10.50			
4-Hydroxypteridine	1.3(+1)	7.89(0)		
2-Hydroxypyridine	1.25(+1)	11.62(0)		
3-Hydroxypyridine	4.80(+1)	8.72(0)		
4-Hydroxypyridine	3.23(+1)	11.09(0)		
2-Hydroxypyridine- <i>N</i> -oxide	−0.62(+1)	5.97(0)		
2-Hydroxypyrimidine	2.24(+1)	9.17(0)		
4-Hydroxypyrimidine	1.85(+1)	8.59(0)		
8-Hydroxyquinazoline	3.41(+1)	8.65(0)		
2-Hydroxyquinoline (20°C)	−0.31(+1)	11.74		
3-Hydroxyquinoline (20°C)	4.30(+1)	8.06(0)		
4-Hydroxyquinoline (20°C)	2.27(+1)	11.25(0)		
5-Hydroxyquinoline (20°C)	5.20(+1)	8.54(0)		
6-Hydroxyquinoline (20°C)	5.17(+1)	8.88(0)		
7-Hydroxyquinoline (20°C)	5.48(+1)	8.85(0)		
8-Hydroxyquinoline (20°C)	4.91(+1)	9.81(0)		
8-Hydroxyquinoline-5-sulfonic acid	4.092(+1)	8.776(0)		
DL -Hydroxysuccinic acid (malic acid)	3.458	5.097		
L -Hydroxysuccinic acid	3.40	5.05		
Hydroxytetracycline	3.27(+1)	7.32(0)	9.11(−1)	
5-Hydroxy-1,2,3,4-tetrazole	3.32			
4-Hydroxy-3-(2'-thiazolyazo)toluene	8.36			
2-Hydroxytoluene	10.33			
3-Hydroxytoluene	10.10			
4-Hydroxytoluene	10.276			
4-Hydroxy- α,α,α -trifluorotoluene	8.675			
1-Hydroxy-2,4,6-trihydroxymethylbenzene	9.56			
Hydroxyuracil	8.64			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
Hydroxyvaline	2.55(+1)	9.77(0)	12.07(−1)	
Hyoscyamine	9.68(+1)			
Hypoxanthene	1.79(+1)	8.91(0)		
Hypoxanthine	5.3			
Imidazole	6.993(+1)	10.58(0)	10.70(0)	
Imidazolidinetrione (parabanic acid)	6.10			
4-(4-Imidazolyl)butanoic acid ($\mu = 0.1$)	4.26(+1)	7.26(0)		
2-(4-Imidazolyl)ethylamine	5.784(+2)	9.756(+1)		
3-(4-Imidazolyl)propanoic acid ($\mu = 0.16$)	3.96(+1)	7.57(0)		
3,3'-Iminobispropanoic acid	4.11(0)	9.61(−1)		
3,3'-Iminobispropylamine (30°C)	8.02(+2)	9.70(+1)		
2,2'-Iminodiacetic acid (diglycine) (30°C, $\mu = 0.1$)	2.54(0)	9.12(−1)		
4-Indanol	10.32			
Indole-3-acetic acid	4.75			
Inosine	ca 1.5(+1)	8.96(0)	12.36	6.92(−3)
Inosine-5'-phosphoric acid	1.54(0)	6.66(−1)		
Inosine-5'-triphosphoric acid [pK_5 7.68(−4)]	—	—	2.2(−2)	
Iodoacetic acid	3.175			
2-Iodoaniline	2.54(+1)			
3-Iodoaniline	3.58(+1)			
4-Iodoaniline	3.82(+1)			
2-Iodobenzoic acid	2.86			
3-Iodobenzoic acid	3.86			
4-Iodobenzoic acid	4.00			
5-Iodohistamine	4.06(+1)	9.20(+1)	11.88(0)	
	(imidazole)	(NH_3^+)	(imino)	
7-Iodo-8-hydroxyquinoline-5-sulfonic acid	2.514	7.417		
Iodomandelic acid	3.264			
Iodomethylphosphoric acid	1.30	6.72		
2-Iodophenol	8.464			
3-Iodophenol	8.879			
4-Iodophenol	9.200			
2-Iodophenoxyacetic acid	3.17			
3-Iodophenoxyacetic acid	3.13			
4-Iodophenoxyacetic acid	3.16			7.06
2-Iodophenylacetic acid	4.038			
3-Iodophenylacetic acid	4.159			
4-Iodophenylacetic acid	4.178			
2-Iodophenylphosphoric acid	1.74			
2-Iodopropanoic acid	3.11			
3-Iodopropanoic acid	4.08			
2-Iodopyridine	1.82(+1)			
3-Iodopyridine	3.25(+1)			
4-Iodopyridine (20°C)	4.02(+1)			
Isoasparagine	2.97(+1)	8.02(0)		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Isobutylacetic acid (18°C)	4.79			
Isobutylamine	10.41(+1)			
Isochlorotetracycline	3.1(+1)	6.7(0)	8.3(−1)	
Isocreatine	2.84(+1)			
Isogluatamine	3.81(+1)	7.88(0)		
Isohistamine ($\mu = 0.1$)	6.036(+2)	9.274(+1)		
L-Isoleucine	2.35(+1)	9.68(0)		
Isolysergic acid	3.33(0)	8.46(NH)		
Isopilocarpine (15°C)	7.18(+1)			
2-(Isopropoxy)benzoic acid (20°C)	4.24			
3-(Isopropoxy)benzoic acid (20°C)	4.15			
4-(Isopropoxy)benzoic acid (20°C)	4.68			
Isopropylamine	10.64(+1)			
N-Isopropylaniline	5.50(+1)			
5-Isopropylbarbituric acid	4.907(+1)			
2-Isopropylbenzene acid	3.64			
4-Isopropylbenzene acid	4.36			
N-Isopropylglycine ($\mu = 0.1$)	2.36(+1)	10.06(0)		
Isopropylmalonic acid	2.94	5.88		
Isopropylmalonic acid mononitrile	2.401			
3-Isopropyl-4-(methylamino)pyridine (20°C)	9.96(+1)			
3-Isopropylpentanedioic acid	4.30	5.51		
4-Isopropylphenylacetic acid	4.391			
Isopropylphosphinic acid	3.56			
Isopropylphosphonic acid	2.66	8.44		
2-Isopropylpyridine	5.83(+1)			
3-Isopropylpyridine (20°C)	5.72(+1)			
4-Isopropylpyridine	6.02(+1)			
DL-Isoproterenol	8.64(+1)			
Isoquinoline	5.40(+1)			
Isoretronecanol	10.83			
L-Isoserine ($\mu = 0.16$)	2.72(+1)	9.25(0)		
Isothiocyanatoacetic acid	6.62			
L-(+)-Lactic acid	3.858			
L-Leucine	2.33(+1)	9.60(0)		
Leucine amide	7.80(+1)			
Leucine, ethyl ester ($\mu = 0.1$)	7.57(+1)			
L-Leucyl-L-asparagine	3.00(+1)	8.12(0)		
L-Leucyl-L-glutamine	2.99(+1)	8.11(0)		
DL-Leucylglycine	3.25(+1)	8.28(0)		
Leucylisoserine (20°C)	3.188(+1)	8.207(0)		
D-Leucyl-L-tyrosine	3.12(+1)	8.38(0)	10.35(−1)	
L-Leucyl-L-tyrosine	3.46(+1)	7.84(0)	10.09(−1)	
Lysergic acid	3.44(+1)	7.68(0)		
L-(+)-Lysine	2.18(+2)	8.94(+1)	10.53(0)	
Lysine, methyl ester ($\mu = 0.1$)	6.965(+1)	10.251(0)		
L-Lysyl-L-alanine	3.22(+1)	7.62(0)	10.70(−1)	
L-Lysyl-D-alanine	3.00(+1)	7.74(0)	10.63(−1)	
Lysylglutamic acid	2.93(+2)	4.47(+1)	7.75(0)	10.50(+1)
L-Lysyl-L-lysine ($\mu = 0.1$)	3.01(+2)	7.53(+1)	10.05(0)	10.01(−1)

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
L-Lysyl-D-lysine ($\mu = 0.1$)	2.85(+2)	7.53(+1)	9.92(0)	10.89(−1)
L-Lysyl-L-lysyl-L-lysine ($\mu = 0.1$)	3.08(+2)	7.34(+1)	9.80(0)	10.54(−1)
L-Lysyl-D-lysyl-L-lysine ($\mu = 0.1$)	2.91(+2)	7.29(+1)	9.79(0)	10.54(−1)
L-Lysyl-D-lysyl-lysine ($\mu = 0.1$)	2.94(+2)	7.15(+1)	9.60(0)	10.38(−1)
α -D-Lyxose	12.11			
Maleic acid	1.910	6.33		
Malonamic acid	3.641(0)			
Malonic acid	2.826	5.696		
Malonitrile (cyanoacetic acid)	2.460			
Mandelic acid	3.411			
D-(+)-Mannose	12.08			
Mercaptoacetic acid (thioglycolic acid)	3.60(0)	10.56(SH)		
2-Mercaptobenzoic acid (20°C)	4.05(0)			
2-Mercaptobutanoic acid	3.53(0)			
Mercaptodiacetic acid	3.32	4.29		
2-Mercaptoethanesulfonic acid (20°C)		9.5(−1)		
2-Mercaptoethanol	9.88			
2-Mercaptoethylamine	8.27(+1)	10.53(0)		
2-Mercaptohistidine	1.84(+1)	8.47(0)	11.4(SH)	
Mercapto-S-phenylacetic acid ($\mu = 0.1$)	3.9			
2-Mercaptopropane ($\mu = 0.1$)	10.86			
3-Mercapto-1,2-propanediol ($\mu = 0.5$)	9.43			
2-Mercaptopropanoic acid	4.32(0)	10.20(SH)		
3-Mercaptopropanoic acid	—	10.84(SH)		
2-Mercaptopyridine (20°C)	−1.07(+1)	10.00(0)		
3-Mercaptopyridine (20°C)	2.26(+1)	7.03(0)		
4-Mercaptopyridine (20°C)	1.43(+1)	8.86(0)		
2-Mercaptoquinoline (20°C)	−1.44(+1)	10.21(0)		
3-Mercaptoquinoline (20°C)	2.33(+1)	6.13(0)		
4-Mercaptoquinoline (20°C)	0.77(+1)	8.83(0)		
Mercaptosuccinic acid	3.30(0)	4.94(−1)	10.94(SH)	
Mesitylenic acid	4.32			
Mesoxaldialdehyde	3.60			
Methacrylic acid	4.66			
Methanethiol	10.70			
DL-Methionine	2.28(+1)	9.21(0)		
2-(N-Methoxyacetamido)pyridine	2.01(+1)			
3-(N-Methoxyacetamido)pyridine	3.52(+1)			
4-(N-Methoxyacetamido)pyridine	4.62(+1)			
Methoxyacetic acid	3.570			
3-Methoxy-D- α -alanine	2.037(+1)	9.176(0)		
2-Methoxyaniline	4.53(+1)			
3-Methoxyaniline	4.20(+1)			
4-Methoxyaniline	5.36(+1)			
2-Methoxybenzoic acid	4.09			
3-Methoxybenzoic acid	4.08			
4-Methoxybenzoic acid	4.49			
N,N-Methoxybenzylamine	9.68(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Methoxycarbonylaniline	2.23(+1)			
3-Methoxycarbonylaniline	3.64(+1)			
4-Methoxycarbonylaniline	2.38(+1)			
Methoxycarbonylmethylamine	7.66(+1)			
2-Methoxycarbonylpyridine	2.21(+1)			
3-Methoxycarbonylpyridine	3.13(+1)			
4-Methoxycarbonylpyridine	3.26(+1)			
<i>trans</i> -2-Methoxycinnamic acid	4.462			
<i>trans</i> -3-Methoxycinnamic acid	4.376			
<i>trans</i> -4-Methoxycinnamic acid	4.539			
2-Methoxyethylamine	9.45(+1)			
2-Methoxy-4-nitrophenylphosphonic acid	1.53	6.96		
2-Methoxyphenol	9.99			
3-Methoxyphenol	9.652			
4-Methoxyphenol	10.20			
(2'-Methoxy)phenoxyacetic acid	3.231			
(3'-Methoxy)phenoxyacetic acid	3.141			
(4'-Methoxy)phenoxyacetic acid	3.213			
4'-Methoxyphenylacetic acid	4.358			
(4-Methoxyphenyl)phosphinic acid (17°C)	2.35			
(2-Methoxyphenyl)phosphonic acid	2.16	7.77		
(4-Methoxyphenyl)phosphonic acid (17°C)	2.4	7.15		
3-(2'-Methoxyphenyl)propanoic acid	4.804			
3-(3'-Methoxyphenyl)propanoic acid	4.654			
3-(4'-Methoxyphenyl)propanoic acid	4.689			
3-Methoxyphenylselenic acid	4.65			
4-Methoxyphenylselenic acid	5.05			
2-Methoxy-4-(2-propenyl)phenol	10.0			
2-Methoxypyridine	3.06(+1)			
3-Methoxypyridine	4.91(+1)			
4-Methoxypyridine	6.47(+1)			
4-Methoxy-2-(2'-thiazoylazo)phenol	7.83			
2-Methylacrylic acid (18°C)	4.66			
<i>N</i> -Methylalanine	2.22(+1)	10.19(0)		
<i>O</i> -Methylallothreonine ($\mu = 0.1$)	1.92(+1)	8.90(0)		
Methylamine	10.62(+1)			
2-(<i>N</i> -Methylamino)benzoic acid	1.93(+1)	5.34(0)		
3-(<i>N</i> -Methylamino)benzoic acid	—	5.10(0)		
4-(<i>N</i> -Methylamino)benzoic acid	—	5.05		
Methylaminodiacetic acid (20°C)	2.146	10.088		
2-(Methylamino)ethanol	9.88(+1)			
2-(2-Methylaminoethyl)pyridine (30°C)	3.58(+2)	9.65(+1)		
2-(Methylaminomethyl)6-methylpyridine ($\mu = 0.5$)	3.03(+2)	9.15(+1)		
2-(Methylaminomethyl)pyridine (30°C)	2.92(+2)	8.82(+1)		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
4-Methylamino-3-methylpyridine (20°C)	9.83(+1)			
(3-Methylamino)phenylphosphonic acid	1.1(+1)	4.72(+1)	7.30(−1)	
(4-Methylamino)phenylphosphonic acid	—	—	7.85(−1)	
3-(Methylamino)pyridine (30°C)	8.70(+1)			
4-(Methylamino)pyridine (20°C)	9.65(+1)			
4-(Methylamino)-2,3,5,6-tetra-methylpyridine (20°C)	10.06(+1)			
<i>N</i> -Methylaniline	4.85(+1)			
Methylarsonic acid (18°C)	3.41	8.18		
1-Methylbarbituric acid	4.35(+1)			
5-Methylbarbituric acid	3.386(+1)			
2-(<i>N</i> -Methylbenzamido)pyridine	1.44(+1)			
3-(<i>N</i> -Methylbenzamido)pyridine	3.66(+1)			
4-(<i>N</i> -Methylbenzamido)pyridine	4.68(+1)			
2-Methylbenzimidazole ($\mu = 0.16$)	6.29(+1)			
2-Methylbenzoic acid (<i>o</i> -toluic acid)	3.90			
3-Methylbenzoic acid	4.269			
4-Methylbenzoic acid	4.362			
<i>N</i> -Methyl-1-benzoyllecgonine	8.65			
Methylbiguanidine	3.00(+2)	11.44(+1)		
2-Methyl-2-butanethiol	11.35			
2-Methylbutanoic acid	4.761			
3-Methylbutanoic acid (20°C)	4.767			
(<i>E</i>)-2-Methyl-2-butendioic acid (mesaconic acid)	3.09	4.75		
3-Methyl-2-butenic acid	5.12			
(<i>E</i>)-2-Methyl-2-butenic acid (tiglic acid)	4.96			
(<i>Z</i>)-2-Methyl-2-butenic acid (angelic acid)	4.30			
4-Methylcarboxylphenol	8.47			
(<i>E</i>)-2-Methylcinnamic acid	4.500			
(<i>E</i>)-3-Methylcinnamic acid	4.442			
(<i>E</i>)-4-Methylcinnamic acid	4.564			
1-Methylcyclohexane-1-carboxylic acid	5.13			
<i>cis</i> -2-Methylcyclohexane-1-carboxylic acid	5.03			
<i>trans</i> -2-Methylcyclohexane-1-carboxylic acid	5.73			
<i>cis</i> -3-Methylcyclohexane-1-carboxylic acid	4.88			
<i>trans</i> -3-Methylcyclohexane-1-carboxylic acid	5.02			
<i>cis</i> -4-Methylcyclohexane-1-carboxylic acid	5.04			
<i>trans</i> -4-Methylcyclohexane-1-carboxylic acid	4.89			
2-Methylcyclohexyl-1,1-diacetic acid	3.53	6.89		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
3-Methylcyclohexyl-1,1-diacetic acid	3.49	6.08		
4-Methylcyclohexyl-1,1,1-diacetic acid	3.49	6.10		
3-Methylcyclopentyl-1,1-diacetic acid	3.79	6.74		
S-Methyl-L-cysteine	8.97			
N-Methylcytidine	3.88			
5-Methylcytidine	4.21			
N-Methyl-2'-deoxycytidine	3.97			
5-Methyl-2'-deoxycytidine	4.33			
2-Methyl-3,5-dinitrobenzoic acid	2.97			
5-Methyldipropylenetriamine (30°C)	6.32(+3)	9.19(+2)	10.33(+1)	
2,2'-Methylenebis(4-chlorophenol)	7.6	11.5		
2,2'-Methylenebis(4,6-dichlorophenol)	5.6	10.56		
Methylenebis(thioacetic acid) (18°C)	3.310	4.345		
3,3'-(Methylenedithio)dialanine	2.200(+1)	8.16(0)		
Methylenesuccinic acid	3.85	5.45		
N-Methylethylamine	4.23(+1)			
N-Methylethylenediamine	6.86(+1)	10.15(+1)		
α -Methylglucoside	13.71			
3-Methylglutaric acid	4.24	5.41		
N-Methylglycine (sarcosine)	2.12(+1)	10.20(0)		
5-Methyl-2,4-heptanedione	8.52(enol); 9.10(keto)			
5-Methyl-2,4-hexanedione	8.66(enol); 9.31(keto)			
5-Methyl-4-hexenoic acid	4.80			
3-Methylhistamine	5.80(+1)	9.90(0)		
1-Methylhistidine	1.69	6.48	8.85	
2-Methylhistidine (18°C)	1.7	7.2	9.5	
2-Methyl-8-hydroxyquinoline ($\mu = 0.005$)	4.58(+1)	11.71(0)		
4-Methyl-8-hydroxyquinoline	4.67(+1)	11.62(0)		
1-Methylimidazole	7.06(+1)			
4-Methylimidazole	7.55(+1)			
N-Methyliminodiacetic acid	2.15	10.09		
S-Methylisothiourea	9.83(+1)			
O-Methylisourea	9.72(+1)			
Methylmalonic acid	3.07	5.87		
2-(N-Methylmethanesulfonamido)pyridine	1.73(+1)			
3-(N-Methylmethanesulfonamido)pyridine	3.94(+1)			
4-(N-Methylmethanesulfonamido)pyridine	5.14(+1)			
2-Methyl-6-methylaminopyridine (20°C)	3.17(+1)	8.84(0)		
3-Methyl-4-methylaminopyridine (20°C)	—	9.84(0)		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
4-Methyl-2,2'-(4-methylpyridyl)pyridine	5.32(+1)			
N-Methylmorpholine	7.13(+1)			
2-Methyl-1-naphthoic acid	3.11			
N-Methyl-1-naphthylamine	3.70(+1)			
2-Methyl-4-nitrobenzoic acid	1.86			
2-Methyl-6-nitrobenzoic acid	1.87			
1-Methyl-2-nitroterephthalic acid	3.11			
4-Methyl-2-nitroterephthalic acid	1.82			
3-Methylpentanedioic acid	4.25	5.41		
3-Methylpentane-2,4-dione	10.87			
2-Methylpentanoic acid	4.782			
3-Methylpentanoic acid	4.766			
4-Methylpentanoic acid	4.845			
cis-3-Methyl-2-pentenoic acid	5.15			
trans-3-Methyl-2-pentenoic acid	5.13			
4-Methyl-2-pentenoic acid	4.70			
4-Methyl-3-pentenoic acid	4.60			
6-Methyl-1,10-phenanthroline	5.11(+1)			
(2-Methylphenoxy)acetic acid	3.227			
(3-Methylphenoxy)acetic acid	3.203			
(4-Methylphenoxy)acetic acid	3.215			
(2-Methylphenyl)acetic acid (18°C)	4.35			
(4-Methylphenyl)acetic acid	4.370			
5-Methyl-5-phenylbarbituric acid	8.011(0)			
3-(2-Methylphenyl)propanoic acid	4.66			
3-(3-Methylphenyl)propanoic acid	4.677			
3-(4-Methylphenyl)propanoic acid	4.684			
1-Methyl-2-phenylpyrrolidine	8.80			
5-Methyl-1-phenyl-1,2,3-triazole-4-carboxylic acid	3.73			
Methylphosphinic acid	3.08			
Methylphosphonic acid	2.38	7.74		
3-Methyl-o-phthalic acid	3.18			
4-Methyl-o-phthalic acid	3.89			
N-Methylpiperazine ($\mu = 0.1$)	4.94(+2)	9.09(+1)		
2-Methylpiperazine	5.62(+2)	9.60(+1)		
N-Methylpiperidine	10.19(+1)			
2-Methylpiperidine	10.95(+1)			
3-Methylpiperidine	11.07(+1)			
4-Methylpiperidine ($\mu = 0.5$)	11.23(+1)			
2-Methyl-1,2-propanediamine	6.178(+2)	9.420(+1)		
2-Methyl-2-propanethiol	11.2			
2-Methylpropanoic acid	4.853			
2-Methyl-2-propylamine	10.682(+1)			
2-Methyl-2-propylglutaric acid	3.626			
2-Methylpyridine	5.96(+1)			
3-Methylpyridine	5.68(+1)			
4-Methylpyridine	6.00(+1)			
Methyl 4-pyridinecarboxylate	3.26(+1)			
6-Methylpyridine-2-carboxylic acid	5.83			
2-Methylpyridine-1-oxide	1.029(+1)			
3-Methylpyridine-1-oxide	10.921(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
4-Methylpyridine-1-oxide	1.258(+1)			
O-Methylpyridoxal ($\mu = 0.16$)	4.74			
Methyl-2-pyridyl ketoxime	9.97			
1-Methyl-2-(3-pyridyl)pyrrolidine	3.41	7.94		
1-Methylpyrrolidine	10.46(+1)			
1-Methyl-3-pyrroline	9.88(+1)			
5-Methylquinoline	4.62(+1)			
Methylsuccinic acid	4.13	5.64		
Methylsulfonylacetic acid	2.36			
3-Methylsulfonylaniline	2.68(+1)			
4-Methylsulfonylaniline	1.48(+1)			
3-Methylsulfonylbenzoic acid	3.52			
4-Methylsulfonylbenzoic acid	3.64			
4-Methylsulfonyl-3,5-dimethylphenol	8.13			
3-Methylsulfonylphenol	9.33			
4-Methylsulfonylphenol	7.83			
1-Methyl-1,2,3,4-tetrahydro-3-pyridinecarboxylic acid (arecaidine; isoguvacine)	9.07			
5-Methyl-1,2,3,4-tetrazole	3.32			
2-Methylthiazole ($\mu = 0.1$)	3.40(+1)			
4-Methylthiazole ($\mu = 0.1$)	3.16(+1)			
5-Methylthiazole ($\mu = 0.1$)	3.03(+1)			
Methylthioacetic acid	3.72			
4-Methylthioaniline	4.40(+1)			
2-Methylthioethylamine (30°C)	9.18(+1)			
Methylthioglycolic acid	7.68			
3-(S-Methylthio)phenol	9.53			
4-(S-Methylthio)phenol	9.53			
2-Methylthiopyridine (20°C)	3.59(+1)			
3-Methylthiopyridine (20°C)	4.42(+1)			
4-Methylthiopyridine (20°C)	5.94(+1)			
5-Methylthio-1,2,3,4-tetrazole	4.00(+1)			
O-Methylthreonine	2.02(+1)	9.00(0)		
O-Methyltyrosine	2.21(+1)	9.35(0)		
1-Methylxanthine	7.70	12.0		
3-Methylxanthine	8.10	11.3		
7-Methylxanthine	8.33	ca 13		
9-Methylxanthine	6.25			
Morphine (20°C)	7.87(+1)	9.85(0)		
Morpholine	8.492(+1)			
2-(N-Morpholino)ethanesulfonic acid (MES) (20°C)	6.15			
3-(N-Morpholino)-2-hydroxypropanesulfonic acid (37°C)	6.75			
3-(N-Morpholino)propanesulfonic acid (20°C)	7.20			
Murexide	0.0	9.20	10.50	
Myosmine	5.26			
1-Naphthalenecarboxylic acid (1-naphthoic acid)	3.695			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2-Naphthalenecarboxylic acid	4.161			
1-Naphthol (20°C)	9.30			
2-Naphthol (20°C)	9.57			
Naphthoquinone monoxime	8.01			
1-Naphthylacetic acid	4.236			
2-Naphthylacetic acid	4.256			
1-Naphthylamine	3.92(+1)			
2-Naphthylamine	4.11(+1)			
1-Naphthylarsonic acid	3.66	8.66		
1-Naphthysulfonic acid	0.57			
Narceine (15°C)	3.5(+1)	9.3		
Narcotine	6.18(+1)			
Nicotine	3.15(+1)	7.87(0)		
Nicotyrine	4.76(+1)			
Nitrilotriacetic acid (NTA) (20°C)	1.65	2.94	10.33	
Nitroacetic acid	1.68			
2-Nitroaniline	−0.28(+1)			
3-Nitroaniline	2.46(+1)			
4-Nitroaniline	1.01(+1)			
2-Nitrobenzene-1,4-dicarboxylic acid	1.73			
3-Nitrobenzene-1,2-dicarboxylic acid	1.88			
4-Nitrobenzene-1,2-dicarboxylic acid	2.11			
2-Nitrobenzoic acid	2.18			
3-Nitrobenzoic acid	3.46			
4-Nitrobenzoic acid	3.441			
<i>trans</i> -2-Nitrocinnamic acid	4.15			
<i>trans</i> -3-Nitrocinnamic acid	4.12			
<i>trans</i> -4-Nitrocinnamic acid	4.05			
Nitroethane	8.57			
2-Nitrohydroquinone	7.63	10.06		
<i>N</i> -Nitroiminodiacetic acid	2.21	3.33		
3-Nitromesitol	8.984			
Nitromethane	10.12			
1-Nitro-6,7-phenanthroline ($\mu = 0.2$)	3.23(+1)			
5-Nitro-1,10-phenanthroline	3.232(+1)			
6-Nitro-1,10-phenanthroline	3.23(+1)			
2-Nitrophenol	7.222			
3-Nitrophenol	8.360			
4-Nitrophenol	7.150			
(2-Nitrophenoxy)acetic acid	2.896			
(3-Nitrophenoxy)acetic acid	2.951			
(4-Nitrophenoxy)acetic acid	2.893			
2-Nitrophenylacetic acid	4.00			
3-Nitrophenylacetic acid	3.97			
4-Nitrophenylacetic acid	3.85			
2-Nitrophenylarsonic acid	3.37	8.54		
3-Nitrophenylarsonic acid	3.41	7.80		
4-Nitrophenylarsonic acid	2.90	7.80		
7-(4-Nitrophenylazo)-8-hydroxy-5-quinolinesulfonic acid	3.14(0)	7.495(−1)		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
3-Nitrophenylphosphonic acid	1.30	6.27		
4-Nitrophenylphosphonic acid	1.24	6.23		
3-(2'-Nitrophenyl)propanoic acid	4.504			
3-(4'-Nitrophenyl)propanoic acid	4.473			
3-Nitrophenylselenic acid	4.07			
4-Nitrophenylselenic acid	4.00			
1-Nitropropane	8.98			
2-Nitropropane	7.675			
2-Nitropropanoic acid	3.79			
2-Nitropyridine ($\mu = 0.02$)	-2.06(+1)			
3-Nitropyridine ($\mu = 0.02$)	0.79(+1)			
4-Nitropyridine ($\mu = 0.02$)	1.23(+1)			
<i>N</i> -Nitrosoiminodiacetic acid	2.28	3.38		
4-Nitrosophenol	6.48			
Nitrourea	4.15(+1)			
1,9-Nonanedioic acid (azelaic acid)	4.53	5.40		
Nonanoic acid (pelargonic acid)	4.95			
DL -Norleucine	2.335(+1)	9.834(0)		
Novocaine	8.85(+1)			
2,2,3,3,4,4,5,5-Octafluoropentanoic acid	2.65			
1,8-Octanedioic acid (suberic acid)	4.512	5.404		
Octanoic acid (caprylic acid)	4.895			
Octopine- DD	1.35	2.30	8.68	11.25
Octopine- LD	1.40	2.30	8.72	11.34
Octylamine	10.65(+1)			
L -(+)-Ornithine	1.94(+2)	8.65(+1)	10.76(0)	
Oxalic acid	1.271	4.272		
3,6-Oxaoctanedioic acid ($\mu = 1.0$)	3.055	3.676		
Oxoacetic acid	3.46			
2-Oxabutanedioic acid (oxaloacetic acid)	2.56	4.37		
2-Oxobutanoic acid	2.50			
5-Oxohexanoic acid (5-ketohexanoic acid) (18°C)	4.662			
3-Oxo-1,5-pentanedioic acid	3.10			
4-Oxopentanoic acid (levulinic acid)	4.59			
2-Oxopropanoic acid (pyruvic acid)	2.49			
Oxytetracycline	3.10(+1)	7.26	9.11	
Papaverine	5.90(+1)			
Pentamethylenebis(thioacetic acid) (18°C)	3.485	4.413		
3,3-Pentamethylenepentanedioic acid	3.49	6.96		
1,5-Pentanediamine	10.05(+2)	10.916(+1)		
2,4-Pentanedione	8.24(enol); 8.95(keto)			
1-Pentanoic acid (valeric acid)	4.842			
2-Pentenoic acid	4.70			
3-Pentenoic acid	4.52			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
4-Pentenoic acid	4.677			
Pentylarsonic acid	4.14	9.07		
<i>N</i> -Pentylveratramine	7.28(+1)			
Perhydrodiphenic acid (20°C)	4.96	6.68		
Perlolidine (18°C)	4.01	11.39		
Peroxyacetic acid	8.20			
1,7-Phenanthroline	4.30(+1)			
1,10-Phenanthroline	4.857(+1)			
6,7-Phenanthroline	4.857(+1)			
Phenazine	1.2(+1)			
Phenethylthioacetic acid	3.795			
Phenol	9.99			
Phenol-3-phosphoric acid	1.78	7.03	10.2	
Phenol-4-phosphoric acid	1.99	7.25	9.9	
Phenolphthalein	9.4			
3-Phenolsulfonic acid	—	9.05(−1)		
Phenosulsulfonephthalein	7.9			
Phenoxyactic acid	3.171			
2-Phenoxybenzoic acid	3.53			
3-Phenoxybenzoic acid	3.95			
4-Phenoxybenzoic acid	4.52			
5-Phenoxy-1,2,3,4-tetrazole	3.49(+1)			
Phenylacetic acid	4.312			
L -3-Phenyl- α -alanine	1.83(+1)	9.12(0)		
3-Phenyl- α -alanine, methyl ester	7.05(+1)			
Phenylalanylarginine ($\mu = 0.01$)	2.66(+1)	7.57(0)	12.40(−1)	
Phenylalanylglycine ($\mu = 0.01$)	3.10(+1)	7.71(0)		
7-Phenylazo-8-hydroxy-5-quinolinesulfonic acid	3.41(0)	7.850(−1)		
5-Phenylbarbituric acid	2.544(+1)			
2-Phenyl-2-benzylsuccinic acid	3.69	6.47		
1-Phenylbiguanide	2.13(+2)	10.76(+1)		
4-Phenylbutanoic acid	4.757			
Phenylbutazone	4.5(+1)			
2-Phenylenediamine	<2(+2)	4.47(+1)		
3-Phenylenediamine	2.65(+2)	4.88(+1)		
4-Phenylenediamine	3.29(+2)	6.08(+1)		
2-Phenylethylamine	9.83(+1)			
β -Phenylethylboronic acid	10.0			
DL - α -Phenylglycine	1.83(+1)	4.39(0)		
Phenylguanidine	10.77(+1)			
Phenylhydrazine	5.20(+1)			
2-Phenyl-3-hydroxypropanoic acid	3.53			
3-Phenyl-3-hydroxypropanoic acid	4.40			
Phenyliminodiacetic acid (20°C)	2.40	4.98		
Phenylmalonic acid	2.58	5.03		
Phenylmethanethiol	10.70			
2-Phenyl-2-phenethylsuccinic acid (20°C)	3.74	6.52		
2-Phenylphenol	9.55			
3-Phenylphenol	9.63			
4-Phenylphenol	9.55			
Phenylphosphinic acid (17°C)	2.1			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Phenylphosphonic acid	1.83	7.07		
O-Phenylphosphorylserine	2.13(+1)	8.79		
O-Phenylphosphorylserylglycine	3.18(+1)	6.95(0)		
O-Phenylphosphoryl-L-seryl-L-leucine	3.16(+1)	7.12(0)		
N-Phenylpiperazine ($\mu = 0.1$)	8.71(+1)			
2-Phenylpropanoic acid	4.38			
3-Phenylpropanoic acid (35°C)	4.664			
3-Phenyl-1-propylamine	10.39(+1)			
Phenylpropynoic acid (35°C)	2.269			
Phenylselenic acid	4.79			
Phenylselenoacetic acid ($\mu = 0.1$)	3.75			
β -Phenylserine ($\mu = 0.16$)	8.79(0)			
Phenylsuccinic acid (20°C)	3.78	5.55		
Phenylsulfonylacetic acid	2.66			
Phenylsulfonylacetic acid	2.44			
5-Phenyl-1,2,3,4-tetrazole	4.38(+1)			
1-Phenyl-1,2,3-triazole-4-carboxylic acid	2.88			
1-Phenyl-1,2,3-triazole-4,5-dicarboxylic acid	2.13	4.93		
Phosphoramidic acid	3.08	8.63		
O-Phosphorylethanolamine	5.838(+1)	10.638(0)		
O-Phosphorylserylglycine	3.13	5.41	8.01	
O-Phosphoryl-L-seryl-L-leucine	3.11	5.47	8.26	
Phosphoserine	2.08	5.65	9.74	
Phthalamide	3.79(0)			
Phthalazine	3.47(+1)			
o-Phthalic acid	2.950	5.408		
Phthalimide	9.90(0)			
Physostigmine	1.76(+1)	7.88(0)		
Picric acid (2,4,6-trinitrophenol) (18°C)	0.419			
Pilocarpine	1.3(+1)	6.85(0)		
Piperazine	5.333(+2)	9.781(+1)		
1,4-Piperazinebis(ethanesulfonic acid) (20°C)	6.80			
Piperazine-2-carboxylic acid	1.5	5.41	9.53	
Piperidine	11.123(+1)			
2-Piperidinecarboxylic acid	2.12(+1)	10.75(0)		
3-Piperidinecarboxylic acid	3.35(+1)	10.64(0)		
4-Piperidinecarboxylic acid	3.73(+1)	10.72(0)		
1-(2-Piperidinyl)-2-propanone (15°C)	9.45			
Piperine (15°C)	1.98(+1)			
Proline	1.99(+1)	10.96(0)		
1,2-Propanediamine	6.607(+2)	9.702(+1)		
1,3-Propanediamine	8.49(+2)	10.47(+1)		
1-Propanethiol	10.86			
1,2,3-Propanetriamine	3.72(+3)	7.95(+2)	9.59(+1)	
1,2,3-Propanetricarboxylic acid	3.67	4.87	6.38	
Propanoic acid	4.874			
Propenoic acid	4.247			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (Continued)

Substance	pK_1	pK_2	pK_3	pK_4
<i>N</i> -Propionylglycine	3.718(0)			
2-Propoxybenzoic acid (20°C)	4.24			
3-Propoxybenzoic acid (20°C)	4.20			
4-Propoxybenzoic acid (20°C)	4.78			
<i>N</i> -Propylalanine	2.21(+1)	10.19(0)		
Propylamine	10.568(+1)			
Propylarsonic acid (18°C)	4.21	9.09		
Propylenimine	8.18(+1)			
<i>N</i> -Propylglycine ($\mu = 0.1$)	2.38(+1)	10.03(0)		
<i>L</i> -Propylglycine	3.19(+1)	8.97(0)		
Propylmalonic acid	2.97	5.84		
Propylphosphinic acid	3.46			
Propylphosphonic acid	2.49	8.18		
2-Propylpyridine	6.30(+1)			
<i>N</i> -Propylveratramine	7.20(+1)			
2-Propynoic acid	1.887			
Pseudoecgonine	9.70			
Pseudoisocyanine ($\mu = 0.2$)	4.59(+2)			
Pseudotropine	9.86(+1)			
Pteroylglutamic acid	8.26			
Purine	2.52(+1)	8.92(0)		
Pyrazine	0.6(+1)			
Pyrazinecarboxamide	0.5(+1)			
Pyrazole	2.61(+1)			
Pyridazine	2.33(+1)			
Pyridine	5.17(+1)			
Pyridine- d_5	5.83(+1)			
2-Pyridinealdoxime	3.56(+1)	10.17(0)		
3-Pyridinealdoxime	4.07(+1)	10.39(0)		
4-Pyridinealdoxime	4.73(+1)	10.03(0)		
2-Pyridinecarbaldehyde	3.84(+1)			
3-Pyridinecarbaldehyde	3.80(+1)			
4-Pyridinecarbaldehyde	4.74(+1)			
3-Pyridinecarbamide (nicotin- amide)	3.33(+1)			
3-Pyridinecarbonitrile	1.35(+1)			
Pyridine-2-carboxylic acid (picol- inic acid)	1.01(+1)	5.29(0)		
Pyridine-3-carboxylic acid (nico- tinic acid)	2.07(+1)	4.75(0)		
Pyridine-4-carboxylic acid (isoni- cotic acid)	1.84(+1)	4.86(0)		
Pyridine-2,3-dicarboxylic acid	2.36(+1)	7.08(0)		
Pyridine-2,4-dicarboxylic acid	2.23(+1)	7.02(0)		
Pyridine-2,6-dicarboxylic acid	2.16(+1)	6.92(0)		
Pyridine-1-oxide	0.688(+1)			
Pyridoxal	4.20(+1)	8.66(ring OH)		
Pyridoxal-5-phosphate ($\mu = 0.15$)	<2.5	4.14	6.20	8.69
Pyridoxamine ($\mu = 0.1$)	3.37(+2)	8.01(+1)	10.13(ring OH)	
Pyridoxamine-5-phosphate ($\mu =$ 0.15; pK_5 10.92)	2.5	3.69	5.76	8.61

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Pyridoxine (vitamin B ₆) (18°C)	5.00(+1)	8.96(ring OH)		
3-(2'-Pyridyl)alanine	1.37(+2)	4.02(+1)	9.22(0)	
3-(3'-Pyridyl)alanine	1.77(+2)	4.64(+1)	9.10(0)	
2-(2'-Pyridyl)benzimidazole ($\mu = 0.16$)	5.58(+1)			
2-(2'-Pyridyl)imidazole ($\mu = 0.005$)	8.98(+1)			
4-(2'-Pyridyl)imidazole ($\mu = 0.1$)	5.49(+1)			
Pyrimidine	1.30(+1)			
2,4(1 <i>H</i> ,3 <i>H</i>)-Pyrimidinedione (uracil)	0.6(+1)	9.46(0)		
2,4,5,6(1 <i>H</i> ,3 <i>H</i>)-Pyrimidinetetrone-5-oxime	4.57(0)			
Pyrocatecholsulfonephthaleine	7.82	9.76	11.73	
Pyroxilidine	11.11(+1)			
Pyrrole-1-carboxylic acid	4.45			
Pyrrole-2-carboxylic acid	4.45			
Pyrrole-3-carboxylic acid	4.453			
Pyrrolidine	11.305(+1)			
Pyrrolidine-2-carboxylic acid (proline)	1.952(+1)	10.640(0)		
2-[2-(<i>N</i> -Pyrrolidinyl)ethyl]pyridine	3.60(+2)	9.39(+1)		
3-[2-(<i>N</i> -Pyrrolidinyl)ethyl]pyridine	4.28(+2)	9.28(+1)		
4-[2-(<i>N</i> -Pyrrolidinyl)ethyl]pyridine	4.65(+2)	9.27(+1)		
2-(1-Pyrrolidinylmethyl)pyridine	2.54(+1)	8.56(+1)		
3-(1-Pyrrolidinylmethyl)pyridine	3.14(+2)	8.36(+1)		
4-(1-Pyrrolidinylmethyl)pyridine	3.38(+2)	8.16(+1)		
3-Pyrroline	-0.27(+1)			
Quinidine	4.0(+1)	8.54(0)		
Quinine	4.11(+1)	8.52(0)		
Quinoline	4.80(+1)			
Quinoxaline	0.72(+1)			
D -Raffinose	12.74			
Riboflavin (vitamin B ₂) ($\mu = 0.01$)	ca -0.2	9.69		
α - D -Ribofuranose	12.11			
D -Ribose-5'-phosphonic acid	—	6.70(-1)	13.05(-2)	
D -Saccharic acid	5.00(0)			
Saccharin (<i>o</i> -benzoic sulfimide)	2.32			
Sarcosine	2.12(+1)	10.20(0)		
Sarcosine amide	8.35(+1)			
Sarcosine dimethylamide	8.86(+1)			
Sarcosine methylamide	8.28(+1)			
Sarcosylglycine ($\mu = 0.16$)	3.15(+1)	8.56(0)		
Sarcosylleucine	3.15(+1)	8.67(0)		
Sarcosylsarcosine	2.92(+1)	9.15(0)		
Sarcosylserine	3.17(+1)	8.63(0)		
3-Selenosemicarbazide ($\mu = 0.1$)	0.8(+1)			
Semicarbazide ($\mu = 0.1$)	3.53(+1)			
L -Serine	2.21(+1)	9.15(0)	13.6	

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
Serine, methyl ester ($\mu = 0.1$)	7.03(+1)			
Serylglycine ($\mu = 0.15$)	2.10(+1)	7.33(0)		
L-Seryl-L-leucine	3.08(+1)	7.45(0)		
Solanine	7.34(+1)			
D-Sorbitol (17.5°C)	13.60			
L-(−)-Sorbose (18°C)	11.55			
Sparteine	4.49(+1)	11.76(0)		
Spinaceamine ($\mu = 0.1$)	4.895(+2)	8.90(+1)		
Spinacine	1.649(+2)	4.936(+1)	8.663(0)	
L-Strychnine (15°C)	2.50	8.20		
Succinamic acid (succinic acid monoamide)	4.39(0)			
Succinic acid	4.207	5.635		
DL-Succinimide	9.623			
β -(4'-Sulfaminophenyl)alanine	1.99(+1)	8.64(0)	10.26(−1)	
3-Sulfamylbenzoic acid	3.54			
4-Sulfamylbenzoic acid	3.47			
4-Sulfamylphenylphosphoric acid	1.42	6.38	10.0	
Sulfanilamide	10.43(+1)			
Sulfoacetic acid	—	4.0		
3-Sulfobenzoic acid	—	3.78		
4-Sulfobenzoic acid	—	3.72		
3-Sulfophenol	0.39	9.07		
4-Sulfophenol	0.58	8.70		
2-Sulfopropanoic acid	1.99			
5-Sulfosalicyclic acid	2.49	12.00		
Sylvic acid	7.62			
D-Tartaric acid	3.036	4.366		
meso-Tartaric acid	3.22	4.81		
Tetracycline ($\mu = 0.005$)	3.30(+1)	7.68	9.69	
Tetradehydrohimbine	10.59(+1)			
Tetraethylenepentamine [$\mu = 0.1$; pK_s 9.67(+1)]	2.98(+5)	4.72(+4)	8.08(+3)	9.10(+2)
1,4,5,6-Tetrahydro-1,2-dimethylpyridine	11.38(+1)			
1,4,5,6-Tetrahydro-2-methylpyridine	9.53(+1)			
cis-Tetrahydronaphthalene-2,3-dicarboxylic acid (20°C)	3.98	6.47		
trans-Tetrahydronaphthalene-2,3-dicarboxylic acid (20°C)	4.00	5.70		
5,6,7,8-Tetrahydro-1-naphthol	10.28			
5,6,7,8-Tetrahydro-2-naphthol	10.48			
Tetrahydroserpentine	10.55(+1)			
2,3,5,6-Tetramethylbenzoic acid	3.415			
Tetramethylenebis(thioacetic acid) (18°C)	3.463	4.423		
Tetramethylenediamine	9.22(+2)	10.75(+1)		
N,N,N',N'-Tetramethylethylenediamine	2.20(+2)	6.35(+1)		
2,3,5,6-Tetramethyl-4-methylaminopyridine	0.07(+1)			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2,2,6,6-Tetramethylpiperidine ($\mu = 0.5$)	1.24(+1)			
2,3,5,6-Tetramethylpyridine (20°C)	7.90(+1)			
Tetramethylsuccinic acid	3.50	7.28		
1,2,3,4-Tetrazole	4.90			
Thebaine	7.95(+1)			
2-Thenoyltrifluoroacetone	5.70(0)			
Theobromine	0.68(+1)	7.89		
Theophylline	<1(+1)	8.80		
Thiazoline	2.53(+1)			
Thioacetic acid	3.33			
<i>o</i> -Thiocresol	6.64			
<i>m</i> -Thiocresol	6.58			
<i>p</i> -Thiocresol	6.52			
Thiocyanatoacetic acid	2.58			
2,2'-Thiodiacetic acid	3.32	4.29		
4,4'-Thiodibutanoic acid (18°C)	4.351	5.275		
3,3'-Thiodipropanoic acid (18°C)	4.085	5.075		
3-Thio-S-methylcarbazine ($\mu = 0.1$)	7.563(+1)			
1-Thionylcarboxylic acid	3.53			
2-Thionylcarboxylic acid	4.10			
2-Thiophenecarboxylic acid (30°C)	3.529			
3-Thiophenecarboxylic acid (3-thenoic acid)	4.10			
Thiophenol	6.50			
3-Thiosemicarbazide ($\mu = 0.1$)	1.5(+1)			
3-Thiosemicarbazide-1,1-diacetic acid (30°C)	2.94	4.07		
Thiourea	2.03(+1)			
Thorin	3.7	8.3	11.8	
Thymidine	9.79	12.85		
<i>p</i> -Toluenesulfinic acid	1.7			
Toluhydroquinone	10.03	11.62		
<i>o</i> -Toluidine	4.45(+1)			
<i>m</i> -Toluidine	4.71(+1)			
<i>p</i> -Toluidine	5.08(+1)			
<i>o</i> -Tolylacetic acid (18°C)	4.36			
<i>p</i> -Tolylacetic acid (18°C)	4.36			
<i>o</i> -Tolylarsonic acid	3.82	8.85		
<i>m</i> -Tolylarsonic acid	3.82	8.60		
<i>p</i> -Tolylarsonic acid	3.70	8.68		
<i>o</i> -Tolylphosphonic acid	2.10	7.68		
<i>m</i> -Tolylphosphonic acid	1.88	7.44		
<i>p</i> -Tolylphosphonic acid	1.84	7.33		
3-Tolylselenic acid	4.80			
4-Tolylselenic acid	4.88			
Triacetylmethane	5.81			
Triallylamine	8.31(+1)			
1,3,5-Triazine-2,4,6-triol	7.20	11.10		
1 <i>H</i> -1,2,3-Triazole	—	9.26		
1 <i>H</i> -1,2,4-Triazole	2.386(+1)	9.972		
1,2,3-Triazole-4-carboxylic acid	3.22	8.73		

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
1,2,3-Triazole-4,5-dicarboxylic acid	1.86	5.90	9.30	
1,2,4-Triazolidine-3,5-dione (urazole)	5.80			
Tribromoacetic acid	−0.147			
2,4,6-Tribromobenzoic acid	1.41			
Trichloroacetic acid	0.52			
Trichloroacrylic acid	1.15			
3,3,3-Trichlorolactic acid	2.34			
Trichloromethylphosphonic acid	1.63	4.81		
2,4,5-Trichlorophenol	7.37			
3,4,5-Trichlorophenol	7.839			
Tricine (20°C)	8.15			
Triethanolamine	7.76(+1)			
Triethylamine	10.72(+1)			
Triethylenediamine	4.18(+2)	8.19(+1)		
Triethylenetetramine (20°C)	3.32(+4)	6.67(+3)	9.20(+2)	9.92(+1)
Triethylsuccinic acid	2.74			
Trifluoroacetic acid	0.50			
Trifluoroacrylic acid	1.79			
4,4,4-Trifluoro-2-aminobutanoic acid	1.600(+1)	8.169(0)		
4,4,4-Trifluoro-3-aminobutanoic acid	2.756(+1)	5.822(0)		
4,4,4-Trifluorobutanoic acid	4.16			
α,α,α -Trifluoro- <i>m</i> -cresol	8.950			
4,4,4-Trifluorocrotonic acid	3.15			
5,5,5-Trifluoroleucine	2.045(+1)	8.942(0)		
3-(Trifluoromethyl)aniline	3.5(+1)			
4-(Trifluoromethyl)aniline	2.6(+1)			
3-Trifluoromethylphenol	8.950			
5-Trifluoromethyl-1,2,3,4-tetrazole	1.70			
6,6,6-Trifluoronorleucine	2.164(+1)	9.463(0)		
5,5,5-Trifluoronorvaline	2.042(+1)	8.916(0)		
5,5,5-Trifluoropentanoic acid	4.50			
3,3,3-Trifluoropropanoic acid	3.06			
4,4,4-Trifluorothreonine	1.554(+1)	7.822(0)		
4,4,4-Trifluorovaline	1.537(+1)	8.098(0)		
1,2,3-Trihydroxybenzene (pyrogallol)	9.03(0)	11.63(−1)		
1,3,5-Trihydroxybenzene (phloroglucinol)	8.45(0)	8.88(−1)		
2,4,6-Trihydroxybenzoic acid	1.68(0)			
3,4,5-Trihydroxybenzoic acid	4.19(0)	8.85(−1)		
3,4,5-Trihydroxycyclohex-1-ene-1-carboxylic acid [D -(−)-shikimic acid]	4.15			
2,4,6-Tri(hydroxymethyl)phenol	9.56			
Triisobutylamine	10.42(+1)			
Trimethylamine	9.80(+1)			
3-(Trimethylamino)phenol	8.06			
4-(Trimethylamino)phenol	8.35			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
2,4,6-Trimethylaniline	4.38(+1)			
2,4,6-Trimethylbenzoic acid	3.448			
Trimethylenebis(thioacetic acid) (18°C)	3.435	5.383		
2,3,4-Trimethylphenol	10.59			
2,4,5-Trimethylphenol	10.57			
2,4,6-Trimethylphenol	10.88			
3,4,5-Trimethylphenol	10.25			
2,3,6-Trimethylpyridine ($\mu = 0.5$)	7.60(+1)			
2,4,6-Trimethylpyridine	7.43(+1)			
2,4,6-Trimethylpyridine-1-oxide	1.990(+1)			
3-(Trimethylsilyl)benzoic acid	4.089			
4-(Trimethylsilyl)benzoic acid	4.192			
2,4,5-Trimethylthiazole ($\mu = 0.1$)	4.55			
2,4,6-Trinitroaniline (picramide)	-10.23(+1)			
2,4,6-Trinitrobenzene acid	0.654			
2,2,2-Trinitroethanol	2.36			
Trinitromethane (20°C)	0.17			
Triphenylacetic acid	3.96			
Tripropylamine	10.66(+1)			
Tris(2-hydroxyethyl)amine	7.762(+1)			
Tri(hydroxymethyl)aminomethane (TRIS)	8.08(+1)			
2-[Tris(hydroxymethyl)methyl amino]-1-ethanesulfonic acid (TES)	7.50			
3-[Tris(hydroxymethyl)methyl amino]-1-propanesulfonic acid (TAPS) (20°C)	8.4			
N-[Tris(hydroxymethyl)methyl]- glycine (tricine)	2.023(+1)	8.135		
Tris(trimethylsilyl)amine	4.70(+1)			
Trithiocarbonic acid (20°C)	2.64			
Tropacocaine (15°C)	9.88(+1)			
3-Tropanol (tropine)	10.33(+1)			
Trypsin ($\mu = 0.1$)	6.25			
L-Tryptophan	2.38(+1)	9.39(0)		
DL-Tyrosine	2.18(+1)	9.11(0)	10.6(OH)	
Tyrosine amide	7.48	9.89		
Tyrosine, ethyl ester	7.33	9.80		
Tyrosylarginine ($\mu = 0.01$)	2.65(+1)	7.39(0)	9.36(-1)	11.62(-2)
Tyrosyltyrosine	3.52(+1)	7.68(0)	9.80(-1)	10.26(-2)
α -Ureidobutanoic acid	3.886(0)			
γ -Ureidobutanoic acid	4.683(0)			
β -Ureidopropanoic acid	4.487(0)			
Uric acid	5.40	5.53		
Uridine	9.30			
Uridine-5'-diphosphoric acid	7.16			
Uridine-5'-phosphoric acid (5'-uri- dylic acid)	6.63			
Uridine-5'-triphosphoric acid	7.58			

TABLE 8.8 pK_a Values of Organic Materials in Water at 25°C (*Continued*)

Substance	pK_1	pK_2	pK_3	pK_4
DL -Valine	2.32(+1)	9.61(0)		
L -Valine	2.296(+1)	9.79(0)		
Valine amide ($\mu = 0.2$)	8.00			
L -Valine, methyl ester	7.49(+1)			
L -Valylglycine	3.23(+1)	8.00(0)		
Vetramine	7.49(+1)			
Veratrine	8.85(+1)			
Vinylmethylamine	9.69(+1)			
2-Vinylpyridine	4.98(+1)			
4-Vinylpyridine	5.62(+1)			
Vitamin B ₁₂	7.64(+1)			
Xanthine (40°C)	0.68(+1)			
Xanthosine	<2.5(+1)	5.67(0)	12.00(−1)	
Xylenol Orange [pK_5 10.46(−4); pK_6 12.28(−5)]	—	2.58(−1)	3.23(−2)	6.37(−3)
D -(+)-Xylose	12.15(0)			
Zincon	—	4	7.85	15

TABLE 8.9 Selected Equilibrium Constants in Aqueous Solution at Various Temperatures

Abbreviations Used in the Table

- (+ 1), protonated cation
- (− 2), doubly ionized anion
- (0), neutral molecule
- pK_{auto} , negative logarithm (base 10) of autoprotolysis constant
- (− 1), singly ionized anion
- pK_{sp} , negative logarithm (base 10) of solubility product

Substance	Temperature, °C									
	0	5	10	15	20	25	30	35	40	50
Acetic acid (0)	4.780	4.770	4.762	4.758	4.757	4.756	4.757	4.762	4.769	4.787
DL-N-Acetylalanine (+ 1)		3.699	3.699	3.703	3.708	3.715	3.725	3.733	3.745	3.774
β-Acetylaminopropionic (+ 1)		4.479	4.465	4.465	4.449	4.445	4.444	4.443	4.445	4.457
N-Acetylglycine (+ 1)		3.682	3.676	3.673	3.667	3.670	3.673	3.678	3.685	3.706
α-Alanine										
(+ 1)	2.42		2.39		2.35	2.34	2.33	2.33	2.33	2.33
(0)	10.59		10.29		10.01	9.87	9.74	9.62	9.49	9.26
2-Aminobenzenesulfonic acid (0), pK ₂	2.633	2.591	2.556	2.521	2.448	2.459	2.431	2.404	2.380	2.338
3-Aminobenzenesulfonic acid (0), pK ₂	4.075	4.002	3.932	3.865	3.799	3.738	3.679	3.622	3.567	3.464
4-Aminobenzenesulfonic acid (0), pK ₂	3.521	3.457	3.398	3.338	3.283	3.227	3.176	3.126	3.079	2.989
3-Aminobenzoic acid (0)					4.90	4.79	4.75		4.68	4.60
4-Aminobenzoic acid (0)					4.95	4.85	4.90		4.95	5.10
2-Aminobutyric acid										
(+ 1)			2.334			2.286		2.289 ^{37.5°C}		2.297
(0)			10.530			9.380		9.518 ^{37.5°C}		9.234
4-Aminobutyric acid										
(+ 1)			4.057	4.046	4.038	4.031	4.027	4.025	4.027	4.032
(0)			11.026	10.867	10.706	10.556	10.409	10.269	10.114	9.874
2-Aminoethylsulfonic acid (0)			9.452	9.316	9.186	9.061	8.940	8.824	8.712	9.499

TABLE 8.9 Selected Equilibrium Constants in Aqueous Solution at Various Temperatures (*Continued*)

Substance	Temperature, °C									
	0	5	10	15	20	25	30	35	40	50
2-Amino-3-methylpentanoic acid (+ 1)	2.365 ^{1°C}		2.338 ^{12.5°C}			2.320		2.317 ^{37.5°C}		2.332
(0)	10.460 ^{1°C}		10.100 ^{12.5°C}			9.758		9.439 ^{37.5°C}		9.157
2-Amino-2-methyl- 1,3-propanediol	9.612	9.433	9.266	9.104	8.951	8.801	8.659	8.519	8.385	8.132
2-Amino-2-methylpropionic acid (+ 1)	2.419 ^{1°C}		2.380 ^{12.5°C}			2.357		2.351 ^{37.5°C}		2.356
(0)	10.960 ^{1°C}		10.580 ^{12.5°C}			10.205		9.872 ^{37.5°C}		9.561
2-Aminopentanoic acid (+ 1)	2.376 ^{1°C}		2.347			2.318			2.309	2.313
(0)	10.508 ^{1°C}			10.154 ^{12.5°C}		9.808		9.490 ^{37.5°C}		9.198
3-Aminopropionic acid (+ 1)	3.656	3.627		3.583		3.551		3.524	3.517	
(0)	11.000	10.830		10.526		10.235		9.963	9.842	
4-Aminopyridine (+ 1)	9.873	9.704	9.549	9.398	9.252	9.114	8.978	8.846	8.717	8.477
Ammonium ion (+ 1)	10.081	9.904	9.731	9.564	9.400	9.245	9.093	8.947	8.805	8.539
Arginine (+ 1)	1.914	1.885	1.870	1.849	1.837	1.823	1.814	1.801	1.800	1.787
(0)	9.718	9.563	9.407	9.270	9.123	8.994	8.859	8.739	8.614	8.385
Barbituric acid (+ 1)				3.969	3.980	4.02	4.00	4.008	4.017	4.032
(0)				8.493	8.435	8.372	8.302	8.227	8.147	7.974
Benzoic acid (0)		4.231	4.220	4.215	4.206	4.204	4.203	4.207	4.219	4.223
Boric acid (0)	9.508	9.439	9.380	9.327	9.280	9.236	9.197	9.161	9.132	9.080
Bromoacetic acid (0)				2.875	2.887	2.902	2.918	2.936		
3-Bromobenzoic acid (0)				3.818	3.813	3.810	3.808	3.810	3.813	
4-Bromobenzoic acid (0)				4.011	4.005	3.99	4.001	4.001	4.003	
Bromopropynoic acid (0)			1.786	1.814	1.839	1.855	1.879	1.900	1.919	

3- <i>tert</i> -Butylbenzoic acid (0)				4.266	4.231	4.199	4.170	4.143	4.119	
4- <i>tert</i> -Butylbenzoic acid (0)				4.463	4.425	4.389	4.354	4.320	4.287	
2-Butynoic acid (0)			2.618	2.626	2.611	2.620	2.618	2.621	2.631	
Butyric acid (0)	4.806	4.804	4.803	4.805	4.810	4.817	4.827	4.840	4.854	4.885
DL - <i>N</i> -Carbamoylalanine (+ 1)		3.898	3.894	3.891	3.890	3.892	3.896	3.902	3.908	3.931
<i>N</i> -Carbamoylglycine (+ 1)		3.911	3.900	3.889	3.879	3.876	3.874	3.873	3.875	3.888
Carbon dioxide + water										
(0)	6.577	6.517	6.465	6.429	6.382	6.352	6.327	6.309	6.296	6.285
(− 1)	10.627	10.558	10.499	10.431	10.377	10.329	10.290	10.250	10.220	10.172
Chloroacetic acid (0)				2.845	2.856	2.867	2.883	2.900		
3-Chlorobenzoic acid (0)				3.838	3.831	3.83	3.825	3.826	3.829	
4-Chlorobenzoic acid (0)				4.000	3.991	3.986	3.981	3.980	3.981	
Chloropropynoic acid (0)			1.766	1.796	1.820	1.845	1.864	1.879	1.893	
Citric acid										
(0)	3.220	3.200	3.176	3.160	3.142	3.128	3.116	3.109	3.099	3.095
(− 1)	4.837	4.813	4.797	4.782	4.769	4.761	4.755	4.751	4.750	4.757
(− 2)	6.393	6.386	6.383	6.384	6.388	6.396	6.406	6.423	6.439	6.484
Cyanoacetic acid (0)		2.445	2.447	2.452	2.460	2.460	2.482	2.496	2.511	
2-Cyano-2-methylpropionic acid										
(0)		2.342	2.360	2.379	2.400	2.422	2.446	2.471	2.498	
5,5-Diethylbarbituric acid (0)	8.40	8.30	8.22	8.169	8.094	8.020	7.948	7.877	7.808	7.673
Diethylmalonic acid										
(0)			2.129	2.136	2.144	2.151	2.160	2.172	2.187	
(− 1)			7.400	7.401	7.408	7.417	7.428	7.441	7.457	
2,3-Dimethylbenzoic acid (0)				3.663	3.687	3.771	3.726	3.762	3.788	
2,4-Dimethylbenzoic acid (0)				4.154	4.187	4.217	4.244	4.268	4.290	
2,5-Dimethylbenzoic acid (0)				3.911	3.954	3.990	4.020	4.045	4.065	
2,6-Dimethylbenzoic acid (0)				3.234	3.304	3.362	3.409	3.445	3.472	
3,5-Dimethylbenzoic acid (0)				4.292	4.299	4.302	4.304	4.306	4.306	
<i>N,N'</i> -Dimethylethyleneamine-										
<i>N,N'</i> -diacetic acid										
(0)	6.294		6.169		6.047		5.926		5.803	
(− 1)	10.446		10.268		10.068		9.882		9.684	
<i>N,N</i> -Dimethylglycine (0)		10.34		10.14		9.94		9.76		
3,5-Dinitrobenzoic acid (0)			2.60		2.73		2.85		2.96	3.07

TABLE 8.9 Selected Equilibrium Constants in Aqueous Solution at Various Temperatures (*Continued*)

Substance	Temperature, °C									
	0	5	10	15	20	25	30	35	40	50
2-Ethylbutyric acid (0)	4.623		4.664		4.710	4.751	4.758		4.812	4.869
5-Ethyl-5-phenylbarbituric acid (0)				7.592	7.517	7.445	7.377	7.311	7.248	7.130
Fluoroacetic acid (0)				2.555	2.571	2.586	2.604	2.624		
Formic acid (0)	3.786	3.772	3.762	3.757	3.753	3.751	3.752	3.758	3.766	3.782
2-Furancarboxylic acid (0)						3.164	3.200	3.216	3.239	
Glucose-1-phosphate (0)		6.506	6.500	6.499	6.500	6.504	6.510	6.519	6.531	6.561
Glycerol-1-phosphoric acid (− 1)		6.642	6.641	6.643	6.648	6.656	6.666	6.679	6.695	6.733
Glycerol-2-phosphoric acid (0)		1.223	1.245	1.271	1.301	1.335	1.372	1.413	1.457	1.554
(− 1)		6.657	6.650	6.646	6.646	6.650	6.657	6.666	6.679	6.712
Glycine (+ 1)			2.397	2.380	2.36	2.351	2.34	2.33	2.327	2.32
(0)		10.34	10.193	10.044	9.91	9.780	9.65	9.53	9.412	9.19
Glycolic acid (0)	3.875		3.844 ^{12.5°C}			3.831		3.833 ^{37.5°C}		3.849
Glycylasparagine (+ 1)		2.968	2.958	2.952	2.943	2.942	2.942	2.944	2.947	2.959
N-Glycylglycine (+ 1)	3.201					3.126				3.159
			8.594 ^{12.5°C}			8.252		7.948 ^{37.5°C}		7.668
Hexanoic acid (0)	4.840		4.839		4.849		4.865		4.890	4.920
Hydrogen cyanide (0)			9.63	9.49	9.36	9.21	9.11	8.99	8.88	
Hydrogen peroxide (0)	12.23			11.86	11.75	11.65	11.55	11.45		11.21
Hydrogen sulfide (0)		7.33	7.24	7.13	7.05	6.97	6.90	6.82	6.79	6.69
(− 1)		13.5		13.2		12.90	12.75	12.6		
4-Hydroxybenzoic acid (0)				4.596	4.586	4.582	4.577	4.576	4.578	
Hydroxylamine (0)				6.186	6.063	5.948		5.730		
2-Hydroxy-1-naphthoic acid (0)					3.29		3.24		3.19	3.26
(− 1)					9.68		9.65		9.61	9.58

4-Hydroxyproline (+ 1) (0)	1.900 ^{1°C} 10.274 ^{1°C}		1.850 ^{12.5°C} 9.958 ^{12.5°C}			1.818 9.662		1.798 ^{37.5°C} 9.394 ^{37.5°C}		1.796 9.138
2-Hydroxypropionic acid (0)	3.880	3.873	3.868	3.861	3.857	3.858	3.861	3.867	3.873	3.895
DL-2-Hydroxysuccinic acid (0) (− 1)	3.537 5.119	3.520 5.108	3.494 5.098	3.482 5.096	3.472 5.096	3.458 5.097	3.452 5.099	3.446 5.104	3.444 5.117	3.445 5.149
Hypobromous acid (0)				8.83		8.60		8.47	8.37 ^{45°C}	
Hypochlorous acid (0)	7.82	7.75	7.69	7.63	7.58	7.54	7.50	7.46		7.05
Imidazole (+ 1)	7.581	7.467	7.334	7.216	7.103	6.993	6.887	6.784	6.685	6.497
Iodoacetic acid (0)				3.143	3.158	3.175	3.193	3.213		
DL-Isoleucine (+ 1) (0)	2.365 10.460		2.338 ^{12.5°C} 10.100 ^{12.5°C}			2.318 9.758		2.317 ^{37.5°C} 9.439 ^{37.5°C}		2.332 9.157
Isopropylmalonic acid, mononitrile (0)		2.299	2.320	2.343	2.365	2.401	2.427	2.452	2.481	
Lactic acid (0)	3.880	3.873	3.868	3.862	3.857	3.858	3.861	3.867	3.873	3.895
Lead sulfate, pK _{sp}	8.01			7.87		7.80		7.73		7.63
DL-Leucine (+ 1) (0)	2.383 ^{1°C} 10.458 ^{1°C}		2.348 ^{12.5°C} 10.095 ^{1.5°C}			2.328 9.744		2.327 ^{37.5°C} 9.434 ^{37.5°C}		2.333 9.142
Malonic acid (− 1)	5.670	5.665	5.667	5.673	5.683	5.696	5.710	5.730	5.753	5.803
Mannose (0)			12.45			12.08			11.81	
Mercury(I) chloride, pK _{sp}			18.65	18.48	18.27	17.88		16.79		
Methanol (solvent), pK _{auto}		17.12		16.84		16.71	16.65	16.53		
Methylamine (+ 1)	11.496		11.130		10.787	10.62	10.466		10.161	9.876
Methylaminodiacetic acid (0) (− 1)	2.138 10.474		2.142 10.287		2.146 10.088		2.150 9.920		2.154 9.763	
3-Methylbenzoic acid (0)				4.303	4.285	4.269	4.256	4.244	4.235	
4-Methylbenzoic acid (0)				4.390	4.376	4.362	4.349	4.336	4.322	
3-Methylbutyric acid (0)	4.726		4.742		4.767		4.794		4.831	4.871
4-Methylpentanoic acid (0)	4.827		4.827		4.837		4.853		4.879	4.908

TABLE 8.9 Selected Equilibrium Constants in Aqueous Solution at Various Temperatures (*Continued*)

Substance	Temperature, °C									
	0	5	10	15	20	25	30	35	40	50
5-Methyl-5-phenylbarbituric acid (0)				8.104	8.057	8.011	7.966	7.922	7.879	7.797
2-Methylpropionic acid (0)	4.825		4.827		4.840	4.853	4.886		4.918	4.955
2-Methyl-2-propylamine (+ 1)		11.439	11.240	11.048	10.862	10.682	10.511	10.341		
Nitric acid (0)	− 1.65					− 1.38				− 1.20
Nitrilotriacetic acid (0)	1.69		1.65		1.65		1.66		1.67	
(− 1)	2.95		2.95		2.94		2.96		2.98	
(− 2)	10.59		10.45		10.33		10.23			
4-Nitrobenzoic acid (0)				3.448	3.444	3.441	3.441	3.442	3.445	
Nitrous acid (0)				3.244	3.177	3.138		3.100		
DL-Norleucine (+ 1)	2.394		2.356 ^{12.5°C}			2.335		2.324 ^{37.5°C}		2.328
(0)	10.564		10.190 ^{12.5°C}			9.834		9.513 ^{37.5°C}		9.224
Oxalic acid (− 1)	4.210	4.216	4.227	4.240	4.254	4.272	4.295	4.318	4.349	4.409
2,4-Pentanedione (0)	9.07					8.95			8.90	
Pentanoic acid (0)	4.823		4.763		4.835	4.842	4.851		4.861	4.906
Phenylalanine (0)			9.75			9.31			8.96	
Phosphoric acid (0)	2.056	2.073	2.088	2.107	2.127	2.148	2.171	2.196	2.224	2.277
(− 1)	7.313	7.282	7.254	7.231	7.213	7.198	7.189	7.185	7.181	7.183
o-Phthalic acid (0)	2.925	2.927	2.931	2.937	2.943	2.950	2.958	2.967	2.978	3.001
(− 1)	5.432	5.418	5.410	5.405	5.405	5.408	5.416	5.427	5.442	5.485
Piperidine (+ 1)	11.963	11.786	11.613	11.443	11.280	11.123	10.974	10.818	10.670	10.384
Proline (+ 1)	2.011		1.964 ^{12.5°C}			1.952		1.950 ^{37.5°C}		1.958
(0)	11.296		10.972 ^{12.5°C}			10.640		10.342 ^{37.5°C}		10.064
Propenoic acid (0)				4.267	4.250	4.247	4.249	4.267	4.301	

N-Propionylglycine (+ 1)		3.728	3.723	3.718	3.716	3.718	3.721	3.725	3.731	3.750
Propynoic acid (0)			1.791	1.829	1.867	1.887	1.940	1.932	1.963	
Pyrrolidine (+ 1)	12.17	11.98	11.81	11.63	11.43	11.30	11.15	10.99	10.84	11.56
Serine										
(+ 1)	2.296 ^{1°C}		2.232 ^{12.5°C}			2.186		2.154 ^{37.5°C}		2.132
(0)	9.880 ^{1°C}		9.542 ^{12.5°C}			9.208		8.904 ^{37.5°C}		8.628
Silver bromide, pK _{sp}		13.33		12.83	12.57	12.30	12.07	11.83	11.61	11.19
Silver chloride, pK _{sp}		10.595		10.152		9.749		9.381	9.21	8.88
Succinic acid										
(0)	4.285	4.263	4.245	4.232	4.218	4.207	4.198	4.191	4.188	4.186
(− 1)	5.674	5.660	5.649	5.642	5.639	5.635	6.541	5.647	5.654	5.680
Sulfuric acid (− 1)	1.778	1.812 ^{4.3°C}		1.894		1.987	2.05	2.095	2.17	2.246
Sulfurous acid (0)	1.63		1.74			1.89		1.98		2.12
D-Tartaric acid										
(0)	3.118	3.095	3.075	3.057	3.044	3.036	3.025	3.019	3.018	3.021
(− 1)	4.426	4.407	4.391	4.381	4.372	4.366	4.365	4.367	4.372	4.391
2,3,5,6-Tetramethylbenzoic acid				3.310	3.367	3.415	3.453	3.483	3.505	
(0)										
Threonine										
(+ 1)	2.200 ^{1°C}		2.132 ^{12.5°C}			2.088		2.070 ^{37.5°C}		2.055
(0)	9.748 ^{1°C}		9.420 ^{12.5°C}			9.100		8.812 ^{37.5°C}		8.548
o-Toluidine (0)				4.58	4.495	4.45	4.345	4.28	4.20	
1,2,4-Triazole										
(+ 1)				2.451	2.418	2.386	2.327			
(0)				10.205	10.083	9.972	9.768			
3,4,5-Trihydroxybenzoic acid (0)					4.19		4.30		4.38	4.53
Tris(2-hydroxyethyl)amine (+ 1)	8.290	8.173	8.067	7.963	7.861	7.762	7.666	7.570	7.477	7.299
2,4,6-Trimethylbenzoic acid (0)				3.325	3.391	3.448	3.498	3.541	3.577	
3-Trimethylsilylbenzoic acid (0)				4.142	4.116	4.089	4.060	4.029	3.996	
4-Trimethylsilylbenzoic acid (0)				4.270	4.230	4.192	4.155	4.119	4.084	
β-Ureidopropionic acid (0)		4.514	4.505	4.497	4.490	4.487	4.486	4.486	4.488	4.500
DL-Valine										
(+ 1)	2.320		2.297 ^{12.5°C}			2.296		2.292 ^{37.5°C}		2.310
(0)	10.413		10.064 ^{12.5°C}			9.719		9.405 ^{37.5°C}		9.124

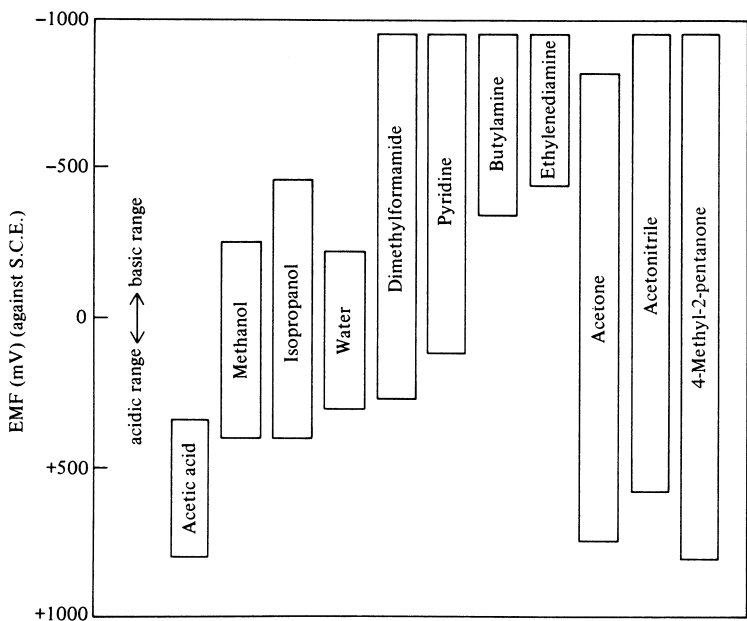


FIGURE 8.1 Approximate potential ranges in nonaqueous solvents.

TABLE 8.10 Properties of Common Acid-Base Solvents

Solvent	Potential Span, mV	− log K _s	Dielectric Constant, 25°C
Acetic acid	400	14.5	6.1(20°)
Acetic anhydride	800	14.5	20.7(20°)
Acetone	1600		20.7
Acetonitrile	1600	26.5	37.5(20°)
Ammonia (at − 50°C)		33	22(− 33°)
<i>n</i> -Butanol	900		17.1
<i>n</i> -Butylamine	500		4.88(20°)
Chlorobenzene	1500		5.62
<i>N,N</i> -Dimethylformamide	1300	18.0	36.71
Dimethylsulfoxide		17.3	46.6
Ethanol	800	19.1	24.55
Ethanolamine		5.1	37.7
Ethyl acetate	1500		6.02
Ethylenediamine	500	15.3	14.2(20°)
Formic acid	200	6.2	58.5
Methanol	800	16.7	32.7
4-Methyl-2-pentanone (methyl isobutyl ketone)	1600	25.0	13.1(20°)
Nitromethane	1000		35.8(30°)
2-Propanol	900		19.92
Pyridine	1000		12.3
Sulfuric acid		3.85	101
Water	800	14.0	78.3

TABLE 8.11 pK_a Values for Proton-Transfer Reactions in Nonaqueous Solvents

Acid	Methanol	Ethanol	Other Solvents
Acetic acid	9.52	10.32	11.4 ^a , 9.75 ^d
<i>p</i> -Aminobenzoic acid	10.25		
Ammonium ion	10.7		6.40 ^b
Anilinium ion	6.0	5.70	
Benzoic acid		10.72	10.0 ^a
Bromocresol purple	11.3	11.5	
Bromocresol green	9.8	10.65	
Bromophenol blue	8.9	9.5	
Bromothymol blue	12.4	13.2	
Di- <i>n</i> -butylammonium ion			10.3 ^a
<i>o</i> -Chloroanilinium ion	3.4		
Cyanoacetic acid		7.49	
2,5-Dichloroanilinium ion			9.48 ^b
Dimethylaminoazobenzene		5.2	6.32 ^b
<i>N,N'</i> -Dimethylanilinium ion		4.37	
Formic acid		9.15	
Hydrobromic acid			5.5 ^c
Hydrochloric acid			8.55 ^b , 8.9 ^c
Methyl orange	3.8	3.4	
Methyl red (acid range)	4.1	3.55	
(alkaline range)	9.2	10.45	
Methyl yellow	3.4	3.55	
Neutral red	8.2	8.2	
<i>o</i> -Nitrobenzoic acid	7.6		
<i>m</i> -Nitrobenzoic acid	8.3		
<i>p</i> -Nitrobenzoic acid	8.4		
Perchloric acid			4.87 ^b
Phenol	14.0		
Phenol red	12.8	13.4	
Phthalic acid, pK_2	11.65		11.5 ^d , 6.10 ^d (pK_1)
Picric acid	3.8	3.8	8.9 ^c
Pyridinium ion			6.1 ^b
Salicylic acid	8.7	7.9	
Stearic acid	10.0		
Succinic acid, pK_2	11.4		
Sulfuric acid, pK_1			7.24 ^{b,c}
Tartaric acid, pK_2	9.9		
Thymol blue (alkaline range)	14.0	15.2	
(acid range)	4.7	5.35	
Thymolbenzein (acid range)	3.5		
(alkaline range)	13.1		
<i>p</i> -Toluenesulfonic acid			8.44 ^b
<i>p</i> -Toluidinium ion		6.24	
Tribenzylammonium ion			5.40 ^b
Tropeoline 00	2.2		
Urea (protonated cation)			6.96 ^b
Veronal	12.6		

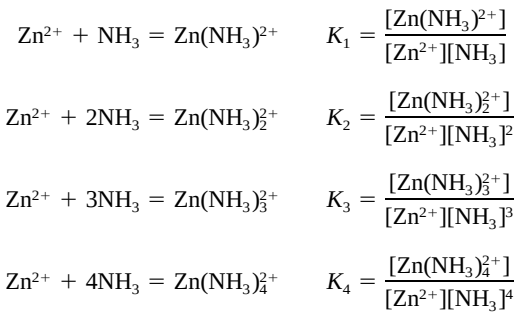
^a Dimethylsulfoxide. ^b Glacial acetic acid. ^c Acetonitrile. ^d Acetone + 10% water.

8.2.2 Formation Constants of Metal Complexes

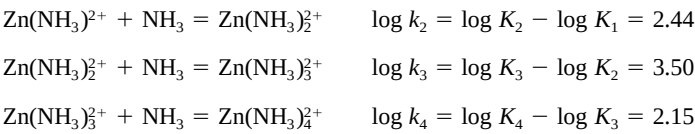
Each value listed in Tables 8.12 and 8.13 is the logarithm of the overall formation constant for the cumulative binding of a ligand L to the central metal cation M , viz.:

	Cumulative formation constant	Stepwise stability constants
$M + L = ML$	K_1	k_1
$M + 2L = ML_2$	K_2	$k_1 k_2$
.....		
$M + nL = ML_n$	K_n	$k_1 k_2 \cdots k_n$

As an example, the entries in Table 8.12 for the zinc ammine complexes represent these equilibria:



If the stepwise stability or formation constants of the reactions are desired, for the first step $\log K_1 = \log k_1 = 2.37$. For the second and succeeding steps the equilibria and corresponding constants are as follows:



The reverse of the association or formation reactions would represent the dissociation or instability constant for the systems, i.e., $-\log K_f = \log K_{\text{instab}}$.

The data in the tables generally refer to temperatures of about 20 to 25°C. Most of the values in Table 8.12 refer to zero ionic strength, but those in Table 8.13 often refer to a finite ionic strength.

TABLE 8.12 Cumulative Formation Constants for Metal Complexes with Inorganic Ligands

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$	$\log K_5$	$\log K_6$
Ammonia						
Cadmium	2.65	4.75	6.19	7.12	6.80	5.14
Cobalt(II)	2.11	3.74	4.79	5.55	5.73	5.11
Cobalt(III)	6.7	14.0	20.1	25.7	30.8	35.2
Copper(I)	5.93	10.86				
Copper(II)	4.31	7.98	11.02	13.32	12.86	
Iron(II)	1.4	2.2				
Manganese(II)	0.8	1.3				
Mercury(II)	8.8	17.5	18.5	19.28		
Nickel	2.80	5.04	6.77	7.96	8.71	8.74
Platinum(II)						35.3
Silver(I)	3.24	7.05				
Zinc	2.37	4.81	7.31	9.46		
Bromide						
Astatine	2.51 [AtBr]					
Bismuth(III)	4.30	5.55	5.89	7.82		9.70
Bromine	1.24 [Br ₃]					
Cadmium	1.75	2.34	3.32	3.70		
Cerium(III)	0.42					
Copper(I)		5.89				
Copper(II)	0.30					
Gold(I)		12.46				
Indium	1.30	1.88	2.48			
Iodine	2.64 [IBr]					
Iron(III)	-0.30	-0.50				
Lead	1.2	1.9		1.1		
Mercury(II)	9.05	17.32	19.74	21.00		
Palladium(II)				13.1		
Platinum(II)				20.5		
Rhodium(III)		14.3	16.3	17.6	18.4	17.2
Scandium	2.08	3.08				
Silver(I)	4.38	7.33	8.00	8.73		
Thallium(I)	0.93					
Thallium(III)	9.7	16.6	21.2	23.9	29.2	31.6
Tin(II)	1.11	1.81	1.46			
Uranium(IV)	0.18					
Yttrium	1.32					
Chloride						
Americium(III)	1.17					
Antimony(III)	2.26	3.49	4.18	4.72		
Bismuth(III)	2.44	4.7	5.0	5.6		
Cadmium	1.95	2.50	2.60	2.80		
Cerium(III)	0.48					
Copper(I)		5.5	5.7			
Copper(II)	0.1	-0.6				
Curium(III)	1.17					
Gold(III)		9.8				
Indium	1.42	2.23	3.23			
Iron(II)	0.36					
Iron(III)	1.48	2.13	1.99	0.01		
Lead	1.62	2.44	1.70	1.60		
Manganese(II)	0.96					
Mercury(II)	6.74	13.22	14.07	15.07		

TABLE 8.12 Cumulative Formation Constants for Metal Complexes with Inorganic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4	log K_5	log K_6
Palladium(II)	6.1	10.7	13.1	15.7		
Platinum(II)		11.5	14.5	16.0		
Plutonium(III)	1.17					
Silver(I)	3.04	5.04		5.30		
Thallium(I)	0.52					
Thallium(III)	8.14	13.60	15.78	18.00		
Thorium	1.38	0.38				
Tin(II)	1.51	2.24	2.03	1.48		
Tin(IV)						4
Uranium(IV)	0.8					
Uranium(VI)	0.22					
Zinc	0.43	0.61	0.53	0.20		
Zirconium	0.9	1.3	1.5	1.2		
Cyanide						
Cadmium	5.48	10.60	15.23	18.78		
Copper(I)		24.0	28.59	30.30		
Gold(I)		38.3				
Iron(II)						35
Iron(III)						42
Mercury(II)				41.4		
Nickel				31.3		
Silver(I)		21.1	21.7	20.6		
Zinc				16.7		
Fluoride						
Aluminum	6.10	11.15	15.00	17.75	19.37	19.84
Beryllium	5.1	8.8	12.6			
Cerium(III)	3.20					
Chromium(III)	4.41	7.81	10.29			
Gadolinium	3.46					
Gallium	5.08					
Indium	3.70	6.25	8.60	9.70		
Iron(III)	5.28	9.30	12.06			
Lanthanum	2.77					
Magnesium	1.30					
Manganese(II)	5.48					
Plutonium(III)	6.77					
Scandium						17.3
Thallium(I)	0.1					
Thallium(III) [TlO ⁺]	6.44					
Thorium	7.65	13.46	17.97			
Titanium(IV) [TiO ²⁺]	5.4	9.8	13.7	18.0		
Uranium(VI)	4.59	7.93	10.47	11.84		
Yttrium	4.81	8.54	12.14			
Zirconium	8.80	16.12	21.94			
Hydroxide						
Aluminum	9.27			33.03		
Antimony(III)		24.3	36.7	38.3		
Arsenic [as AsO ⁺]	14.33	18.73	20.60	21.20		
Beryllium	9.7	14.0	15.2			
Bismuth(III)	12.7	15.8		35.2		
Cadmium	4.17	8.33	9.02	8.62		
Cerium(III)	14.6					
Cerium(IV)	13.28	26.46				

TABLE 8.12 Cumulative Formation Constants for Metal Complexes with Inorganic Ligands (*Continued*)

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$	$\log K_5$	$\log K_6$
Chromium(III)	10.1	17.8		29.9		
Copper(II)	7.0	13.68	17.00	18.5		
Dysprosium	5.2					
Erbium(III)	5.4					
Gadolinium	4.6					
Gallium	11.0	21.7		34.3	38.0	40.3
Indium	9.9	19.8		28.7		
Iodine	9.49	11.24				
Iron(II)	5.56	9.77	9.67	8.58		
Iron(III)	11.87	21.17	29.67			
Lanthanum	3.3					
Lead(II)	7.82	10.85	14.58			61.0
Lutetium	6.6					
Magnesium	2.58					
Manganese(II)	3.90		8.3			
Neodymium	5.5					
Nickel	4.97	8.55	11.33			
Praseodymium	4.30					
Plutonium(III)	7.0					
Plutonium(IV)	12.39					
Plutonium [as PuO_2^{2+}]	8.3	16.6	20.9			
Samarium(III)	4.8					
Scandium	8.9					
Tellurium(IV)			41.6	53.0	64.8	72.0
Thallium(III)	12.86	25.37				
Titanium(III)	12.71					
Uranium(IV)	13.3				41.2	
Uranium(VI) [as UO_2^{2+}]	9.5	22.80		32.4		
Vanadium(III)	11.1	21.6				
Vanadium(IV) [as VO^{2+}]	8.6		[25.8 for $\text{V}_2\text{O}_4(\text{OH})^-$]			
Vanadium(V) [as VO^{3+}]		25.2		46.2	58.5	
Yttrium	5.0					
Zinc	4.40	11.30	14.14	17.66		
Zirconium	14.3	28.3	41.9	55.3		
Iodide						
Bismuth	3.63			14.95	16.80	18.80
Cadmium	2.10	3.43	4.49	5.41		
Copper(I)		8.85				
Indium	1.00	2.26				
Iodine	2.89	5.79				
Iron(III)	1.88					
Lead	2.00	3.15	3.92	4.47		
Mercury(II)	12.87	23.82	27.60	29.83		
Silver	6.58	11.74	13.68			
Thallium(I)	0.72	0.90	1.08			
Thallium(III)	11.41	20.88	27.60	31.82		
Iodate						
Barium	1.05					
Calcium	0.89					
Magnesium	0.72					
Strontium	1.00					
Thorium	2.88	4.79	7.15			

TABLE 8.12 Cumulative Formation Constants for Metal Complexes with Inorganic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4	log K_5	log K_6
Nitrate						
Barium	0.92					
Beryllium	1.62					
Bismuth(III)	1.26					
Cadmium	0.40					
Calcium	0.28					
Cerium(III)	1.04	2.55				
Curium(III)	0.57					
Hafnium	0.92	2.43	4.32	6.40	8.48	10.29
Iron(III)	1.0					
Lanthanum	0.26	0.69	1.27			
Lead	1.18					
Mercury(II)	0.35					
Neodymium	0.52	1.18				
Neptunium(IV)	0.38					
Plutonium(III)	0.77	1.93	3.09			
Plutonium(IV)	0.54					
Strontium	0.82					
Thallium(I)	0.33					
Thallium(III)	0.92					
Thorium	0.78	1.89	2.89	3.63		
Uranium(IV)	0.20	0.37				
Uranium(VI)	0.34	0.45				
Ytterbium	0.45	1.30	2.42			
Zirconium [as ZrO^{2+}]		1.91		3.54		
Pyrophosphate						
Barium	4.6					
Calcium	4.6					
Cadmium	5.6					
Copper(II)	6.7	9.0				
Lead		5.3				
Magnesium	5.7					
Nickel	5.8	7.4				
Strontium	4.7					
Yttrium		9.7				
Zirconium		6.5				
Sulfate						
Cerium(III)	3.40					
Erbium	3.58					
Gadolinium	3.66					
Holmium	3.58					
Indium	1.78	1.88	2.36			
Iron(III)	2.03	2.98				
Lanthanum	3.64					
Neodymium	3.64					
Nickel	2.4					
Plutonium(IV)	3.66					
Praseodymium	3.62					
Samarium	3.66					
Thorium	3.32	5.50				
Uranium(IV)	3.24	5.42				
Uranium(VI)	1.70	2.45	3.30			

TABLE 8.12 Cumulative Formation Constants for Metal Complexes with Inorganic Ligands (*Continued*)

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$	$\log K_5$	$\log K_6$
Yttrium	3.47					
Ytterbium	3.58					
Zirconium	3.79	6.64	7.77			
Sulfite						
Copper(I)	7.5	8.5	9.2			
Mercury(II)		22.66				
Silver	5.30	7.35				
Thiocyanate						
Bismuth	1.15	2.26	3.41	4.23		
Cadmium	1.39	1.98	2.58	3.6		
Chromium(III)	1.87	2.98				
Cobalt(II)	-0.04	-0.70	0	3.00		
Copper(I)	12.11	5.18				
Gold(I)		23		42		
Indium	2.58	3.00	4.63			
Iron(III)	2.95	3.36				
Mercury(II)		17.47		21.23		
Nickel	1.18	1.64	1.81			
Ruthenium(III)	1.78					
Silver		7.57	9.08	10.08		
Thallium(I)	0.80					
Uranium(IV)	1.49	2.11				
Uranium(VI)	0.76	0.74	1.18			
Vanadium(III)	2.0					
Vanadium(IV)	0.92					
Zinc	1.62					
Thiosulfate						
Cadmium	3.92	6.44				
Copper(I)	10.27	12.22	13.84			
Iron(III)	2.10					
Lead		5.13	6.35			
Mercury(II)		29.44	31.90	33.24		
Silver	8.82	13.46				

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands

Temperature is 25°C and ionic strengths are approaching zero unless indicated otherwise: (a) At 20°C, (b) at 30°C, (c) 0.1 M uni-univalent salt, (d) 1.0 M uni-univalent salt, (e) 2.0 M uni-univalent salt present.

	log K_1	log K_2	log K_3	log K_4
Acetate				
Ag(I)	0.73	0.64		
Ba(II)	0.41			
Ca(II)	0.6			
Cd(II)	1.5	2.3	2.4	
Ce(III)	1.68	2.69	3.13	3.18
Co(II)	1.5	1.9		
Cr(III)	1.80	4.72		
Cu(II) <i>a</i>	2.16	3.20		
Fe(II) <i>c</i>	3.2	6.1	8.3	
Fe(III) <i>a,d</i>	3.2			
In(III)	3.50	5.95	7.90	9.08
Hg(II)		8.43		
La(III) <i>a,e</i>	1.56	2.48	2.98	2.95
Mg(II)	0.8			
Mn(II)	9.84	2.06		
Ni(II)	1.12	1.81		
Pb(II)	2.52	4.0	6.4	8.5
Rare earths <i>a,e</i>	1.6–1.9	2.8–3.0	3.3–3.7	
Sr(II)	0.44			
Tl(III)				15.4
UO ₂ (II) <i>a,e</i>	2.38	4.36	6.34	
Y(III) <i>a,e</i>	1.53	2.65	3.38	
Zn(II)	1.5			
Acetylacetone				
Al(III) <i>b</i>	8.6	15.5		
Be(II)	7.8	14.5		
Cd(II)	3.84	6.66		
Ce(III)	5.30	9.27	12.65	
Cr(II)	5.9	11.7		
Co(II)	5.40	9.54		
Cu(II)	8.27	16.34		
Dy(III) <i>b</i>	6.03	10.70	14.04	
Er(III) <i>b</i>	5.99	10.67	14.09	
Eu(III) <i>b</i>	5.87	10.35	13.64	
Fe(II)	5.07	8.67		
Fe(III)	11.4	22.1	26.7	
Ga(III)	9.5	17.9	23.6	
Gd(III) <i>b</i>	5.90	10.38	13.79	
Hf(IV)	8.7	15.4	21.8	28.1
Ho(III)	6.05	10.73	14.13	
In(III)	8.0	15.1		
La(III) <i>b</i>	5.1	8.90	11.90	
Lu(III) <i>b</i>	6.23	11.00	13.63	
Mg(II)	3.65	6.27		
Mn(II)	4.24	7.35		
Mn(III)			3.86	
Nd(III)	5.6	9.9	13.1	
Ni(II) <i>a</i>	6.06	10.77	13.09	

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Pd(II) <i>b</i>	16.2	27.1		
Pr(III) <i>b</i>	5.4	9.5	12.5	
Pu(IV) <i>c</i>	10.5	19.7	28.1	34.1
Sc(III) <i>b</i>	8.0	15.2		
Sm(III) <i>b</i>	5.9	10.4		
Tb(III) <i>b</i>	6.02	10.63	14.04	
Th(IV)	8.8	16.2	22.5	26.7
Tm(IV) <i>b</i>	6.09	10.85	14.33	
U(IV) <i>a,c</i>	8.6	17.0	23.4	29.5
UO ₂ (II) <i>b</i>	7.74	14.19		
VO(II)	8.68	15.79		
V(II)	5.4	10.2	14.7	
Y(III) <i>b</i>	6.4	11.1	13.9	
Yb(III) <i>b</i>	6.18	11.04	13.64	
Zn(II) <i>b</i>	4.98	8.81		
Zr(IV)	8.4	16.0	23.2	30.1
Alizarin red				
Cr(VI)	4.7			
Cu(II)	4.1			
Hf(IV)		10.4		
Mo(VI)		9.6		
Pb(II)	6.0			
Th(IV)		8.24		
UO ₂ (II)	4.22			
V(V)		8.6		
W(VI)		7.8		
Arsenazo				
Hf(IV)	10.07			
Zr(IV)	12.95			
Aurintricarboxylic acid				
Be(II)	4.54			
Cu(II)	4.1	8.81		
Fe(III)	4.68			
Th(IV)	5.04			
UO ₂ (II)	4.77			
Benzoylacetone (75% dioxane)				
Ba(II)		9.4		
Be(II)	12.59	24.01		
Cd(II)	7.79	14.36		
Ce(III)	10.09	19.42	27.04	
Co(II)	9.42	17.83		
Cu(II)	12.05	23.01		
La(III)	6.33	11.66	16.78	
Mg(II)	7.69	14.09		
Mn(II)	8.66	15.78		
Ni(II)	9.58	18.00		
Pb(II)	8.84	16.35		
Pr(III)	7.02	13.62	18.74	
UO ₂ (II)	12.15	23.27		
Y(III)	8.24	14.98	20.57	
Zn(II)	9.62	17.90		

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

		log K_1		log K_2		log K_3		log K_4	
Calmagite Ca Mg		6.05 8.05							
	Complex of HL^{2-} Anion			Complex of L^{3-} Anion			Complex of H_2L^-		
	log K_1		log K_2	log K_1		log K_2	log K_3		
Citric acid									
Ag	7.1								
Al	7.0			20.0					
Ba	2.98								
Be	4.52								
Ca	4.68								
Cd	3.98			11.3					
Ce(III)		6.18			9.65		3.2		
Co(II)	4.8			12.5					
Cu(II)	4.35			14.2					
Eu(III)		6.46			9.80				
Fe(II)	3.08			15.5					
Fe(III)	12.5			25.0					
La		6.97			9.45		6.22		
Mg	3.29								
Mn(II)	3.67								
Nd(III)		6.32			9.70				
Ni	5.11			14.3					
Pb	6.50								
Pr							3.4		
Ra	2.36								
Sr	2.8								
Tl(I)	1.04								
UO ₂	8.5	10.8							
Y							3.6		
Yb					8				
Zn	4.71			11.4					
	log K_1		log K_2		log K_3				
1,2-Diaminocyclohexane-N,N,N',N'-tetraacetic acid									
Al c	17.63								
Ba c	8.64								
Ca c	12.3								
Cd c	19.88								
Ce(III) c	16.76								
Co(II) c	19.57								
Cu(II) c	21.95								
Dy(III) c	19.69								
Er(III) c	20.20								
Eu(III) c	18.77								
Fe(III) c	27.48								
Ga c	22.91								

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$
Gd <i>c</i>	18.80			
Hg(II) <i>c</i>	24.4			
Ho <i>c</i>	19.89			
La <i>c</i>	16.35			
Lu <i>c</i>	21.51			
Mg <i>c</i>	10.41			
Mn(II) <i>c</i>	17.43			
Nd <i>c</i>	17.69			
Ni <i>c</i>	19.4			
Pb <i>c</i>	20.33			
Pr <i>c</i>	17.23			
Sm(III) <i>c</i>	18.63			
Sr <i>c</i>	8.92			
Tb <i>c</i>	19.30			
Tm <i>c</i>	20.46			
VO(II) <i>c</i>	19.40			
Y <i>c</i>	19.41			
Yb <i>c</i>	20.80			
Zn <i>c</i>	18.6			
Dibenzoylmethane (75% dioxane)				
Ba	6.10	11.50		
Be	13.62	26.03		
Ca	7.17	13.55		
Cd	8.67	16.63		
Ce(III)	10.99	21.53	30.38	
Co(II)	10.35	20.05		
Cu(II)	12.98	24.98		
Cs	3.42			
Fe(II)	11.15	21.50		
K	3.67			
Li	5.95			
Mg	8.54	16.21		
Mn(II)	9.32	17.79		
Na	4.18			
Ni	10.83	20.72		
Pb	9.75	18.79		
Rb	3.52			
Sr	6.40	12.10		
Zn	10.23	19.65		
	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_f$ [MHL]
4,5-Dihydroxybenzene-1,3-disulfonic acid (Tiron)				
Al	19.02	31.10	33.5	
Ba	4.10			14.6
Ca	5.80			14.8
Cd <i>d</i>	7.69	13.29		
Ce(III)		3.75		
Co(II) <i>d</i>	8.19	14.41		15.7
Cu(II) <i>d</i>	12.76	23.73		18.1

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_f [MHL]
Fe(III) <i>a,c</i>	20.7	35.9	46.9	22.6
La	12.9			18.6 [La(OH)L]
Mg <i>a,c</i>	6.86			14.6
Mn(II) <i>c</i>	8.6			
Ni <i>a,c</i>	8.56	14.90		15.6
Pb <i>d</i>	11.95	18.28		
Sr <i>c</i>	4.55			
UO ₂ (II) <i>c</i>	15.90			
VO(II)	15.88			
Zn <i>d</i>	9.00	16.91		15.9
	log K_1	log K_2	log K_f [M ₂ L ₃]	
2,3-Dimercaptopropan-1-ol (BAL)				
Fe(II)	15.8			
Fe(III)	30.6 [Fe(OH)L]		28	
Mn(II)	5.23	10.43		
Ni		22.78		
Zn	13.48	23.3	40.6	
	log K_1	log K_2	log K_3	log K_4
Dimethylglyoxime (50% dioxane)				
Cd	5.7	10.7		
Co(II)	9.80	18.94		
Cu(II)	12.00	33.44		
Fe(II)		7.25		
La	6.6	12.5		
Ni	11.16			
Pb	7.3			
Zn	7.7	13.9		
2,2'-Dipyridyl				
Ag	3.65	7.15		
Cd	4.26	7.81	10.47	
Co(II)	5.73	11.57	17.59	
Cr(II)	4.5	10.5	14.0	
Cu(I)		14.2		
Cu(II)	8.0	13.60	17.08	
Fe(II)	4.36	8.0	17.45	
Hg(II)	9.64	16.74	19.54	
Mg	0.5			
Mn(II) <i>d</i>	4.06	7.84	11.47	
Ni	6.80	13.26	18.46	
Pb	3.0			
Ti(III)			25.28	
V(II)	4.9	9.6	13.1	
Zn	5.30	9.83	13.63	
Eriochrome Black T				
Ca	5.4			
Mg	7.0			
Zn	13.5	20.6		

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Ethanolamine				
Ag	3.29	6.92		16.48
Cu(II)		6.68		
Hg(II)	8.51	17.32		
Ethylenediamine				
Ag	4.70	7.70		
Cd <i>a</i>	5.47	10.09	12.09	
Co(II)	5.91	10.64	13.94	
Co(III)	18.7	34.9	48.69	
Cr(II)	5.15	9.19		
Cu(I)		10.8		
Cu(II)	10.67	20.00	21.0	
Fe(II)	4.34	7.65	9.70	
Hg(II)	14.3	23.3		
Mg	0.37			
Mn(II)	2.73	4.79	5.67	
Ni	7.52	13.84	18.33	
Pd(II)		26.90		
V(II)	4.6	7.5	8.8	
Zn	5.77	10.83	14.11	
Ethylenediamine-<i>N, N, N', N'</i>-tetraacetic acid				
Ag	7.32			
Al	16.11			
Am(III)	18.18			
Ba	7.78			
Be	9.3			
Bi	22.8			
Ca	11.0			
Cd	16.4			
Ce(III)	16.80			
Cf(III)	19.09			
Cm(III)	18.45			
Co(II)	16.31			
Co(III)	36			
Cr(II)	13.6			
Cr(III)	23			
Cu(II)	18.7			
Dy	18.0			
Er	18.15			
Eu(III)	17.99			
Fe(II)	14.33			
Fe(III)	24.23			
Ga	20.25			
Gd	17.2			
Hg(II)	21.80			
Ho	18.1			
In	24.95			
La	16.34			
Li	2.79			
Lu	19.83			
Mg	8.64			
Mn(II)	13.8			
Mo(V)	6.36			

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Na	1.66			
Nd	16.6			
Ni	18.56			
Pb	18.3			
Pd(II)	18.5			
Pm(III)	17.45			
Pr	16.55			
Pu(III)	18.12			
Pu(IV)	17.66			
Pu(VI)	17.66			
Ra	7.4			
Sc	23.1			
Sm	16.43			
Sn(II)	22.1			
Sr	8.80			
Tb	17.6			
Th	23.2			
Ti(III)	21.3			
TiO(II)	17.3			
Tl(III)	22.5			
Tm	19.49			
U(IV)	17.50			
V(II)	12.70			
V(III)	25.9			
VO(II)	18.0			
V(V)	18.05			
Y	18.32			
Yb	18.70			
Zn	16.4			
Zr	19.40			
Glycine				
Ag	3.41	6.89		
Ba	0.77			
Be		4.95		
Ca	1.38			
Cd	4.74	8.60		
Co(II)	5.23	9.25	10.76	
Cu(II)	8.60	15.54	16.27	
Dy		12.2		
Er		12.7		
Fe(II) <i>a</i>	4.3	7.8		
Fe(III) <i>a,d</i>	10.0			
Gd		11.9		
Hg(II)	10.3	19.2		
La		11.2		
Mg	3.44	6.46		
Mn(II)	3.6	6.6		
Ni	6.18	11.14	15	
Pb	5.47	8.92		
Pd(II)	9.12	17.55		
Pr		11.5		
Sm		11.7		

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Sr	0.91			
Y		12.5		
Yb		13.0		
Zn	5.52	9.96		
<i>N'</i>-(2-Hydroxyethyl)ethylenediamine-<i>N,N,N'</i>-triacetic acid				
Ba c	5.54			
Ca c	8.43			
Cd c	13.0			
Ce(III) c	14.11			
Co(II) c	14.4			
Cu(II) c	17.40			
Dy c	15.30			
Er c	15.42			
Eu(III) c	15.35			
Fe(II) c	11.6			
Fe(III) c	19.8			
Gd c	15.22			
Hg(II) c	20.1			
Ho c	15.32			
La c	13.46			
Lu c	15.88			
Mg c	5.78			
Mn(II) c	10.7			
Nd c	14.86			
Ni c	17.0			
Pb c	15.5			
Pr c	14.61			
Sm c	15.28			
Sr c	6.92			
Tb c	15.32			
Th c	18.5			
Tm c	15.59			
Y c	14.65			
Yb c	15.88			
Zn c	14.5			
8-Hydroxy-2-methylquinoline (50% dioxane)				
Cd	9.00	9.00	16.60	
Ce(III)	7.71			
Co(II)	9.63	18.50		
Cu(II)	12.48	24.00		
Fe(II)	8.75	17.10		
Mg	5.24	9.64		
Mn(II)	7.44	13.99		
Ni	9.41	17.76		
Pb	10.30	18.50		
UO ₂ (II)	9.4	17		
Zn	9.82	18.72		
8-Hydroxyquinoline-5-sulfonic acid				
Ba	2.31			
Ca	3.52			
Cd	7.70	14.20		
Ce(III)	6.05	11.05	14.95	

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$
Co(II)	8.11	15.05	20.41	32.04
Cu(II)	11.92	21.87		
Er	7.16	13.34	18.56	
Fe(II)	8.4	15.7	21.75	
Fe(III)	11.6	22.8	35.65	
Gd	6.64	12.37	17.27	
La	5.63	10.13	13.83	
Mg	4.79	8.19		
Mn(II)	5.67	10.72		
Nd	6.3	11.6	16.0	
Ni	9.57	18.27	22.9	
Pb	8.53	16.13		
Pr	6.17	11.37	15.67	
Sm	6.58	12.28	17.04	
Sr	2.75			
Th	9.56	18.29	25.92	
UO ₂ (II)	8.52	15.67		
Zn	8.65	16.15		
Lactic acid				
Ba	0.64			
Ca	1.42			
Cd	1.70			
Ce(III) <i>a,c</i>	2.76	4.73	5.96	
Co(II)	1.90			
Cu(II)	3.02	4.85		
Er	2.77	5.11	6.70	
Eu(III)	2.53	4.60	5.88	
Fe(III)	7.1			
Gd	2.53	4.63	5.91	
Ho	2.71	4.97	6.55	
La <i>a,c</i>	2.60	4.34	5.64	
Li	0.20			
Mg	1.37			
Mn(II)	1.43			
Nd	2.47	4.37	5.60	
Ni	2.22			
Pb	2.40	3.80		
Pr <i>a,c</i>	2.85	4.90	6.10	
Rare earths <i>a,c</i>	2.8–3.0	4.9–5.4	6.1–7.8	
Sm	2.56	4.58	5.90	
Sr	0.98			
Tb	2.61	4.73	6.01	
Y	2.53	4.70	6.12	
Yb	2.85	5.27	7.96	
Zn	2.20	3.75		
Nitrilotriacetic acid				
Al	>10			
Ba <i>a</i>	5.88			
Ca	7.60	11.61		
Cd <i>c</i>	9.80	15.2		
Ce(III) <i>c</i>	10.83	18.67		

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Co(II) <i>c</i>	10.38	14.5		
Cr(III)	>10			
Cu(II) <i>c</i>	13.10			
Dy <i>c</i>	11.74	21.15		
Er <i>c</i>	12.03	21.29		
Eu(III) <i>c</i>	11.52	20.70		
Fe(II) <i>c</i>	8.84			
Fe(III) <i>c</i>	15.87	24.32		
Gd <i>c</i>	11.54	20.80		
Hg(II)	12.7			
Ho <i>c</i>	11.90	21.25		
In	15			
La <i>c</i>	10.36	17.60		
Li <i>a</i>	3.28			
Lu <i>c</i>	12.49	21.91		
Mg <i>c</i>	5.36	10.2		
Mn(II)	8.60	11.1		
Na	2.15			
Nd <i>c</i>	11.26	19.73		
Ni	11.26	16.0		
Pb <i>a,c</i>	11.8			
Pr <i>c</i>	11.07	19.25		
Sm(III) <i>c</i>	11.53	20.53		
Sr	6.73			
Tb <i>c</i>	11.59	20.97		
Tl(I)	3.44			
Th <i>c</i>	12.4			
Tm <i>c</i>	12.22	21.45		
Y <i>c</i>	11.48	20.43		
Yb <i>c</i>	12.40	21.69		
Zn <i>c</i>	10.45	13.45		
Zr <i>c</i>	20.8			
1-Nitroso-2-naphthol (75% dioxane)				
Ag	7.74			
Cd	6.18	11.38		
Co(II)	10.67	22.81		
Cu(II)	12.52	23.37		
Mg	6.2	10.60		
Nd	9.5	17.7	25.6	
Ni	10.75	21.29	28.09	
Pb	9.73	17.31		
Pr	9.04	17.06	23.85	
Th <i>c</i>	8.50	16.13	24.03	30.29
Y	9.02	17.74	25.04	
Zn	9.32	17.02		
Zr	3.6			
Oxalate				
Ag	2.41			
Al	7.26	13.0	16.3	
Am(III)		9.8		[Am(HL) ₄] 11.0]
Ba	2.31			

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Be	4.90			
Ca	3.0			
Cd	3.52	5.77		
Ce(III)	6.52	10.5	11.3	
Co(II)	4.79	6.7	9.7	
Co(III)			~20	
Cu(II)	6.16	8.5		
Er	4.82	8.21	10.03	
Fe(II)	2.9	4.52	5.22	
Fe(III)	9.4	16.2	20.2	
Gd	7.04			
Hg(II)		6.98		
Mg	3.43	4.38		
Mn(II)	3.97	5.80		
Mn(III) <i>e</i>	9.98	16.57	19.42	
Mo(III)	3.38			
Mo(VI)				[MoO ₃ (L) ²⁻ 13.0]
Nd	7.21	11.5	> 14	
Ni	5.3	7.64	~8.5	
NpO ₂ (II)	3.30	7.07		
Pb		6.54		
Pu(III)	9.31	18.70	28	
Pu(IV)	8.74	16.91	23.39	27.50
PuO ₂ (II)		11.4		
Sr	2.54			
Th				24.48
TiO(II)	2.67			
Tl(I)	2.03			
UO ₂ (II)		10.57		
VO(II)		9.80		
V(II)	~2.7			
Y	6.52	10.10	11.47	
Yb	7.30	11.7	> 14	
Zn	4.89	7.60	8.15	
Zr	9.80	17.14	20.86	21.15
1,10-Phenanthroline				
Ag	5.02	12.07		
Ca	0.7			
Cd	5.93	10.53	14.31	
Co(II)	7.25	13.95	19.90	
Cu(II)	9.08	15.76	20.94	
Fe(II)	5.85	11.45	21.3	
Fe(III)	6.5	11.4	23.5	
Hg(II)		19.65	23.35	
Mg	1.2			
Mn(II)	3.88	7.04	10.11	
Ni	8.80	17.10	24.80	
Pb	4.65	7.5	9	
VO(II)	5.47	9.69		
Zn	6.55	12.35	17.55	

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Phthalic acid				
Ba	2.33			
Ca	2.43			
Cd	2.5			
Co(II)	1.81	4.51		
Cu(II)	3.46	4.83		
La		7.74		
Ni	2.14			
Pb <i>d</i>	3.4			
UO ₂ (II)	4.38			
Zn	2.2			
Piperidine				
Ag	3.30	6.48		
Hg(II)	8.70	17.44		
Pt(II)			log K_5 5.7	log K_6 8.2
Propylene-1,2-diamine				
Cd <i>b,c</i>		9.97	12.12	
Co(II) <i>d</i>	5.42	11.47	14.72	
Cu(II) <i>c</i>	6.41	20.06		
Hg(II) <i>c</i>	10.78	23.53	23.25	
Ni <i>d</i>	7.43	13.62	17.89	
Zn <i>b,c</i>	5.89	10.87	12.57	
Pyridine				
Ag	1.97	4.35		
Cd	1.40	1.95	2.27	2.50
Co(II)	1.14	1.54		
Cu(I)		3.34	4.51	5.44
Cu(II)	2.59	4.33	5.93	log K_6 6.89
Fe(II)	0.71		log K_5 7.00	6.54
Hg(II)	5.1	10.0	10.4	log K_6 10.2
Mn(II)	1.92	2.77	3.37	3.50
VO(II)	-1.70			
Zn	1.41	1.11	1.61	1.93
Pyridine-2,6-dicarboxylic acid				
Ba <i>a,d</i>	3.46			
Ca <i>a,d</i>	4.6	7.2		
Cd <i>a,d</i>	5.7	10.0		
Ce(III) <i>a,d</i>	8.34	14.42	18.80	
Co(II) <i>a,d</i>	7.0	12.5		
Cu(II) <i>a,d</i>	9.14	16.52		
Dy <i>a,d</i>	8.69	16.19	22.14	
Er <i>a,d</i>	8.77	16.39	22.14	
Eu(III) <i>a,d</i>	8.84	15.98	21.00	
Fe(II) <i>a,d</i>	5.71	10.36		
Fe(III) <i>a,d</i>	10.91	17.13		
Gd <i>a,d</i>	8.74	16.06	21.83	
Ho <i>a,d</i>	8.72	16.23	22.08	
La <i>a,d</i>	7.98	13.79	18.06	
Lu <i>a,d</i>	9.03	16.80	21.48	

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Hg(II) <i>a,d</i>	20.28			
Mg <i>a,d</i>	2.7			
Mn(II) <i>a,d</i>	5.01	8.49		
Nd <i>a,d</i>	8.78	15.60	20.66	
Ni <i>a,d</i>	6.95	13.50		
Pb <i>a,d</i>	8.70	10.60		
Pr <i>a,d</i>	8.63	15.10	19.94	
Sm <i>a,d</i>	8.86	15.88	21.23	
Sr <i>a,d</i>	3.89			
Tb <i>a,d</i>	8.68	16.11	22.03	
Tm <i>a,d</i>	8.83	16.54	22.04	
Y <i>a,d</i>	8.46	15.73	21.34	
Yb <i>a,d</i>	8.85	16.61	21.83	
Zn <i>a,d</i>	6.35	11.88		
1-(2-Pyridylazo)-2-naphthol (PAN)				
Co(II)	> 12			
Cu(II)	16			
Mn(II)	8.5	16.4		
Ni	12.7	25.3		
Tl(III)	2.29			
Zn	11.2	21.7		
		log K_f [ML]	log K_f [MHL]	log K_f [M(HL) ₂]
4-(2-Pyridylazo)resorcinal (PAR)				
Co(II)			> 12	
Cu(II)	10.3			
Mn(II)			9.7	18.9
Ni			13.2	26.0
Sc	4.8			
Tl(III)	4.23			
Zn			12.4	23.5
		log K_f [ML]	log K_f [M ₂ L]	log K_f [MHL]
Pyrocatechol-3,5-disulfonate (Pyrocatechol Violet)				
Al	19.13		4.95	
Bi	27.07		5.25	
Cd	8.13			5.86
Co(II)	9.01			6.53
Cu(II)	16.47			11.18
Ga	22.18	4.65		
In	18.10	4.81		
Mg	4.42	4.6		3.66
Mn(II)	7.13			5.36
Ni	9.35	4.38		6.85
Pb	13.25			10.19
Th	23.36	4.42		
Zn	10.41	6.21		7.21
Zr	27.40	4.18		

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$
8-Quinolinol				
Ba	2.07			
Be	3.36			
Ca (75% dioxane)	7.3	13.2		
Cd	7.2	13.4		
Ce(III) (50% dioxane)	9.15	17.13		
Co(II)	9.1	17.2		
Cu(II)	12.2	23.4		
Fe(II)	8.58	16.93	22.23	
Fe(III)	12.3	23.6	33.9	
La	5.85	16.95		
Mg (50% dioxane)	6.38	11.81		
Mn(II) (50% dioxane)	8.28	15.45		
Ni (50% dioxane)	11.44	21.38		
Pb (50% dioxane)	10.61	18.70		
Sm	6.84		19.50	
Sr	2.89	6.08		
Th	10.45	20.40	29.85	38.80
UO ₂ (II) (50% dioxane)	11.25	20.89		
V(II)	12.8	23.6		
VO(II)	10.97	20.19		
Y	8.15	14.90	20.25	
Zn (50% dioxane)	9.96	18.86		
		$\log K_f \text{ [MHL}^+]$		$\log K_f \text{ [M(HL)}_2]$
Salicylaldoxime				
Ba		0.53		3.72
Be		<7		
Ca		0.92		3.72
Cd		<4.4		
Co(II)				8.13
Cu(II)				8.13
Mg		0.64		4.10
Ni				3.77
Sr				3.77
Zn		<5.2		
	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$
Salicylic acid				
Al	14.11			
Be	17.4			
Cd	5.55			
Ce(III)	2.66			
Co(II)	6.72	11.42		
Cr(II)	8.4	15.3		
Cu(II)	10.60	18.45		
Fe(II)	6.55	11.25		
Fe(III) <i>a,c</i>	16.48	28.12	36.80	
La	2.64			

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
Mg (75% dioxane)	4.7			
Mn(II)	5.90	9.80		
Nd	2.70			
Ni	6.95	11.75		
Pr	2.68			
Th	4.25	7.60	10.05	11.60
TiO(II)	6.09			
UO ₂ (II)	13.4			
V(II)	6.3			
Zn	6.85			
Succinic acid				
Ba	2.08			
Be	3.08			
Ca	2.0			
Cd	2.2			
Co(II)	2.22			
Cu(II)	3.33			
Fe(III)	7.49			
Hg(II)		7.28		
La	3.96			
Mg	1.20			
Mn(II)	2.26			
Nd	8.1			
Ni	2.36			
Pb	2.8			
Ra	1.0			
Sr	1.06			
Zn	1.6			
5-Sulfosalicylic acid				
Al c	13.20	22.83	28.89	
Be c	11.71	20.81		
Cd c	16.68	29.08		
Co(II) c	6.13	9.82		
Cr(II) c	7.1	12.9		
Cr(III) c	9.56			
Cu(II) c	9.52	16.45		
Fe(II) c	5.90			
Fe(III) c	14.64	25.18	32.12	
La c	9.11			
Mn(II) c	5.24	8.24		
NbO(III) c	4.0	7.7		
Ni c	6.42	10.24		
UO ₂ (II) c	11.14	19.20		
Zn c	6.05	10.65		
Tartaric acid				
Ba		1.62		
Bi			8.30	
Ca	2.98	9.01		
Cd	2.8			
Co(II)	2.1			
Cu(II)	3.2	5.11	4.78	6.51
				log K_f 19.14 [Cu(OH) ₂ L ²⁻]

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	$\log K_1$	$\log K_2$	$\log K_3$	$\log K_4$
Eu(III)	4.98	8.11		
Fe(III)	7.49			
La	3.06			
Mg		1.36		
Nd	9.0			
Pb	3.78		4.7	$\log K_f$ 14.1 [Pb(OH) ₂ L ²⁻]
Ra	1.24			
Sr	1.60			
Zn	2.68	8.32		
Thioglycolic acid				
Ce(III) <i>a,c</i>	1.99	3.03		
Co(II)	5.84	12.15		
Fe(II)		10.92		
Hg(II)		43.82		
La <i>a,c</i>	1.98	2.98		
Mn(II)	4.38	7.56		
Pb	8.5			
Ni	6.98	13.53		
Rare earths <i>a,c</i>	1.9–2.1	3.0–3.3		
Y <i>a,c</i>	1.91	3.19		
Zn	7.86	15.04		
Thiourea				
Ag	7.4	13.1		
Bi				$\log K_6$ 11.9
Cd	0.6	1.6	2.6	4.6
Cu(I)			13	15.4
Hg(II)		22.1	24.7	26.8
Pb	1.4	3.1	4.7	8.3
Ru(III)	1.21		0.72	
Thoron				
Th		10.15		
Triethanolamine				
Ag	2.30	3.64		
Co(II)	1.73			
Cu(II)	4.30			
Hg(II)	6.90	13.08		
Ni	2.7			
Zn	2.00			
Triethylenetetramine (Trien)				
Ag	7.7			
Cd	10.75	13.9		
Co(II)	11.0			
Cu(II)	20.4			
Fe(II)	7.8			
Fe(III)	21.9			
Hg(II)	25.26			
Mn(II)	4.9			
Ni	14.0			
Pb	10.4			
Zn	11.9			

TABLE 8.13 Cumulative Formation Constants for Metal Complexes with Organic Ligands (*Continued*)

	log K_1	log K_2	log K_3	log K_4
1,1,1-Trifluoro-3-2'-Thenoylacetone (TTA)				
Ba		10.6		
Cu(II)	6.55	13.0		
Fe(III)	6.9			
Ni	10.0			
Pr	9.53			
Pu(III)	9.53			
Pu(IV)	8.0			
Th	8.1			
U(IV)	7.2			
Zr	3.03 [as ZrL^{3+}]			
Xylenol orange				
Bi	5.52			
Fe(III)	5.70			
Hf	6.50			
Tl(III)	4.90			
Zn	6.15			
Zr	7.60			
Zincon				
Zn	13.1			

8.3 BUFFER SOLUTIONS

8.3.1 Standard Reference pH Buffer Solutions

The assigned values of pH_s , according to the Bates-Guggenheim convention [*Pure Applied Chem.* **1**:163 (1960)], for the primary standard solutions prepared from salts issued by the National Institute for Science and Technology (NIST, US) (U.S.) are given in Table 8.14. These are smoothed values. The ionic strength of these reference solutions is 0.1 or less. Strictly speaking the NIST scale uses a molality concentration system; however, values are given in molarity units for convenience.

As a result of a variable liquid-junction potential, the measured pH may be expected to differ seriously from the pa_H determined from cells without a liquid junction in solutions of high acidity or high alkalinity. Merely to affirm the proper functioning of the glass electrode at the extreme ends of the pH scale, two secondary standards are included in Table 8.14. In addition, values for a 0.1 *m* solution of HCl are given to extend the pH scale up to 275°C [see R. S. Greeley, *Anal. Chem.* **32**:1717 (1960)]:

$t, ^\circ C:$	25	60	90	125	150	175	200	225–275
pH:	1.10	1.11	1.12	1.13	1.14	1.15	1.16	1.2

Uncertainties in the values are ± 0.03 pH unit from 25 to 90°C, ± 0.05 pH unit from 125 to 200°C, and ± 0.1 pH unit from 225 to 275°C.

TABLE 8.14 National Bureau of Standards (U.S.) Reference pH Buffer Solutions

Temperature °C	Secondary standard 0.05 M K tetraoxalate	KH tartrate (saturated at 25°C)	0.05 M KH ₂ citrate	0.05 M KH phthalate	0.025 M KH ₂ PO ₄ , 0.025 M Na ₂ HPO ₄	0.0087 M KH ₂ PO ₄ , 0.0302 M Na ₂ HPO ₄	0.01 M Na ₂ B ₄ O ₇	0.025 M NaHCO ₃ , 0.025 M Na ₂ CO ₃	Secondary standard Ca(OH) ₂ (saturated at 25°C)
0	1.666		3.860	4.003	6.984	7.534	9.464	10.317	13.423
5	1.668		3.840	3.999	6.951	7.500	9.395	10.245	13.207
10	1.638		3.820	3.997	6.923	7.472	9.332	10.179	13.003
15	1.642		3.802	3.998	6.900	7.448	9.276	10.118	12.810
20	1.644		3.788	4.002	6.881	7.429	9.225	10.062	12.627
25	1.646	3.557	3.776	4.005	6.865	7.413	9.180	10.012	12.454
30	1.648	3.552	3.766	4.011	6.853	7.400	9.139	9.966	12.289
35		3.549	3.759	4.018	6.844	7.389	9.102	9.925	12.133
38	1.649	3.548	3.756	4.030	6.840	7.384	9.088	9.910	12.043
40	1.650	3.547	3.753	4.035	6.838	7.380	9.068	9.889	11.984
45		3.547		4.047	6.834	7.373	9.038		11.841
50	1.653	3.549	3.749	4.050	6.833	7.367	9.011	9.828	11.705
55		3.554		4.075	6.834		8.985		11.574
60	1.660	3.560		4.081	6.836		8.962		11.449
70	1.671	3.580		4.116	6.845		8.921		
80	1.689	3.609		4.164	6.859		8.885		
90	1.72	3.650		4.205	6.877		8.850		
95	1.73	3.674		4.227	6.886		8.833		
Dilution value ΔpH _{1/2}	+0.186	+0.049	0.024	+0.052	+0.080	+0.070	+0.01	0.079	−0.28

Source: R. G. Bates, *J. Res. Natl. Bur. Stand. (U.S.)*, **66A**:179 (1962) and B. R. Staples and R. G. Bates, *J. Res. Natl. Bur. Stand. (U.S.)*, **73A**:37 (1969).

Note: The uncertainty is ±0.003 in pH in the range 0–50°C, rising to ±0.02 above 70°C.

TABLE 8.15 Compositions of Standard pH Buffer Solutions [National Bureau of Standards (U.S.)]
Air weight of material per liter of buffer solution.

Standard	Weight, g
$\text{KH}_2(\text{C}_2\text{O}_4)_2 \cdot 2\text{H}_2\text{O}$, 0.05 <i>M</i>	12.61
Potassium hydrogen tartrate, about 0.034 <i>M</i>	Saturated at 25°C
Potassium hydrogen phthalate, 0.05 <i>M</i>	10.12
Phosphate:	
KH_2PO_4 , 0.025 <i>M</i>	3.39
Na_2HPO_4 , 0.025 <i>M</i>	3.53
Phosphate:	
KH_2PO_4 , 0.008665 <i>M</i>	1.179
Na_2HPO_4 , 0.03032 <i>M</i>	4.30
$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, 0.01 <i>M</i>	3.80
Carbonate:	
NaHCO_3 , 0.025 <i>M</i>	2.10
Na_2CO_3 , 0.025 <i>M</i>	2.65
$\text{Ca}(\text{OH})_2$, about 0.0203 <i>M</i>	Saturated at 25°C

The buffer values for the NBS reference pH buffer solutions are given below:

		0.05 <i>M</i>	0.05 <i>M</i>	0.025 <i>M</i>	0.0087 <i>M</i>		0.025 <i>M</i>
	KH	KH ₂	KH	KH ₂ PO ₄ ,	KH ₂ PO ₄ ,		NaHCO ₃ ,
Buffer solution	tartrate	citrate	phthalate	0.25 <i>M</i>	0.0302 <i>M</i>	0.01 <i>M</i>	0.025 <i>M</i>
				Na ₂ HPO ₄	Na ₂ HPO ₄	Na ₂ B ₄ O ₇	Na ₂ CO ₃
Buffer value β	0.027	0.034	0.016	0.029	0.016	0.020	0.029

For the secondary pH reference standards, the buffer value is 0.070 for potassium tetroxalate and 0.09 for calcium hydroxide.

To prepare the standard pH buffer solutions recommended by the National Bureau of Standards (U.S.), the indicated weights of the pure materials in Table 8.15 should be dissolved in water of specific conductivity not greater than 5 micromhos. The tartrate, phthalate, and phosphates can be dried for 2 h at 100°C before use. Potassium tetroxalate and calcium hydroxide need not be dried. Fresh-looking crystals of borax should be used. Before use, excess solid potassium hydrogen tartrate and calcium hydroxide must be removed. Buffer solutions pH 6 or above should be stored in plastic containers and should be protected from carbon dioxide with soda-lime traps. The solutions should be replaced within 2 to 3 weeks, or sooner if formation of mold is noticed. A crystal of thymol may be added as a preservative.

8.3.2 Standards for pH Measurement of Blood and Biological Media

Blood is a well-buffered medium. In addition to the NBS phosphate standard of 0.025 *M* (pH_s = 6.480 at 38°C), another reference solution containing the same salts, but in the molal ratio 1 : 4, has an ionic strength of 0.13. It is prepared by dissolving 1.360 g of KH₂PO₄ and 5.677 g of Na₂HPO₄ (air weights) in carbon dioxide-free water to make 1 liter of solution. The pH_s is 7.416 ± 0.004 at 37.5 and 38°C.

The compositions and pH_s values of *tris*(hydroxymethyl)aminomethane, covering the pH range 7.0 to 8.9, are listed in Table 8.16.

TABLE 8.16 Composition and pH Values of Buffer Solutions

Values based on the conventional activity pH scale as defined by the National Bureau of Standards (U.S.) and pertain to a temperature of 25°C [Ref: Bower and Bates, *J. Research Natl. Bur. Standards (U.S.)*, **55**:197 (1955) and Bates and Bower, *Anal. Chem.*, **28**:1322 (1956)]. Buffer value is denoted by column headed β .

25 ml 0.2M KCl + x ml 0.2M HCl, Diluted to 100 ml			50 ml 0.1M KH Phthalate + x ml 0.1M HCl, Diluted to 100 ml			50 ml 0.1M KH Phthalate + x ml 0.1M NaOH, Diluted to 100 ml		
pH	x	β	pH	x	β	pH	x	β
1.00	67.0	0.31	2.20	49.5		4.20	3.0	0.017
1.20	42.5	0.34	2.40	42.2	0.036	4.40	6.6	0.020
1.40	26.6	0.19	2.60	35.4	0.033	4.60	11.1	0.025
1.60	16.2	0.077	2.80	28.9	0.032	4.80	16.5	0.029
1.80	10.2	0.049	3.00	22.3	0.030	5.00	22.6	0.031
2.00	6.5	0.030	3.20	15.7	0.026	5.20	28.8	0.030
2.20	3.9	0.022	3.40	10.4	0.023	5.40	34.1	0.025
			3.60	6.3	0.018	5.60	38.8	0.020
			3.80	2.9	0.015	5.80	42.3	0.015
50 ml 0.1M KH ₂ PO ₄ + x ml 0.1M NaOH, Diluted to 100 ml			50 ml 0.1M Tris(hydroxy- methyl)aminomethane + x ml 0.1M HCl, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.028$ $I = 0.001x$			50 ml of a Mixture 0.1M with Respect to Both KCl and H ₃ BO ₃ + x ml 0.1M NaOH, Diluted to 100 ml		
pH	x	β	pH	x	β	pH	x	β
5.80	3.6		7.00	46.6		8.00	3.9	
6.00	5.6	0.010	7.20	44.7	0.012	8.20	6.0	0.011
6.20	8.1	0.015	7.40	42.0	0.015	8.40	8.6	0.015
6.40	11.6	0.021	7.60	38.5	0.018	8.60	11.8	0.018
6.60	16.4	0.027	7.80	34.5	0.023	8.80	15.8	0.022
6.80	22.4	0.033	8.00	29.2	0.029	9.00	20.8	0.027
7.00	29.1	0.031	8.20	22.9	0.031	9.20	26.4	0.029
7.20	34.7	0.025	8.40	17.2	0.026	9.40	32.1	0.027
7.40	39.1	0.020	8.60	12.4	0.022	9.60	36.9	0.022
7.60	42.4	0.013	8.80	8.5	0.016	9.80	40.6	0.016
7.80	44.5	0.009	9.00	5.7		10.00	43.7	0.014
8.00	46.1					10.20	46.2	
50 ml 0.025M Borax + x ml 0.1M HCl, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.008$ $I = 0.025$			50 ml 0.025M Borax + x ml 0.1M NaOH, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.008$ $I = 0.001(25 + x)$			50 ml 0.05M NaHCO ₃ + x ml 0.1M NaOH, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.009$ $I = 0.001(25 + 2x)$		
pH	x	β	pH	x	β	pH	x	β
8.00	20.5		9.20	0.9		9.60	5.0	
8.20	19.7	0.010	9.40	3.6	0.026	9.80	6.2	0.014
8.40	16.6	0.012	9.60	11.1	0.022	10.00	10.7	0.016
8.60	13.5	0.018	9.80	15.0	0.018	10.20	13.8	0.015
8.80	9.4	0.023	10.00	18.3	0.014	10.40	16.5	0.013

TABLE 8.16 Composition and pH Values of Buffer Solutions (Continued)

50 ml 0.025 <i>M</i> Borax + <i>x</i> ml 0.1 <i>M</i> HCl, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.008$ $I = 0.025$			50 ml 0.025 <i>M</i> Borax + <i>x</i> ml 0.1 <i>M</i> NaOH, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.008$ $I = 0.001(25 + x)$			50 ml 0.05 <i>M</i> NaHCO ₃ + <i>x</i> ml 0.1 <i>M</i> NaOH, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.009$ $I = 0.001(25 + 2x)$		
pH	<i>x</i>	β	pH	<i>x</i>	β	pH	<i>x</i>	β
9.00	4.6	0.026	10.20	20.5	0.009	10.60	19.1	0.012
9.10	2.0		10.40	22.1	0.007	10.80	21.2	0.009
			10.60	23.3	0.005	11.00	22.7	
50 ml 0.05 <i>M</i> Na ₂ HPO ₄ + <i>x</i> ml 0.1 <i>M</i> NaOH, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.025$ $I = 0.001(77 + 2x)$				25 ml 0.2 <i>M</i> KCl + <i>x</i> ml 0.2 <i>M</i> NaOH, Diluted to 100 ml $\Delta\text{pH}/\Delta t \approx -0.033$ $I = 0.001(50 + 2x)$				
pH	<i>x</i>	β	pH		<i>x</i>	β		
11.00	4.1	0.009	12.00		6.0	0.028		
11.20	6.3	0.012	12.20		10.2	0.048		
11.40	9.1	0.017	12.40		16.2	0.076		
11.60	13.5	0.026	12.60		25.6	0.12		
11.80	19.4	0.034	12.80		41.2	0.21		
11.90	23.0	0.037	13.00		66.0	0.30		

The phosphate-succinate system gives the values of p*H*_s shown below:

Molality KH ₂ PO ₄	=	Molality Na ₂ HC ₆ H ₅ O ₇	p <i>H</i> _s	$\Delta(\text{pH}_s/\Delta t)$
0.005			6.251	− 0.000 86 deg ^{−1}
0.010			6.197	− 0.000 71
0.015			6.162	
0.020			6.131	
0.025			6.109	− 0.000 4

TABLE 8.17 Standard Reference Values pH_s^* for the Measurement of Acidity in 50 Weight Percent Methanol-Water

Temperature, °C	0.02 <i>m</i> HOAc, 0.02 <i>m</i> NaOAc, 0.02 <i>m</i> NaCl	0.02 <i>m</i> NaHSuc, 0.02 <i>m</i> NaCl	0.02 <i>m</i> KH_2PO_4 , 0.02 <i>m</i> Na_2HPO_4 , 0.02 <i>m</i> NaCl
10	5.560	5.806	7.937
15	5.549	5.786	7.916
20	5.543	5.770	7.898
25	5.540	5.757	7.884
30	5.540	5.748	7.872
35	5.543	5.743	7.863
40	5.550	5.741	7.858

OAc = acetate Suc = succinate

Reference: R. G. Bates, *Anal Chem.*, **40**(6):35A (1968).**TABLE 8.18** pH^* Values for Buffer Solutions in Alcohol-Water Solvents at 25°C*Liquid-junction potential not included.*

Solvent Composition (weight per cent alcohol)	0.01 <i>M</i> $\text{H}_2\text{C}_2\text{O}_4$, 0.01 <i>M</i> $\text{NH}_4\text{HC}_2\text{O}_4$	0.01 <i>M</i> H_2Suc , 0.01 <i>M</i> LiHSuc	0.01 <i>M</i> HSal , 0.01 <i>M</i> NaSal
	Methanol-Water Solvents		
0	2.15	4.12	
10	2.19	4.30	
20	2.25	4.48	
30	2.30	4.67	
40	2.38	4.87	
50	2.47	5.07	
60	2.58	5.30	
70	2.76	5.57	
80	3.13	6.01	
90	3.73	6.73	
92	3.90	6.92	
94	4.10	7.13	
96	4.39	7.43	
98	4.84	7.89	
99	5.20	8.23	
100	5.79	8.75	7.53
	Ethanol-Water Solvents		
0	2.15	4.12	
30	2.32	4.70	
50	2.51	5.07	
71.9	2.98	5.71	
100			8.32

Suc = succinate Sal = salicylate

8.3.3 Buffer Solutions Other Than Standards

The range of the buffering effect of a single weak acid group is approximately one pH unit on either side of the pK_a . The ranges of some useful buffer systems are collected in Table 8.19. After all the components have been brought together, the pH of the resulting solution should be determined at the temperature to be employed with reference to standard reference solutions. Buffer components should be compatible with other components in the system under study; this is particularly significant for buffers employed in biological studies. Check tables of formation constants to ascertain whether metal-binding character exists.

TABLE 8.19 pH Values of Biological and Other Buffers for Control Purposes

Materials	Acronym	pK_a	pH range
<i>p</i> -Toluenesulfonate and <i>p</i> -toluenesulfonic acid		1.7	1.1–3.3
Glycine and HCl		2.35	1.0–3.7
Citrate and HCl		3.13	1.3–4.7
Formate and HCl		3.71	2.8–4.6
Succinate and borax		4.21, 5.64	3.0–5.8
Phenyl acetate and HCl		4.31	3.5–5.0
Acetate and acetic acid		4.76	3.7–5.6
Succinate and succinic acid		4.21, 5.64	4.8–6.3
2-(<i>N</i> -Morpholino)ethanesulfonic acid	MES	6.1	5.5–6.7
Bis(2-hydroxyethyl)iminotris(hydroxymethyl)methane	BIS-TRIS	6.5	5.8–7.2
KH ₂ PO ₄ and borax		2.2, 7.2; 9	5.8–9.2
<i>N</i> -(2-Acetamido)-2-iminodiacetic acid	ADA	6.6	6.0–7.2
2-[(2-Amino-2-oxoethyl)amino]ethanesulfonic acid	ACES	6.8	6.1–7.5
Piperazine- <i>N,N'</i> -bis(2-ethanesulfonic acid)	PIPES	6.8	6.1–7.5
3-(<i>N</i> -Morpholino)-2-hydroxypropanesulfonic acid	MOPSO	6.9	6.2–7.6
1,3-Bis[tris(hydroxymethyl)methylamino]propane	BIS-TRIS PROPANE	6.8, 9.0	6.3–9.5
KH ₂ PO ₄ and Na ₂ HPO ₄		7.2	6.1–7.5
<i>N,N</i> -Bis(2-hydroxyethyl)-2-aminoethanesulfonic acid	BES	7.1	6.4–7.8
3-(<i>N</i> -Morpholino)propanesulfonic acid	MOPS	7.2	6.5–7.9
<i>N</i> -(2-Hydroxyethyl)piperazine- <i>N'</i> -(2-ethanesulfonic acid)	HEPES	7.5	6.8–8.2
<i>N</i> -Tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid	TES	7.5	6.8–8.2
3-[<i>N,N</i> -Bis(2-hydroxyethyl)amino]-2-hydroxypropanesulfonic acid	DIPSO	7.6	7.0–8.2
3-[<i>N</i> -tris(hydroxymethyl)methylamino]-2-hydroxypropanesulfonic acid	TAPSO	7.6	7.0–8.2
5,5-Diethylbarbiturate (veronal) and HCl		8.0	7.0–8.5
Tris(hydroxymethyl)aminoethane	TRIZMA	8.1	7.0–9.1
<i>N</i> -(2-hydroxyethyl)piperazine- <i>N'</i> -(2-hydroxypropanesulfonic acid)	HEPPSO	7.8	7.1–8.5
Piperazine- <i>N,N'</i> -bis(2-hydroxypropanesulfonic acid)	POPSO	7.8	7.2–8.5
Triethanolamine	TEA	7.8	6.9–8.5
<i>N</i> -Tris(hydroxymethyl)methylglycine	TRICINE	8.1	7.4–8.8
Borax and HCl			7.6–8.9
<i>N,N</i> -Bis(2-hydroxyethyl)glycine	BICINE	8.3	7.6–9.0
<i>N</i> -Tris(hydroxymethyl)methyl-3-aminopropanesulfonic acid	TAPS	8.4	7.7–9.1
3-[(1,1-Dimethyl-2-hydroxyethyl)-2-hydroxypropanesulfonic acid	AMPSO	9.0	8.3–9.7
Ammonia (aqueous) and NH ₄ Cl		9.2	8.3–9.2
2-(<i>N</i> -Cyclohexylamino)-2-hydroxy-1-propanesulfonic acid	CHES	9.3	8.6–10.0

TABLE 8.19 pH Values of Biological and Other Buffers for Control Purposes (*Continued*)

Materials			Acronym	pK _a	pH range
Glycine and NaOH				9.7	8.2–10.1
Ethanolamine (2-aminoethanol) and HCl				9.5	8.6–10.4
3-(Cyclohexylamino)-2-hydroxy-1-propanesulfonic acid			CAPSO	9.6	8.9–10.3
2-Amino-2-methyl-1-propanol			AMP	9.7	9.0–10.5
Carbonate and hydrogen carbonate				10.3	9.2–11.0
Borax and NaOH					9.4–11.1
3-(Cyclohexylamino)-1-propanesulfonic acid			CAPS	10.4	9.7–11.1
Na ₂ HPO ₄ and NaOH				11.9	11.0–12.0

x mL of 0.2M Sodium Acetate (27.199 g NaOAc · 3H ₂ O per liter) plus y mL of 0.2M Acetic Acid			x mL of 0.1M KH ₂ PO ₄ (13.617 g · L ⁻¹) plus y mL of 0.05M Borax Solution (19.404 g Na ₂ B ₄ O ₇ · 10H ₂ O per Liter)					
pH	NaOAc, mL	Acetic Acid, mL	pH	KH ₂ PO ₄ , mL	Borax, mL	pH	KH ₂ PO ₄ , mL	Borax, mL
3.60	7.5	92.5	5.80	92.1	7.9	7.60	51.7	48.3
3.80	12.0	88.0	6.00	87.7	12.3	7.80	49.2	50.8
4.00	18.0	82.0	6.200	83.0	17.0	8.00	46.5	53.5
4.20	26.5	73.5	6.40	77.8	22.2	8.20	43.0	57.0
4.40	37.0	63.0	6.60	72.2	27.8	8.40	38.7	61.3
4.60	49.0	51.0	6.80	66.7	33.3	8.60	34.0	66.0
4.80	60.0	40.0	7.00	62.3	37.7	8.80	27.6	72.4
5.00	70.5	29.5	7.20	58.1	41.9	9.00	17.5	82.5
5.20	79.0	21.0	7.40	55.0	45.0	9.20	5.0	95.0
5.40	85.5	14.5						
5.60	90.5	9.5						

x mL of Veronal (20.6 g Na Diethylbarbiturate per Liter) plus y mL of 0.1M HCl			x mL of 0.2M Aqueous NH ₃ Solution plus y mL of 0.2M NH ₄ Cl (10.699 g · L ⁻¹)			x mL of 0.1M Citrate (21.0 g Citric Acid Monohydrate + 200 mL 1M NaOH per Liter) plus y mL of 0.1M NaOH		
pH	Veronal, mL	HCl, mL	pH	Aq NH ₃ , mL	NH ₄ Cl, mL	pH	Citrate, mL	NaOH, mL
7.00	53.6	46.4	8.00	5.5	94.5	5.10	90.0	10.0
7.20	55.4	44.6	8.20	8.5	91.5	5.30	80.0	20.0
7.40	58.1	41.9	8.40	12.5	87.5	5.50	71.0	29.0
7.60	61.5	38.5	8.60	18.5	81.5	5.70	67.0	33.0
7.80	66.2	33.8	8.80	26.0	74.0	5.90	62.0	38.0
8.00	71.6	28.4	9.00	36.0	64.0			
8.20	76.9	23.1	9.25	50.0	50.0			
8.40	82.3	17.7	9.40	58.5	41.5			
8.60	87.1	12.9	9.60	69.0	31.0			
8.80	90.8	9.2	9.80	78.0	22.0			
9.00	93.6	6.4	10.00	85.0	15.0			

TABLE 8.19 pH Values of Biological and Other Buffers for Control Purposes (*Continued*)

x mL of 0.2M NaOH Added to 100 mL of Stock Solution (0.04M Acetic Acid, 0.04M H ₃ PO ₄ , and 0.04M Boric Acid)								
pH	NaOH, mL		pH	NaOH, mL		pH	NaOH, mL	
1.81	0.0		4.10	25.0		6.80	50.0	
1.89	2.5		4.35	27.5		7.00	52.5	
1.98	5.0		4.56	30.0		7.24	55.0	
2.09	7.5		4.78	32.5		7.54	57.5	
2.21	10.0		5.02	35.0		7.96	60.0	
2.36	12.5		5.33	37.5		8.36	62.5	
2.56	15.0		5.72	40.0		8.69	65.0	
2.87	17.5		6.09	42.5		8.95	67.5	
3.29	20.0		6.37	45.0		9.15	70.0	
3.78	22.5		6.59	47.5		9.37	72.5	
x mL of 0.1M HCl plus y mL of 0.1M Glycine (7.505 g Glycine + 5.85 g NaCl per Liter)			x mL of 0.1M HCl plus y mL of 0.1M Citrate (21.008 g Citric Acid Monohydrate + 200 ml 1M NaOH per Liter)			x mL of 0.05M Succinic Acid (5.90 g · L ⁻¹) plus y mL of Borax Solution (19.404 g Na ₂ B ₄ O ₇ · 10H ₂ O per Liter)		
pH	HCl, mL	Glycine, mL	pH	HCl, mL	Citrate, mL	pH	Succinic Acid, mL	Borax, mL
1.20	84.0	16.0	3.50	52.8	47.2	3.60	90.5	9.5
1.40	71.0	29.0	3.60	51.3	48.7	3.80	86.3	13.7
1.60	61.8	38.2	3.80	48.6	51.4	4.00	82.2	17.8
1.80	55.2	44.8	4.00	43.8	56.2	4.20	77.8	22.2
2.00	49.1	50.9	4.20	38.6	61.4	4.40	73.8	26.2
2.20	42.7	57.3	4.40	34.6	65.4	4.60	70.0	30.0
2.40	36.5	63.5	4.60	24.3	75.7	4.80	66.5	33.5
2.60	30.3	69.7	4.80	11.0	89.0	5.00	63.2	36.8
2.80	24.0	76.0				5.20	60.5	39.5
3.00	17.8	82.2				5.40	57.9	42.1
3.30	10.8	89.2				5.60	55.7	44.3
3.60	6.0	94.0				5.80	54.0	46.0
x mL of 0.2M Na ₂ HPO ₄ · 2H ₂ O (35.599 g · L ⁻¹) plus y mL of 0.1M Citric Acid (19.213 g · L ⁻¹)								
pH	Na ₂ HPO ₄ , mL	Citric Acid, mL	pH	Na ₂ HPO ₄ , mL	Citric Acid, mL	pH	Na ₂ HPO ₄ , mL	Citric Acid, mL
2.20	2.00	98.00	4.20	41.40	58.60	6.20	66.10	33.90
2.40	6.20	93.80	4.40	44.10	55.90	6.40	69.25	30.75
2.60	10.90	89.10	4.60	46.75	53.25	6.60	72.75	27.25
2.80	15.85	84.15	4.80	49.30	50.70	6.80	77.25	22.75
3.00	20.55	79.45	5.00	51.50	48.50	7.00	82.35	17.65
3.20	24.70	75.30	5.20	53.60	46.40	7.20	86.95	13.05
3.40	28.50	71.50	5.40	55.75	44.25	7.40	90.85	9.15
3.60	32.20	67.80	5.60	58.00	42.00	7.60	93.65	6.35
3.80	35.50	64.50	5.80	60.45	39.55	7.80	95.75	4.25
4.00	38.55	61.45	6.00	63.15	36.85	8.00	97.25	2.75

When there are two or more acid groups per molecule, or a mixture is composed of several overlapping acids, the useful range is larger. Universal buffer solutions consist of a mixture of acid groups which overlap such that successive pK_a values differ by 2 pH units or less. The Prideaux-Ward mixture comprises phosphate, phenyl acetate, and borate plus HCl and covers the range from 2 to 12 pH units. The McIlvaine buffer is a mixture of citric acid and Na_2HPO_4 that covers the range from pH 2.2 to 8.0. The Britton-Robinson system consists of acetic acid, phosphoric acid, and boric acid plus NaOH and covers the range from pH 4.0 to 11.5. A mixture composed of Na_2CO_3 , NaH_2PO_4 , citric acid, and 2-amino-2-methyl-1,3-propanediol covers the range from pH 2.2 to 11.0.

General directions for the preparation of buffer solutions of varying pH but fixed ionic strength are given by Bates.* Preparation of McIlvaine buffered solutions at ionic strengths of 0.5 and 1.0 and Britton-Robinson solutions of constant ionic strength have been described by Elving et al.† and Frugoni,‡ respectively.

* Bates, *Determination of pH, Theory and Practice*, Wiley, New York, 1964, pp. 121–122.

† Elving, Markowitz, and Rosenthal, *Anal. Chem.*, **28**:1179 (1956).

‡ Frugoni, *Gazz. Chim. Ital.*, **87**:L403 (1957).

8.4 REFERENCE ELECTRODES

TABLE 8.20 Potentials of Reference Electrodes in Volts as a Function of Temperature

Liquid-junction potential included.

Temp., °C	0.1M KCl Calomel*	1.0M KCl Calomel*	3.5M KCl Calomel*	Satd. KCl Calomel*	1.0M KCl Ag/AgCl†	1.0M KBr Ag/AgBr‡	1.0M KI Ag/AgI§
0	0.3367	0.2883		0.25918	0.23655	0.08128	−0.14637
5					0.23413	0.07961	−0.14719
10	0.3362	0.2868	0.2556	0.25387	0.23142	0.07773	−0.14822
15	0.3361			0.2511	0.22857	0.07572	−0.14942
20	0.3358	0.2844	0.2520	0.24775	0.22557	0.07349	−0.15081
25	0.3356	0.2830	0.2501	0.24453	0.22234	0.07106	−0.15244
30	0.3354	0.2815	0.2481	0.24118	0.21904	0.06856	−0.15405
35	0.3351			0.2376	0.21565	0.06585	−0.15590
38	0.3350		0.2448	0.2355			
40	0.3345	0.2782	0.2439	0.23449	0.21208	0.06310	−0.15788
45					0.20835	0.06012	−0.15998
50	0.3315	0.2745		0.22737	0.20449	0.05704	−0.16219
55					0.20056		
60	0.3248	0.2702		0.2235	0.19649		
70					0.18782		
80				0.2083	0.1787		
90					0.1695	0.0251	

* Bates et al., *J. Research Natl. Bur. Standards*, **45**, 418 (1950).

† Bates and Bower, *J. Research Natl. Bur. Standards*, **53**, 283 (1954).

‡ Hetzer, Robinson and Bates, *J. Phys. Chem.*, **66**, 1423 (1962).

§ Hetzer, Robinson and Bates, *J. Phys. Chem.*, **68**, 1929 (1964).

TABLE 8.20 Potentials of Reference Electrodes in Volts as a Function of Temperature (*Continued*)

Temp., °C	125	150	175	200	225	250	275
1.0M KCl Ag/AgCl*	0.1330	0.1032	0.0708	0.0348	−0.0051	−0.054	−0.090
1.0M KBr Ag/AgBr†	−0.0048	−0.0312	−0.0612	−0.0951			

* Greeley et al., *J. Phys. Chem.*, **64**, 652 (1960).

† Towns et al., *J. Phys. Chem.*, **64**, 1861 (1960).

The values of several additional reference electrodes at 25°C are listed:

Ag/AgCl, satd. KCl	0.198
Ag/AgCl, 01M KCl	0.288
Hg/HgO, 1.0M NaOH	0.140
Hg/HgO, 0.1M NaOH	0.165
Hg/Hg ₂ SO ₄ , satd. K ₂ SO ₄ (22°C)	0.658
Hg/Hg ₂ SO ₄ , satd. KCl	0.655

TABLE 8.21 Potentials of Reference Electrodes (in Volts) at 25°C for Water–Organic Solvent Mixtures
Electrolyte solution of 1M HCl.

Solvent, wt %	Methanol, Ag/AgCl	Ethanol, Ag/AgCl	2-Propanol, Ag/AgCl	Acetone, Ag/AgCl	Dioxane, Ag/AgCl	Ethylene glycol, Ag/AgCl	Methanol, calomel	Dioxane, calomel
5			0.2180	0.2190		0.2190		
10	0.2153	0.2146	0.2138	0.2156		0.2160		
20	0.2090	0.2075	0.2063	0.2079	0.2031	0.2101	0.255	0.2501
30		0.2003				0.2036		
40	0.1968	0.1945		0.1859		0.1972	0.243	
45					0.1635			0.2104
50		0.1859		0.158				
60	0.1818	0.173				0.1807		
70		0.158			0.0659		0.216	0.1126
80	0.1492	0.136						
82					−0.0614			−0.0014
90	0.1135	0.096		−0.034				
94.2	0.0841							
98		0.0215						
99							0.103	
100	−0.0099	−0.0081		−0.53				

8.4.1 Electrometric Measurement of pH

The pH value is defined for an aqueous solution in an operational (arbitrary but reproducible) manner according to the Bates-Guggenheim convention:

$$\text{pH}_x = \text{pH}_s + \frac{E_x - E_s}{2.3026RT/F}$$

where R is the gas constant per mole, T is the temperature on the absolute scale, and F is the faraday. The pH_x of the unknown medium is calculated from that of an accepted standard (pH_s) and the measured difference in the emf (E) of the electrode combination when the standard solution is removed from the cell and replaced by the unknown. The double vertical line marks a liquid junction. Electrodes as fabricated exhibit variations in the reproducibility of the reference electrode, in the liquid-junction potential, and, with glass electrodes, in the asymmetry potential. These differences are all eliminated in the standardizing procedure with standard reference pH buffers. (See R. G. Bates, *Determination of pH, Theory and Practice*, Wiley, New York, 1964.)

Electrode reversible to hydrogen ions	Standard reference buffer or unknown solution	Salt bridge (KCl. 3.5 <i>M</i> or saturated)	Reference electrode
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An electrometric pH-measurement system consists of (1) pH-responsive electrode, (2) reference electrode, and (3) potential-measuring device—some form of high-impedance electronic voltmeter for glass-electrode combinations and this or a potentiometer arrangement for other pH-responsive electrodes. Electronic pH meters are simply voltmeters with scale divisions in pH units which are equivalent to the values of $2.3026RT/F$ (in mV) per pH unit. Values of this function at several temperatures are given in Table 8.22. There is no compensation incorporated in the meter for the changes in pH of the test solution as a function of temperature. Reliability of an indicator–reference electrode combination must be ascertained by standardization of the pH meter with one standard buffer and checking the pH response by immersing the combination in a second and different reference buffer.

The temperature compensator on a pH meter varies the instrument definition of a pH unit from 54.20 mV at 0°C to perhaps 66.10 mV at 60°C. This permits one to measure the pH of the sample (and reference buffer standard) at its actual temperature and thus avoid error due to dissociation equilibria and to junction potentials which have significant temperature coefficients.

TABLE 8.22 Values of $2.3026RT/F$ at Several Temperatures

In millivolts.

t °C	Value	t °C	Value	t °C	Value	t °C	Value
0	54.197	25	59.157	50	64.118	80	70.070
5	55.189	30	60.149	55	65.110	85	71.062
10	56.181	35	61.141	60	66.102	90	72.054
15	57.173	38	61.737	65	67.094	95	73.046
18	57.767	40	62.133	70	68.086	100	74.038
20	58.165	45	63.126	75	69.078		

8.5 INDICATORS

TABLE 8.23 Indicators for Aqueous Acid-Base Titrations

This table lists some selected indicators. The pH range or transition interval given in the third column may vary appreciably from one observer to another, and, in addition, it is affected by ionic strength, temperature, and illumination; consequently only approximate values can be given. They should be considered to refer to solutions having low ionic strengths and a temperature of about 25°C. In the fourth column the pK_a ($-\log K_a$) of the indicator as determined spectrophotometrically is listed. In the fifth column the wavelength of maximum absorption is given first for the acidic and then for the basic form of the indicator, and the same order is followed in giving the colors in the sixth column. The abbreviations used to describe the colors of the two forms of the indicator are as follows:

B, blue
V, violet
Y, yellow
O, orange
C, colorless

G, green
P, purple
R, red
O-Br, orange-brown

Indicator	Chemical name	pH range	pK_a	$\lambda_{\max}$, nm	Color change
Cresol red (acid range)	<i>o</i> -Cresolsulfonephthalein	0.2–1.8			R-Y
Cresol purple (acid range)	<i>m</i> -Cresolsulfonephthalein	1.2–2.8	1.51	533, —	R-Y
Thymol blue (acid range)	Thymolsulfonephthalein	1.2–2.8	1.65	544, 430	R-Y
Tropeolin OO	Diphenylamino- <i>p</i> -benzene sodium sulfonate	1.3–3.2	2.0	527, —	R-Y
2,6-Dinitrophenol	2,6-Dinitrophenol	2.4–4.0	3.69		C-Y
2,4-Dinitrophenol	2,4-Dinitrophenol	2.5–4.3	3.90		C-Y
Methyl yellow	Dimethylaminoazobenzene	2.9–4.0	3.3	508, —	R-Y
Methyl orange	Dimethylaminoazobenzene sodium sulfonate	3.1–4.4	3.40	522, 464	R-O
Bromophenol blue	Tetrabromophenolsulfonephthalein	3.0–4.6	3.85	436, 592	Y-BV
Bromocresol green	Tetrabromo- <i>m</i> -cresolsulfonephthalein	4.0–5.6	4.68	444, 617	Y-B
Methyl red	<i>o</i> -Carboxybenzeneazo-dimethylaniline	4.4–6.2	4.95	530, 427	R-Y
Chlorophenol red	Dichlorophenolsulfonephthalein	5.4–6.8	6.0	—, 573	Y-R
Bromocresol purple	Dibromo- <i>o</i> -cresolsulfonephthalein	5.2–6.8	6.3	433, 591	Y-P
Bromophenol red	Dibromophenolsulfonephthalein	5.2–6.8		—, 574	Y-R
<i>p</i> -Nitrophenol	<i>p</i> -Nitrophenol	5.3–7.6	7.15	320, 405	C-Y
Bromothymol blue	Dibromothymolsulfonephthalein	6.2–7.6	7.1	433, 617	Y-B
Neutral red	Aminodimethylaminotolu-phenazonium chloride	6.8–8.0	7.4		R-Y
Phenol red	Phenolsulfonephthalein	6.4–8.0	7.9	433, 558	Y-R
<i>m</i> -Nitrophenol	<i>m</i> -Nitrophenol	6.4–8.8	8.3	—, 570	C-Y

TABLE 8.23 Indicators for Aqueous Acid-Base Titrations (*Continued*)

Indicator	Chemical name	pH range	pK_a	$\lambda_{\max}$, nm	Color change
Cresol red	<i>o</i> -Cresolsulfonephthalein	7.2–8.8	8.2	434, 572	Y-R
<i>m</i> -Cresol purple	<i>m</i> -Cresolsulfonephthalein	7.6–9.2	8.32	—, 580	Y-P
Thymol blue	Thymolsulfonephthalein	8.0–9.6	8.9	430, 596	Y-B
Phenolphthalein	Phenolphthalein	8.0–10.0	9.4	—, 553	C-R
α -Naphtholbenzein	α -Naphtholbenzein	9.0–11.0			Y-B
Thymolphthalein	Thymolphthalein	9.4–10.6	10.0	—, 598	C-B
Alizarin Yellow R	5-(<i>p</i> -Nitrophenylazo)-salicyclic acid, Na salt	10.0–12.0	11.16		Y-V
Tropeolin O	<i>p</i> -Sulfobenzenazo-resorcinol	11.0–13.0			Y-O-Br
Nitramine	2,4,6-Trinitrophenyl-methylnitroamine	10.8–13.0			C-O-Br

TABLE 8.24 Mixed Indicators

Mixed indicators give sharp color changes and are especially useful in titrating to a given titration exponent (pI).

The information given in this table is from the two-volume work *Volumetric Analysis* by Kolthoff and Stenger, published by Interscience Publishers, Inc., New York, 1942 and 1947, and reproduced with their permission.

Composition of Indicator Solution	pI	Color		Notes
		Acid	Alkaline	
1 part 0.1% methyl yellow in alc.	*	Blue-violet	Green	Still green at pH 3.4, blue-violet at 3.2†
1 part 0.1% methylene blue in alc.				
1 part 0.14% xylene cyanol FF in alc.	*	Violet	Green	Color is gray at pH 3.8
1 part 0.1% methyl orange in aq.				
1 part 0.1% methyl orange in aq.	*	Violet	Green	Good indicator, especially in artificial light
1 part 0.25% indigo carmine in aq.				
1 part 0.1% methyl orange in aq.		Violet	Green	
1 part 0.1% aniline blue in aq.				
1 part 0.1% bromcresol green sodium salt in aq.		Orange	Blue-green	Yellow at pH 3.5, greenish yellow at 4.0, weakly green at 4.3
1 part 0.02% methyl orange in aq.				
3 parts 0.1% bromcresol green in alc.		Wine-red	Green	Very sharp color change†
1 part 0.2% methyl red in alc.				
1 part 0.2% methyl red in alc.	*	Red-violet	Green	Color is red-violet at pH 5.2, a dirty blue at 5.4, and a dirty green at 5.6
1 part 0.1% methylene blue in alc.				
1 part 0.1% chlorphenol red sodium salt in aq.		Green	Violet	Pale violet at pH 5.8
1 part 0.1% aniline blue in water				
1 part 0.1% bromcresol green sodium salt in aq.		Yellow-green	Blue-violet	Blue-green at pH 5.4, blue at 5.8, blue with a touch of violet at 6.0, blue-violet at 6.2
1 part 0.1% chlorphenol red sodium salt in aq.				
1 part 0.1% bromcresol purple sodium salt in aq.		Yellow	Violet-blue	Yellow-violet at pH 6.2, violet at 6.6, blue-violet at 6.8
1 part 0.1% bromthymol blue sodium salt in aq.				
2 parts 0.1% bromthymol blue sodium salt in aq.		Violet	Blue	
1 part 0.1% azolitmin in aq.				

1 part 0.1% neutral red in alc.	*		Violet-blue	Green	Violet blue at pH 7.0†
1 part 0.1% methylene blue in alc.		7.0			
1 part 0.1% neutral red in alc.			Rose	Green	Dirty green at pH 7.4, pale rose at 7.2, clear rose at 7.0
1 part 0.1% bromthymol blue in alc.		7.2			
2 parts 0.1% cyanine in 50% alc.			Yellow	Violet	Orange at pH 7.2, beautiful violet at 7.4, color fades on standing
1 part 0.1% phenol red in 50% alc.		7.3			
1 part 0.1% bromthymol blue sodium salt in aq.			Yellow	Violet	Dirty green at pH 7.2, pale violet at 7.4, strong violet at 7.6†
1 part 0.1% phenol red sodium salt in aq.		7.5			
1 part 0.1% cresol red sodium salt in aq.			Yellow	Violet	Rose at pH 8.2, distinctly violet at 8.4†
3 parts 0.1% thymol blue sodium salt in aq.		8.3			
2 parts 0.1% α -naphtholphthalein in alc.	*		Pale rose	Violet	Pale violet at pH 8.2, strong violet at 8.4
1 part 0.1% cresol red in alc.		8.3			
1 part 0.1% α -naphtholphthalein in alc.			Pale rose	Violet	Pale green at pH 8.6, violet at 9.0
3 parts 0.1% phenolphthalein in alc.		8.9			
1 part 0.1% phenolphthalein in alc.			Green	Violet	Pale blue at pH 8.8, violet at 9.0
2 parts 0.1% methyl green in alc.		8.9			
1 part 0.1% thymol blue in 50% alc.			Yellow	Violet	From yellow thru green to violet†
3 parts 0.1% phenolphthalein in 50% alc.		9.0			
1 part 0.1% phenolphthalein in alc.			Colorless	Violet	Rose at pH 9.6, violet at 10; sharp color change
1 part 0.1% thymolphthalein in alc.		9.9			
1 part 0.1% phenolphthalein in alc.					
2 parts 0.2% Nile blue in alc.		10.0	Blue	Red	Violet at pH 10†
2 parts 0.1% thymolphthalein in alc.					
1 part 0.1% alizarin yellow in alc.		10.2	Yellow	Violet	Sharp color change
2 parts 0.2% Nile blue in aq.					
1 part 0.1% alizarin yellow in alc.		10.8	Green	Red-brown	

* Keep in a dark bottle. † Excellent indicator.

TABLE 8.25 Fluorescent Indicators

Name	pH Range	Color Change Acid to Base	Indicator Solution
Benzoflavine	− 0.3 to 1.7	Yellow to green	1
3,6-Dihydroxyphthalimide	0 to 2.4	Blue to green	1
	6.0 to 8.0	Green to yellow/green	
Eosin (tetrabromofluorescein)	0 to 3.0	Non-fl to green	4, 1%
4-Ethoxyacridone	1.2 to 3.2	Green to blue	1
3,6-Tetramethyldiaminoxanthone	1.2 to 3.4	Green to blue	1
Esculin	1.5 to 2.0	Weak blue to strong blue	
Anthranilic acid	1.5 to 3.0	Non-fl to light blue	2 (50% ethanol)
	4.5 to 6.0	Light blue to dark blue	
	12.5 to 14	Dark blue to non-fl	
3-Amino-1-naphthoic acid	1.5 to 3.0	Non-fl to green	2 (as sulfate in 50% ethanol)
	4.0 to 6.0	Green to blue	
	11.6 to 13.0	Blue to non-fl	
1-Naphthylamino-6-sulfonamide (also the 1-, 7-)	1.9 to 3.9	Non-fl to green	3
	9.6 to 13.0	Green to non-fl	
2-Naphthylamino-6-sulfonamide (also the 2-, 8-)	1.9 to 3.9	Non-fl to dark blue	3
	9.6 to 13.0	Dark blue to non-fl	
1-Naphthylamino-5-sulfonamide	2.0 to 4.0	Non-fl to yellow/orange	3
	9.5 to 13.0	Yellow/orange to non-fl	
1-Naphthoic acid	2.5 to 3.5	Non-fl to blue	4
Salicylic acid	2.5 to 4.0	Non-fl to dark blue	4 (0.5%)
Phloxin BA extra (tetrachlorotetrabromofluorescein)	2.5 to 4.0	Non-fl to dark blue	2
Erythrosin B (tetraiodofluorescein)	2.5 to 4.0	Non-fl to light green	4 (0.2%)
2-Naphthylamine	2.8 to 4.4	Non-fl to violet	1
Magdala red	3.0 to 4.0	Non-fl to purple	
<i>p</i> -Aminophenylbenzenesulfonamide	3.0 to 4.0	Non-fl to light blue	3
2-Hydroxy-3-naphthoic acid	3.0 to 6.8	Blue to green	4 (0.1%)
Chromotropic acid	3.1 to 4.4	Non-fl to light blue	4 (5%)
1-Naphthionic acid	3 to 4	Non-fl to blue	4
	10 to 12	Blue to yellow-green	
1-Naphthylamine	3.4 to 4.8	Non-fl to blue	1
5-Aminosalicylic acid	3.1 to 4.4	Non-fl to light green	1 (0.2% fresh)
Quinine	3.0 to 5.0	Blue to weak violet	1 (0.1%)
	9.5 to 10.0	Weak violet to non-fl	
<i>o</i> -Methoxybenzaldehyde	3.1 to 4.4	Non-fl to green	4 (0.2%)
<i>o</i> -Phenylenediamine	3.1 to 4.4	Green to non-fl	5
<i>p</i> -Phenylenediamine	3.1 to 4.4	Non-fl to orange/yellow	5
Morin (2',4',3,5,7-pentahydroxyflavone)	3.1 to 4.4	Non-fl to green	6 (0.2%)
	8 to 9.8	Green to yellow/green	
Thioflavine S	3.1 to 4.4	Dark blue to light blue	6 (0.2%)
Fluorescein	4.0 to 4.5	Pink/green to green	4 (1%)
Dichlorofluorescein	4.0 to 6.6	Blue green to green	1
β -Methylesculetin	4.0 to 6.2	Non-fl to blue	1
	9.0 to 10.0	Blue to light green	
Quininic acid	4.0 to 5.0	Yellow to blue	6 (satd)
β -Naphthoquinoline	4.4 to 6.3	Blue to non-fl	3
Resorufin (7-oxyphenoxazone)	4.4 to 6.4	Yellow to orange	

Indicator solutions: 1, 1% solution in ethanol; 2, 0.1% solution in ethanol; 3, 0.05% solution in 90% ethanol; 4, sodium or potassium salt in distilled water; 5, 0.2% solution in 70% ethanol; 6, distilled water.

Reference: G.F. Kirkbright, "Fluorescent Indicators," Chap. 9 in *Indicators*, E. Bishop (ed.), Pergamon Press, Oxford, 1972.

TABLE 8.25 Fluorescent Indicators (*Continued*)

Name	pH Range	Color Change Acid to Base	Indicator Solution
Acridine	5.2 to 6.6	Green to violet	2
3,6-Dihydroxyxanthone	5.4 to 7.6	Non-fl to blue/violet	1
5,7-Dihydroxy-4-methylcoumarin	5.5 to 5.8	Light blue to dark blue	
3,6-Dihydroxyphthalic acid dinitrile	5.8 to 8.2	Blue to green	1
1,4-Dihydroxybenzenedisulfonic acid	6 to 7	Non-fl to light blue	4 (0.1%)
Luminol	6 to 7	Non-fl to blue	
2-Naphthol-6-sulfonic acid	5–7 to 8–9	Non-fl to blue	4
Quinoline	6.2 to 7.2	Blue to non-fl	6 (satd)
1-Naphthol-5-sulfonic acid	6.5 to 7.5	Non-fl to green	6 (satd)
Umbelliferone	6.5 to 8.0	Non-fl to blue	
Magnesium-8-hydroxyquinolate	6.5 to 7.5	Non-fl to yellow	6 (0.1% in 0.01 <i>M</i> HCl)
Orcinaurine	6.5 to 8.0	Non-fl to green	6 (0.03%)
Diazo brilliant yellow	6.5 to 7.5	Non-fl to blue	
Coumaric acid	7.2 to 9.0	Non-fl to green	1
β -Methylumbelliferone	>7.0	Non-fl to blue	2 (0.3%)
Harmin	7.2 to 8.9	Blue to yellow	
2-Naphthol-6,8-disulfonic acid	7.5 to 9.1	Blue to light blue	4
Salicylaldehyde semicarbazone	7.6 to 8.0	Yellow to blue	2
1-Naphthol-2-sulfonic acid	8.0 to 9.0	Dark blue to light blue	4
Salicylaldehyde acetylhydrazone	8.3	Non-fl to green/blue	2
Salicylaldehyde thiosemicarbazone	8.4	Non-fl to blue/green	2
1-Naphthol-4-sulfonic acid	8.2	Dark blue to light blue	4
Naphthol AS	8.2 to 10.3	Non-fl to yellow/green	4
2-Naphthol	8.5 to 9.5	Non-fl to blue	2
Acridine orange	8.4 to 10.4	Non-fl to yellow/green	1
Orcinsulfonephthalein	8.6 to 10.0	Non-fl to yellow	
2-Naphthol-3,6-disulfonic acid	9.0 to 9.5	Dark blue to light blue	4
Ethoxyphenylnaphthostilbazonium chloride	9 to 11	Green to non-fl	1
<i>o</i> -Hydroxyphenylbenzothiazole	9.3	Non-fl to blue green	2
<i>o</i> -Hydroxyphenylbenzoxazole	9.3	Non-fl to blue/violet	2
<i>o</i> -Hydroxyphenylbenzimidazole	9.9	Non-fl to blue/violet	2
Coumarin	9.5 to 10.5	Non-fl to light green	
6,7-Dimethoxyisoquinoline-1-carboxylic acid	9.5 to 11.0	Yellow to blue	0.1% in glycerine/ ethanol/water in 2:2:18 ratio
1-Naphthylamino-4-sulfonamide	9.5 to 13.0	Dark blue to white/blue	3

TABLE 8.26 Selected List of Oxidation-Reduction Indicators

Name	Reduction Potential (30°C) in Volts at		Suitable pH Range	Color Change Upon Oxidation
	pH = 0	pH = 7		
Bis(5-bromo-1,10-phenanthroline) ruthenium(II) dinitrate	1.41*			Red to faint blue
Tris(5-nitro-1,10-phenanthroline) iron(II) sulfate	1.25*			Red to faint blue
Iron(II)-2,2',2''-tripyridine sulfate	1.25*			Pink to faint blue
Tris(4,7-diphenyl-1,10-phenanthroline) iron(II) disulfate	1.13 (4.6 M H ₂ SO ₄)* 0.87 (1.0 M H ₂ SO ₄)*			Red to faint blue
<i>o,m'</i> -Diphenylaminedicarboxylic acid	1.12			Colorless to blue-violet
Setopaline	1.06 (<i>trans</i>)†			Yellow to orange
<i>p</i> -Nitrodiphenylamine	1.06			Colorless to violet
Tris(1,10-phenanthroline)-iron(II) sulfate	1.06 (1.00 M H ₂ SO ₄)* 1.00 (3.0 M H ₂ SO ₄)* 0.89 (6.0 M H ₂ SO ₄)*			Red to faint blue
Setoglaurine O	1.01 (<i>trans</i>)†			Yellow-green to yellow-red
Xylene cyanole FF	1.00 (<i>trans</i>)†			Yellow-green to pink
Erioglaurine A	1.00 (<i>trans</i>)†			Green-yellow to bluish red
Eriogreen	0.99 (<i>trans</i>)†			Green-yellow to orange
Tris(2,2'-bipyridine)-iron(II) hydrochloride	0.97*			Red to faint blue
2-Carboxydiphenylamine [<i>N</i> -phenyl- anthranilic acid]	0.94			Colorless to pink
Benzidine dihydrochloride	0.92			Colorless to blue
<i>o</i> -Toluidine	0.87			Colorless to blue
Bis(1,10-phenanthroline)-osmium(II) perchlorate	0.859 (0.1 M H ₂ SO ₄)			Green to pink
Diphenylamine-4-sulfonate (Na salt)	0.85			Colorless to violet

3,3'-Dimethoxybenzidine dihydrochloride [o-dianisidine]	0.85			Colorless to red
Ferrocypen	0.81			Yellow to violet
4'-Ethoxy-2,4-diaminoazobenzene	0.76			Red to pale yellow
N,N-Diphenylbenzidine	0.76			Colorless to violet
Diphenylamine	0.76			Colorless to violet
N,N-Dimethyl-p-phenylenediamine	0.76			Colorless to red
Variamine blue B hydrochloride	0.712‡	0.310	1.5–6.3	Colorless to blue
N-Phenyl-1,2,4-benzenetriamine	0.70			Colorless to red
Bindschedler's green	0.680‡	0.224	2–9.5	
2,6-Dichloroindophenol (Na salt)	0.668‡	0.217	6.3–11.4	Colorless to blue
2,6-Dibromophenolindophenol	0.668‡	0.216	7.0–12.3	Colorless to blue
Brilliant cresyl blue [3-amino-9-dimethyl- amino-10-methylphenoxyazine chloride]	0.583	0.047	0–11	Colorless to blue
Iron(II)-tetrapyridine chloride	0.59			Red to faint blue
Thionine [Lauth's violet]	0.563‡	0.064	1–13	Colorless to violet
Starch (soluble potato, I ₃ ⁻ present)	0.54			Colorless to blue
Gallocyanine (25°C)		0.021		Colorless to violet-blue
Methylene blue	0.532‡	0.011	1–13	Colorless to blue
Nile blue A [aminonaphthodiethylamino- phenoxazine sulfate]	0.406‡	–0.119	1.4–12.3	Colorless to blue
Indigo-5,5',7,7'-tetrasulfonic acid (Na salt)	0.365‡	–0.046	<9	Colorless to blue
Indigo-5,5',7-trisulfonic acid (Na salt)	0.332‡	–0.081	<9	Colorless to blue
Indigo-5,5'-disulfonic acid (Na salt)	0.291‡	–0.125	<9	Colorless to blue
Phenosafranine	0.280‡	–0.252	1–11	Colorless to violet-blue
Indigo-5-monosulfonic acid (Na salt)	0.262‡	–0.157	<9	Colorless to blue
Safranin T	0.24‡	–0.289	1–12	Colorless to violet-blue
Bis(dimethylglyoximate)-iron(II) chloride	0.155		6–10	Red to colorless
Induline scarlet	0.047‡	–0.299	3–8.6	Colorless to red
Neutral red		–0.323	2–11	Colorless to red-violet

* Transition point is at higher potential than the tabulated formal potential because the molar absorptivity of the reduced form is very much greater than that of the oxidized form.

† *Trans* = first noticeable color transition; often 60 mV less than E°

‡ Values of E° are obtained by extrapolation from measurements in weakly acid or weakly alkaline systems.

8.6 ELECTRODE POTENTIALS

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C

Standard potentials are tabulated except when a solution composition is stated; the latter are formal potentials and the concentrations are in mol/liter.

Half-reaction	Standard or formal potential	Solution composition
Actinium		
$\text{Ac}^{3+} + 3e^- = \text{Ac}$	-2.13	
Aluminum		
$\text{Al}^{3+} + 3e^- = \text{Al}$	-1.676	
$\text{AlF}_6^{3-} + 3e^- = \text{Al} + 6\text{F}^-$	-2.07	
$\text{Al}(\text{OH})_4^- + 3e^- = \text{Al} + 4\text{OH}^-$	-2.310	
Americium		
$\text{AmO}_2^{2+} + 4\text{H}^+ + 2e^- = \text{Am}^{4+} + 2\text{H}_2\text{O}$	1.20	
$\text{AmO}_2^{2+} + e^- = \text{AmO}_2^+$	1.59	
$\text{AmO}_2^+ + 4\text{H}^+ + e^- = \text{Am}^{4+} + 2\text{H}_2\text{O}$	0.82	
$\text{AmO}_2^+ + 4\text{H}^+ + 2e^- = \text{Am}^{3+} + 2\text{H}_2\text{O}$	1.72	
$\text{Am}^{4+} + e^- = \text{Am}^{3+}$	2.62	
$\text{Am}^{4+} + 4e^- = \text{Am}$	-0.90	
$\text{Am}^{3+} + 3e^- = \text{Am}$	-2.07	
Antimony		
$\text{Sb}(\text{OH})_4^- + 2e^- = \text{SbO}_2^- + 2\text{OH}^- + 2\text{H}_2\text{O}$	-0.465	1 NaOH
$\text{SbO}_2^- + 2\text{H}_2\text{O} + 3e^- = \text{Sb} + 4\text{OH}^-$	0.639	1 NaOH
$\text{Sb} + 3\text{H}_2\text{O} + 3e^- = \text{SbH}_3 + 3\text{OH}^-$	-1.338	1 NaOH
$\text{Sb}_2\text{O}_5 + 6\text{H}^+ + 4e^- = 2\text{SbO}^+ + 3\text{H}_2\text{O}$	0.605	
$\text{Sb}_2\text{O}_5 + 4\text{H}^+ + 4e^- = \text{Sb}_2\text{O}_3 + 2\text{H}_2\text{O}$	0.699	
$\text{Sb}_2\text{O}_5 + 2\text{H}^+ + 2e^- = \text{Sb}_2\text{O}_4 + \text{H}_2\text{O}$	1.055	
$\text{Sb}_2\text{O}_4 + 2\text{H}^+ + 2e^- = \text{Sb}_2\text{O}_3 + \text{H}_2$	0.342	
$\text{SbO}^+ + 2\text{H}^+ + 3e^- = \text{Sb} + \text{H}_2\text{O}$	0.204	
$\text{Sb} + 3\text{H}^+ + 3e^- = \text{SbH}_3$	-0.510	
Arsenic		
$\text{H}_3\text{AsO}_4 + 2\text{H}^+ + 2e^- = \text{HAsO}_2 + 2\text{H}_2\text{O}$	0.560	
$\text{HAsO}_2 + 3\text{H}^+ + 3e^- = \text{As} + 2\text{H}_2\text{O}$	0.240	
$\text{As} + 3\text{H}^+ + 3e^- = \text{AsH}_3$	-0.225	
$\text{AsO}_4^{3-} + 2\text{H}^+ + 2e^- = \text{AsO}_2^- + 4\text{OH}^-$	-0.67	
$\text{AsO}_2^- + 2\text{H}_2\text{O} + 3e^- = \text{As} + 4\text{OH}^-$	-0.68	
$\text{As} + 3\text{H}_2\text{O} + 3e^- = \text{AsH}_3 + 3\text{OH}^-$	-1.37	
Astatine		
$\text{HAtO}_3 + 4\text{H}^+ + 4e^- = \text{HAtO} + 2\text{H}_2$	ca. 1.4	
$2\text{HAtO} + 2\text{H}^+ + 2e^- = \text{At}_2 + 2\text{H}_2\text{O}$	ca. 0.7	
$\text{At}_2 + 2e^- = 2\text{At}^-$	0.20	
Barium		
$\text{BaO}_2 + 4\text{H}^+ + 2e^- = \text{Ba}^{2+} + 2\text{H}_2\text{O}$	2.365	
$\text{Ba}^{2+} + 2e^- = \text{Ba}$	-2.92	

Source: A. J. Bard, R. Parsons, and J. Jordan (eds.), *Standard Potentials in Aqueous Solution* (prepared under the auspices of the International Union of Pure and Applied Chemistry), Marcel Dekker, New York, 1985; G. Charlot et al., *Selected Constants: Oxidation-Reduction Potentials of Inorganic Substances in Aqueous Solution*, Butterworths, London, 1971.

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Berkelium		
$\text{Bk}^{4+} + 4e^- = \text{Bk}$	-1.05	
$\text{Bk}^{4+} + e^- = \text{Bk}^{3+}$	1.67	
$\text{Bk}^{3+} + 3e^- = \text{Bk}$	-2.01	
Beryllium		
$\text{Be}^{2+} + 2e^- = \text{Be}$	-1.99	
Bismuth		
Bi_2O_4 (bismuthate) + $4\text{H}^+ + 2e^- = 2\text{BiO}^+ + 2\text{H}_2\text{O}$	1.59	
$\text{Bi}^{3+} + 3e^- = \text{Bi}$	0.317	
$\text{Bi} + 3\text{H}^+ + 3e^- = \text{BiH}_3$	-0.97	
$\text{BiCl}_4^- + 3e^- = \text{Bi} + 4\text{Cl}^-$	0.199	
$\text{BiBr}_4^- + 3e^- = \text{Bi} + 4\text{Br}^-$	0.168	
$\text{BiOCl} + 2\text{H}^+ + 3e^- = \text{Bi} + \text{H}_2\text{O} + \text{Cl}^-$	0.170	
Boron		
$\text{B}(\text{OH})_3 + 3\text{H}^+ + 3e^- = \text{B} + 3\text{H}_2\text{O}$	-0.890	
$\text{BO}_2^- + 6\text{H}_2\text{O} + 8e^- = \text{BH}_3^- + 8\text{OH}^-$	-1.241	
$\text{B}(\text{OH})_4^- + 3e^- = \text{B} + 4\text{OH}^-$	-1.811	
Bromine		
$\text{BrO}_4^- + 2\text{H}^+ + 2e^- = \text{BrO}_3^- + \text{H}_2\text{O}$	1.853	
$\text{BrO}_3^- + 6\text{H}^+ + 6e^- = \text{Br}^- + 3\text{H}_2\text{O}$	1.478	
$\text{BrO}_3^- + 5\text{H}^+ + 4e^- = \text{HBrO} + 2\text{H}_2\text{O}$	1.444	
$2\text{BrO}_3^- + 12\text{H}^+ + 10e^- = \text{Br}_2 + 6\text{H}_2\text{O}$	1.5	
$2\text{HBrO} + 2\text{H}^+ + 2e^- = \text{Br}_2 + 2\text{H}_2\text{O}$	1.604	
$\text{HBrO} + \text{H}^+ + 2e^- = \text{Br}^- + \text{H}_2\text{O}$	1.341	
$\text{BrO}^- + \text{H}_2\text{O} + 2e^- = \text{Br}^- + 2\text{OH}^-$	0.76	
$\text{Br}_3^- + 2e^- = 3\text{Br}^-$	1.050	
$\text{Br}_2(\text{aq}) + 2e^- = 2\text{Br}^-$	1.087	1 NaOH
Cadmium		
$\text{Cd}^{2+} + 2e^- = \text{Cd}$	-0.403	
$\text{Cd}^{2+} + \text{Hg} + 2e^- = \text{Cd}(\text{Hg})$	-0.352	
$\text{CdCl}_4^{2-} + 2e^- = \text{Cd} + 4\text{Cl}^-$	-0.453	
$\text{Cd}(\text{CN})_4^{2-} + 2e^- = \text{Cd} + 4\text{CN}^-$	-0.943	
$\text{Cd}(\text{NH}_3)_4^{2+} + 2e^- = \text{Cd} + 4\text{NH}_3$	-0.622	
$\text{Cd}(\text{OH})_4^{2-} + 2e^- = \text{Cd} + 4\text{OH}^-$	-0.670	
Calcium		
$\text{CaO}_2 + 4\text{H}^+ + 2e^- = \text{Ca}^{2+} + \text{H}_2\text{O}$	2.224	
$\text{Ca}^{2+} + 2e^- = \text{Ca}$	-2.84	
$\text{Ca} + 2\text{H}^+ + 2e^- = \text{CaH}_2$	0.776	
Californium		
$\text{Cf}^{3+} + 3e^- = \text{Cf}$	-1.93	
$\text{Cf}^{3+} + e^- = \text{Cf}^{2+}$	-1.6	
$\text{Cf}^{2+} + 2e^- = \text{Cf}$	-2.1	
Carbon		
$\text{CO}_2 + 2\text{H}^+ + 2e^- = \text{CO} + \text{H}_2\text{O}$	-0.106	
$\text{CO}_2 + 2\text{H}^+ + 2e^- = \text{HCOOH}$	-0.20	
$2\text{CO}_2 + 2\text{H}^+ + 2e^- = \text{H}_2\text{C}_2\text{O}_4$	-0.481	
$\text{C}_2\text{O}_4^{2-} + 2\text{H}^+ + 2e^- = 2\text{HCOO}^-$	0.145	
$\text{HCOOH} + 2\text{H}^+ + 2e^- = \text{HCHO} + \text{H}_2\text{O}$	0.034	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
$\text{C}_2\text{N}_2 + 2\text{H}^+ + 2\text{e}^- = 2\text{HCN}$	0.373	
$\text{HCNO} + 2\text{H}^+ + 2\text{e}^- = \text{CO} + \text{H}_2\text{O}$	0.330	
$\text{HCHO} + 2\text{H}^+ + 2\text{e}^- = \text{CH}_3\text{OH}$	0.2323	
$\text{CNO}^- + \text{H}_2\text{O} + 2\text{e}^- = \text{CN}^- + 2\text{OH}^-$	-0.97	
Cerium		
$\text{Ce(IV)} + \text{e}^- = \text{Ce(III)}$	1.70	1 HClO_4
	1.61	1 HNO_3
	1.44	0.5 H_2SO_4
	1.28	1 HCl
$\text{Ce}^{3+} + 3\text{e}^- = \text{Ce}$	-2.34	
Cesium		
$\text{Cs}^+ + \text{e}^- = \text{Cs}$	-2.923	
$\text{Cs}^+ + \text{Hg} + \text{e}^- = \text{Cs(Hg)}$	-1.78	
Chlorine		
$\text{ClO}_4^- + 2\text{H}^+ + 2\text{e}^- = \text{ClO}_3^- + \text{H}_2\text{O}$	1.201	
$2\text{ClO}_4^- + 16\text{H}^+ + 14\text{e}^- = \text{Cl}_2 + 8\text{H}_2\text{O}$	1.392	
$\text{ClO}_4^- + 8\text{H}^+ + 8\text{e}^- = \text{Cl}^- + 4\text{H}_2\text{O}$	1.388	
$\text{ClO}_3^- + 2\text{H}^+ + \text{e}^- = \text{ClO}_2(\text{g}) + \text{H}_2\text{O}$	1.175	
$\text{ClO}_3^- + 3\text{H}^+ + 2\text{e}^- = \text{HClO}_2 + \text{H}_2\text{O}$	1.181	
$2\text{ClO}_3^- + 12\text{H}^+ + 10\text{e}^- = \text{Cl}_2 + 6\text{H}_2\text{O}$	1.468	
$\text{ClO}_3^- + 6\text{H}^+ + 6\text{e}^- = \text{Cl}^- + 3\text{H}_2\text{O}$	1.45	
$\text{ClO}_2(\text{g}) + \text{H}^+ + \text{e}^- = \text{HClO}_2$	1.188	
$\text{HClO}_2 + 2\text{H}^+ + 2\text{e}^- = \text{HClO} + \text{H}_2\text{O}$	1.64	
$\text{HClO}_2 + 3\text{H}^+ + 4\text{e}^- = \text{Cl}^- + 2\text{H}_2\text{O}$	1.584	
$2\text{HClO}_2 + 6\text{H}^+ + 6\text{e}^- = \text{Cl}_2(\text{g}) + 4\text{H}_2\text{O}$	1.659	
$2\text{ClO}^- + 2\text{H}_2\text{O} + 2\text{e}^- = \text{Cl}_2(\text{g}) + 4\text{OH}^-$	0.421	1 NaOH
$\text{ClO}^- + \text{H}_2\text{O} + 2\text{e}^- = \text{Cl}^- + 2\text{OH}^-$	0.890	1 NaOH
$\text{Cl}_3^- + 2\text{e}^- = 3\text{Cl}^-$	1.415	
$\text{Cl}_2(\text{aq}) + 2\text{e}^- = 2\text{Cl}^-$	1.396	
Chromium		
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- = 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.36	
	1.15	0.1 H_2SO_4
	1.03	1 HClO_4
$\text{CrO}_4^{2-} + 4\text{H}_2\text{O} + 3\text{e}^- = \text{Cr(OH)}_4^- + 4\text{OH}^-$	-0.13	1 NaOH
$\text{Cr}^{3+} + \text{e}^- = \text{Cr}^{2+}$	-0.424	
$\text{Cr}^{3+} + 3\text{e}^- = \text{Cr}$	-0.74	
$\text{Cr}^{2+} + 2\text{e}^- = \text{Cr}$	0.90	
Cobalt		
$\text{CoO}_2 + 4\text{H}^+ + \text{e}^- = \text{Co}^{3+} + 2\text{H}_2\text{O}$	1.416	
$\text{Co(H}_2\text{O)}_6^{3+} + \text{e}^- = \text{Co(H}_2\text{O)}_6^{2+}$	1.92	
$\text{Co(NH}_3)_6^{3+} + \text{e}^- = \text{Co(NH}_3)_6^{2+}$	0.058	7 NH_3
$\text{Co(OH)}_3 + \text{e}^- = \text{Co(OH)}_2 + \text{OH}^-$	0.17	
$\text{Co(en)}_3^{3+} + \text{e}^- = \text{Co(en)}_3^{2+}$ [en = ethylenediamine]	-0.2	0.1 en
$\text{Co(CN)}_6^{3-} + \text{e}^- = \text{Co(CN)}_6^{2-} + \text{CN}^-$	-0.8	0.8 KOH
$\text{Co}^{2+} + 2\text{e}^- = \text{Co}$	-0.277	
$\text{Co(NH}_3)_6^{2+} + 2\text{e}^- = \text{Co} + 6\text{NH}_3$	-0.422	
$[\text{Co(CO)}_4]_2 + 2\text{e}^- = 2\text{Co(CO)}_4^-$	-0.40	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Copper		
$\text{Cu}^{2+} + 2e^- = \text{Cu}$	0.340	
$\text{Cu}^{2+} + e^- = \text{Cu}^+$	0.159	
$\text{Cu}^+ + e^- = \text{Cu}$	0.520	
$\text{Cu}^{2+} + \text{Cl}^- + e^- = \text{CuCl}$	0.559	
$\text{Cu}^{2+} + 2\text{Br}^- + e^- = \text{CuBr}_2^-$	0.52	1 KBr
$\text{Cu}^{2+} + \text{I}^- + e^- = \text{CuI}$	0.86	
$\text{Cu}^{2+} + 2\text{CN}^- + e^- = \text{Cu(CN)}_2^-$	1.12	
$\text{Cu(NH}_3)_3^+ + e^- = \text{Cu(NH}_3)_2^+ + 2\text{NH}_3$	0.10	1 NH_3
$\text{Cu(en)}_2^+ + e^- = \text{Cu(en)}^+ + \text{en}$	-0.35	
$\text{Cu(CN)}_2^- + e^- = \text{Cu} + 2\text{CN}^-$	-0.44	
$\text{CuCl}_2^- + e^- = \text{Cu} + 3\text{Cl}^-$	0.178	1 HCl
$\text{Cu(NH}_3)_2^+ + e^- = \text{Cu} + 2\text{NH}_3$	-0.100	
Curium		
$\text{Cm}^{4+} + e^- = \text{Cm}^{3+}$	3.2	1 HClO_4
$\text{Cm}^{3+} + 3e^- = \text{Cm}$	-2.06	
Dysprosium		
$\text{Dy}^{3+} + 3e^- = \text{Dy}$	-2.29	
$\text{Dy}^{3+} + e^- = \text{Dy}^{2+}$	-2.5	
$\text{Dy}^{2+} + 2e^- = \text{Dy}$	-2.2	
Einsteinium		
$\text{Es}^{3+} + 3e^- = \text{Es}$	-2.0	
$\text{Es}^{3+} + e^- = \text{Es}^{2+}$	-1.5	
$\text{Es}^{2+} + 2e^- = \text{Es}$	-2.2	
Erbium		
$\text{Er}^{3+} + 3e^- = \text{Er}$	-2.32	
Europium		
$\text{Eu}^{3+} + 3e^- = \text{Eu}$	-1.99	
$\text{Eu}^{3+} + e^- = \text{Eu}^{2+}$	-0.35	
$\text{Eu}^{2+} + 2e^- = \text{Eu}$	-2.80	
Fermium		
$\text{Fm}^{3+} + 3e^- = \text{Fm}$	-1.96	
$\text{Fm}^{3+} + e^- = \text{Fm}^{2+}$	-1.15	
$\text{Fm}^{2+} + 2e^- = \text{Fm}$	-2.37	
Fluorine		
$\text{F}_2 + 2\text{H}^+ + 2e^- = 2\text{HF}$	3.053	
$\text{F}_2 + \text{H}^+ + 2e^- = \text{HF}_2^-$	2.979	
$\text{F}_2 + 2e^- = 2\text{F}^-$	2.87	
$\text{OF}_2 + 3\text{H}^+ + 4e^- = \text{HF}_2^- + \text{H}_2\text{O}$	2.209	
Francium		
$\text{Fr}^+ + e^- = \text{Fr}$	ca. -2.9	
Gadolinium		
$\text{Gd}^{3+} + 3e^- = \text{Gd}$	-2.28	
Gallium		
$\text{Ga}^{3+} + 3e^- = \text{Ga}$	-0.529	
$\text{Ga}^{3+} + e^- = \text{Ga}^{2+}$	-0.65	
$\text{Ga}^{2+} + 2e^- = \text{Ga}$	-0.45	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Germanium		
$\text{GeO}_2(\text{tetr}) + 2\text{H}^+ + 2\text{e}^- = \text{GeO}(\text{yellow}) + \text{H}_2\text{O}$	-0.255	
$\text{GeO}_2(\text{tetr}) + 4\text{H}^+ + 2\text{e}^- = \text{Ge}^{2+} + 2\text{H}_2\text{O}$	-0.210	
$\text{GeO}_2(\text{hex}) + 4\text{H}^+ + 2\text{e}^- = \text{Ge}^{2+} + 2\text{H}_2\text{O}$	-0.132	
$\text{H}_2\text{GeO}_3 + 4\text{H}^+ + 4\text{e}^- = \text{Ge} + 3\text{H}_2\text{O}$	0.012	
$\text{Ge}^{4+} + 2\text{e}^- = \text{Ge}^{2+}$	0.0	
$\text{Ge}^{2+} + 2\text{e}^- = \text{Ge}$	0.247	
$\text{GeO} + 2\text{H}^+ + 2\text{e}^- = \text{Ge} + \text{H}_2\text{O}$	-0.255	
$\text{Ge} + 4\text{H}^+ + 4\text{e}^- = \text{GeH}_4$	-0.29	
Gold		
$\text{Au}^{3+} + 3\text{e}^- = \text{Au}$	1.52	
$\text{Au}^{3+} + 2\text{e}^- = \text{Au}^+$	1.36	
$\text{Au}^+ + \text{e}^- = \text{Au}$	1.83	
$\text{AuCl}_4^- + 2\text{e}^- = \text{AuCl}_2^- + 2\text{Cl}^-$	0.926	
$\text{AuBr}_4^- + 2\text{e}^- = \text{AuBr}_2^- + 2\text{Br}^-$	0.802	
$\text{Au}(\text{SCN})_4^- + 2\text{e}^- = \text{Au}(\text{SCN})_2^- + 2\text{SCN}^-$	0.623	
$\text{AuBr}_4^- + 3\text{e}^- = \text{Au} + 4\text{Br}^-$	0.854	
$\text{AuCl}_4^- + 3\text{e}^- = \text{Au} + 4\text{Cl}^-$	1.002	
$\text{Au}(\text{SCN})_4^- + 3\text{e}^- = \text{Au} + 4\text{SCN}^-$	0.662	
$\text{Au}(\text{OH})_3 + 3\text{H}^+ + 3\text{e}^- = \text{Au} + 3\text{H}_2\text{O}$	1.45	
$\text{AuBr}_2^- + \text{e}^- = \text{Au} + 2\text{Br}^-$	0.960	
$\text{AuCl}_2^- + \text{e}^- = \text{Au} + 2\text{Cl}^-$	1.15	
$\text{AuI}_2^- + \text{e}^- = \text{Au} + 2\text{I}^-$	0.576	
$\text{Au}(\text{CN})_2^- + \text{e}^- = \text{Au} + 2\text{CN}^-$	-0.596	
$\text{Au}(\text{SCN})_2 + \text{e}^- = \text{Au} + 2\text{SCN}^-$	0.69	
Hafnium		
$\text{Hf}^{4+} + 4\text{e}^- = \text{Hf}$	-1.70	
$\text{HfO}_2 + 4\text{H}^+ + 4\text{e}^- = \text{Hf} + 2\text{H}_2\text{O}$	-1.57	
Holmium		
$\text{Ho}^{3+} + 3\text{e}^- = \text{Ho}$	-2.23	
Hydrogen		
$2\text{H}^+ + 2\text{e}^- = \text{H}_2$	0.0000	
$2\text{D}^+ + 2\text{e}^- = \text{D}_2$	0.029	
$2\text{H}_2\text{O} + 2\text{e}^- = \text{H}_2 + 2\text{OH}^-$	-0.828	
Indium		
$\text{In}^{3+} + 3\text{e}^- = \text{In}$	-0.338	
$\text{In}^{3+} + 2\text{e}^- = \text{In}^+$	-0.444	
$\text{In}^+ + \text{e}^- = \text{In}$	-0.126	
Iodine		
$\text{H}_5\text{IO}_6 + \text{H}^+ + 2\text{e}^- = \text{IO}_3^- + 3\text{H}_2\text{O}$	1.603	
$\text{IO}_3^- + 5\text{H}^+ + 4\text{e}^- = \text{HIO} + 2\text{H}_2\text{O}$	1.14	
$\text{HIO}_3 + 5\text{H}^+ + 2\text{Cl}^- + 4\text{e}^- = \text{ICl}_2^- + 3\text{H}_2\text{O}$	1.214	
$2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- = \text{I}_2(\text{c}) + 3\text{H}_2\text{O}$	1.195	
$\text{IO}_3^- + 3\text{H}_2\text{O} + 6\text{e}^- = \text{I}^- + 6\text{OH}^-$	0.257	
$2\text{IBr}_2^- + 2\text{e}^- = \text{I}_2\text{Br}^- + 3\text{Br}^-$	0.821	
$2\text{IBr}_2^- + 2\text{e}^- = \text{I}_2(\text{c}) + 4\text{Br}^-$	0.874	
$2\text{IBr} + 2\text{e}^- = \text{I}_2\text{Br}^- + \text{Br}^-$	0.973	
$2\text{IBr} + 2\text{e}^- = \text{I}_2 + 2\text{Br}^-$	1.02	
$2\text{ICl} + 2\text{e}^- = \text{I}_2(\text{c}) + 2\text{Cl}^-$	1.20	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
$2\text{ICl}_2^- + 2e^- = \text{I}_2(\text{c}) + 4\text{Cl}^-$	1.07	
$2\text{ICN} + 2\text{H}^+ + 2e^- = \text{I}_2(\text{c}) + 2\text{HCN}$	0.695	
$2\text{ICN} + 2\text{H}^+ + 2e^- = \text{I}_2(\text{aq}) + 2\text{HCN}$	0.609	
$2\text{HIO} + 2\text{H}^+ + 2e^- = \text{I}_2 + 2\text{H}_2\text{O}$	1.45	
$\text{HIO} + \text{H}^+ + 2e^- = \text{I}^- + \text{H}_2\text{O}$	0.985	
$\text{I}_3^- + 2e^- = 3\text{I}^-$	0.536	
$\text{I}_2(\text{aq}) + 2e^- = 2\text{I}^-$	0.621	
$\text{I}_2(\text{c}) + 2e^- = 2\text{I}^-$	0.5355	
Iridium		
$\text{IrBr}_6^{2-} + e^- = \text{IrBr}_6^{3-}$	0.805	
$\text{IrCl}_6^{2-} + e^- = \text{IrCl}_6^{3-}$	0.867	
$\text{IrI}_6^{2-} + e^- = \text{IrI}_6^{3-}$	0.49	
$\text{IrO}_2 + 4\text{H}^+ + e^- = \text{Ir}^{3+} + 2\text{H}_2\text{O}$	0.223	
$\text{IrO}_2 + 4\text{H}^+ + 4e^- = \text{Ir} + 2\text{H}_2\text{O}$	0.935	1 H_2SO_4
$\text{Ir}^{3+} + 3e^- = \text{Ir}$	1.156	
$\text{IrCl}_6^{2-} + 4e^- = \text{Ir} + 6\text{Cl}^-$	0.835	
$\text{IrCl}_6^{3-} + 3e^- = \text{Ir} + 6\text{Cl}^-$	0.77	
Iron		
$\text{FeO}_4^{2-} + 8\text{H}^+ + 3e^- = \text{Fe}^{3+} + 4\text{H}_2\text{O}$	2.2	
$\text{FeO}_4^{2-} + 2\text{H}_2\text{O} + 3e^- = \text{FeO}_2^- + 4\text{OH}^-$	0.55	10 NaOH
$\text{Fe}^{3+} + e^- = \text{Fe}^{2+}$	0.771	
	0.70	1 HCl
	0.67	0.5 H_2SO_4
	0.44	0.3 H_3PO_4
$\text{Fe}(\text{CN})_6^{3-} + e^- = \text{Fe}(\text{CN})_6^{4-}$	0.361	
	0.71	1 HCl
$\text{Fe}(\text{EDTA})^- + e^- = \text{Fe}(\text{EDTA})^{2-}$	0.12	0.1 EDTA, pH 4–6
$\text{Fe}(\text{OH})_4^- + e^- = \text{Fe}(\text{OH})_3$	–0.73	1 NaOH
$\text{Fe}^{2+} + 2e^- = \text{Fe}$	–0.44	
$[\text{Fe}(\text{CO})_4]_3 + 6e^- = 3\text{Fe}(\text{CO})_2^-$	–0.70	
Lanthanum		
$\text{La}^{3+} + 3e^- = \text{La}$	–2.38	
Lawrencium		
$\text{Lr}^{3+} + 3e^- = \text{Lr}$	–2.0	
Lead		
$\text{Pb}^{4+} + 2e^- = \text{Pb}^{2+}$	1.65	
$\text{PbO}_2(\text{alpha}) + \text{SO}_4^{2-} + 4\text{H}^+ + 2e^- = \text{PbSO}_4 + 2\text{H}_2\text{O}$	1.690	
$\text{PbO}_2 + 4\text{H}^+ + 2e^- = \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.46	
$\text{PbO}_2 + 2\text{H}^+ + 2e^- = \text{PbO} + \text{H}_2\text{O}$	0.28	
$\text{PbO}_2^- + \text{H}_2\text{O} + 2e^- = \text{HPbO}_2^- + 3\text{OH}^-$	0.3	2 NaOH
$\text{Pb}^{2+} + 2e^- = \text{Pb}$	–0.126	
$\text{HPbO}_2^- + \text{H}_2\text{O} + 2e^- = \text{Pb} + 3\text{OH}^-$	–0.54	
$\text{PbHPO}_4 + 2e^- = \text{Pb} + \text{HPO}_4^{2-}$	–0.465	
$\text{PbSO}_4 + 2e^- = \text{Pb} + \text{SO}_4^{2-}$	–0.356	
$\text{PbF}_2 + 2e^- = \text{Pb} + 2\text{F}^-$	–0.344	
$\text{PbCl}_2 + 2e^- = \text{Pb} + 2\text{Cl}^-$	–0.268	
$\text{PbBr}_2 + 2e^- = \text{Pb} + 2\text{Br}^-$	–0.280	
$\text{PbI}_2 + 2e^- = \text{Pb} + 2\text{I}^-$	–0.365	
$\text{Pb} + 2\text{H}^+ + 2e^- = \text{PbH}_2$	–1.507	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Lithium		
$\text{Li}^+ + e^- = \text{Li}$	-3.040	
$\text{Li}^+ + \text{Hg} + e^- = \text{Li}(\text{Hg})$	-2.00	
Lutetium		
$\text{Lu}^{3+} + 3e^- = \text{Lu}$	-2.30	
Magnesium		
$\text{Mg}^{2+} + 2e^- = \text{Mg}$	-2.356	
$\text{Mg}(\text{OH})_2 + 2e^- = \text{Mg} + 2\text{OH}^-$	-2.687	
Manganese		
$\text{MnO}_4^- + e^- = \text{MnO}_4^{2-}$	0.56	
$\text{MnO}_4^- + 4\text{H}^+ + 3e^- = \text{MnO}_2(\text{beta}) + 2\text{H}_2\text{O}$	1.70	
$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3e^- = \text{MnO}_2 + 4\text{OH}^-$	0.60	
$\text{MnO}_4^- + 8\text{H}^+ + 5e^- = \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51	
$\text{MnO}_4^{2-} + e^- = \text{MnO}_4^{3-}$	0.27	
$\text{MnO}_4^{2-} + 2\text{H}_2\text{O} + 2e^- = \text{MnO}_2 + 4\text{OH}^-$	0.62	
$\text{MnO}_3^- + 2\text{H}_2\text{O} + e^- = \text{MnO}_2 + 4\text{OH}^-$	0.96	
$\text{MnO}_2 + 4\text{H}^+ + e^- = \text{Mn}^{3+} + 2\text{H}_2\text{O}$	0.95	
$\text{MnO}_2(\text{beta}) + 4\text{H}^+ + 2e^- = \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.23	
$\text{Mn}^{3+} + e^- = \text{Mn}^{2+}$	1.5	
$\text{Mn}(\text{H}_2\text{P}_2\text{O}_7)_3^{3-} + 2\text{H}^+ + e^- = \text{Mn}(\text{H}_2\text{P}_2\text{O}_7)_2^{2-} + \text{H}_4\text{P}_2\text{O}_7$	1.15	0.4 $\text{H}_2\text{P}_2\text{O}_7^{2-}$
$\text{Mn}(\text{CN})_6^{3-} + e^- = \text{Mn}(\text{CN})_6^{4-}$	-0.24	1.5 NaCN
$\text{Mn}^{2+} + 2e^- = \text{Mn}$	-1.17	
Mendelevium		
$\text{Md}^{3+} + 3e^- = \text{Md}$	-1.7	
$\text{Md}^{3+} + e^- = \text{Md}^{2+}$	-0.15	
$\text{Md}^{2+} + 2e^- = \text{Md}$	-2.4	
Mercury		
$2\text{Hg}_2^{2+} + 2e^- = \text{Hg}_2^{2+}$	0.911	
$2\text{HgCl}_2 + 2e^- = \text{Hg}_2\text{Cl}_2 + 2\text{Cl}^-$	0.63	
$\text{Hg}_2^{2+} + 2e^- = \text{Hg}(\text{lq})$	0.8535	
$\text{HgO}(\text{c,red}) + 2\text{H}^+ + 2e^- = \text{Hg} + \text{H}_2\text{O}$	0.926	
$\text{Hg}_2^{2+} + 2e^- = 2\text{Hg}$	0.7960	
$\text{Hg}_2\text{F}_2 + 2e^- = 2\text{Hg} + 2\text{F}^-$	0.656	
$\text{Hg}_2\text{Cl}_2 + 2e^- = 2\text{Hg} + 2\text{Cl}^-$	0.2682	
$\text{Hg}_2\text{Br}_2 + 2e^- = 2\text{Hg} + 2\text{Br}^-$	0.1392	
$\text{Hg}_2\text{I}_2 + 2e^- = 2\text{Hg} + 2\text{I}^-$	-0.0405	
$\text{Hg}_2\text{SO}_4 + 2e^- = 2\text{Hg} + \text{SO}_4^{2-}$	0.614	
Molybdenum		
$\text{MoO}_4^{2-} + 4\text{H}_2\text{O} + 6e^- = \text{Mo} + 8\text{OH}^-$	-0.913	
$\text{H}_2\text{MoO}_4 + 6\text{H}^+ + 6e^- = \text{Mo} + 4\text{H}_2\text{O}$	0.114	
$\text{H}_2\text{MoO}_4 + 2\text{H}^+ + 2e^- = \text{MoO}_2 + 2\text{H}_2\text{O}$	0.646	
$\text{MoO}_2 + 4\text{H}^+ + 4e^- = \text{Mo} + 2\text{H}_2\text{O}$	-0.152	
$\text{H}_2\text{MoO}_4 + 6\text{H}^+ + 3e^- = \text{Mo}^{3+} + 4\text{H}_2\text{O}$	0.428	
$\text{Mo}(\text{CN})_6^{3-} + e^- = \text{Mo}(\text{CN})_6^{4-}$	0.725	
$\text{Mo}^{3+} + 3e^- = \text{Mo}$	-0.2	
Neodymium		
$\text{Nd}^{3+} + 3e^- = \text{Nd}$	-2.32	
$\text{Nd}^{3+} + e^- = \text{Nd}^{2+}$	-2.6	
$\text{Nd}^{2+} + 2e^- = \text{Nd}$	-2.2	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Neptunium		
$\text{NpO}_3^+ + 2\text{H}^+ + e^- = \text{NpO}_2^{2+} + \text{H}_2\text{O}$	2.04	
$\text{NpO}_2^{2+} + e^- = \text{NpO}_2^+$	1.34	
$\text{NpO}_2^{2+} + 4\text{H}^+ + 2e^- = \text{Np}^{4+} + 2\text{H}_2\text{O}$	0.95	
$\text{Np}^{4+} + e^- = \text{Np}^{3+}$	0.18	
$\text{Np}^{4+} + 4e^- = \text{Np}$	-1.30	
$\text{Np}^{3+} + 3e^- = \text{Np}$	-1.79	
Nickel		
$\text{NiO}_2^- + 4\text{H}^+ + 2e^- = \text{NiO}_2 + 2\text{H}_2\text{O}$	1.8	
$\text{NiO}_2 + 4\text{H}^+ + 2e^- = \text{Ni}^{2+} + 2\text{H}_2\text{O}$	1.593	
$\text{NiO}_2 + 2\text{H}_2\text{O} + 2e^- = \text{Ni}(\text{OH})_2 + 2\text{OH}^-$	0.490	
$\text{Ni}(\text{CN})_4^{2-} + e^- = \text{Ni}(\text{CN})_3^- + \text{CN}^-$	-0.401	
$\text{Ni}^{2+} + 2e^- = \text{Ni}$	-0.257	
$\text{Ni}(\text{OH})_2 + 2e^- = \text{Ni} + 2\text{OH}^-$	-0.72	
$\text{Ni}(\text{NH}_3)_6^{2+} + 2e^- = \text{Ni} + 6\text{NH}_3$	-0.49	
Niobium		
$\text{Nb}_2\text{O}_5 + 10\text{H}^+ + 4e^- = 2\text{Nb}^{3+} + 5\text{H}_2\text{O}$	-0.1	
$\text{Nb}_2\text{O}_5 + 10\text{H}^+ + 10e^- = 2\text{Nb} + 5\text{H}_2\text{O}$	-0.65	
$\text{Nb}^{3+} + 3e^- = \text{Nb}$	-1.1	
Nitrogen		
$2\text{NO}_3^- + 4\text{H}^+ + 2e^- = \text{N}_2\text{O}_4 + 2\text{H}_2\text{O}$	0.803	
$\text{NO}_3^- + 3\text{H}^+ + 2e^- = \text{HNO}_2 + \text{H}_2\text{O}$	0.94	
$\text{N}_2\text{O}_4 + 2\text{H}^+ + 2e^- = 2\text{HNO}_2$	1.07	
$\text{HNO}_2 + \text{H}^+ + e^- = \text{NO} + \text{H}_2\text{O}$	0.996	
$2\text{HNO}_2 + 4\text{H}^+ + 4e^- = \text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}$	1.297	
$2\text{HNO}_2 + 4\text{H}^+ + 4e^- = \text{H}_2\text{N}_2\text{O}_2 + 2\text{H}_2\text{O}$	0.86	
$2\text{NO} + 2\text{H}^+ + 2e^- = \text{H}_2\text{N}_2\text{O}_2$	0.71	
$2\text{NO} + 2\text{H}^+ + 2e^- = \text{N}_2\text{O} + \text{H}_2\text{O}$	1.59	
$\text{H}_2\text{N}_2\text{O}_2 + 6\text{H}^+ + 4e^- = 2\text{HONH}_3^+$	0.496	
$\text{N}_2\text{O} + 2\text{H}^+ + 2e^- = \text{N}_2 + \text{H}_2\text{O}$	1.77	
$\text{N}_2\text{O} + 6\text{H}^+ + \text{H}_2\text{O} + 4e^- = 2\text{HONH}_3^+$	-0.05	
$\text{N}_2 + 2\text{H}_2\text{O} + 4\text{H}^+ + 2e^- = 2\text{HONH}_3^+$	-1.87	
$\text{N}_2 + 5\text{H}^+ + 4e^- = \text{N}_2\text{H}_5^+$	-0.23	
$\text{HONH}_3^+ + 2\text{H}^+ + 2e^- = \text{NH}_4^+ + \text{H}_2\text{O}$	1.35	
$2\text{HONH}_3^+ + \text{H}^+ + 2e^- = \text{N}_2\text{H}_5^+ + 2\text{H}_2\text{O}$	1.41	
$\text{N}_2\text{H}_5^+ + 3\text{H}^+ + 2e^- = 2\text{NH}_4^+$	1.275	
$3\text{N}_2 + 2\text{H}^+ + 2e^- = 2\text{HN}_3$	-3.40	
Nobelium		
$\text{No}^{3+} + 3e^- = \text{No}$	-1.2	
$\text{No}^{3+} + e^- = \text{No}^{2+}$	1.4	
$\text{No}^{2+} + 2e^- = \text{No}$	-2.5	
Osmium		
$\text{OsO}_4(\text{aq}) + 4\text{H}^+ + 4e^- = \text{OsO}_2 \cdot 2\text{H}_2\text{O} + 2\text{H}_2\text{O}$	0.964	
$\text{OsO}_4(\text{c, yellow}) + 8\text{H}^+ + 8e^- = \text{Os} + 4\text{H}_2\text{O}$	0.85	
$\text{OsO}_2 + 4\text{H}^+ + 4e^- = \text{Os} + 2\text{H}_2\text{O}$	0.687	
$\text{OsCl}_6^{2-} + e^- = \text{OsCl}_6^{3-}$	0.45	
$\text{OsBr}_6^{2-} + e^- = \text{OsBr}_6^{3-}$	0.35	
Oxygen		
$\text{O}_3 + 2\text{H}^+ + 2e^- = \text{O}_2 + \text{H}_2\text{O}$	2.075	
$\text{O}_3 + \text{H}_2\text{O} + 2e^- = \text{O}_2 + 2\text{OH}^-$	1.240	1 NaOH

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (Continued)

Half-reaction	Standard or formal potential	Solution composition	
$O_2 + 4H^+ + 4e^- = 2H_2O$	1.229	1 NaOH	
$O_2 + 2H^+ + 2e^- = H_2O$	0.695		
$O_2 + H_2O + 2e^- = HO_2^- + OH^-$	-0.076		
$H_2O_2 + 2H^+ + 2e^- = 2H_2O$	1.763		
$HO_2^- + H_2O + 2e^- = 3OH^-$	0.867		
$O_2 + 2H_2O + 4e^- = 4OH^-$	0.401		
Palladium			
$PdO_3 + 2H^+ + 2e^- = PdO_2 + H_2O$	2.030	1 HCl	
$PdCl_6^{2-} + 2e^- = PdCl_4^{2-} + 2Cl^-$	1.470		
$PdBr_6^{2-} + 2e^- = PdBr_4^{2-} + 2Br^-$	0.99		
$PdI_6^{2-} + 2e^- = PdI_4^{2-} + 2I^-$	0.48		
$Pd^{2+} + 2e^- = Pd$	0.915		
$PdCl_4^{2-} + 2e^- = Pd + 4Cl^-$	0.62		
$PdBr_4^{2-} + 2e^- = Pd + 4Br^-$	0.49	1 NH ₃ 1 KCN	
$Pd(NH_3)_4^{2+} + 2e^- = Pd + 4NH_3$	0.0		
$Pd(CN)_4^{2-} + 2e^- = Pd + 4CN^-$	-1.35		
Phosphorus			
$H_3PO_4 + 2H^+ + 2e^- = H_3PO_3 + H_2O$	-0.276		
$2H_3PO_4 + 2H^+ + 2e^- = H_4P_2O_6 + 2H_2O$	-0.933		
$H_4P_2O_6 + 2H^+ + 2e^- = 2H_3PO_3$	0.380		
$H_3PO_3 + 2H^+ + 2e^- = HPH_2O_2 + H_2O$	-0.499		
$HPH_2O_2 + H^+ + e^- = P + 2H_2O$	-0.365		
$H_3PO_3 + 3H^+ + 3e^- = P + 3H_2O$	-0.502		
$2P(\text{white}) + 4H^+ + 4e^- = P_2H_4$	-0.100		
$P_2H_4 + 2H^+ + 2e^- = 2PH_3$	-0.006		
$P(\text{white}) + 3H^+ + 3e^- = PH_3$	-0.063		
Platinum			
$PtO_3 + 2H^+ + 2e^- = PtO_2 + H_2O$	2.0	1 KBr 1 KI	
$PtO_2 + 2H^+ + 2e^- = PtO + H_2O$	1.045		
$PtCl_6^{2-} + 2e^- = PtCl_4^{2-} + 2Cl^-$	0.726		
$PtBr_6^{2-} + 2e^- = PtBr_4^{2-} + 2Br^-$	0.613		
$PtI_6^{2-} + 2e^- = PtI_4^{2-} + 2I^-$	0.321		
$Pt^{2+} + 2e^- = Pt$	1.188		
$PtCl_4^{2-} + 2e^- = Pt + 4Cl^-$	0.758	1 H ₃ PO ₄ 1 HF	
$PtBr_4^{2-} + 2e^- = Pt + 4Br^-$	0.698		
Plutonium			
$PuO_3^{2+} + e^- = PuO_2^+$	1.02		
$PuO_2^{2+} + 4H^+ + 2e^- = Pu^{4+} + 2H_2O$	1.04		
$Pu^{4+} + e^- = Pu^{3+}$	1.01		
	0.80		
	0.50		
$Pu^{4+} + 4e^- = Pu$	-1.25		
$Pu^{3+} + 3e^- = Pu$	-2.00		
Polonium			
$PoO_2 + 4H^+ + 2e^- = Po^{2+} + 2H_2O$	1.1	ca. -1.0	
$Po^{4+} + 4e^- = Po$	0.73		
$Po^{2+} + 2e^- = Po$	0.37		
$Po + 2H^+ + 2e^- = H_2Po$			

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Potassium		
$K^+ + e^- = K$	-2.924	
$K^+ + Hg + e^- = K(Hg)$	ca. -1.9	
Praseodymium		
$Pr^{4+} + e^- = Pr^{3+}$	3.2	
$Pr^{3+} + e^- = Pr$	-2.35	
Promethium		
$Pm^{3+} + 3e^- = Pm$	-2.42	
Protoactinium		
$PaOOH^{2+} + 3H^+ + e^- = Pa^{4+} + 2H_2O$	-0.10	
$PaOOH^{2+} + 3H^+ + 5e^- = Pa + 2H_2O$	-1.19	
$Pa^{4+} + 4e^- = Pa$	-1.46	
Radium		
$Ra^{2+} + 2e^- = Ra$	-2.916	
Rhenium		
$ReO_4^- + 2H^+ + e^- = ReO_3 + H_2O$	0.768	
$ReO_4^- + 4H^+ + 3e^- = ReO_2 + 2H_2O$	0.51	
$ReO_4^- + 2H_2O + 3e^- = ReO_2 + 4OH^-$	-0.594	
$ReO_4^- + 6Cl^- + 8H^+ + 3e^- = ReCl_6^{3-} + 4H_2O$	0.12	
$2ReO_4^- + 10H^+ + 8e^- = Re_2O_3 + 5H_2O$	-0.808	
$ReO_3 + 2H^+ + 2e^- = ReO_2 + H_2O$	0.63	
$ReO_2 + 4H^+ + 4e^- = Re + 2H_2O$	0.22	
$ReCl_6^{3-} + 4e^- = Re + 6Cl^-$	0.51	
$Re + e^- = Re^-$	-0.10	
Rhodium		
$RhO_2 + 4H^+ + e^- = Rh^{3+} + 2H_2O$	1.881	
$Rh^{3+} + 3e^- = Rh$	0.76	
$RhCl_3^- + 3e^- = Rh + 6Cl^-$	0.5	
Rubidium		
$Rb^+ + e^- = Rb$	-2.924	
$Rb^+ + Hg + e^- = Rb(Hg)$	-1.81	
Ruthenium		
$RuO_4 + e^- = RuO_4^-$	0.89	
$RuO_4 + 4H^+ + 4e^- = RuO_2 + 2H_2O$	1.4	
$RuO_4 + 8H^+ + 8e^- = Ru + 4H_2O$	1.04	
$RuO_4^- + e^- = RuO_4^{2-}$	0.593	
$RuO_4^{2-} + 4H^+ + 2e^- = RuO_2 + 2H_2O$	2.0	
$RuO_2 + 4H^+ + 4e^- = Ru + 2H_2O$	0.68	
$Ru(H_2O)_6^{3+} + e^- = Ru(H_2O)_6^{2+}$	0.249	
$Ru(NH_3)_6^{3+} + e^- = Ru(NH_3)_6^{2+}$	0.10	
$Ru(CN)_6^{3-} + e^- = Ru(CN)_6^{4-}$	0.86	
$Ru^{3+} + e^- = Ru^{2+}$	0.249	
Samarium		
$Sm^{3+} + 3e^- = Sm$	-2.30	
$Sm^{3+} + e^- = Sm^{2+}$	-1.55	
$Sm^{2+} + 2e^- = Sm$	-2.67	
Scandium		
$Sc^{3+} + 3e^- = Sc$	-2.03	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Selenium		
$\text{SeO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- = \text{H}_2\text{SeO}_3 + \text{H}_2\text{O}$	1.151	1 NaOH
$\text{H}_2\text{SeO}_3 + 4\text{H}^+ + 4\text{e}^- = \text{Se} + 3\text{H}_2\text{O}$	0.74	
$\text{Se}(\text{c}) + 2\text{H}^+ + 2\text{e}^- = \text{H}_2\text{Se}(\text{aq})$	-0.115	
$\text{Se} + \text{H}^+ + 2\text{e}^- = \text{HSe}^-$	-0.227	
$\text{Se} + 2\text{e}^- = \text{Se}^{2-}$	-0.670	
Silicon		
$\text{SiO}_2(\text{quartz}) + 4\text{H}^+ + 4\text{e}^- = \text{Si} + 2\text{H}_2\text{O}$	-0.909	1 NaOH
$\text{SiO}_2 + 2\text{H}^+ + 2\text{e}^- = \text{SiO} + \text{H}_2\text{O}$	-0.967	
$\text{SiO}_2 + 8\text{H}^+ + 8\text{e}^- = \text{SiH}_4 + 2\text{H}_2\text{O}$	-0.516	
$\text{SiF}_6^{2-} + 4\text{e}^- = \text{Si} + 6\text{F}^-$	-1.37	
$\text{SiO} + 2\text{H}^+ + 2\text{e}^- = \text{Si} + \text{H}_2\text{O}$	-0.808	
$\text{Si} + 4\text{H}^+ + 4\text{e}^- = \text{SiH}_4(\text{g})$	-0.143	
Silver		
$\text{AgO}^+ + 2\text{H}^+ + \text{e}^- = \text{Ag}^{2+} + \text{H}_2\text{O}$	1.360	
$\text{Ag}_2\text{O}_3 + 2\text{H}^+ + 2\text{e}^- = 2\text{AgO} + \text{H}_2\text{O}$	1.569	
$\text{Ag}_2\text{O}_3 + \text{H}_2\text{O} + 2\text{e}^- = 2\text{AgO} + 2\text{OH}^-$	0.739	
$\text{Ag}_2\text{O}_3 + 6\text{H}^+ + 4\text{e}^- = 2\text{Ag}^+ + 3\text{H}_2\text{O}$	1.670	
$\text{Ag}^{2+} + \text{e}^- = \text{Ag}^+$	1.980	
$\text{AgO} + 2\text{H}^+ + \text{e}^- = \text{Ag}^+ + \text{H}_2\text{O}$	1.772	
$\text{Ag}^+ + \text{e}^- = \text{Ag}$	0.7991	
$\text{Ag}_2\text{SO}_4 + 2\text{e}^- = 2\text{Ag} + \text{SO}_4^{2-}$	0.653	
$\text{Ag}_2\text{C}_2\text{O}_4 + 2\text{e}^- = 2\text{Ag} + \text{C}_2\text{O}_4^{2-}$	0.47	
$\text{Ag}_2\text{CrO}_4 + 2\text{e}^- = 2\text{Ag} + \text{CrO}_4^{2-}$	0.447	
$\text{Ag}(\text{NH}_3)_2^+ + \text{e}^- = \text{Ag} + 2\text{NH}_3$	0.373	
$\text{AgCl} + \text{e}^- = \text{Ag} + \text{Cl}^-$	0.2223	
$\text{AgBr} + \text{e}^- = \text{Ag} + \text{Br}^-$	0.071	
$\text{AgCN} + \text{e}^- = \text{Ag} + \text{CN}^-$	-0.017	
$\text{AgI} + \text{e}^- = \text{Ag} + \text{I}^-$	-0.152	
$\text{Ag}(\text{CN}) + \text{e}^- = \text{Ag} + 2\text{CN}^-$	-0.31	
$\text{AgSCN} + \text{e}^- = \text{Ag} + \text{SCN}^-$	0.09	
$\text{Ag}_2\text{S} + 2\text{e}^- = 2\text{Ag} + \text{S}^{2-}$	-0.71	
Sodium		
$\text{Na}^+ + \text{e}^- = \text{Na}$	-2.713	
$\text{Na}^+ + \text{Hg} + \text{e}^- = \text{Na}(\text{Hg})$	-1.84	
Strontium		
$\text{SrO}_2 + 4\text{H}^+ + 2\text{e}^- = \text{Sr}^{2+}$	2.33	
$\text{Sr}^{2+} + 2\text{e}^- = \text{Sr}$	-2.89	
Sulfur		
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^- = 2\text{SO}_4^{2-}$	1.96	
$\text{S}_2\text{O}_8^{2-} + 2\text{H}^+ + 2\text{e}^- = 2\text{HSO}_4^-$	2.08	
$2\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- = \text{S}_2\text{O}_6^{2-} + 2\text{H}_2\text{O}$	-0.25	
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- = \text{SO}_2(\text{aq}) + \text{H}_2\text{O}$	0.158	
$\text{SO}_4^{2-} + \text{H}_2\text{O} + 2\text{e}^- = \text{SO}_3^{2-} + 2\text{OH}^-$	-0.936	
$\text{S}_2\text{O}_6^{2-} + 4\text{H}^+ + 2\text{e}^- = 2\text{H}_2\text{SO}_3$	0.569	
$\text{S}_2\text{O}_6^{2-} + 2\text{e}^- = 2\text{SO}_3^{2-}$	0.037	
$2\text{HSO}_3^- + 2\text{H}^+ + 2\text{e}^- = \text{S}_2\text{O}_4^{2-} + 2\text{H}_2\text{O}$	0.099	
$2\text{SO}_3^{2-} + 2\text{H}_2\text{O} + 2\text{e}^- = \text{S}_2\text{O}_4^{2-} + 4\text{OH}^-$	-1.13	
$4\text{H}_2\text{SO}_3 + 4\text{H}^+ + 6\text{e}^- = \text{S}_4\text{O}_6^{2-} + 6\text{H}_2\text{O}$	0.507	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
$4\text{HSO}_3^- + 8\text{H}^+ + 6e^- = \text{S}_4\text{O}_6^{2-} + 6\text{H}_2\text{O}$	0.577	
$2\text{SO}_2(\text{aq}) + 2\text{H}^+ + 4e^- = \text{S}_2\text{O}_3^{2-} + \text{H}_2\text{O}$	0.400	
$2\text{SO}_3^{2-} + 3\text{H}_2\text{O} + 4e^- = \text{S}_2\text{O}_3^{2-} + 6\text{OH}^-$	-0.576	1 NaOH
$\text{SO}_3^{2-} + 3\text{H}_2\text{O} + 4e^- = \text{S} + 6\text{OH}^-$	-0.59	1 NaOH
$\text{S}_4\text{O}_6^{2-} + 2e^- = 2\text{S}_2\text{O}_3^{2-}$	0.080	
$\text{S}_2\text{O}_3^{2-} + 6\text{H}^+ + 4e^- = 2\text{S} + 3\text{H}_2\text{O}$	0.5	
$\text{SF}_4(\text{g}) + 4e^- = \text{S} + 4\text{F}^-$	0.97	
$\text{S}_2\text{Cl}_2(\text{g}) + 2e^- = 2\text{S} + 2\text{Cl}^-$	1.19	
$\text{S} + \text{H}^+ + 2e^- = \text{HS}^-$	0.287	
$\text{S} + 2\text{H}^+ + 2e^- = \text{H}_2\text{S}(\text{aq})$	0.144	
$\text{S} + 2\text{H}^+ + 2e^- = \text{H}_2\text{S}(\text{g})$	0.174	
$\text{S} + 2e^- = \text{S}^{2-}$	-0.407	
Tantalum		
$\text{Ta}_2\text{O}_5 + 10\text{H}^+ + 10e^- = 2\text{Ta} + 5\text{H}_2\text{O}$	-0.81	
$\text{TaF}_6^{2-} + 5e^- = \text{Ta} + 7\text{F}^-$	-0.45	
Technetium		
$\text{TcO}_4^- + 4\text{H}^+ + 3e^- = \text{TcO}_2 + 2\text{H}_2\text{O}$	0.738	
$\text{TcO}_4^- + 2\text{H}^+ + e^- = \text{TcO}_3 + \text{H}_2\text{O}$	0.700	
$\text{TcO}_4^- + e^- = \text{TcO}_4^{2-}$	0.569	
$\text{TcO}_4^- + 8\text{H}^+ + 7e^- = \text{Tc} + 4\text{H}_2\text{O}$	0.472	
$\text{TcO}_4^{2-} + 4\text{H}^+ + 2e^- = \text{TcO}_2 + 2\text{H}_2\text{O}$	1.39	
$\text{TcO}_2 + 4\text{H}^+ + 4e^- = \text{Tc} + 2\text{H}_2\text{O}$	0.272	
$\text{Tc} + e^- = \text{Tc}^-$	ca. -0.5	
Tellurium		
$\text{H}_2\text{TeO}_4 + 6\text{H}^+ + 2e^- = \text{Te}^{4+} + 4\text{H}_2\text{O}$	0.929	
$\text{H}_2\text{TeO}_4 + 2\text{H}^+ + 2e^- = \text{TeO}_2(\text{c}) + 2\text{H}_2\text{O}$	1.02	
$\text{TeO}_4^{2-} + 2\text{H}^+ + 2e^- = \text{TeO}_3^{2-} + \text{H}_2\text{O}$	0.897	
$\text{TeOOH}^+ + 3\text{H}^+ + 4e^- = \text{Te} + 2\text{H}_2\text{O}$	0.559	
$\text{H}_2\text{TeO}_3 + 4\text{H}^+ + 4e^- = \text{Te} + 3\text{H}_2\text{O}$	0.589	
$\text{TeO}_3^{2-} + 6\text{H}^+ + 4e^- = \text{Te} + 3\text{H}_2\text{O}$	0.827	
$\text{TeO}_3^{2-} + 3\text{H}_2\text{O} + 4e^- = \text{Te} + 6\text{OH}^-$	-0.415	
$\text{TeO}_2(\text{c}) + 4\text{H}^+ + 4e^- = \text{Te} + 2\text{H}_2\text{O}$	0.521	
$\text{Te} + 2\text{H}^+ + 2e^- = \text{H}_2\text{Te}(\text{aq})$	-0.740	
$\text{Te} + \text{H}^+ + 2e^- = \text{HTe}^-$	-0.817	
$\text{Te}^{2-} + 2\text{H}^+ + 2e^- = 2\text{HTe}^-$	-0.794	
Terbium		
$\text{Tb}^{3+} + 3e^- = \text{Tb}$	-2.31	
Thallium		
$\text{Tl}^{3+} + 2e^- = \text{Tl}^+$	1.25	1 HClO_4
	0.77	1 HCl
$\text{Tl}^{3+} + 3e^- = \text{Tl}$	0.72	
$\text{Tl}^+ + e^- = \text{Tl}$	-0.336	
$\text{TlCl} + e^- = \text{Tl} + \text{Cl}^-$	-0.557	
$\text{TlBr} + e^- = \text{Tl} + \text{Br}^-$	-0.658	
$\text{TlI} + e^- = \text{Tl} + \text{I}^-$	-0.752	
Thorium		
$\text{Th}^{4+} + 4e^- = \text{Th}$	-1.83	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Thullium		
$\text{Tm}^{3+} + 3e^- = \text{Tm}$	-2.32	
Tin		
$\text{Sn}^{4+} + 2e^- = \text{Sn}^{2+}$	0.154	
$\text{SnCl}_6^{2-} + 2e^- = \text{SnCl}_4^{2-} + 2\text{Cl}^-$	0.14	
$\text{SnO}_3^{2-} + 6\text{H}^+ + 2e^- = \text{Sn}^{2+} + 3\text{H}_2\text{O}$	0.849	
$\text{SnF}_6^{2-} + 4e^- = \text{Sn} + 6\text{F}^-$	-0.200	
$\text{Sn}^{2+} + 2e^- = \text{Sn}$	-0.1375	
$\text{SnCl}_4^{2-} + 2e^- = \text{Sn} + 4\text{Cl}^-$	-0.19	1 HCl
$\text{HSnO}_2 + \text{H}_2\text{O} + 2e^- = \text{Sn} + 3\text{OH}^-$	-0.91	
$\text{Sn} + 4\text{H}^+ + 4e^- = \text{SnH}_4$	-1.07	
Titanium		
$\text{TiO}^{2+} + 2\text{H}^+ + e^- = \text{Ti}^{3+} + \text{H}_2\text{O}$	-0.10	
$\text{TiO}^{2+} + 2\text{H}^+ + 4e^- = \text{Ti} + \text{H}_2\text{O}$	-0.86	
$\text{Ti}^{3+} + e^- = \text{Ti}^{2+}$	-0.37	
$\text{Ti}^{3+} + 3e^- = \text{Ti}$	-1.21	
$\text{Ti}^{2+} + 2e^- = \text{Ti}$	-1.63	
Tungsten		
$2\text{WO}_3 + 2\text{H}^+ + 2e^- = \text{W}_2\text{O}_5 + \text{H}_2\text{O}$	-0.029	
$\text{WO}_3 + 6\text{H}^+ + 6e^- = \text{W} + 3\text{H}_2\text{O}$	-0.090	
$\text{WO}_4^{2-} + 4\text{H}_2\text{O} + 6e^- = \text{W} + 8\text{OH}^-$	-1.074	
$\text{WO}_4^{2-} + 2\text{H}_2\text{O} + 2e^- = \text{WO}_2 + 4\text{OH}^-$	-1.259	
$\text{W}_2\text{O}_5 + 2\text{H}^+ + 2e^- = 2\text{WO}_2 + \text{H}_2\text{O}$	-0.031	
$\text{W}(\text{CN})_8^{3-} + e^- = \text{W}(\text{CN})_8^{4-}$	0.457	
$\text{WO}_2 + 4\text{H}^+ + 4e^- = \text{W} + 2\text{H}_2\text{O}$	-0.119	
$\text{WO}_2 + 2\text{H}_2\text{O} + 4e^- = \text{W} + 4\text{OH}^-$	-0.982	
Uranium		
$\text{UO}_2^{2+} + e^- = \text{UO}_2^+$	0.16	
$\text{UO}_2^{2+} + 4\text{H}^+ + 2e^- = \text{U}^{4+} + 2\text{H}_2\text{O}$	0.27	
$\text{UO}_2^+ + 4\text{H}^+ + e^- = \text{U}^{4+} + 2\text{H}_2\text{O}$	0.38	
$\text{U}^{4+} + e^- = \text{U}^{3+}$	-0.52	
$\text{U}^{4+} + 4e^- = \text{U}$	-1.38	
$\text{U}^{3+} + 3e^- = \text{U}$	-1.66	
Vanadium		
$\text{VO}_2^+ + 2\text{H}^+ + e^- = \text{VO}^{2+} + \text{H}_2\text{O}$	1.000	
$\text{VO}_2^+ + 4\text{H}^+ + 2e^- = \text{V}^{3+} + 2\text{H}_2\text{O}$	0.668	
$\text{VO}_2^+ + 4\text{H}^+ + 3e^- = \text{V}^{2+} + 2\text{H}_2\text{O}$	0.361	
$\text{VO}_2^+ + 4\text{H}^+ + 5e^- = \text{V} + 4\text{H}_2\text{O}$	-0.236	
$\text{VO}^{2+} + 2\text{H}^+ + e^- = \text{V}^{3+} + \text{H}_2\text{O}$	0.337	
$\text{V}^{3+} + e^- = \text{V}^{2+}$	-0.255	
$\text{V}^{2+} + 2e^- = \text{V}$	-1.13	
Xenon		
$\text{H}_4\text{XeO}_6 + 2\text{H}^+ + 2e^- = \text{XeO}_3 + 3\text{H}_2\text{O}$	2.42	
$\text{HXeO}_6^{3-} + 2\text{H}_2\text{O} + e^- = \text{HXeO}_4 + 4\text{OH}^-$	0.9	
$\text{XeO}_3 + 6\text{H}^+ + 2\text{F}^- + 4e^- = \text{XeF}_2 + 3\text{H}_2\text{O}$	1.6	
$\text{XeO}_3 + 6\text{H}^+ + 6e^- = \text{Xe}(\text{g}) + 3\text{H}_2\text{O}$	2.10	
$\text{XeF}_2 + e^- = \text{XeF} + \text{F}^-$	0.9	
$\text{XeF}_2 + 2\text{H}^+ + 2e^- = \text{Xe}(\text{g}) + 2\text{HF}$	2.64	
$\text{XeF} + e^- = \text{Xe}(\text{g}) + \text{F}^-$	3.4	

TABLE 8.27 Potentials of the Elements and Their Compounds at 25°C (*Continued*)

Half-reaction	Standard or formal potential	Solution composition
Ytterbium		
$\text{Yb}^{3+} + e^- = \text{Yb}^{2+}$	-1.05	
$\text{Yb}^{2+} + 2e^- = \text{Yb}$	-2.8	
$\text{Yb}^{3+} + 3e^- = \text{Yb}$	-2.22	
Yttrium		
$\text{Y}^{3+} + 3e^- = \text{Y}$	-2.37	
Zinc		
$\text{Zn}^{2+} + 2e^- = \text{Zn}$	-0.7626	
$\text{Zn}(\text{NH}_3)_4^{2+} + 2e^- = \text{Zn} + 4\text{NH}_3$	-1.04	
$\text{Zn}(\text{CN})_4^{2-} + 2e^- = \text{Zn} + 4\text{CN}^-$	-1.34	
$\text{Zn}(\text{tartrate})_4^{6-} + 2e^- = \text{Zn} + 4(\text{tartrate})^{2-}$	-1.15	
$\text{Zn}(\text{OH})_4^{2-} + 2e^- = \text{Zn} + 4\text{OH}^-$	-1.285	
Zirconium		
$\text{Zr}^{4+} + 4e^- = \text{Zr}$	-1.55	
$\text{ZrO}_2 + 4\text{H}^+ + 4e^- = \text{Zr} + 2\text{H}_2\text{O}$	-1.45	

TABLE 8.28 Potentials of Selected Half-Reactions at 25°C

A summary of oxidation-reduction half-reactions arranged in order of decreasing oxidation strength and useful for selecting reagent systems.

Half-reaction	E° , volts
$\text{F}_2(\text{g}) + 2\text{H}^+ + 2e^- = 2\text{HF}$	3.053
$\text{O}_3 + \text{H}_2\text{O} + 2e^- = \text{O}_2 + 2\text{OH}^-$	1.246
$\text{O}_3 + 2\text{H}^+ + 2e^- = \text{O}_2 + \text{H}_2\text{O}$	2.075
$\text{Ag}^{2+} + e^- = \text{Ag}^+$	1.980
$\text{S}_2\text{O}_8^{2-} + 2e^- = 2\text{SO}_4^{2-}$	1.96
$\text{HN}_3 + 3\text{H}^+ + 2e^- = \text{NH}_4^+ + \text{N}_2$	1.96
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2e^- = 2\text{H}_2\text{O}$	1.763
$\text{Ce}^{4+} + e^- = \text{Ce}^{3+}$	1.72
$\text{MnO}_4^- + 4\text{H}^+ + 3e^- = \text{MnO}_2(\text{c}) + 2\text{H}_2\text{O}$	1.70
$2\text{HClO} + 2\text{H}^+ + 2e^- = \text{Cl}_2 + \text{H}_2\text{O}$	1.630
$2\text{HBrO} + 2\text{H}^+ + 2e^- = \text{Br}_2 + \text{H}_2\text{O}$	1.604
$\text{H}_5\text{IO}_6 + \text{H}^+ + 2e^- = \text{IO}_3^- + 3\text{H}_2\text{O}$	1.603
$\text{NiO}_2 + 4\text{H}^+ + 2e^- = \text{Ni}^{2+} + 2\text{H}_2\text{O}$	1.593
$\text{Bi}_2\text{O}_4(\text{bismuthate}) + 4\text{H}^+ + 2e^- = 2\text{BiO}^+ + 2\text{H}_2\text{O}$	1.59
$\text{MnO}_4^- + 8\text{H}^+ + 5e^- = \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51
$2\text{BrO}_3^- + 12\text{H}^+ + 10e^- = \text{Br}_2 + 6\text{H}_2\text{O}$	1.478
$\text{PbO}_2 + 4\text{H}^+ + 2e^- = \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.468
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6e^- = 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.36
$\text{Cl}_2 + 2e^- = 2\text{Cl}^-$	1.3583
$2\text{HNO}_2 + 4\text{H}^+ + 4e^- = \text{N}_2\text{O} + 3\text{H}_2\text{O}$	1.297
$\text{N}_2\text{H}_5^+ + 3\text{H}^+ + 2e^- = 2\text{NH}_4^+$	1.275
$\text{MnO}_2 + 4\text{H}^+ + 2e^- = \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.23
$\text{O}_2 + 4\text{H}^+ + 4e^- = 2\text{H}_2\text{O}$	1.229
$\text{ClO}_4^- + 2\text{H}^+ + 2e^- = \text{ClO}_3^- + \text{H}_2\text{O}$	1.201

TABLE 8.28 Potentials of Selected Half-Reactions at 25°C (*Continued*)

Half-reaction	E° , volts
$2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- = \text{I}_2 + 3\text{H}_2\text{O}$	1.195
$\text{N}_2\text{O}_4 + 2\text{H}^+ + 2\text{e}^- = 2\text{HNO}_3$	1.07
$2\text{ICl}_2^- + 2\text{e}^- = 4\text{Cl}^- + \text{I}_2$	1.07
$\text{Br}_2(\text{lq}) + 2\text{e}^- = 2\text{Br}^-$	1.065
$\text{N}_2\text{O}_4 + 4\text{H}^+ + 4\text{e}^- = 2\text{NO} + 2\text{H}_2\text{O}$	1.039
$\text{HNO}_2 + \text{H}^+ + \text{e}^- = \text{NO} + \text{H}_2\text{O}$	0.996
$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- = \text{NO} + 2\text{H}_2\text{O}$	0.957
$\text{NO}_3^- + 3\text{H}^+ + 2\text{e}^- = \text{HNO}_2 + \text{H}_2\text{O}$	0.94
$2\text{Hg}^{2+} + 2\text{e}^- = \text{Hg}_2^{2+}$	0.911
$\text{Cu}^{2+} + \text{I}^- + \text{e}^- = \text{CuI}$	0.861
$\text{OsO}_4(\text{c}) + 8\text{H}^+ + 8\text{e}^- = \text{Os} + 4\text{H}_2\text{O}$	0.84
$\text{Ag}^+ + \text{e}^- = \text{Ag}$	0.7991
$\text{Hg}_2^{2+} + 2\text{e}^- = 2\text{Hg}$	0.7960
$\text{Fe}^{3+} + \text{e}^- = \text{Fe}^{2+}$	0.771
$\text{H}_2\text{SeO}_3 + 4\text{H}^+ + 4\text{e}^- = \text{Se} + 3\text{H}_2\text{O}$	0.739
$\text{HN}_3 + 11\text{H}^+ + 8\text{e}^- = 2\text{NH}_4^+$	0.695
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- = \text{H}_2\text{O}_2$	0.695
$\text{Ag}_2\text{SO}_4 + 2\text{e}^- = 2\text{Ag} + \text{SO}_4^{2-}$	0.654
$\text{Cu}^{2+} + \text{Br}^- + \text{e}^- = \text{CuBr}(\text{c})$	0.654
$\text{Au}(\text{SCN})_4^- + 3\text{e}^- = \text{Au} + 4\text{SCN}^-$	0.636
$2\text{HgCl}_2 + 2\text{e}^- = \text{Hg}_2\text{Cl}_2(\text{c}) + 2\text{Cl}^-$	0.63
$\text{Sb}_2\text{O}_5 + 6\text{H}^+ + 4\text{e}^- = 2\text{SbO}^+ + 3\text{H}_2\text{O}$	0.605
$\text{H}_3\text{AsO}_4 + 2\text{H}^+ + 2\text{e}^- = \text{HAsO}_2 + 2\text{H}_2\text{O}$	0.560
$\text{TeOOH}^+ + 3\text{H}^+ + 4\text{e}^- = \text{Te} + 2\text{H}_2\text{O}$	0.559
$\text{Cu}^{2+} + \text{Cl}^- + \text{e}^- = \text{CuCl}(\text{c})$	0.559
$\text{I}_3^- + 2\text{e}^- = 3\text{I}^-$	0.536
$\text{I}_2 + 2\text{e}^- = 2\text{I}^-$	0.536
$\text{Cu}^+ + \text{e}^- = \text{Cu}$	0.53
$4\text{H}_2\text{SO}_3 + 4\text{H}^+ + 6\text{e}^- = \text{S}_4\text{O}_6^{2-} + 6\text{H}_2\text{O}$	0.507
$\text{Ag}_2\text{CrO}_4 + 2\text{e}^- = 2\text{Ag} + \text{CrO}_4^{2-}$	0.449
$2\text{H}_2\text{SO}_3 + 2\text{H}^+ + 4\text{e}^- = \text{S}_2\text{O}_3^{2-} + 3\text{H}_2\text{O}$	0.400
$\text{UO}_2^{2+} + 4\text{H}^+ + \text{e}^- = \text{U}^{4+} + 2\text{H}_2\text{O}$	0.38
$\text{Fe}(\text{CN})_6^{3-} + \text{e}^- = \text{Fe}(\text{CN})_6^{4-}$	0.361
$\text{Cu}^{2+} + 2\text{e}^- = \text{Cu}$	0.340
$\text{VO}^{2+} + 2\text{H}^+ + \text{e}^- = \text{V}^{3+} + \text{H}_2\text{O}$	0.337
$\text{BiO}^+ + 2\text{H}^+ + 3\text{e}^- = \text{Bi} + \text{H}_2\text{O}$	0.32
$\text{UO}_2^{2+} + 4\text{H}^+ + 2\text{e}^- = \text{U}^{4+} + 2\text{H}_2\text{O}$	0.27
$\text{Hg}_2\text{Cl}_2(\text{c}) + 2\text{e}^- = 2\text{Hg} + 2\text{Cl}^-$	0.2676
$\text{AgCl} + \text{e}^- = \text{Ag} + \text{Cl}^-$	0.2223
$\text{SbO}^+ + 2\text{H}^+ + 3\text{e}^- = \text{Sb} + \text{H}_2\text{O}$	0.212
$\text{CuCl}_3^- + \text{e}^- = \text{Cu} + 3\text{Cl}^-$	0.178
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- = \text{H}_2\text{SO}_3 + \text{H}_2\text{O}$	0.158
$\text{Sn}^{4+} + 2\text{e}^- = \text{Sn}^{2+}$	0.15
$\text{S} + 2\text{H}^+ + 2\text{e}^- = \text{H}_2\text{S}$	0.144
$\text{Hg}_2\text{Br}_2(\text{c}) + 2\text{e}^- = 2\text{Hg} + 2\text{Br}^-$	0.1392
$\text{CuCl} + \text{e}^- = \text{Cu} + \text{Cl}^-$	0.121
$\text{TiO}^{2+} + 2\text{H}^+ + \text{e}^- = \text{Ti}^{3+} + \text{H}_2\text{O}$	0.100
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^- = 2\text{S}_2\text{O}_3^{2-}$	0.08
$\text{AgBr} + \text{e}^- = \text{Ag} + \text{Br}^-$	0.0711
$\text{HCOOH} + 2\text{H}^+ + 2\text{e}^- = \text{HCHO} + \text{H}_2\text{O}$	0.056
$\text{CuBr} + \text{e}^- = \text{Cu} + \text{Br}^-$	0.033
$2\text{H}^+ + 2\text{e}^- = \text{H}_2$	0.0000
$\text{Hg}_2\text{I}_2 + 2\text{e}^- = 2\text{Hg} + 2\text{I}^-$	-0.0405

TABLE 8.28 Potentials of Selected Half-Reactions at 25°C (*Continued*)

Half-reaction	E° , volts
$\text{Pb}^{2+} + 2e^- = \text{Pb}$	-0.125
$\text{Sn}^{2+} + 2e^- = \text{Sn}$	-0.136
$\text{AgI} + e^- = \text{Ag} + \text{I}^-$	-0.1522
$\text{N}_2 + 5\text{H}^+ + 4e^- = \text{N}_2\text{H}_5^+$	-0.225
$\text{V}^{3+} + e^- = \text{V}^{2+}$	-0.255
$\text{Ni}^{2+} + 2e^- = \text{Ni}$	-0.257
$\text{Co}^{2+} + 2e^- = \text{Co}$	-0.277
$\text{Ag}(\text{CN})_2^- + e^- = \text{Ag} + 2\text{CN}^-$	-0.31
$\text{PbSO}_4 + 2e^- = \text{Pb} + \text{SO}_4^{2-}$	-0.3505
$\text{Cd}^{2+} + 2e^- = \text{Cd}$	-0.4025
$\text{Cr}^{3+} + e^- = \text{Cr}^{2+}$	-0.424
$\text{Fe}^{2+} + 2e^- = \text{Fe}$	-0.44
$\text{H}_3\text{PO}_3 + 2\text{H}^+ + 2e^- = \text{H}_2\text{PO}_2 + \text{H}_2\text{O}$	-0.499
$2\text{CO}_2 + 2\text{H}^+ + 2e^- = \text{H}_2\text{C}_2\text{O}_4$	-0.49
$\text{U}^{4+} + e^- = \text{U}^{3+}$	-0.52
$\text{Zn}^{2+} + 2e^- = \text{Zn}$	-0.7626
$\text{Mn}^{2+} + 2e^- = \text{Mn}$	-1.18
$\text{Al}^{3+} + 3e^- = \text{Al}$	-1.67
$\text{Mg}^{2+} + 2e^- = \text{Mg}$	-2.356
$\text{Na}^+ + e^- = \text{Na}$	-2.714
$\text{K}^+ + e^- = \text{K}$	-2.925
$\text{Li}^+ + e^- = \text{Li}$	-3.045
$3\text{N}_2 + 2\text{H}^+ + 2e^- = 2\text{HN}_3$	-3.10

TABLE 8.29 Overpotentials for Common Electrode Reactions at 25°C

The overpotential is defined as the difference between the actual potential of an electrode at a given current density and the reversible electrode potential for the reaction.

Electrode	Current Density, A/cm ²					
	0.001	0.01	0.1	0.5	1.0	5.0
	Overpotential, volts					
Liberation of H₂ from 1M H₂SO₄						
Ag	0.097	0.13	0.3		0.48	0.69
Al	0.3	0.83	1.00		1.29	
Au	0.017		0.1		0.24	0.33
Bi	0.39	0.4			0.78	0.98
Cd		1.13	1.22		1.25	
Co		0.2				
Cr		0.4				
Cu			0.35		0.48	0.55
Fe		0.56	0.82		1.29	
Graphite	0.002		0.32		0.60	0.73
Hg	0.8	0.93	1.03		1.07	
Ir	0.0026	0.2				
Ni	0.14	0.3			0.56	0.71
Pb	0.40	0.4			0.52	1.06
Pd	0	0.04				
Pt (smooth)	0.0000	0.16	0.29		0.68	
Pt (platinized)	0.0000	0.030	0.041		0.048	0.051
Sb		0.4				
Sn		0.5	1.2			
Ta		0.39	0.4			
Zn	0.48	0.75	1.06		1.23	
Liberation of O₂ from 1M KOH						
Ag	0.58	0.73	0.98		1.13	
Au	0.67	0.96	1.24		1.63	
Cu	0.42	0.58	0.66		0.79	
Graphite	0.53	0.90	1.09		1.24	
Ni	0.35	0.52	0.73		0.85	
Pt (smooth)	0.72	0.85	1.28		1.49	
Pt (platinized)	0.40	0.52	0.64		0.77	
Liberation of Cl₂ from saturated NaCl solution						
Graphite			0.25	0.42	0.53	
Platinized Pt	0.006		0.026	0.05		
Smooth Pt	0.008	0.03	0.054	0.161	0.236	
Liberation of Br₂ from saturated NaBr solution						
Graphite		0.002	0.027	0.16	0.33	
Platinized Pt		0.002	0.012	0.069	0.21	
Smooth Pt		0.002	0.006*	0.26	0.38†	
Liberation of I₂ from saturated NaI solution						
Graphite	0.002	0.014	0.097			
Platinized Pt		0.006	0.032		0.196	
Smooth Pt		0.003	0.03	0.12	0.22	

* At 0.23 A/cm². † At 0.72 A/cm².

The overpotential required for the evolution of O₂ from dilute solutions of HClO₄, HNO₃, H₃PO₄ or H₂SO₄ onto smooth platinum electrodes is approximately 0.5 V.

TABLE 8.30 Half-Wave Potentials of Inorganic Materials*All values are in volts vs. the saturated calomel electrode.*

Element	$E_{1/2}$, volts	Solvent system
Aluminum 3+	-0.5	0.2M acetate, pH 4.5–4.7, plus 0.07% azo dye Pontochrome Violet SW; reduction wave of complexed dye is 0.2 V more negative than that of the free dye.
Antimony 3+ to 0	-0.15 -0.31(1) -0.8 -1.0; -1.2 -1.26 -1.32	1M HCl 1M HNO ₃ (or 0.5M H ₂ SO ₄) 0.5M tartrate, pH 4.5 0.5M tartrate, pH 9 (waves not distinct) 1M NaOH; also anodic wave (3+ to 5+) at -0.45 0.5M tartrate plus 0.1M NaOH
5+	0.0; -0.257	6M HCl. First wave (5+ to 3+) starts at the oxidation potential of Hg; second wave is 3+ to 0.
5+ to 0	-0.35	1M HCl plus 4M KBr
Arsenic 3+ to 5+	-0.26	0.5M KOH (anodic wave); only suitable wave
3+	-0.8; -1.0 -0.7; -1.0	0.1M HCl; ill-defined waves 0.5M H ₂ SO ₄ (or 1M HNO ₃)
Barium 2+ to 0	-1.94	0.1M (C ₂ H ₅) ₄ NI
Bismuth 3+ to 0	-0.025(15) -0.09 -0.29 -0.7 -1.0	1M HNO ₃ (or 0.5M H ₂ SO ₄) 1M HCl 0.5M tartrate, pH 4.5 0.5M tartrate (pH 9), wave not well-developed 0.5M tartrate plus 0.1M NaOH, poor wave
Bromine 5+ to 1-	-1.75 0.13	0.1M alkali chlorides (or 0.1M NaOH) 0.05M H ₂ SO ₄
0 to 1- Br ⁻	0.0 0.1	Wave (anodic) starts at zero; Hg ₂ Br ₂ forms Oxidation of Hg to form mercury(I) bromide
Cadmium 2+ to 0	-0.60 -0.64 -0.81	0.1M KCl, or 0.5M H ₂ SO ₄ , or 1M HNO ₃ 0.5M tartrate at pH 4.5 or 9 1M NH ₄ Cl plus 1M NH ₃
Calcium 2+ to 0	-2.22 -2.13	0.1M (C ₂ H ₅) ₄ NCl 0.1M (C ₂ H ₅) ₄ NCl in 80% ethanol
Cerium 3+ to 0	-1.97	0.02M alkali sulfate
Cesium 1+ to 0	-2.05	0.1M (C ₂ H ₅) ₄ NOH in 50% ethanol
Chlorine Cl ⁻	0.25	Oxidation of Hg to form Hg ₂ Cl ₂
Chromium 6+ to 3+ 3+ to 0 3+ to 2+	-0.85 -0.35; -1.70 -0.95	CrO ₄ ²⁻ to CrO ₂ ⁻ in 0.1 to 1M NaOH 1M NH ₄ Cl—NH ₃ buffer (pH 8–9); 3+ to 2+ to 0 0.1M pyridine–0.1M pyridinium chloride

TABLE 8.30 Half-Wave Potentials of Inorganic Materials (*Continued*)

Element	$E_{1/2}$, volts	Solvent system
2+ to 0	-1.54	1M KCl
2+ to 3+	-0.40	1M KCl (anodic wave)
Cobalt		
3+ to 0	-0.5; -1.3	1M NH ₄ Cl plus 1M NH ₃ ; 3+ to 2+ to 0
2+ to 0	-1.07	0.1M pyridine plus pyridinium chloride
	-1.03	Neutral 1M potassium thiocyanate
	-1.4	Co(H ₂ O) ₆ ²⁺ in noncomplexing systems
3+ to 2+	0.0	1M sodium oxalate in acetate buffer (pH 5); diffusion current measured between 0 and -0.1 V
Copper		
2+ to 0	0.04	0.1M KNO ₃ , 0.1M NH ₄ ClO ₄ , or 1M Na ₂ SO ₄
	-0.085	0.1M Na ₄ P ₂ O ₇ plus 0.2M Na acetate, pH 4.5
	-0.09	0.5M Na tartrate, pH 4.5
	-0.20	0.1M potassium oxalate, pH 5.7 to 10
	-0.22	0.5M potassium citrate, pH 7.5
	-0.4	0.5M Na tartrate plus 0.1M NaOH (pH 12)
	-0.568	0.1M KNO ₃ plus 1M ethylenediamine
2+	0.04; -0.22	1M KCl; consecutive waves: 2+ to 1+ to 0
	-0.02; -0.39	0.1M KSCN; consecutive waves: 2+ to 1+ to 0
	0.05; -0.25	0.1M pyridine plus 0.1M pyridinium chloride; consecutive waves: 2+ to 1+ to 0
	-0.24; -0.50	1M NH ₄ Cl plus 1M NH ₃ ; consecutive waves
Gallium		
3+ to 0	-1.1	Not more than 0.001M HCl or wave masked by hydrogen wave which immediately follows
Germanium		
2+ to 0	-0.45	6M HCl; prior reduction with HPH ₂ O ₂ to 2+
Gold		
3+ to 1+	0	1M KCN; wave starts at 0 V
1+ to 0	-1.4	Au(CN) ₂ ⁻ wave best for analytical purposes
Indium		
3+ to 0	-0.60	1M KCl In Na acetate, pH 3.9 to 4.2
Iodine		
IO ₄ ⁻	0.36	First wave at pH 0 (shifts to -0.08 at pH 12); second wave corresponds to iodate reduction
IO ₃ ⁻	-0.075	0.2M KNO ₃ (shifts -0.13 V/pH unit increase)
	-0.305	0.1M hydrogen phthalate, pH 3.2
	-0.500	0.1M acetate plus 0.1M KCl, pH 4.9
	-0.650	0.1M citrate, pH 5.95
	-1.050	0.2M phosphate, pH 7.10
	-1.20	0.05M borax + 0.1M KCl, pH 9.2; or NaOH plus 0.1M KCl, pH 13.0
0 to 1-	0.0	Wave starts from zero in acid media; Hg ₂ I ₂ formed
1-	-0.1	Oxidation of Hg to form Hg ₂ I ₂
Iron		
3+	-0.44; -1.52	1M (NH ₄) ₂ CO ₃ ; two waves; 3+ to 2+ to 0
	-0.17; -1.50	0.5M Na tartrate, pH 5.8; two waves; 3+ to 2+ to 0
	-0.9; -1.5	0.1 to 5M KOH plus 8% mannitol; 3+ to 2+ to 0

TABLE 8.30 Half-Wave Potentials of Inorganic Materials (*Continued*)

Element	$E_{1/2}$, volts	Solvent system
3+ to 2+	-0.13 -0.27 -0.28 -1.46(2) -1.36	0.1M EDTA plus 2M Na acetate, pH 6-7 0.2M Na oxalate, pH 7.9 or less 0.5M Na citrate, pH 6.5 1M NH_4ClO_4 0.1M KHF_2 , pH 4 or less
2+ to 3+	-0.28 -0.27 -0.17 -1.36	0.5M Na citrate, pH 6.5 0.2M Na oxalate, pH 7.9 or less 0.5M Na tartrate, pH 5.8 0.1M KHF_2 , pH 4 or less
Lead		
2+ to 0	-0.405 -0.435 -0.49(1) -0.72 -0.75	1M HNO_3 1M KCl (or HCl) 0.5M Na tartrate, pH 4.5 or 9 1M KCN 1M KOH or 0.5M Na tartrate plus 0.1M NaOH
Lithium		
1+ to 0	-2.31	0.1M $(\text{C}_2\text{H}_5)_4\text{NOH}$ in 50% ethanol
Magnesium		
2+ to 0	-2.2	0.1M $(\text{C}_2\text{H}_5)_4\text{NCl}$ (poorly defined wave)
Manganese		
2+ to 0	-1.65 -1.55 -1.33	1M NH_4Cl plus 1M NH_3 1M KCNS 1.5M KCN
Molybdenum		
6+	-0.26; -0.63	0.3M HCl, two waves: 6+ to 5+ to 3+
Nickel		
2+ to 0	-0.70 -0.78 -1.09 -1.1 -1.36	1M KSCN 1M KCl plus 0.5M pyridine 1M NH_4Cl plus 1M NH_3 $\text{Ni}(\text{H}_2\text{O})_6^{2+}$ in NH_4ClO_4 or KNO_3 $\text{Ni}(\text{CN})_4^{2-}$ in 1M KCN (alkaline media)
Niobium		
5+ to 3+	-0.80(4)	1M HNO_3
Nitrogen		
Nitrate	-1.45	0.017M LaCl_3 (reduced to hydroxylamine)
HNO_2	-0.77	0.1M HCl
C_2N_2	-1.2; -1.55	0.1M Na acetate, two waves
Oxamic acid	-1.55	0.1M Na acetate
Cyanide	-0.45	0.1M NaOH; anodic wave starts at -0.45
Thiocyanate	0.18	Anodic wave; neutral or weakly alkaline medium
Osmium		
OsO_4	0.0; -0.41; -1.16	Sat'd $\text{Ca}(\text{OH})_2$. Three waves: first starts at 0; second wave is OsO_4^{2-} to Os(V); and third wave is Os(V) to Os(III)
Oxygen		
O_2	-0.05; -0.9	Buffer solutions of pH 1 to 10. Two waves: O_2 to H_2O_2 , and H_2O_2 to H_2O . Second wave extends from -0.5 to -1.3
H_2O_2	-0.9	Very extended wave (see above); sharper in presence of Aerosol OT

TABLE 8.30 Half-Wave Potentials of Inorganic Materials (*Continued*)

Element	$E_{1/2}$, volts	Solvent system
Palladium 2+ to 0	-0.31 -0.64 -0.72	1M pyridine plus 1M KCl 0.1M ethylenediamine plus 1M KCl 1M NH_4Cl plus 1M NH_3
Potassium 1+ to 0	-2.10	0.1M $(\text{C}_2\text{H}_5)_4\text{NOH}$ in 50% ethanol
Rhenium 7+ to 4+ 4+ to 3+	-0.44 -0.51	2M HCl or (better) 4M HClO_4 ReCl_6^{2-} ion in 1M HCl
Rhodium 3+ to 2+	-0.41	1M pyridine plus 1M KCl
Rubidium 1+ to 0	-1.99	0.1M $(\text{C}_2\text{H}_5)_4\text{NOH}$ in 50% ethanol
Scandium 3+ to 0	-1.80	0.1M LiCl, KCl, or BaCl_2
Selenium 4+ to 2- 2-	-1.44 -1.54 -0.49 -0.94	1M NH_4Cl plus NH_3 , pH 8.0 Same system adjusted to pH 9.5 Anodic wave at pH 0 due to HgSe Anodic wave at pH 12 (0.01M NaOH)
Silver 1+ to 0 1+ to 0	-0.3	Wave starts at oxidation potential of Hg 0.0014M $\text{KAg}(\text{CN})_2$ without excess cyanide
Sodium 1+ to 0	-2.07	0.1M $(\text{C}_2\text{H}_5)_4\text{NOH}$ in 50% ethanol
Strontium 2+ to 0	-2.11	0.1M $(\text{C}_2\text{H}_5)_4\text{NI}$, water or 80% ethanol
Sulfur SO_2 $\text{S}_2\text{O}_4^{2-}$ $\text{S}_2\text{O}_3^{2-}$ 0 to 2- HS^-	-0.38 -0.43 -0.15 -0.50 -0.76	1M HNO_3 (or other strong acid); 4+ to 2+ 0.5M $(\text{NH}_4)_2\text{HPO}_4$ plus 1M NH_3 (anodic wave) 1M strong acid; anodic mercury wave 90% methanol, 9.5% pyridine, 0.5% HCl (pH 6) 0.1M NaOH (anodic mercury wave)
Tellurium 4+ to 0 4+ to 2- 2- to 0	-0.4 -0.63 -1.22 -0.72 -0.08	Citrate buffer, pH 1.6 (second of two waves) Ammoniacal buffer, pH 9.4 0.1M NaOH 1M HCl (true anodic reversible wave) 1M NaOH (same as above; intermediate values at pH 1 to 13)
Thallium 3+ to 0	-0.48	1M KCl, KNO_3 , K_2SO_4 , KOH, or NH_3
Tin 4+ to 2+ 2+ to 0 2+ to 4+	-0.25; -0.52 -0.59 -1.22 -0.28 -0.73	4M NH_4Cl + 1M HCl; two waves: 4+ to 2+ to 0 0.5M tartrate, pH 4.3 1M NaOH (stannite ion to tin) 0.5M Na tartrate, pH 4.3 (anodic wave) 1M NaOH (stannite ion to stannate ion)

TABLE 8.30 Half-Wave Potentials of Inorganic Materials (*Continued*)

Element	$E_{1/2}$, volts	Solvent system
Titanium 4+ to 3+	-0.173 -1.22	0.1M K ₂ C ₂ O ₄ plus 1M H ₂ SO ₄ 0.4M tartrate, pH 6.5
Tungsten 6+	0.0; -0.64	6M HCl; two waves: first wave starts at zero and is W(VI) to W(V), the second wave is W(V) to W(III)
Uranium 6+	-0.180; -0.92	UO ₂ ²⁺ to UO ₂ ⁺ , then U ³⁺ in 0.02M HCL
Vanadium 5+ to 4+ to 2+	-0.97; -1.26	1M NH ₄ Cl plus 1M NH ₃ and 0.08M Na ₂ SO ₃
4+ to 2+	-0.98	0.05M H ₂ SO ₄
3+ to 2+	-0.55	0.5M H ₂ SO ₄
4+ to 5+	-0.32	1M NH ₄ Cl, 1M NH ₃ , and 0.08M Na ₂ SO ₃
4+ to 5+	0.76	0.05M H ₂ SO ₄ ; anodic wave starting from zero
2+ to 3+	-0.55	0.5M H ₂ SO ₄ ; anodic wave
Zinc 2+ to 0	-0.995 -1.01 -1.15 -1.23 -1.33 -1.53	0.1M KCl 0.1M KSCN 0.5M tartrate, pH 9 0.5M tartrate, pH 4.5 1M NH ₄ Cl plus 1M NH ₃ 1M NaOH

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C

The solvent systems in this table are listed below:

- A, acetonitrile and a perchlorate salt such as LiClO₄ or a tetraalkyl ammonium salt
- B, acetic acid and an alkali acetate, often plus a tetraalkyl ammonium iodide
- C, 0.05 to 0.175*M* tetraalkyl ammonium halide and 75% 1,4-dioxane
- D, buffer plus 50% ethanol (EtOH)

Abbreviations Used in the Table
Bu, butyl Me, methyl
Et, ethyl MeOH, methanol
EtOH, ethanol PrOH, propanol
M, molar

Compound	Solvent system	<i>E</i> _{1/2}
Unsaturated aliphatic hydrocarbons		
Acrylonitrile	C but 30% EtOH	− 1.94
Allene	C	− 2.29
1,3-Butadiene	A	− 2.03
	C	− 2.59
1,3-Butadiyne	C	− 1.89
1-Buten-2-yne	C	− 2.40
1,4-Cyclohexadiene	A	− 1.6
Cyclohexene	A	− 1.89
1,3,5,7-Cyclooctatetraene	B	− 1.42
	C	− 1.51
Diethyl fumarate	B, pH 4.0	− 0.84
Diethyl maleate	B, pH 4.0	− 0.95
2,3-Dimethyl-1,3-butadiene	A	− 1.83
Dimethylfulvene	C	− 1.89
Diphenylacetylene	C	− 2.20
1,1-Diphenylethylene	B	− 1.52
	C	− 2.19
Ethyl methacrylate	0.1 <i>N</i> LiCl+ 25% EtOH	− 1.9
2-Methyl-1,3-butadiene	A	− 1.84
2-Methyl-1-butene	A	− 1.97
1-Piperidino-4-cyano-4-phenyl-1,3-butadiene	LiClO ₄ in dimethylformamide	− 0.16
<i>trans</i> -Stilbene	B	− 1.51
Tetrakis(dimethylamino)ethylene	A	− 0.75
Aromatic hydrocarbons		
Acenaphthene	A	− 0.95
	B	− 1.36
	C	− 2.58
Anthracene	A	− 0.84
	B	− 1.20
	C	− 1.94
Azulene	A	− 0.71
	C	− 1.66, − 2.26, − 2.56

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Aromatic hydrocarbons (continued)		
1,2-Benzanthracene	C	− 2.03, − 2.54
2,3-Benzanthracene	A	− 0.54, − 1.20
Benzene	A	− 2.08
1,2-Benzo[<i>a</i>]pyrene	A	− 0.76
Biphenyl	A	− 1.48
	B	− 1.91
	C	− 2.70
Chrysene	A	− 1.22
1,2,5,6-Dibenzanthracene	A	− 1.00, − 1.26
1,2-Dihydronaphthalene	C	− 2.57
9,10-Dimethylantracene	A	− 0.65
2,3-Dimethylnaphthalene	A	− 1.08, − 1.34
9,10-Diphenylantracene	A	− 0.92
Fluorene	A	− 1.25
	B	− 1.65
	C	− 2.65
Hexamethylbenzene	A	− 1.16
	B	− 1.52
Indan	A	− 1.59, − 2.02
Indene	A	− 1.23
	C	− 2.81
1-Methylnaphthalene	A	− 1.24
	B	− 1.53
	C	− 2.46
2-Methylnaphthalene	A	− 1.22
	B	− 1.55
	C	− 2.46
Naphthalene	A	− 1.34
	B	− 1.72
Pentamethylbenzene	A	− 1.28
	B	− 1.62
Phenanthrene	A	− 1.23
	B	− 1.68
	C	− 2.46, − 2.71
Phenylacetylene	C	− 2.37
Pyrene	A	− 1.06, − 1.24
<i>trans</i> -Stilbene	B	− 1.51
	C	− 2.26
Styrene	C	− 2.35
1,2,3,5-Tetramethylbenzene	A	− 1.50, − 1.99
1,2,4,5-Tetramethylbenzene	A	− 1.29
Tetraphenylethylene	C	− 2.05
1,4,5,8-Tetraphenylnaphthalene	A	− 1.39
Toluene	A	− 1.98
1,2,3-Trimethylbenzene	A	− 1.58
1,2,4-Trimethylbenzene	A	− 1.41

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Aromatic hydrocarbons (continued)		
1,3,5-Trimethylbenzene	A	− 1.50
	B	− 1.90
Triphenylene	A	− 1.46, − 1.55
Triphenylmethane	C	− 1.01, − 1.68, − 1.96
<i>o</i> -Xylene	A	− 1.58, − 2.04
<i>m</i> -Xylene	A	− 1.58
<i>p</i> -Xylene	A	− 1.56
Aldehydes		
Acetaldehyde	B, pH 6.8–13	− 1.89
Benzaldehyde	McIlvaine buffer, pH 2.2	− 0.96, − 1.32
Bromoacetaldehyde	pH 8.5	− 0.40
	pH 9.8	− 1.58, − 1.82
Chloroacetaldehyde	Ammonia buffer, pH 8.4	− 1.06, − 1.66
Cinnamaldehyde	Buffer + EtOH, pH 6.0	− 0.9, − 1.5, − 1.7
Crotonaldehyde	B, pH 1.3–2.0	− 0.92
	Ammonia buffer, pH 8.0	− 1.30
Dichloroacetaldehyde	Ammonia buffer, pH 8.4	− 1.03, − 1.67
3,7-Dimethyl-2,6-octadienal	0.1 M Et ₄ NI	− 1.56, − 2.22
Formaldehyde	0.05 M KOH+0.1 M KCl, pH 12.7	− 1.59
2-Furaldehyde	pH 1–8	− 0.86–0.07 pH
	pH 10	− 1.43
Glucose	Phosphate buffer, pH 7	− 1.55
Glyceraldehyde	Britton-Robinson buffer, pH 5.0	− 1.47
	Britton-Robinson buffer, pH 8.0	− 1.55
Glycolaldehyde	0.1 M KOH, pH 13	− 1.70
Glyoxal	B, pH 3.4	− 1.41
4-Hydroxybenzaldehyde	Britton-Robinson buffer, pH 1.8	− 1.16
	Britton-Robinson buffer, pH 6.8	− 1.45
4-Hydroxy-2-methoxybenzaldehyde	McIlvaine buffer, pH 2.2	− 1.05
	McIlvaine buffer, pH 5.0	− 1.16, − 1.36
	McIlvaine buffer, pH 8.0	− 1.47
<i>o</i> -Methoxybenzaldehyde	Britton-Robinson buffer, pH 1.8	− 1.02
	Britton-Robinson buffer, pH 6.8	− 1.49
<i>p</i> -Methoxybenzaldehyde	Britton-Robinson buffer, pH 1.8	− 1.17
	Britton-Robinson buffer, pH 6.8	− 1.48
Methyl glyoxal	A, pH 4.5	− 0.83
<i>m</i> -Nitrobenzaldehyde	Buffer +10% EtOH, pH 2.0	− 0.28, − 1.20
Phthalaldehyde	Buffer, pH 3.1	− 0.64, − 1.07
	Buffer, pH 7.3	− 0.89, − 1.29
2-Propenal (acrolein)	pH 4.5	− 1.36
	pH 9.0	− 1.1
Propionaldehyde	0.1 M LiOH, pH 13	− 1.93
Pyrrole-2-carbaldehyde	0.1 M HCl+50% EtOH	− 1.25

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Aldehydes (<i>continued</i>)		
Salicylaldehyde	McIlvaine buffer, pH 2.2 McIlvaine buffer, pH 5.0 McIlvaine buffer, pH 8.0	− 0.99, − 1.23 − 1.20, − 1.30 − 1.32
Trichloroacetaldehyde	Ammonia buffer, pH 8.4 0.1 M KCl + 50% EtOH	− 1.35, − 1.66 − 1.55
Ketones		
Acetone	B, pH 9.3 C	− 1.52 − 2.46
Acetophenone	D + McIlvaine buffer, pH 4.9 D + McIlvaine buffer, pH 7.2 D + McIlvaine buffer, pH 1.3	− 1.33 − 1.58 − 1.08
7 <i>H</i> -Benz[<i>de</i>]anthracen-7-one	0.1 N H ₂ SO ₄ + 75% MeOH	− 0.96
Benzil	D + McIlvaine buffer, pH 1.3 D + McIlvaine buffer, pH 4.9	− 0.27 − 0.50
Benzoin	D + McIlvaine buffer, pH 1.3 D + McIlvaine buffer, pH 8.6	− 0.90 − 1.49
Benzophenone	D + McIlvaine buffer, pH 1.3 D + McIlvaine buffer, pH 8.6	− 0.94 − 1.36
Benzoylacetone	Buffer, pH 2.6 Buffer, pH 5.3 and pH 7.6 Buffer, pH 9.7	− 1.60 − 1.68 − 1.72
Bromoacetone	0.1 M LiCl	− 0.29
2,3-Butanedione	0.1 M HCl	− 0.84
3-Buten-2-one	0.1 M KCl	− 1.42
Butyrophenone	0.1 M NH ₄ Cl + 50% EtOH	− 1.55
D-Carvone	0.1 M Et ₄ Ni + 80% EtOH	− 1.71
Chloroacetone	0.1 M LiCl	− 1.18
Coumarin	McIlvaine buffer, pH 2.0 McIlvaine buffer, pH 5.0	− 0.95 − 1.11, − 1.44
Cyclohexanone	C	− 2.45
<i>cis</i> -Dibenzoylethylene	D, pH 1	− 0.30
<i>trans</i> -Dibenzoylethylene	D, pH 11 D, pH 1	− 0.62, − 1.65 − 0.12
Dibenzoylmethane	D, pH 11 D, pH 1.3	− 0.57, − 1.52 − 0.59
9,10-Dihydro-9-oxoanthracene	D, pH 11.3	− 1.30, − 1.62
1,5-Diphenyl-1,5-pentanedione	D, pH 2.0	− 0.93
1,5-Diphenylthiocarbazone	A D, pH 7.0	− 2.10 − 0.6
Flavanone	Acetate buffer + Me ₂ NOH + 50% 2-PrOH, pH 6.1 Acetate buffer + Me ₂ NOH + 50% 2-PrOH, pH 9.6	− 1.30 − 1.51
Fluorescein	Acetate buffer, pH 2.0 Phthalate buffer, pH 5.0	− 0.50 − 0.65
Fructose	Borate buffer, pH 10.1 0.02 M LiCl	− 1.18, − 1.44 − 1.76

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Ketones (continued)		
Girard derivatives of aliphatic ketones	pH 8.2	− 1.52
<i>o</i> -Hydroxyacetophenone	D, pH 5	− 1.36
<i>p</i> -Hydroxyacetophenone	D, pH 5	− 1.46
1,2,3-Indantrione (ninhydrin)	Britton-Robinson buffer, pH 2.5	− 0.67, − 0.83
	Britton-Robinson buffer, pH 4.5	− 0.73, − 1.01
	Britton-Robinson buffer, pH 6.8	− 0.10, − 0.90, − 1.20
	Britton-Robinson buffer, pH 9.2	− 1.35
α -Ionone	C	− 1.59, − 2.08
Isatin	Phosphate buffer+citrate buffer, pH 2.9	− 0.3, − 0.5
	Phosphate buffer+citrate buffer, pH 4.3	− 0.3, − 0.5, − 0.8
	Phosphate buffer+citrate buffer, pH 5.4	− 0.8
4-Methyl-3,5-heptadien-2-one	A	− 0.64
4-Methyl-2,6-heptanedione	A	− 1.28
4-Methyl-3-penten-2-one	D+McIlvaine buffer, pH 1.3	− 1.01
	D+McIlvaine buffer, pH 11.3	− 1.60
4-Phenyl-3-buten-2-one	D, pH 1.3	− 0.72
	D, pH 8.6	− 1.27
Phthalide	0.1 <i>M</i> Bu ₄ NI+50% dioxane	− 0.20
Phthalimide	pH 4.2	− 1.1, − 1.5
	pH 9.7	− 1.2, − 1.4
Pulegone	C	− 1.74
Quinalizarin	Phosphate buffer+1% EtOH, pH 8.0	− 0.56
Testosterone	D+Britton-Robinson buffer, pH 2.6	− 1.20
	D+Britton-Robinson buffer, pH 5.8	− 1.40
	D+Britton-Robinson buffer, pH 8.8	− 1.53, − 1.79
Quinones		
Anthraquinone	Acetate buffer+40% dioxane, pH 5.6	− 0.51
	Phosphate buffer+40% dioxane, pH 7.9	− 0.71
<i>o</i> -Benzoquinone	Britton-Robinson buffer, pH 7.0	+ 0.20
	Britton-Robinson buffer, pH 9.0	+ 0.08
2,3-Dimethylnaphthoquinone	D, pH 5.4	− 0.22
1,2-Naphthoquinone	Phosphate buffer, pH 5.0	− 0.03
	Phosphate buffer, pH 7.0	− 0.13
1,4-Naphthoquinone	Britton-Robinson buffer, pH 7.0	− 0.07
	Britton-Robinson buffer, pH 9.0	− 0.19

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Acids		
Acetic acid	A	-2.3
Acrylic acid	pH 5.6	-0.85
Adenosine-5'-phosphoric acid	HClO ₄ +KClO ₄ , pH 2.2	-1.13
4-Aminobenzenesulfonic acid	0.05 M Me ₄ NI	-1.58
3-Aminobenzoic acid	pH 5.6	-0.67
Anthranilic acid	pH 5.6	-0.67
Ascorbic acid	Britton-Robinson buffer, pH 3.4	+0.17
	Britton-Robinson buffer, pH 7.0	-0.06
Barbituric acid	Borate buffer, pH 9.3	-0.04
Benzoic acid	A	-2.1
Benzoylformic acid	Britton-Robinson buffer, pH 2.2	-0.48
	Britton-Robinson buffer, pH 5.5	-0.85, -1.26
	Britton-Robinson buffer, pH 7.2	-0.98, -1.25
	Britton-Robinson buffer, pH 9.2	-1.25
Bromoacetic acid	pH 1.1	-0.54
2-Bromopropionic acid	pH 2.0	-0.39
Crotonic acid	C	-1.94
Dibromoacetic acid	pH 1.1	-0.03, -0.59
Dichloroacetic acid	pH 8.2	-1.57
5,5-Diethylbarbituric acid	Borate buffer, pH 9.3	0.00
Flavanol	D, pH 5.6	-1.25
	D, pH 7.7	-1.40
Folic acid	Britton-Robinson buffer, pH 4.6	-0.73
Formic acid	0.1 M KCl	-1.66
Fumaric acid	HCl+KCl, pH 2.6	-0.83
	Acetate buffer, pH 4.0	-0.93
	Acetate buffer, pH 5.9	-1.20
2,4-Hexadienedioic acid	Acetate buffer, pH 4.5	-0.97
Iodoacetic acid	pH 1	-0.16
Maleic acid	Britton-Robinson buffer, pH 2.0	-0.70
	Britton-Robinson buffer, pH 4.0	-0.97
	Britton-Robinson buffer, pH 6.0	-1.11, -1.30
	Britton-Robinson buffer, pH 10.0	-1.51
Mercaptoacetic acid	B, pH 6.8	-0.38
Methacrylic acid	D+0.1 M LiCl	-1.69
Nitrobenzoic acids	Buffer+10% EtOH, pH 2.0	-0.2, -0.7
Oxalic acid	B, pH 5.4-6.1	-1.80
2-Oxo-1,5-pentanedioic acid	HCl+KCl, pH 1.8	-0.59
	Ammonia buffer, pH 8.2	-1.30
2-Oxopropionic acid	Britton-Robinson buffer, pH 5.6	-1.17
	Britton-Robinson buffer, pH 6.8	-1.22, -1.53
	Britton-Robinson buffer, pH 9.7	-1.51
Phenolphthalein	Phthalate buffer, pH 2.5	-0.67
	Phthalate buffer, pH 4.7	-0.80
	D, pH 9.6	-0.98, -1.35
Picric acid	pH 4.2	-0.34
	pH 11.7	-0.36, -0.56, -0.96

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
<i>Acids (continued)</i>		
1,2,3-Propenetricarboxylic acid	pH 7.0	− 2.1
Trichloroacetic acid	Ammonia buffer, pH 8.2	− 0.84, − 1.57
	Phosphate buffer, pH 10.4	− 0.9, − 1.6
3,4,5-Trihydroxybenzoic acid	Phosphate buffer, pH 2.9	+ 0.50
	Phosphate buffer, pH 8.8	+ 0.1
<i>p</i> -Aminophenol	Britton-Robinson buffer, pH 6.3	+ 0.14
	Britton-Robinson buffer, pH 8.6	− 0.04
	Britton-Robinson buffer, pH 12.0	− 0.16
<i>o</i> -Chlorophenol	pH 5.6	− 0.63
<i>m</i> -Chlorophenol	pH 5.6	− 0.73
<i>p</i> -Chlorophenol	pH 5.6	− 0.65
<i>o</i> -Cresol	pH 5.6	− 0.56
<i>m</i> -Cresol	pH 5.6	− 0.61
<i>p</i> -Cresol	pH 5.6	− 0.54
1,2-Dihydroxybenzene	pH 5.6	− 0.35
1,3-Dihydroxybenzene	pH 5.6	− 0.61
1,4-Dihydroxybenzene	pH 5.6	− 0.23
<i>o</i> -Methoxyphenol	pH 5.6	− 0.46
<i>m</i> -Methoxyphenol	pH 5.6	− 0.62
<i>p</i> -Methoxyphenol	pH 5.6	− 0.41
1-Naphthol	A	− 0.74
2-Naphthol	A	− 0.82
1,2,3-Trihydroxybenzene	Britton-Robinson buffer, pH 3.1	+ 0.35
	Britton-Robinson buffer, pH 6.5	+ 0.10
	Britton-Robinson buffer, pH 9.5	− 0.10
<i>Halogen compounds</i>		
Bromobenzene	A	− 1.98
	C	− 2.32
1-Bromobutane	C	− 2.27
Bromoethane	C	− 2.08
Bromomethane	C	− 1.63
1-Bromonaphthalene (also 2-bromonaphthalene)	A	− 1.55, − 1.60
3-Bromo-1-propene	C	− 1.29
<i>p</i> -Bromotoluene	A	− 1.72
Carbon tetrachloride	C	− 0.78, − 1.71
Chlorobenzene	A	− 2.07
Chloroform	C	− 1.63
Chloromethane	C	− 2.23
3-Chloro-1-propene	C	− 1.91
α -Chlorotoluene	C	− 1.81
<i>p</i> -Chlorotoluene	A	− 1.76
<i>N</i> -Chloro- <i>p</i> -toluenesulfonamide	0.5 M K ₂ SO ₄	− 0.13
9,10-Dibromoanthracene	A	− 1.15, − 1.47
<i>p</i> -Dibromobenzene	C	− 2.10
1,2-Dibromobutane	D + 1% Na ₂ SO ₃	− 1.45

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Halogen compounds (continued)		
Dibromoethane	C	-1.48
meso-2,3-Dibromosuccinic acid	Acetate buffer, pH 4.0	-0.23, -0.89
Dichlorobenzenes	C	-2.5
Dichloromethane	C	-1.60
Diiodomethane	C	-1.12, -1.53
Hexabromobenzene	C	-0.8, -1.5
Hexachlorobenzene	C	-1.4, -1.7
Iodobenzene	A	-1.72
Iodoethane	C	-1.67
Iodomethane	A	-2.12
	C	-1.63
Tetrabromomethane	C	-0.3, -0.75, -1.49
Tetraiodomethane	C	-0.45, -1.05, -1.46
Tribromomethane	C	-0.64, -1.47
α,α,α -Trichlorotoluene	C	-0.68, -1.65, -2.00
Nitro and nitroso compounds		
1,2-Dinitrobenzene	Phthalate buffer, pH 2.5	-0.12, -0.32, -1.26
	Borate buffer, pH 9.2	-0.38, -0.74
1,3-Dinitrobenzene	Phthalate buffer, pH 2.5	-0.17, -0.29
	Borate buffer, pH 9.2	-0.46, -0.68
1,4-Dinitrobenzene	Phthalate buffer, pH 2.5	-0.12, -0.33
	Borate buffer, pH 9.2	-0.35, -0.80
Methyl nitrobenzoates	Buffer+10% EtOH, pH 2.0	-0.20 to -0.25 -0.68 to -0.74
<i>p</i> -Nitroacetophenone	Britton-Robinson buffer, pH 2.2	-0.16, -0.61, -1.09
	Britton-Robinson buffer, pH 10.0	-0.51, -1.40, -1.73
<i>o</i> -Nitroaniline	0.03 <i>M</i> LiCl+0.02 <i>M</i> benzoic acid in EtOH	-0.88
<i>m</i> -Nitroaniline	Britton-Robinson buffer, pH 4.3	-0.3, -0.8
	Britton-Robinson buffer, pH 7.2	-0.5
	Britton-Robinson buffer, pH 9.2	-0.7
<i>p</i> -Nitroaniline	pH 2.0	-0.36
	Acetate buffer, pH 4.6	-0.5
<i>o</i> -Nitroanisole	Buffer+10% EtOH, pH 2.0	-0.29, -0.58
<i>p</i> -Nitroanisole	Buffer+10% EtOH, pH 2.0	-0.35, -0.64
1-Nitroanthraquinone	Britton-Robinson buffer, pH 7.0	-0.16
Nitrobenzene	HCl+KCl+8% EtOH, pH 0.5	-0.16, -0.76
	Phthalate buffer, pH 2.5	-0.30
	Borate buffer, pH 9.2	-0.70
Nitroresols	Britton-Robinson buffer, pH 2.2	-0.2 to -0.3
	Britton-Robinson buffer, pH 4.5	-0.4 to -0.5
	Britton-Robinson buffer, pH 8.0	-0.6
Nitroethane	Britton-Robinson buffer+30% MeOH, pH 1.8	-0.7
	Britton-Robinson buffer+30% MeOH, pH 4.6	-0.8

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Nitro and nitroso compounds (continued)		
2-Nitrohydroquinone	Phosphate buffer+citrate buffer, pH 2.1	-0.2
	Phosphate buffer+citrate buffer, pH 5.2	-0.4
	Phosphate buffer+citrate buffer, pH 8.0	-0.5
Nitromethane	Britton-Robinson buffer+30% MeOH, pH 1.8	-0.8
	Britton-Robinson buffer+30% MeOH, pH 4.6	-0.85
o-Nitrophenol	Britton-Robinson buffer+10% EtOH, pH 2.0	-0.23
	Britton-Robinson buffer+10% EtOH, pH 4.0	-0.4
	Britton-Robinson buffer+10% EtOH, pH 8.0	-0.65
	Britton-Robinson buffer+10% EtOH, pH 10.0	-0.80
m-Nitrophenol	Britton-Robinson buffer+10% EtOH, pH 2.0	-0.37
	Britton-Robinson buffer+10% EtOH, pH 4.0	-0.40
	Britton-Robinson buffer+10% EtOH, pH 8.0	-0.64
	Britton-Robinson buffer+10% EtOH, pH 10.0	-0.76
p-Nitrophenol	Britton-Robinson buffer+10% EtOH, pH 2.0	-0.35
	Britton-Robinson buffer+10% EtOH, pH 4.0	-0.50
	Britton-Robinson buffer+10% EtOH, pH 8.0	-0.82
	Britton-Robinson buffer+30% MeOH, pH 1.8	-0.73
1-Nitropropane	Britton-Robinson buffer+30% MeOH, pH 8.6	-0.88
	Britton-Robinson buffer+30% MeOH, pH 8.0	-0.95
	McIlvaine buffer, pH 2.1	-0.53
2-Nitropropane	McIlvaine buffer, pH 5.1	-0.81
Nitrosobenzene	McIlvaine buffer, pH 6.0	-0.03
	McIlvaine buffer, pH 8.0	-0.14
1-Nitroso-2-naphthol	D+buffer, pH 4.0	+0.02
	D+buffer, pH 7.0	-0.20
	D+buffer, pH 9.0	-0.31
N-Nitrosophenylhydroxylamine	pH 2.0	-0.84
o-Nitrotoluene	Phthalate buffer, pH 2.5	-0.35, -0.66
	Phthalate buffer, pH 7.4	-0.60, -1.06

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Nitro and nitroso compounds (continued)		
<i>m</i> -Nitrotoluene (also <i>p</i> -nitrotoluene)	Phthalate buffer, pH 2.5 Phthalate buffer, pH 7.4	− 0.30, − 0.53 − 0.58, − 1.06
Tetranitromethane	pH 12.0	− 0.41
1,3,5-Trinitrobenzene	Phthalate buffer, pH 4.1 Borate buffer, pH 9.2	− 0.20, − 0.29, − 0.34 − 0.34, − 0.48, − 0.65
Heterocyclic compounds containing nitrogen		
Acridine	D, pH 8.3	− 0.80, − 1.45
Cinchonine	B, pH 3	− 0.90
2-Furanmethanol	Britton-Robinson buffer, pH 2.0 Britton-Robinson buffer, pH 5.8	− 0.96 − 1.38, − 1.70
2-Hydroxyphenazine	Britton-Robinson buffer, pH 4.0	− 0.24
8-Hydroxyquinoline	B, pH 5.0 Phosphate buffer, pH 8.0	− 1.12 − 1.18, − 1.71
3-Methylpyridine	D+0.1 <i>M</i> LiCl	− 1.76
4-Methylpyridine	D+0.1 <i>M</i> LiCl	− 1.87
Phenazine	Phosphate buffer+citrate buffer, pH 7.0	− 0.36
Pyridine	Phosphate buffer+citrate buffer, pH 7.0	− 1.75
Pyridine-2-carboxylic acid	B, pH 4.1	− 1.10
	B, pH 9.3	− 1.48, − 1.94
Pyridine-3-carboxylic acid	0.1 <i>M</i> HCl	− 1.08
Pyridine-4-carboxylic acid	Britton-Robinson buffer, pH 6.1	− 1.14
	pH 9.0	− 1.39, − 1.68
Pyrimidine	Citrate buffer, pH 3.6 Ammonia buffer, pH 9.2	− 0.92, − 1.24 − 1.54
Quinoline-8-carboxylic acid	pH 9	− 1.11
Quinoxaline	Phosphate buffer+citrate buffer, pH 7.0	− 0.66, − 1.52
Azo, hydrazine, hydroxylamine, and oxime compounds		
Azobenzene	D, pH 4.0	− 0.20
	D, pH 7.0	− 0.50
Azoxybenzene	Buffer+20% EtOH, pH 6.3	− 0.30
Benzoin 1-oxime	Buffer, pH 2.0	− 0.88
	Buffer, pH 5.6	− 1.08
	Buffer, pH 8.2	− 1.67
Benzoylhydrazine	0.13 <i>M</i> NaOH, pH 13.0	− 0.30
Dimethylglyoxime	Ammonia buffer, pH 9.6	− 1.63
Hydrazine	Britton-Robinson buffer, pH 9.3	− 0.09
Hydroxylamine	Britton-Robinson buffer, pH 4.6 Britton-Robinson buffer, pH 9.2	− 1.42 − 1.65

TABLE 8.31 Half-Wave Potentials (vs. Saturated Calomel Electrode) of Organic Compounds at 25°C
(Continued)

Compound	Solvent system	$E_{1/2}$
Azo, hydrazine, hydroxylamine, and oxime compounds (<i>continued</i>)		
Oxamide	Acetate buffer	−1.55
Phenylhydrazine	McIlvaine buffer, pH 2	+0.19
	0.13 M NaOH, pH 13.0	−0.36
Phenylhydroxylamine	McIlvaine buffer+10% EtOH, pH 2	−0.68
	McIlvaine buffer+10 EtOH, pH 4–10	−0.33 0.061 pH
Salicylaldoxime	Phosphate buffer, pH 5.4	−1.02
Thiosemicarbazide	Borate buffer, pH 9.3	−0.26
Thiourea	0.1 M sulfuric acid	+0.02
Indicators and dyestuffs		
Brilliant Green	HCl+KCl, pH 2.0	−0.2, −0.5
Indigo carmine	pH 2.5	−0.24
Indigo disulfonate	pH 7.0	−0.37
Malachite Green G	HCl+KCl, pH 2.0	−0.2, −0.5
Metanil yellow	Phosphate buffer+1% EtOH, pH 7.0	−0.51
Methylene blue	Britton-Robinson buffer, pH 4.9	−0.15
	Britton-Robinson buffer, pH 9.2	−0.30
Methylene green	Phosphate buffer+1% EtOH, pH 7.0	−0.12
Methyl orange	Phosphate buffer+1% EtOH, pH 7.0	−0.51
Morin	D, pH 7.6	−1.7
Neutral red	Britton-Robinson buffer, pH 2.0	−0.21
	Britton-Robinson buffer, pH 7.0	−0.57
Peroxide		
Ethyl peroxide	0.02 M HCl	−0.2

8.7 CONDUCTANCE

TABLE 8.32 Limiting Equivalent Ionic Conductances in Aqueous Solutions

In $10^{-4} \text{ m}^2 \cdot \text{S} \cdot \text{equiv}^{-1}$ or $\text{mho} \cdot \text{cm}^2 \cdot \text{equiv}^{-1}$.

Ion	Temperature, °C		
	0	18	25
Inorganic cations			
Ag ⁺	33	54.5	61.9
Al ³⁺	29		61
Ba ²⁺	33.6	54.3	63.9
Be ²⁺			45
Ca ²⁺	30.8	51	59.5
Cd ²⁺	28	45.1	54
Ce ³⁺			70
Co ²⁺	28	45	53
Co(NH ₃) ₆ ³⁺			100
Co(ethylenediamine) ₃ ³⁺			74.7
Cr ³⁺			67
Cs ⁺	44	68	77.3
Cu ²⁺	28	45.3	56.6
D ⁺ (deuterium)		213.7	
Dy ³⁺			65.7
Er ³⁺			66.0
Eu ³⁺			67.9
Fe ²⁺	28	45.3	53.5
Fe ³⁺			69
Gd ³⁺			67.4
H ⁺	224.1	315.8	350.1
Hg ₂ ²⁺			68.7
Hg ²⁺			63.6
Ho ³⁺			66.3
K ⁺	40.3	64.6	73.5
La ³⁺	35.0	59.2	69.6
Li ⁺	19.1	33.4	38.69
Mg ²⁺	28.5	46	53.06
Mn ²⁺	27	44.5	53.5
NH ₄ ⁺	40.3	64	73.7
N ₂ H ₅ ⁺ (hydrazinium 1 +)			59
Na ⁺	25.85	43.5	50.11
Nd ³⁺			69.6
Ni ²⁺	28	45	50
Pb ²⁺	37.5	60.5	71
Pr ³⁺			69.6
Ra ²⁺	33	56.6	66.8
Rb ⁺	43.5	67.5	77.8
Sc ³⁺			64.7
Sm ³⁺			68.5
Sr ²⁺	31	51	59.46
Tl ⁺	43.3	66	74.9
Tm ³⁺			65.5
UO ₂ ²⁺			32
Y ³⁺			62
Yb ³⁺			65.2
Zn ²⁺	28	45.0	52.8

TABLE 8.32 Limiting Equivalent Ionic Conductances in Aqueous Solutions (Continued)

Ion	Temperature, °C		
	0	18	25
Inorganic anions			
Au(CN) ₂ ⁻			50
Au(CN) ₄ ⁻			36
B(C ₆ H ₅) ₄ ⁻			21
Br ⁻	43.1	67.6	78.1
Br ₃ ⁻			43
BrO ₃ ⁻	31.0	49.0	55.7
Cl ⁻	41.4	65.5	76.31
ClO ₂ ⁻			52
ClO ₃ ⁻	36	55.0	64.6
ClO ₄ ⁻	37.3	59.1	67.3
CN ⁻			78
CO ₃ ²⁺	36	60.5	69.3
Co(CN) ₆ ³⁻			98.9
CrO ₄ ²⁻	42	72	85
F ⁻		46.6	55.4
Fe(CN) ₆ ⁴⁻			110.4
Fe(CN) ₆ ³⁻			100.9
H ₂ AsO ₄ ⁻			34
HCO ₃ ⁻			44.5
HF ₂ ⁻			75
HPO ₄ ²⁻			33
H ₂ PO ₄ ⁻		28	33
HS ⁻	40	57	65
HSO ₃ ⁻	27		50
HSO ₄ ⁻			50
H ₂ SbO ₄ ⁻			31
I ⁻	42.0	66.5	76.9
IO ₃ ⁻	21.0	33.9	40.5
IO ₄ ⁻		49	54.5
MnO ₄ ⁻	36	53	61.3
MoO ₄ ²⁻			74.5
N ₃ ⁻			69.5
N(CN) ₂ ⁻			54.5
NO ₂ ⁻	44	59	71.8
NO ₃ ⁻	40.2	61.7	71.42
NH ₂ SO ₃ ⁻ (sulfamate)			48.6
OCN ⁻ (cyanate)		54.8	64.6
OH ⁻	117.8	175.8	198
PF ₆ ⁻			56.9
PO ₃ F ²⁻			63.3
PO ₄ ³⁻			69.0
P ₂ O ₇ ⁴⁻			96
P ₃ O ₉ ³⁻			83.6
P ₃ O ₁₀ ⁵⁻			109
ReO ₄ ⁻		46.5	54.9
SCN ⁻ (thiocyanate)	41.7	56.6	66.5
SeCN ⁻			64.7
SeO ₄ ²⁻		65	75.7
SO ₃ ⁻			79.9

TABLE 8.32 Limiting Equivalent Ionic Conductances in Aqueous Solutions (*Continued*)

Ion	Temperature, °C		
	0	18	25
SO_4^{2-}	41	68.3	80.0
$\text{S}_2\text{O}_3^{2-}$			85.0
$\text{S}_2\text{O}_4^{2-}$	34		66.5
$\text{S}_2\text{O}_6^{2-}$			93
$\text{S}_2\text{O}_8^{2-}$			86
WO_4^{2-}	35	59	69.4
Organic cations			
Decylpyridinium ⁺			29.5
Diethylammonium ⁺			42.0
Dimethylammonium ⁺			51.5
Dipropylammonium ⁺			30.1
Dodecylammonium ⁺			23.8
Ethylammonium ⁺			47.2
Ethyltrimethylammonium ⁺			40.5
Isobutylammonium ⁺			38.0
Methylammonium ⁺			58.3
Piperidinium ⁺			37.2
Propylammonium ⁺			40.8
Tetrabutylammonium ⁺			19.5
Tetraethylammonium ⁺			32.6
Tetramethylammonium ⁺			44.9
Tetrapropylammonium ⁺			23.5
Triethylsulfonium ⁺			36.1
Trimethylammonium ⁺			47.2
Trimethylsulfonium ⁺			51.4
Tripropylammonium ⁺			26.1
Organic anions			
Acetate ⁻	20	34	41
Benzoate ⁻			32.4
Bromoacetate ⁻			39.2
Bromobenzoate ⁻			30
Butanoate ⁻			32.6
Chloroacetate ⁻			42.2
<i>m</i> -Chlorobenzoate ⁻			31
<i>o</i> -Chlorobenzoate ⁻			30.5
Citrate(3 ⁻)			70.2
Crotonate ⁻			33.2
Cyanoacetate ⁻			43.4
Cyclohexanecarboxylate ⁻			28.7
Cyclopropane-1,3-dicarboxylate ²⁻			53.4
Decylsulfonate ⁻			26
Dichloroacetate ⁻			38.3
Diethylbarbiturate(2 ⁻)			26.3
Dihydrogencitrate ⁻			30
Dimethylmalonate(2 ⁻)			49.4
3,5-Dinitrobenzoate ⁻			28.3
Dodecylsulfonate ⁻			24
Ethylmalonate ⁻			49.3
Ethylsulfonate ⁻			39.6

TABLE 8.32 Limiting Equivalent Ionic Conductances in Aqueous Solutions (*Continued*)

Ion	Temperature, °C		
	0	18	25
Fluoroacetate [−]			44.4
Fluorobenzoate [−]			33
Formate [−]		47	54.6
Fumarate(2 [−])			61.8
Glutarate(2 [−])			52.6
Hydrogenoxalate (1 [−])			40.2
Iodoacetate [−]			40.6
Lactate(1 [−])			38.8
Malate(2 [−])			58.8
Malonate(1 [−])			63.5
3-Methylbutanoate [−]			32.7
Methylsulfonate [−]			48.8
Naphthylacetate [−]			28.4
1,8-Octanedioate(2 [−])			36
Octylsulfonate [−]			29
Oxalate(2 [−])			74.11
Phenylacetate [−]			30.6
<i>m</i> -Phthalate(2 [−])			54.7
<i>o</i> -Phthalate(2 [−])			52.3
Picrate [−]			30.37
Propanoate [−]			35.8
Propylsulfonate [−]			37.1
Salicylate [−]			36
Succinate(2 [−])			58.8
Tartrate(2 [−])		55	59.6
Trichloroacetate [−]			36.6
Trimethylacetate [−]			31.9

TABLE 8.33 Standard Solutions for Calibrating Conductivity Vessels

The values of conductivity κ are corrected for the conductivity of the water used. The cell constant θ of a conductivity cell can be obtained from the equation

$$\theta = \frac{\kappa R R_{\text{solv}}}{R_{\text{solv}} - R}$$

where R is the resistance measured when the cell is filled with a solution of the composition stated in the table below, and R_{solv} is the resistance when the cell is filled with solvent at the same temperature.

Grams KCl per Kilogram Solution (in vacuo)	Conductivity in $\text{ohm}^{-1} \cdot \text{cm}^{-1}$ at		
	0°C	18°C	25°C
71.135 2	0.065 14 ₄	0.097 79 ₀	0.111 28 ₇
7.419 13	0.007 134 ₄	0.011 161 ₂	0.012 849 ₇
0.745 263*	0.000 773 2 ₆	0.001 219 9 ₂	0.001 408 0 ₈

* Virtually 0.0100 *M*.
From the data of Jones and Bradshaw, *J. Am. Chem. Soc.*, **55**, 1780 (1933). The original data have been converted from (int. ohm)^{−1} cm^{−1}.

TABLE 8.34 Electrical Conductivity of Various Pure Liquids

Liquid	Temp. °C	mhos/cm or $\text{ohm}^{-1} \cdot \text{cm}^{-1}$	Liquid	Temp. °C	mhos/cm or $\text{ohm}^{-1} \cdot \text{cm}^{-1}$
Acetaldehyde	15	1.7×10^{-6}	Epichlorohydrin	25	3.4×10^{-8}
Acetamide	100	$< 4.3 \times 10^{-5}$	Ethyl acetate	25	$< 1 \times 10^{-9}$
Acetic acid	0	5×10^{-9}	Ethyl acetoacetate	25	4×10^{-8}
	25	1.12×10^{-8}	Ethyl alcohol	25	1.35×10^{-9}
Acetic anhydride	0	1×10^{-6}	Ethylamine	0	4×10^{-7}
	25	4.8×10^{-7}	Ethyl benzoate	25	$< 1 \times 10^{-9}$
Acetone	18	2×10^{-8}	Ethyl bromide	25	$< 2 \times 10^{-8}$
	25	6×10^{-8}	Ethylene bromide	19	$< 2 \times 10^{-10}$
Acetonitrile	20	7×10^{-6}	Ethylene chloride	25	3×10^{-8}
Acetophenone	25	6×10^{-9}	Ethyl ether	25	$< 4 \times 10^{-13}$
Acetyl bromide	25	2.4×10^{-6}	Ethylidene chloride	25	$< 1.7 \times 10^{-8}$
Acetyl chloride	25	4×10^{-7}	Ethyl iodide	25	$< 2 \times 10^{-8}$
Alizarin	233	$1.45 \times 10^{-6} (?)$	Ethyl isothiocyanate	25	1.26×10^{-7}
Allyl alcohol	25	7×10^{-6}	Ethyl nitrate	25	5.3×10^{-7}
Ammonia	-79	1.3×10^{-7}	Ethyl thiocyanate	25	1.2×10^{-6}
Aniline	25	2.4×10^{-8}	Eugenol	25	$< 1.7 \times 10^{-8}$
Anthracene	230	3×10^{-10}			
Arsenic tribromide	35	1.5×10^{-6}	Formamide	25	4×10^{-6}
Arsenic trichloride	25	1.2×10^{-6}	Formic acid	18	5.6×10^{-5}
				25	6.4×10^{-5}
Benzaldehyde	25	1.5×10^{-7}	Furfural	25	1.5×10^{-6}
Benzene	...	7.6×10^{-8}			
Benzoic acid	125	3×10^{-9}	Gallium	30	36,800
Benzonitrile	25	5×10^{-8}	Glycerol	25	6.4×10^{-8}
Benzyl alcohol	25	1.8×10^{-6}	Glycol	25	3×10^{-7}
Benzylamine	25	$< 1.7 \times 10^{-8}$	Guaiacol	25	2.8×10^{-7}
Benzyl benzoate	25	$< 1 \times 10^{-9}$			
Bromine	17.2	1.3×10^{-13}	Heptane	...	$< 1 \times 10^{-13}$
Bromobenzene	25	$< 2 \times 10^{-11}$	Hexane	18	$< 1 \times 10^{-18}$
Bromoform	25	$< 2 \times 10^{-8}$	Hydrogen bromide	-80	8×10^{-9}
iso-Butyl alcohol	25	8×10^{-8}	Hydrogen chloride	-96	1×10^{-8}
			Hydrogen cyanide	0	3.3×10^{-6}
Capronitrile	25	3.7×10^{-6}	Hydrogen iodide	B.P.	2×10^{-7}
Carbon disulfide	1	7.8×10^{-18}	Hydrogen sulfide	B.P.	1×10^{-11}
Carbon tetrachloride	18	4×10^{-18}			
Chlorine	-70	$< 1 \times 10^{-16}$	Iodine	110	1.3×10^{-10}
Chloroacetic acid	60	1.4×10^{-6}			
m-Chloroaniline	25	5×10^{-8}	Kerosene	25	$< 1.7 \times 10^{-8}$
Chloroform	25	$< 2 \times 10^{-8}$			
Chlorohydrin	25	5×10^{-7}	Mercury	0	10,629.6
m-Cresol	25	$< 1.7 \times 10^{-8}$	Methyl acetate	25	3.4×10^{-6}
Cyanogen	...	$< 7 \times 10^{-9}$	Methyl alcohol	18	4.4×10^{-7}
Cymene	25	$< 2 \times 10^{-8}$	Methyl ethyl ketone	25	1×10^{-7}
			Methyl iodide	25	$< 2 \times 10^{-8}$
Dichloroacetic acid	25	7×10^{-8}	Methyl nitrate	25	4.5×10^{-6}
Dichlorohydrin	25	1.2×10^{-5}	Methyl thiocyanate	25	1.5×10^{-6}
Diethylamine	-33.5	2.2×10^{-9}			
Diethyl carbonate	25	1.7×10^{-8}	Naphthalene	82	4×10^{-10}
Diethyl oxalate	25	7.6×10^{-7}	Nitrobenzene	0	5×10^{-9}
Diethyl sulfate	25	2.6×10^{-7}	Nitromethane	18	6×10^{-7}
Dimethyl sulfate	0	1.6×10^{-7}	o- or m-Nitrotoluene	25	$< 2 \times 10^{-7}$
			Nonane	25	$< 1.7 \times 10^{-8}$

TABLE 8.34 Electrical Conductivity of Various Pure Liquids (*Continued*)

Liquid	Temp. °C	mhos/cm or ohm ⁻¹ · cm ⁻¹	Liquid	Temp. °C	mhos/cm or ohm ⁻¹ · cm ⁻¹
Oleic acid	15	<2 × 10 ⁻¹⁰	Salicylaldehyde	25	1.6 × 10 ⁻⁷
Pentane	19.5	<2 × 10 ⁻¹⁰	Stearic acid	80	<4 × 10 ⁻¹³
Petroleum	...	3 × 10 ⁻¹³	Sulfonyl chloride, SOCl ₂	25	2 × 10 ⁻⁶
Phenetole	25	<1.7 × 10 ⁻⁸	Sulfur	115	1 × 10 ⁻¹²
Phenol	25	<1.7 × 10 ⁻⁸		130	5 × 10 ⁻¹²
Phenyl isothiocyanate	25	1.4 × 10 ⁻⁶		440	1.2 × 10 ⁻⁷
Phosgene	25	7 × 10 ⁻⁹	Sulfur dioxide	35	1.5 × 10 ⁻⁸
Phosphorus	25	4 × 10 ⁻⁷	Sulfuric acid	25	1 × 10 ⁻²
Phosphorus oxychloride	25	2.2 × 10 ⁻⁶	Sulfuryl chloride, SO ₂ Cl ₂	25	3 × 10 ⁻⁸
Pinene	23	<2 × 10 ⁻¹⁰			
Piperidine	25	<2 × 10 ⁻⁷	Toluene	...	<1 × 10 ⁻¹⁴
Propionaldehyde	25	8.5 × 10 ⁻⁷	<i>o</i> -Toluidine	25	<2 × 10 ⁻⁶
Propionic acid	25	<1 × 10 ⁻⁹	<i>p</i> -Toluidine	100	6.2 × 10 ⁻⁸
Propionitrile	25	<1 × 10 ⁻⁷	Trichloroacetic acid	25	3 × 10 ⁻⁹
<i>n</i> -Propyl alcohol	18	5 × 10 ⁻⁸	Trimethylamine	-33.5	2.2 × 10 ⁻¹⁰
	25	2 × 10 ⁻⁸	Turpentine	...	2 × 10 ⁻¹³
<i>iso</i> -Propyl alcohol	25	3.5 × 10 ⁻⁶	<i>iso</i> -Valeric acid	80	<4 × 10 ⁻¹³
<i>n</i> -Propyl bromide	25	<2 × 10 ⁻⁸			
Pyridine	18	5.3 × 10 ⁻⁸	Water	18	4 × 10 ⁻⁸
Quinoline	25	2.2 × 10 ⁻⁸	Xylene	...	<1 × 10 ⁻¹⁵

TABLE 8.35 Equivalent Conductivities of Electrolytes in Aqueous Solutions at 18°C

The unit of Λ in the table is $\Omega^{-1} \cdot \text{cm}^{-2} \cdot \text{equiv}^{-1}$. The entities to which the equivalent relates are given in the first column.

Electrolyte	Concentration, <i>N</i>										
	0.001	0.005	0.01	0.05	0.1	0.5	1.0	2.0	3.0	4.0	5.0
Acetic acid	41	20.0	14.3	6.48	4.60	2.01	1.32		0.54		0.29
AgNO ₃	113.2	110.0	107.8	99.5	94.3	77.8	67.8	56.0	48.2	42.1	37.2
½Ag ₂ SO ₄	116.3	108.4	102.9								
⅓AlBr ₃ (25°)	132	124	119	103	97						
⅓AlCl ₃	121.1	105.0	93.8			65.0	56.2	44.2	34.7	27.2	
⅓AlI ₃ (25°)	131	124	119	108							
⅓Al(NO ₃) ₃ (25°)	123	115	110	94	88						
⅙Al ₂ (SO ₄) ₃ (25°)	107.2	76.8	60.6								
½Ba(OAc) ₂	85.0	80.4	77.1	65.7	60.2	43.8	34.3				
½Ba(BrO ₃) ₂ (25°)	113.6	106.8	102.7								
½BaCl ₂	115.6	112.3	106.7	96.0	90.8	77.3	70.1	60.3	52.3		
½Ba(NO ₃) ₂	111.7	105.3	101.0	86.8	78.9	56.6	48.4		29.8	23.4	
½Ba(OH) ₂	216	213	207	191	180						
Butyric acid						1.66	0.98	0.46	0.26	0.18	0.11
½Ca(OAc) ₂	79.6	75.0	71.9	60.3	54.0	36.3	26.3				
½CaCl ₂	112.0	106.7	103.4	93.3	88.2	74.9	67.5	58.3	49.7	42.4	35.6
½Ca(NO ₃) ₂	108.5	103.0	99.5	88.4	82.5	65.7	55.9	43.5	35.5	26.0	21.5
½Ca(OH) ₂		233	226								
½CaSO ₄	104.3	86.3	77.4								
½CdBr ₂		86.5	76.3	53.2	44.6	25.3	18.3	12.5	9.1	6.8	5.3
½CdCl ₂		91	83	59	50	30.8	22.4	14.4	9.9	7.1	5.4
½CdI ₂		76.7	65.6	40.1	31.0	18.3	15.4	12.3	9.7	8.0	
½Cd(NO ₃) ₂		100	96	86.4	80.8	63.9	54.5	41.0	31.4	23.7	17.6
½CdSO ₄	97.7	79.7	70.3	49.6	42.2	28.7	23.6	17.7	14.0	11.0	8.35
⅓CeCl ₃ (25°)	137.4		122.1		99.0						
⅙Ce ₂ (C ₂ O ₄) ₃ (25°)	85.5	54	45.8	29							
Chloroacetic acid (25°)					42.9	20.2	13.6	8.1	5.6	4.2	3.3
Citric acid	88.4	54	42.5	22.0	16.1	7.3	5.4				
½CoCl ₂		99.3	95.6	82.3	75.0	51.5	45.3	40.3	35.4	30.5	26.4
⅓CrCl ₃						68.6	56.8	44.8	35.2		

TABLE 8.35 Equivalent Conductivities of Electrolytes in Aqueous Solutions at 18°C (*Continued*)

Electrolyte	Concentration, <i>N</i>										
	0.001	0.005	0.01	0.05	0.1	0.5	1.0	2.0	3.0	4.0	5.0
½CrO ₃ (H ₂ CrO ₄) (25°)	201	195	193	191	186						
CsCl	130.7	127.5	125.2		113.5	104.3	100.3	95.7	85.1		
½Cu(OAc) ₂ (25°)	55.7	50.6	47.2	34.9	28.4						
½CuCl ₂								41.2	31.5	24.5	19.1
½Cu(NO ₃) ₂ (15°)	107.9	97.1	93.7	83.7	78.2	67.5	56.8	45.4	35.3	27.8	21.4
½CuSO ₄	98.5	81.0	71.7	53.6	43.8	30.5	25.6	19.7	16.5		
Dichloroacetic acid (25°)					207.5	119	82	44.6	26.5	16.3	9.6
½FeCl ₂ (25°)	131	125	120	103	93						
½FeCl ₃						66.5	52.9	37.6	28.1	20.5	15.9
½FeSO ₄	82	75	70	54	44.5	30.8	25.8	19.5	15.37		
Formic acid	125.6						5.18	3.68	2.93	2.39	1.92
H ₃ AsO ₄ (1 <i>M</i>) (25°)	308.2	230.0	187.0	103.4	80.4						
H ₃ BO ₃	13.5										
HBr					356	306	282	243	214	179	
HBrO ₃ (25°)	401	387	373	272	156						
HCl	377	373	370	360	351	327	301		215		152.2
HClO ₃					343	317	292	247	207		
HClO ₄ (25°)	413	406	402	392	386	358					
HF		90	60	35.9	31.3	27.0	25.7		24.2		24.0
HI					347	322	297	255	215	179	
HIO ₃	343.3	332.8	323.9		253	175	141	106	87	71	
HNO ₃	375	371	368	357	350	324	310		220		156
H ₃ PO ₄ (1 <i>M</i>)	318	279	255				66		53.1		51.3
HSCN (25°)	399	394	390	377	370						
½H ₂ SO ₄	361	330	308	253	225	205	198		166.8		135.0
½HgCl ₂				1.85	1.23						
½InBr ₃					53.9	37.0	28.7	19.8	14.4	10.1	
KOAc	98.3	95.7	94.0	87.7	83.8	71.6	63.4	50.0	40.7	31.4	24.5
KBr	129.4	126.4	124.4	117.8	114.2	105.4	102.5	98.0	93.3	87.9	
KBrO ₃	109.9	106.9	104.7	97.3	93.0						
½K ₃ citrate		109.9	103	87.8	80.8						

KCl	127.3	124.4	122.4	115.8	112.0	102.4	98.3	92.0	88.9		
KClO ₃	116.9	113.6	111.6	103.7	99.2	85.3					
KClO ₄ (25°)	137.9	134.2	131.5	121.6	115.2						
KCN (15°)						104.2	99.7				
½K ₂ CO ₃	133.0	121.6	115.5	100.7	94.1	77.8	70.7	65.0	55.6	49.2	42.9
½K ₂ C ₂ O ₄	122.4	116.7	112.5	100.8	94.9	80.4	73.7				
½K ₂ CrO ₄					100.5	86.4	79.5	72.0	59.9		
½K ₂ Cr ₂ O ₇					98.2	85.4					
KF	108.9	106.2	104.3	97.7	94.0	82.6	76.0	63.4	56.5	51.7	46.5
⅓K ₃ [Fe(CN) ₆]	163.1	150.7									
¼K ₄ [Fe(CN) ₆]	167.2	146.1	134.8	107.7	97.9						
KHCO ₃ (25°)	115.3	112.2	110.1			86.5	78.9				
KH phthalate	119.3	103.7	99.9	89.3	83.8						
KHS						92.5	91.7	86.4	80.7		69.3
KHSO ₄						21.0	18.4	15.2			
KH ₂ PO ₄ (1 M) (25°)	107.1	100.8	98.0	90.7	85.6	60.0 ¹⁸	45.8 ¹⁸				
KI	128.2	125.3	123.4	117.3	114.0	106.2	103.6	101.3	96.4	89.0	81.2
KIO ₃	96.0	93.2	91.2	84.1	79.7						
KIO ₄ (25°)	124.9	121.2	118.5	106.7	98.1						
KMnO ₄ (25°)	133.3		126.5		113						
KNO ₃	123.6	120.5	118.2	109.9	104.8	89.2	80.5	69.4	61.3		
KOH	234	230	228	219	213	197	184		140.6		105.8
KReO ₄ (25°)	125.1	121.3	118.5	106.4	97.4						
½K ₂ S							135.6	119.7	108.3	97.2	86.1
KSCN	118.6	115.8	113.9	107.7	104.3	95.7	91.6	86.8	74.6		
½K ₂ SO ₄	126.9	120.3	115.8	101.9	94.9	78.5	71.6				
½LaCl ₃ (25°)	137.0	127.5	121.8	106.2	99.1						
⅓La(NO ₃) ₃				86.1	72.1	65.4	54.0	39.1	28.5	19.9	
⅙La ₂ (SO ₄) ₃				25.7	21.5						
Lactic acid	108.9	53.5	39	18.1	13.2						
LiOAc					51.3	37.7	28.9	18.2	11.9	7.2	
LiBr				87.9	84.4	73.9	67.2	57.7		44.2	
LiCl	96.5	93.9	92.1	86.1	82.4	70.7	63.4	53.1	45.3		33.3
LiClO ₄ (25°)	103.4	100.6	98.6	92.2	88.6						
½Li ₂ CO ₃				64.2	59.1						
LiI						75.3	69.2	61.0			
LiIO ₃	65.3	62.9	61.2	55.3	51.5	39.0	31.2	21.4	14.6		

TABLE 8.35 Equivalent Conductivities of Electrolytes in Aqueous Solutions at 18°C (*Continued*)

Electrolyte	Concentration, <i>N</i>										
	0.001	0.005	0.01	0.05	0.1	0.5	1.0	2.0	3.0	4.0	5.0
LiNO ₃	92.9	90.3	88.6	82.7	79.2	68.0	60.8	50.3	34.9	27.3	
LiOH						149.0	134.5	113.5	95.7		
½Li ₂ SO ₄	96.4		86.9	74.7	68.2	50.5	41.3	30.7	23.3	18.1	13.9
½MgCl ₂	106.4	101.3	98.1	88.5	83.4	69.6	61.5	52.3	43.3	35.0	28.0
½Mg(NO ₃) ₂	102.6	97.7	94.7	85.3	80.5	67.0	59.0	47.0	39.8		
½MgSO ₄	99.8	84.5	76.2	56.9	49.7	35.4	28.9	23.0	17.3	12.9	9.3
½MnCl ₂					86.0	68.5	61.0	48.5	38.8	30.2	23.0
½MnSO ₄						27.6	24.4	18.3	14.0	10.5	7.3
NH ₃ (aq)	28.0	13.2	9.6	4.6	3.3	1.35	0.89		0.36		0.20
NH ₄ OAc		92.9	91.4	84.9		60.5	54.7	42.9	34.0	26.5	
NH ₄ Cl	127.3	124.3	122.1	115.2	110.7	101.4	97.0	92.1	88.2	85.0	80.7
NH ₄ F					90.1	74.5	65.7	55.3	47.9	42.2	
NH ₄ I				118.0	115.0	106.0	103.1	100.0		91.4	84.5
NH ₄ NO ₃	124.5		118.0	110.0	106.6	94.5	88.8	85.1		71.9	47.6
NH ₄ SCN					104.3	94.0	89.9	84.7	79.2	74.0	
½(NH ₄) ₂ SO ₄		120.0	116.5		89.0	79.5	73.0	65.0		55.2	
NaOAc	75.2	72.4	70.2	64.2	61.1	49.4	41.2	29.8	21.5	15.3	10.5
NaBr				99.1	96.0	84.6	78.1	69.1		53.0	
NaBrO ₃						61.8	54.5	44.1			
Na <i>n</i> -butyrate (25°)	80.3	77.6	75.8	69.3	65.3						
NaCl	106.5	103.8	102.0	95.7	92.0	80.9	74.3	64.8	56.5	49.4	42.7
NaClO ₄	114.9 ²⁵	111.7 ²⁵	109.6 ²⁵	102.4 ²⁵	98.4 ²⁵	71.7	65.0	55.1	46.0	38.8	
½Na ₂ CO ₃	112	102.5	96.2	80.3	72.9	54.5	45.5	34.5	27.2		
½Na ₂ CrO ₄					82.5	66.4	57.7	46.6	38.3	31.1	
½Na ₂ Cr ₂ O ₇ (25°)		103		98.3	94.9						
NaF	87.8	85.2	83.5	77.0	73.1	60.0	51.9				
¼Na ₄ [Fe(CN) ₆] (25°)		129.6	120.0	97.0	88.2						
Na formate	88.6					61.4	53.7	43.1	34.8	28.2	
NaHCO ₃ (25°)	93.5	90.5	88.4	80.6	76.0						
⅓Na ₂ HPO ₄	58.4		54.0		44.0	33.5	28.0				
NaH ₂ PO ₄	67.9	65.8	64.4	57.8	54.1						
¼Na ₂ H ₂ P ₂ O ₇	41.1	39.4	38.2	34.6	32.5	25.4					
NaI	124.2	121.2	119.2	112.8	108.8	97.5	89.9	78.6	69.9	62.2	

NaIO ₃	75.2	72.6	70.9	64.4	60.5						
½Na ₂ MoO ₄	120.8	113	110								
NaN ₃ (25°)	117.1	113.8	110.5	101.3	95.7		68.0				
NaNO ₂ (25°)							75.9	63.1	53.6		39.7
NaNO ₃	102.9	100.1	98.2	91.4	87.2	74.1	65.9	54.5	46.0	39.0	
NaOH	208	203	200	190	183	172	160		108.0		69.0
Na picrate (25°)	78.6	75.7	73.7	66.3	61.8						
⅓Na ₃ PO ₄	125	122	119	91							
Na propionate (25°)	83.5	80.9	79.1								
½Na ₂ S						117.0	104.3	85.0	71.0	59.0	47.2
NaSCN						74.3	68.9	59.8	50.9	43.7	
½Na ₂ SiO ₃	144	139	136	124	116	88	72	51	38	27	19
½Na ₂ SO ₄	106.7	100.8	96.8	83.9	78.4	59.7	50.8	40.0	33.5		
(mono) Na tartrate	120	81.5	74.8	64.3	60.4						
½Na ₂ WO ₄ (25°)	116.1	109.2	104.8	92.2	85.8						
½NiSO ₄	96.3	79.5	70.8	51.0	43.8	30.4	25.1	19.3	15.1		
½Oxalic acid	180.7		158.2	132.9	116.9	75.9	59.4				
½Pb(NO ₃) ₂	116.1	108.6	103.5	86.3	77.3	53.2	42.0	31.0			
Propionic acid						1.57	1.00	0.54		0.20	
RbCl	130.3	127.4	125.3	117.8	113.9		101.9	97.1	92.7	87.2	
RbOH					220.6		192.0	170.0	148.3		
¼SnCl ₄						204.8	121.7	66.9	47.9	32.7	
½SrCl ₂	114.5	108.9	105.4	94.4	90.2	75.7	68.5	58.7	49.9	42.2	
½Sr(NO ₃) ₂	108.3	102.7	99.0	87.3	80.9	62.7	52.1	38.0	29.3	29.3	16.4
Tartaric acid (15°)							7.03	4.58	3.32	2.48	1.83
¼ThCl ₄						61.0	54.0	44.3	36.3	29.8	
TiCl	128.2	123.7	120.2								
TiF	113.3	108.2	105.4	97.4	92.6	78.8	71.5	62.7			
TiNO ₃	124.7	121.1	118.4	107.9	101.2						
½Ti ₂ SO ₄	127.4	118.4	112.3	92.7	83.1						
Trichloroacetic acid (25°)						273	207	127	79	44	19
½UO ₂ F ₂ (25°)	26.10	12.31	9.17	5.43	4.74	3.75	3.22				
½UO ₂ SO ₄ (25°)	106.5	63.2	49.2	27.6	22.2	14.4	11.6				2.7
⅓YCl ₃ (25°)	129	122	118	109							
½Zn(OAc) ₂ (25°)	83	77	73	58	49						
½ZnCl ₂	107	101	98	87	82	65	55	39.6	29.6	23.2	18.5
½Zn(NO ₃) ₂	120	114	111	100							
½ZnSO ₄	98.4	82.1	73.2	53.0	45.6	32.3	26.6	20.0	15.9	12.0	9.0

TABLE 8.36 Conductivity of Very Pure Water at Various Temperatures and the Equivalent Conductances of Hydrogen and Hydroxyl Ions

Temp., °C	Conductivity, $\mu\text{S} \cdot \text{cm}^{-1}$	Resistivity, $\text{M}\Omega \cdot \text{cm}$	Equivalent conductance, $\text{cm}^2 \cdot \text{ohm}^{-1} \cdot \text{equivalent}^{-1}$	
			λ^0, H^+	λ^0, OH^-
0	0.011 61	86.14	224.1	117.8
5	0.016 61	60.21	250.0	133.6
10	0.023 15	43.21	275.6	149.6
15	0.031 53	31.71	300.9	165.9
18	0.037 54	26.64	315.8	491.6
20	0.042 05	23.78	325.7	182.5
25	0.055 08	18.15	350.1	199.2
30	0.070 96	14.09	374.0	216.1
35	0.090 05	11.10	397.4	233.0
40	0.112 7	8.88	420.0	267.2
45	0.139 3	7.18	442.0	267.2
50	0.170 2	5.88	463.3	284.3
55	0.205 5	4.86	483.8	301.4
60	0.245 7	4.06	503.4	318.5
65	0.291 2	3.43	522.0	335.4
70	0.341 6	2.93	539.7	352.2
75	0.397 8	2.51	556.4	368.8
80	0.459 3	2.18	572.0	385.2
85	0.525 8	1.90	586.4	401.4
90	0.597 7	1.67	599.6	417.3
95	0.675 3	1.48	611.6	432.8
100	0.756 9	1.32	622.2	448.1
150	1.84	0.543		
200	2.99	0.334	824	701
250	3.31	0.302		
300	2.42	0.413	894	821

Source: Data from T. S. Light and S. L. Licht, *Anal. Chem.*, **59**:2327–2330 (1987).

8.7.1 Common Conductance Relations*

Conductivity. The standard unit of conductance is electrolytic conductivity (formerly called specific conductance) κ , which is defined as the reciprocal of the resistance [Ω^{-1}] of a 1-m cube of liquid at a specified temperature [$\Omega^{-1} \cdot \text{m}^{-1}$]. See Table 8.33 and the definition of the cell constant.

In accurate work at low concentrations it is necessary to subtract the conductivity of the pure solvent (Table 8.34) from that of the solution to obtain the conductivity due to the electrolyte.

Resistivity (Specific Resistance)

$$\rho = \frac{1}{\kappa} \quad [\Omega \cdot \text{m}]$$

* SI units are in brackets.

Conductance of an Electrolyte Solution

$$\frac{1}{R} = \kappa \frac{S}{d} \quad [\Omega^{-1}]$$

where S is the surface area of the electrode, or the mean cross-sectional area of the solution [m^2], and d is the mean distance between the electrodes [m].

Equivalent Conductivity

$$\Lambda = \frac{\kappa}{C} \quad [\Omega^{-1} \cdot \text{m}^2 \cdot \text{equiv}^{-1}]$$

In the older literature, C is the concentration in equivalents per liter. The volume of the solution in cubic centimeters per equivalent is equal to $1000/C$, and $\Lambda = 1000 \kappa/C$, the units employed in Table 8.32 [$\Omega^{-1} \cdot \text{cm}^2 \cdot \text{equiv}^{-1}$]. The formula unit used in expressing the concentration must be specified; for example, NaCl , $\frac{1}{2}\text{K}_2\text{SO}_4$, $\frac{1}{3}\text{LaCl}_3$.

The equivalent conductivity of an electrolyte is the sum of contributions of the individual ions. At infinite dilution: $\Lambda^\circ = \lambda_c^\circ + \lambda_a^\circ$, where λ_c° and λ_a° are the ionic conductances of cations and anions, respectively, at infinite dilution (Table 8.35).

Ionic Mobility and Ionic Equivalent Conductivity

$$\lambda_c = Fu_c \quad \text{and} \quad \lambda_a = Fu_a \quad [\Omega^{-1} \cdot \text{m}^2 \cdot \text{equiv}^{-1}]$$

where F is the Faraday constant, and u_c , u_a are the ionic mobilities [$\text{m}^2 \cdot \text{s}^{-1} \cdot \text{V}^{-1}$].

$$\Lambda = \alpha F(u_c + u_a) = \alpha(\lambda_c + \lambda_a)$$

where α is the degree of electrolytic dissociation, Λ/Λ° . The electric mobility u of a species is the magnitude of the velocity in an electric field [$\text{m} \cdot \text{s}^{-1}$] divided by the magnitude of the strength of the electric field E [$\text{V} \cdot \text{m}^{-1}$].

Ostwald Dilution Law

$$K_d = \frac{\alpha^2 C}{1 - \alpha}$$

where K_d is the dissociation constant of the weak electrolyte. In general for an electrolyte which yields n ions:

$$K_d = \frac{C^{(n-1)}\Lambda^n}{\Lambda^\circ(n-1)(\Lambda^\circ - \Lambda)}$$

Transference Numbers or Hittorf Transport Numbers

$$T_c = \frac{\lambda_c}{\lambda_c + \lambda_a} \quad T_a = \frac{\lambda_a}{\lambda_c + \lambda_a} \quad T_c + T_a = 1$$

$$\frac{T_c}{T_a} = \frac{u_c}{u_a} = \frac{\lambda_c}{\lambda_a}$$

$$\lambda_c = T_c \Lambda \quad \lambda_a = T_a \Lambda$$